Lecture Notes

Year 1 Semester 2

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Table of Lectures

LC Classical Mechanics and Relativity 2

Tue 17 Feb 2026 11:00

Lecture 8 - Special Relativity I: Foundations and Time Dilation

1 Introduction

In this lecture:

- Recap from CMR1
- Einstein's two postulates
- Time dilation
- Galilean Transformations

Crucially, there are no special or absolute frames of reference - laws of motion are only sensible when we consider a frame of reference. The only slightly special frame is the one in which we are currently stationary.

Special relativity provides a theory of relative motion between inertial frames of reference. An inertial frame is one which is not accelerating relative to the other frames being considered.

2 Frames of Reference

We will generally consider two related frames of reference, denoted Σ and Σ'. Σ is our "stationary" frame, i.e the frame taken by an observer sat on earth. Σ' is our "moving frame", relative to the stationary observer and is moving with constant speed v.

Coordinates in the stationary frame are (x, y, z, t), while in the moving frame they add a prime, so are given by: (x', y', z', t').

We want to compare observations of position/experience of time/velocity in the moving frame with observations of the same categories in the stationary frame.

The frame in which an object is stationary is denoted its rest frame Σ_0.

3 Galilean Transformations

"Galilean Invariance": The laws of physics are invariant in all inertial (non-accelerating) frames.

"Galilean Transformation": Time is invariant and universal in all frames.

In Classical Mechanics, we treat time as invariant, but this stops being true in special relativity. We can transform between frames in such a manner that time is kept constant, this is a Galilean Transformation, but this is not true by default.

Consider our two reference frames Σ and Σ', where the latter moves with speed v relative to the former. Our coordinates in (x, y) become coordinates in (x', y').

An object moves at speed u' in the x-direction in reference frame Σ'. If we assume that time is invariant (as we're still currently doing classical physics without having introduced special relativity), then:

$$t = t'$$

And as the motion is entirely in the x-axis:

$$y = y'$$

After time t:

$$x = x' + \text{distance moved by } \Sigma' \text{ in time } t \text{ relative to } \Sigma$$

$$x = x' + vt$$

The object velocity is defined as:

$$u = \frac{dx}{dt} = \frac{d}{dt}(x' + vt)$$

$$= \frac{dx'}{dt} + v = u' + v$$

This holds if any only if time is the quantity we treat as being invariant. This agrees with our classical understanding.

3.1 Where does this break down?

Lets assume that the moving object is a photon, moving in Σ' with velocity c', hence, according to the previous derivation:

$$c = c' + v$$

Therefore observers in different frames will measure different values for the speed of light...oh no!

If we assume Galilean invariance, the laws of physics are the same in all reference frames, and yet the speed of light is included as a constant in many laws (i.e. electromagnetism). Therefore observers must measure the same speed of light in all reference frames.

This is a contradiction - we cannot assume that both time and the laws of physics are invariant.

3.2 Experimental Verification

The earth travels around the sun extremely quickly, and the sun is travelling even faster around the galactic centre. Therefore, the earth is moving in space.

If Galilean relativity is correct, we would measure the speed of light in one direction as $c + u$ and measure the speed of light in the other direction as $c - u$.

This was tested by the Michelson-Morley experiment, where incoming light was split by a half-silvered mirror. Half the light travels in one direction, and half the light is split off by 90°. They reflect off a pair of mirrors and are recombined in a splitter to be observed.

If the speed of light was different in different directions, the two beams would be out of phase and we would observe an interference pattern on combination. We would expect to see phase difference that varies with time, i.e. turning from destructive to constructive etc.

This was not observed and the interference pattern generated was constant. Therefore, the results showed no variation in the speed of light.[1]

4 The Solution - Einstein's Special Relativity

To fix this contradiction, Einstein came up with two postulates:

1. The laws of physics are the same in any inertial frame of reference.
   - This is the same as Galilean invariance and is easily believable.
2. The speed of light is constant in every frame of reference.
   - This was revolutionary and is much less intuitive.

The last postulate is difficult to understand intuitively, but makes everything work if we accept it as true!

[1]This is the same idea used in the LIGO experiment to discover gravitational waves, except this was used to show that the path length changed, and not the speed of light. If the path lengths changed, this was due to a gravitational wave (ripple in space time) propagating to the earth and interfering with the measurement.

4.1 Proving Time Dilation

Consider an observer on the earth in frame Σ. This observer watches two rockets both travelling side by side away from the earth in the x-direction with speed v. They are both travelling in x, with a constant y-difference y_0 between them.

Suppose the upper rocket A fires a laser beam directly at B (i.e. directly in the y-direction downwards).

Frame Σ: This is the rest frame of the earth (and the observer on earth) with coordinates (x, y).

Frame Σ': This is the rest frame of the rockets, with coordinates (x', y').

In the rockets rest frame Σ', the rockets are stationary and the time taken for a laser pulse to travel between them is:

$$t_0 = \frac{y_0}{c}$$

Or equivalently:

$$y_0 = ct_0$$

In the earth's rest frame Σ, the observer on the earth doesn't just see the light travelling in the y-direction. It also sees the light moving in the x-direction, as the whole Σ' frame containing the rockets relativistically move away.

Effectively, we have:

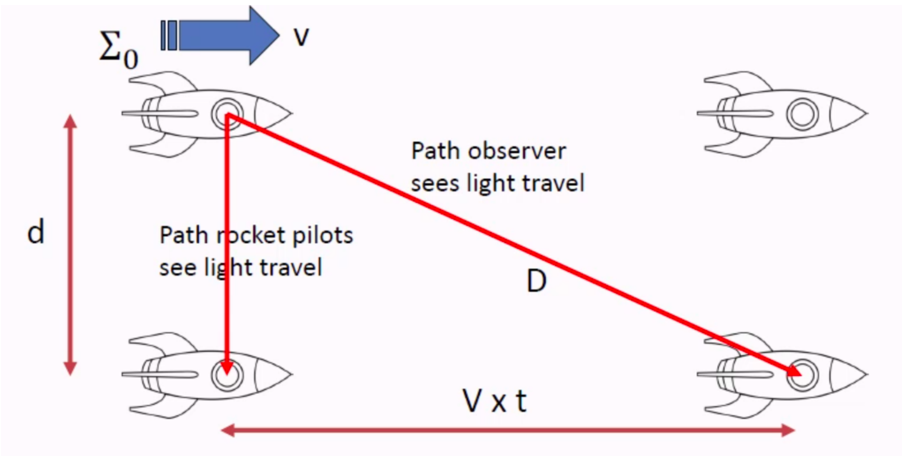


Figure 1.1

B moves distance x horizontally while it waits for the laser to hit it, and the length of the path of the laser beam in time t is given by D. If the time taken for light to reach B in Σ is t, then:

$$D = ct$$

As the speed of light is constant in all frames. As the rockets are moving away:

$$x = vt$$

This gives us a Pythagorean triangle, where:

$$D^2 = x^2 + y^2 = x^2 + y_0^2$$

And substituting in:

$$c^2t^2 = v^2t^2 + c^2t_0^2$$
$$t^2(c^2 - v^2) = t_0^2c^2$$
$$t = t_0\sqrt{\frac{c^2}{c^2 - v^2}}$$

Dividing through by c_2:

$$t = t_0\sqrt{\frac{1}{1-\frac{v^2}{c^2}}}$$

Or:

$$t = t_0\gamma, \qquad \gamma = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}}$$

$\gamma > 1$ for all object speeds that don't exceed the speed of light (which is required), so the time observed in the frame Σ is longer than the "proper time" observed by the rocket.

5 The Twin Paradox

Consider two twins, an astronaut and a physics professor. The astronaut goes on a long trip close to the speed of light and reunites with his brother. Which brother is older?

Special relativity says that because there is no absolute frame of reference, the brother travelling away also sees the same problem in reverse, i.e. he sees the physics professor brother travelling away in the opposite direction at equal and opposite speed, i.e. they are the same age.

In practice, the astronaut is non-inertial and special rel isn't sufficient, so we can't solve it directly.

Thu 19 Feb 2026 15:00

Lecture 9 - Special Relativity II: Length Contraction

1 Proving Length Contraction

Consider a horizontal laser cavity. We fire a laser from the left of the cavity (point A) to the right of the cavity (point B). The laser bounces off a mirror at point B and bounces back.

The cavity has length L_0 as measured in the rest frame Σ_0. In this frame:

$$t_0 - \frac{2L_0}{c}, \qquad \text{Where } t_0 \text{ is the time taken for the A->B->A journey in } \Sigma_0$$

Now suppose the cavity moves at speed v relative to a stationary observer in a different frame (Σ).

The observer sees the cavity as having length L, how long does the A->B->A journey take as observed by a stationary observer in Σ.

$$t = t_{A\to B} + t_{B\to A}$$

The cavity is moving away from the observer, so the laser beam doesn't just travel L when going from $A \to B$. It must travel L plus the distance the whole cavity has moved away in this time (as B is moving away from the pulse). This extra distance is $vt_{A\to B}$, hence the distanced travelled when going from A to B is $L + vt_{A\to B}$.

As c is constant in all frames, we can say:

$$\begin{aligned} c &= \frac{L + vt_{A\to B}}{t_{A\to B}} \\ t_{A\to B}(c - v) &= L \\ t_{A\to B} &= \frac{L}{c - v} \end{aligned}$$

And now on the return journey from $B \to A$, as it travels back towards A, the cavity is moving in the same direction as the light, so A "catches up" to the pulse and results in a smaller required distance to be travelled, $L - vt_{B\to A}$

$$c = \frac{L - vt_{B\to A}}{t_{B\to A}} \implies t_{B\to A} = \frac{L}{c + v}$$

And returning to the total time:

$$\begin{aligned} t &= t_{A\to B} + t_{B\to A} \\ t &= \frac{L}{c - v} + \frac{L}{c + v} \\ &= L\left(\frac{1}{c - v} + \frac{1}{c + v}\right) \\ &= L\left(\frac{c + v + c - v}{(c - v)(c + v)}\right) \\ &= L\frac{2c}{c^2 - v^2} \\ &= \frac{2L}{c}\frac{c^2}{c^2 - v^2} \\ &= \frac{2L}{c}\frac{1}{1 - \frac{v^2}{c^2}} \end{aligned}$$

$$= \frac{2L}{c}\gamma^2$$

We now apply time dilation, which says that $t = \gamma t_0$, hence:

$$\gamma t_0 = \frac{2L}{c}\gamma^2$$

$$t_0 = \frac{2L}{c}\gamma$$

From the rest frame, we know that $t_0 = 2L_0/c$, so:

$$\frac{2L_0}{c} = \frac{2L\gamma}{c}$$

$$L_0 = L\gamma$$

As $\gamma > 1$, $\forall(u < c)$, the proper length measured from the rest frame will always be greater than the length measured by an observer, hence length is contracted for a moving object.

2 Example

In a particle accelerator, protons are accelerated to a measly $0.9c$. These protons pass through a tunnel of length 2km as viewed from the a laboratory rest frame at the accelerator.

Recall that:

$$t = \gamma t_0$$

$$L = \frac{L_0}{\gamma}$$

$$\gamma = \frac{1}{\sqrt{1-\beta^2}}, \qquad \text{where: } \beta = u/c$$

2.1 How long would the journey take, according to the rest frame?

Viewed in the lab frame, and $v = d/t$ so:

$$t = \frac{2 \times 10^3\text{m}}{0.9 \times 3 \times 10^8} = 7.4 \times 10^{-6}\text{s} = 7.4\mu\text{s}$$

No relativity needed!

2.2 How long would the journey take, according to the proton's frame?

We do need to consider relativity here, and use time dilation:

$$t = \gamma t_0$$

$$\gamma = \frac{1}{\sqrt{1-0.9^2}} = 2.3$$

$$t = \gamma t_0 \implies t_0 = \frac{t}{\gamma} = \frac{7.4\mu\text{s}}{2.3} = 3.2\mu\text{s}$$

2.3 How long is the tunnel, according to the proton's frame?

The rest frame now corresponds to the tunnel. It has proper length in its rest frame of 2km.

As viewed by the protons in their rest frame, it is the tunnel that is moving and is rushing towards them at $0.9c$. This is therefore a length contraction problem.

$$L = \frac{L_0}{\gamma} = \frac{2\text{km}}{2.3} = 870\text{m}$$

2.4 Checking Values

To check, we can reuse $v = d/t$ but this time in the proton rest frame.

$$d = 870\text{m}, \quad t = 3.2 \times 10^{-6}\text{s}$$

Hence

$$v = \frac{870\text{m}}{3.2 \times 10^{-6}\text{s}} = 2.7 \times 10^{8}\text{ms}^{-1} = 0.9c \text{ as required!}$$

3 Example II: Cosmic Rays

The highest energy cosmic rays are protons with massive energies $E \sim 10^{20}$eV, compared to the LHC with $E \sim 10^{12}$eV

This corresponds to a $\gamma = 10^{11}$, and our galaxy is $\sim 10^{20}$m across. [1]

As viewed on earth, these cosmic protons travel at $v \approx c$, hence,

$$t \approx \frac{d}{v} \approx \frac{10^{20}}{3 \times 10^{8}} = 3 \times 10^{11}\text{s} \approx 10^{4}\text{years}$$

As viewed by the protons:

$$t_0 = \frac{t}{\gamma} = \frac{3 \times 10^{11}}{10^{11}} \approx 3\text{s}$$

Which is starkly different!

4 Lorentz Transformations

In general, transforming between coordinates in two different frames can get extremely messy, more than can easily be handed in simple applications of length contraction or time dilation.

The Lorentz Transformations provide a general set of coordinate transformations between Σ and Σ', i.e. $(x, y, z, t) \to (x', y', z', t')$.

These transformations must be:

1. Symmetric about a change in sign of u, i.e. the transformation from $\Sigma \to \Sigma'$ with u and the transformation from $\Sigma' \to \Sigma$ with $-u$ must be the same.
2. They must be linear as otherwise a constant speed in one would not be constant in the other (and therefore not inertial frames).
3. If relative velocity is only in the x direction, the transformations must be independent of y and z.

These transformations are (assuming motion only in x):

$$x = \gamma(x' + vt')$$
$$y = y'$$
$$z = z'$$
$$t = \gamma\left(t' + \frac{x'v}{c^2}\right)$$

And:

$$x' = \gamma(x - vt)$$
$$y' = y$$
$$z' = z$$
$$t' = \gamma\left(t - \frac{xv}{c^2}\right)$$

We do not need to know the derivations, but do need to know the results.

[1] $\gamma = E/mc^2$ - will be expanded on in later lectures.

Tue 24 Feb 2026 11:00

Lecture 10 - Special Relativity III: Lorentz Transformations

Recap of the first two relativity lectures:

- Recap from Semester 1 Special Rel:
  - Background
  - Einstein's postulates.
  - Time dilation.
  - Lorentz contraction.
- Galilean transformations and when they fail.
- Lorentz transformations for space and time.

1 Galilean vs. Lorentz Transformations

In a standard setup with two frames, Σ and Σ', where Σ' moves with speed v relative to Σ.

In Σ, we have coordinates (x, y, t), in Σ' we have coordinates $(x', y',')$.

In a Galilean transformations, we kept time invariant across both frames, so $t = t'$. This lead to $x' = x - vt \implies u' = u - v$ but raised a contradiction with the speed of light as observers in different frames would take two measurements of the speed of light.

In the second lecture, we consider the speed of light as being invariant across all frames instead of time to resolve the contradiction. This gave us the Lorentz Transformations:

$$x' = \gamma(x - vt)$$

$$t' = \gamma\left(t = \frac{xv}{c^2}\right)$$

Where $y = y'$, $z =' $.

As we cannot use $u' = u - v$ due to the contradiction with the speed of light, we in this lecture will consider:

- Lorentz transformations for velocity.
- Space-time, Lorentz invariance, causality.

2 Lorentz Transformation for Velocity

We can revisit the question from the first lecture, but in the Lorentz transformation perspective rather than a Galilean perspective.

Again, we have our two standard frames Σ, Σ'. An object is moving with speed u' in frame Σ'. We want to find an expression for the velocity of the object as viewed from Σ, given by u. The prime frame moves at speed v wrt the rest frame and the objects motion is entirely along the x acis.

By definition, we have:

$$u = \frac{dx}{dt} = \frac{dx}{dt'}\frac{dt'}{dt} = \frac{\left(\frac{dx}{dt'}\right)}{\left(\frac{dt'}{dt}\right)} \tag{1}$$

And:

$$\frac{dx}{dt'} = \gamma\left(\frac{dx'}{dt'} + v\right) = \gamma(y' + v) \tag{2}$$

$$\frac{dt}{dt'} = \gamma\left(1 + \frac{v}{c^2}\frac{dx'}{dt'}\right) = \gamma\left(1 + \frac{u'v}{c^2}\right) \tag{3}$$

And substituting (3) and (2) into (1):

$$u = \frac{\gamma(u' + v)}{\gamma\left(1 + \frac{u'v}{c^2}\right)} \implies \boxed{u = \frac{u' + v}{1 + \frac{u'v}{c^2}}}$$

To get the reverse transformation, we invoke forward-backward symmetry, i.e. swap $x \to x'$, $t \to t'$ etc, where $v \to -v'$:

$$u' = \frac{u - v}{1 - \frac{uv}{c^2}}$$

2.1 Limiting Cases

Firstly, we'll look at the non-relativistic case, i.e. where $u, v \ll c$:

$$u = \frac{u' + v}{1 + \frac{u'v}{c^2}} = \frac{u' + v}{1 + \text{small}} \approx u' + v$$

This is the classical Galilean approach, which is what we'd expect for non-relativistic behaviour.

What about an ultra-relativistic limit where $u, u' \to c$ with $v \ll c$. Now we have:

$$u = \frac{u' + \text{small}}{1 + \text{small}} \approx u'$$

As the speed of light is invariant, speeds close to the speed of light do not change between reference frames.

2.2 Example

An observer on a rocket sees a second rocket moving away from them at $v_1 = 0.8c$. Another observer on a planet sees the first rocket moving at $0.7c$. Assuming all motion is in the same direction, how fast does the planetary observer see the second rocket?

- From Σ' (the first rockets rest frame), the second rocket moves with $0.8c$.
- From Σ (the planetary rest frame), the first rocket moves with $0.7c$, and the second rocket moves with some unknown speed u.

We apply the formula:

$$u = \frac{u' + v}{1 + \frac{u'v}{c^2}}$$

In Σ', $u' = 0.8c$. The two frames move with relative velocity equal to the change in speed of the first rocket from $\Sigma \to \Sigma'$: $v = 0.7c$.

In frame Σ we are tying to do:

$$u = \frac{0.8c + 0.7c}{1 + \frac{0.8c \times 0.7c}{c^2}}$$

$$= \frac{1.5c}{1.56} = 0.96c$$

3 Space-Time and Lorentz Invariants

From length contraction and time dilation, we can see that in special relativity, intervals in time and space are not fixed but vary from frame-to-frame.

This means that the order of events may even be different for different observers.

There are some quantities however which are invariant across frames, such as the speed of light. One of these is space-time.

3.1 Space-Time Invariance

As usual, we have Σ and Σ' with relative speed v. The axes of these frames coincide at $t = 0, x = 0$ and $t' = 0, x' = 0$.

At $t = t' = 0$, there is a flash of light at the origin which propagates isotropically.

In Σ, the light forms a spherical shell of radius Δr at time Δt, and in Σ' it reaches a radial distance $\Delta r'$ at time $\Delta t'$.

After time ΔT in Σ we have $\Delta r = c\delta T$. Similarly in Σ' we have $\Delta r' = c\Delta t'$.

Squaring these we have:

$$(\Delta r)^2 = c^2\Delta t^2 = \Delta x^2 + \Delta y^2 + \Delta z^2$$
$$(\Delta r')^2 = c^2\Delta t'^2 = \Delta x'^2 + \Delta y'^2 + \Delta z'^2$$

Hence:

$$c^2\Delta t - \Delta x^2 - \Delta y^2 - \Delta z^2 = 0 = c^2\Delta t'^2 - \Delta x'^2 - \Delta y'^2 - \Delta z'^2$$

This means that, **for light**:

$$c^2\Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2 = 0$$

We call this quantity Δs^2:

What about for objects moving at a speed less than the speed of light? We want to apply the Lorentz transformations to $\Delta s'2$ (again treating the "boost" as being in x):

$$\begin{aligned}
\Delta s'^2 &= c'^2\Delta t'^2 - \Delta x'^2 - \Delta y'^2 - \Delta z'^2 \\
&= c^2\gamma^2\left(\Delta t - \frac{v}{c^2}\Delta x\right)^2 - \gamma^2\left(\Delta x - v\Delta t\right)^2 - \Delta y^2 - \Delta z^2 \\
&= c^2\gamma^2\left(\Delta t^2 + \frac{v^2}{c^4}\Delta x^2 - 2\frac{v}{c^2}\Delta t\Delta x\right) - \gamma^2\left(\Delta x^2 + v^2\Delta t^2 - 2v\Delta x\Delta t\right) - \Delta y^2 - \Delta z^2 \\
&= \Delta t^2\left(c^2\gamma^2 - v^2\gamma^2\right) + \Delta x^2\left(\frac{v^2\gamma^2}{c^2} - \gamma^2\right) + \Delta t\Delta x\left(-2v\gamma^2 + 2v\gamma^2\right) - \Delta y^2 - \Delta z^2 \\
&= \Delta t^2\gamma^2\left(c^2 - v^2\right) + \Delta x^2\gamma^2\left(\frac{v^2}{c^2} - 1\right) - \Delta y^2 - \Delta z^2 \\
&= \Delta t^2\frac{c^2 - v^2}{1 - \frac{v^2}{c^2}} + \Delta x^2\frac{(\frac{v^2}{c^2}) - 1}{1 - \frac{v^2}{c^2}} - \Delta y^2 - \Delta z^2 \\
\Delta s'^2 &= \boxed{c^2\Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2 = \Delta s^2}
\end{aligned}$$

Hence Δs^2 is invariant under the Lorentz transformations regardless of speed.

Tue 03 Mar 2026 11:00

Lecture 12 - Special Relativity IV: More Space-Time

Recap from last lecture:

- Lorentz Transformations for space and time.
- Lorentz Transformations for velocity/velocity addition.
- A Lorentz Invariant: The space-time interval is invariant across frames of reference:

$$\delta S^2 = c^2 \Delta t^2 - \Delta r^2$$

In this lecture:

- Minkowski space and causality.
- Relativistic energy and momentum.

1 Space-Time Invariance

Last lecture we showed that:

$$c^2 \Delta t^2 - \Delta r^2 = c^2 \Delta t'^2 - \Delta r'^2 = \quad (= 0 \text{ for light})$$

And we define the space-time interval Δs as $\Delta S^2 = c^2 \Delta t^2 - r^2$. There is some relationship between space and time that keeps this quantity constant across frames. What does this quantity actually *mean*, however?

In a 3D set of axes, we have:

$$\Delta r^2 = \Delta x^2 + \Delta y^2 + \Delta z^2$$

And we have defined:

$$\Delta s^2 = c^2 \Delta t^2 - \Delta r^2 = c^2 \Delta t^2 - \left(\Delta x^2 + \Delta y^2 + \Delta z^2\right)$$

This looks almost like a 4D space (where one axis is time), but with some strange negative signs appearing? We don't have the equipment yet to properly understand the negative signs, but can be in future courses.

2 Minkowski Space (Space-Time Diagrams)

In 2D, we consider one spatial coordinate and one time coordinate, where the x axis represents our spatial coordinate and the vertical axis is given by ct.

Consider some kind of light-emitting event happening at the origin. This light travels at a fixed speed, so we end up with the path of light at 45 degrees on this space-time diagram:

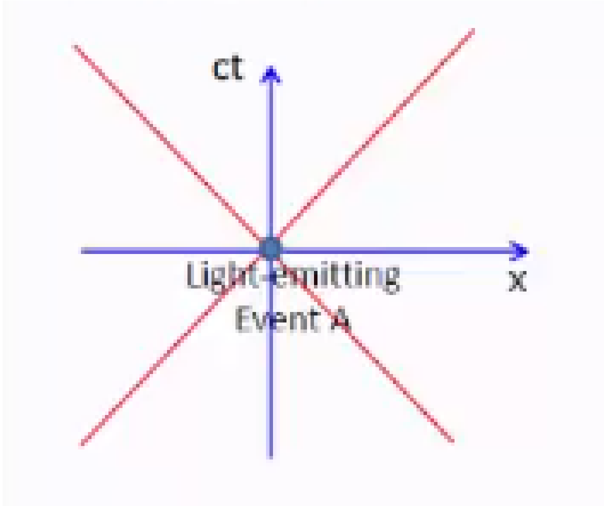


Figure 4.1

These are given by the lines $c^2t^2 = \pm x^2$ and are called “light cones”, as they form cones in higher dimensions. The positive velocity axis is called the “future cone”, the path taken by light emerging from event A. The negative velocity axis is called the “past cone”, and represents the path taken by light emerging at A.

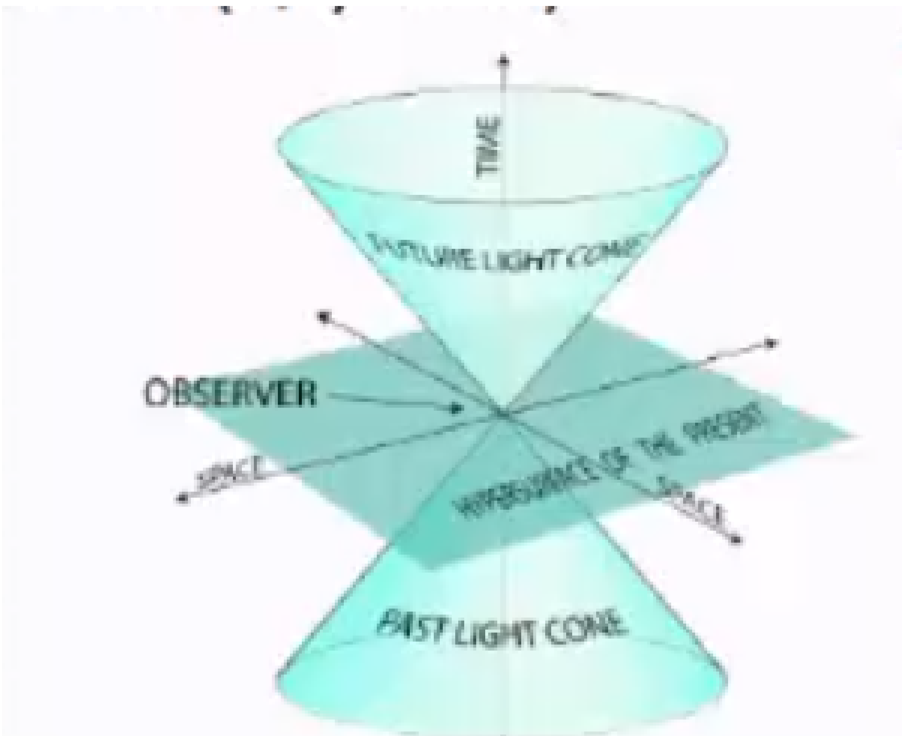


Figure 4.2: Light cones in a 3D space (two spatial, one velocity)

Inside the future cone, events may be caused by the event happening at A. Things travelling at a velocity $< c$ can reach them.

The region inside the past cone gives regions in spacetime where events could have caused A.

Events outside the two cones cannot influence, nor can they be influenced by A. They cannot be causally connected in any way to the event at A.

We now have two events. A is defined as before, and a new event B separated from A by distance $\Delta \underline{r}$ and time Δt.

What is the condition that B lies within the light cone given by A? We require:

$$|\Delta \underline{r}| \leq c|\Delta t|$$

Or in invariant form:

$$\Delta S^2 = c^2\Delta t^2 - \Delta r^2 \geq 0$$

We call ΔS the space-time interval, or invariant distance.

There are three possible cases:

- $\Delta S^2 > 0$, "time-like intervals". This is where $|\Delta \underline{r}| \leq c|\Delta t|$ and B is inside A-s light cone. $\Delta S \in \mathbb{R}$. Here, a signal can pass between A and B and the two events may be causally connected (where A causes B).
- $\Delta S^2 = 0$. Here, $|\Delta \underline{r}| = c|\Delta t|$. This is where B sits on A's light cone, and A/B may be causally connected. A signal can only travel between the two events if the signal is at velocity c. This is called a "light-like interval"
- $\Delta S^2 < 0$, "space-like intervals". This is where $|\Delta \underline{r}| > c|\Delta t|$. $\Delta S = \sqrt{c^2\Delta t^2 - \Delta r^2} \in i\mathbb{R}$. No signal can pass between A and B, not even light. No causal connection is possible.

3 Causality and Time Order

It is possible for observers in two different frames to disagree on the order in which events have taken place. If we have two events with t_1, t_2 in Σ and t'_1, t'_2 in Σ', there are possible cases where:

- $t_2 > t_1$.
- $t'_1 > t'_2$.

This may seem uncomfortable - does this cause a breakdown in causality?

Suppose we have a frame Σ with two events, (x_1, t_1), (x_2, t_2), such that $\Delta t = t_2 - t_1 > 0$.

Frame Σ' moves with speed v relative to Σ and in Σ' the same two events have coordinates (x'_1, t'_1) and (x'_2, t'_2). For the observers to disagree on causality, we have $\Delta t' t'_2 - t'_1 < 0$. We begin by applying the Lorentz Transformation to $\Delta t'$:

$$\Delta t' = \gamma\left(t_2 - \frac{v}{c^2}x_2\right) - \gamma\left(t_1 - \frac{v}{c^2}x_1\right) < 0$$

$$\implies \gamma(t_2 - t_1) + \frac{\gamma v}{c^2}(-x_2 + x_1) < 0$$

$$\Delta t - \frac{v}{c^2}\Delta x < 0$$

I.e. $\Delta t < \frac{v}{c^2}\Delta x$ or $c\Delta t < \frac{v}{c}\Delta x$. We know that $v/c < 1$, hence $c\Delta t < \Delta x$. This means that:

$$\Delta S = \sqrt{c^2\Delta t^2 - \Delta x^2}$$

Which is the square root of a negative number, hence this gives a space-like interval. This means that while we can define two points in space-time where two observers will disagree about their order, the space-time invariant is imaginary hence these events cannot be causally linked. This means that observers disagreeing on event order does not break causality.

4 Simultaneity

Consider a relativistic ladder passing through a stationary garage. The ladder's proper length is longer than the proper length of the garage, so the ladder could not fit inside the garage if both were stationary.

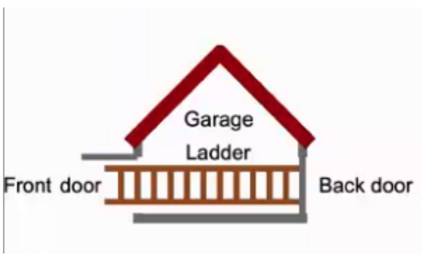


Figure 4.3

However, viewed from the rest frame of the garage, the ladder is length contracted. From this frame, the ladder can momentarily (before smashing into the back door) fit entirely in the garage and it would even be possible to briefly close the front door.

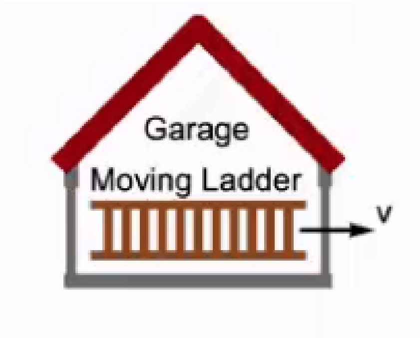


Figure 4.4

From the ladder's rest frame however, the garage is length contracted and not the ladder. Therefore, from this frame, there's no possible way to fit the ladder into the garage.

How can it simultaneously be possible and impossible depending on the rest frame being considered?

4.1 Problem

Let the ladder have proper length $l_0^L = 20\text{m}$ and the garage have proper length $l_0^G = 10\text{m}$. The ladder moves at 0.9c wrt the garage.

$$\gamma = \frac{1}{\sqrt{1-\beta^2}} = \frac{1}{\sqrt{1-0.9^2}} \approx 2.29$$

In the garage's rest frame, the ladder length is $L^L = \frac{20}{2.29} = 8.7\text{m}$. So, from the garage's POV, the ladder fits!

We now consider the length of the garage from the ladder's rest frame: $L^G = \frac{10}{2.29} = 4.4\text{m}$, so from the ladder's POV, the ladder cannot fit! This creates a paradox.

4.2 Solution

We can resolve this problem because the concept of simultaneity is frame dependant. There are two things that happened here:

- A. The front of the ladder reached the back garage door (the first figure)
- B. The back of the ladder reaches the front garage door.

We've been assuming that these events occur at exactly the same time, simultaneously, but this isn't the case.

The time taken between these events has to obey a Lorentz transformation. If they are the same in some frame (i.e. the garage, for the ladder to fit perfectly inside), then $\Delta t_{AB} = 0$. In the rest frame of the ladder:

$$\Delta t'_{AB} = \gamma\left(\Delta t_{AB} - \frac{v}{c^2}\Delta x_{AB}\right) = \frac{\gamma v}{c^2}\Delta x_{AB} \neq 0$$

Hence, if the garage perceives the events as being simultaneous, they cannot be observed as being simultaneous in the ladder's frame.

Tue 10 Mar 2026 11:00

Lecture 14 - Special Relativity V: Relativistic Mechanics

Recap of last lecture:

- Minkowski Space (space-time)
- Time-like and space-like intervals.
- Ensuring that this new model does not violate causality.
- Simultaneity.

1 Relativistic Momentum

We generally use:

$$\boldsymbol{p} = m\boldsymbol{v}$$

But this is just an approximation for $v \ll c$. This would:

- Allow $v > c$ for an object with sufficient momentum.
- Not allow massless objects like photons to have any momentum, but we know this is true.
- Is not obviously conserved.

Instead, we (proof non-examinable) use:

$$p = \gamma m v$$

This is the "true" expression for momentum, but for small speeds it collapses nicely to $p = mv$ as $\gamma \to 1$ so we don't notice in standard mechanics.

This doesn't resolve the issue with massless photons yet, but does fix the first point!

2 Relativistic Kinetic and Total Energy

By definition, we have:

$$F = \frac{dp}{dt}$$

Now, momentum has an extra factor of γ. Suppose we have a massive object which is initially stationary. We apply a force to that object, in order to give it a kinetic energy. This KE is denoted K_A at the point where it reaches speed v_A under the effect of the force. We assume that all work done is entirely transferred to kinetic energy stores.

$$K_A = \int_0^A F\,dx = \int_0^A \frac{dp}{dt}\,dx$$

And we can use the chain rule to say:

$$dx = \frac{dx}{dt}dt \implies K_A = \int_0^A \frac{dp}{dt}\frac{dx}{dt}\,dt \tag{1}$$

We can also say that:

$$\frac{d}{dt}(pv) = p\frac{dv}{dt} + v\frac{dp}{dt}$$

So in eqn. 1 we have:

$$K_A = \int_0^A \left[\frac{d}{dt}(pv)\right] - p\frac{dv}{dt}\,dt$$

Tue 24 Mar 2026 11:00

Lecture IDK - Special Relativity VII: Kinematics Examples, Doppler and Natural Units

1 Example I

As viewed in a lab frame, a cosmic ray proton of energy E_p hits a stationary proton. What is the minimim threshold energy of E_p that allows the reaction:

$$p + p \to p + p + \pi$$

To take place, given $m_p = 938\text{MeV/c}^2$ and $m_\pi = 104\text{MeV/c}^2$?

Let the cosmic ray proton be proton one with E_p, p_p and the stationary proton be proton 2 with m_p.

We can consider energy and momentum in the lab frame, initially (before the collision):

$$(\Sigma E)^{\text{lab}}_{\text{initial}} = E_p + M_p c^2 \qquad \text{and} \qquad (\Sigma p)^{\text{lab}}_{\text{initial}} = p_p \tag{1}$$

For a threshold energy, the produced particles have no relative motion (i.e. at rest in the centre of mass frame). This means that all available relativistic energy goes into particle production with no excess.

Hence in the centre of mass frame finally, we have two stationary protons and one stationary pion. This is the lowest energy state we can have in the centre of mass frame, with all these particles still existing.

$$(\Sigma E)^{\text{CMS}}_{\text{final}} = (2m_p + m_\pi)c^2 \qquad \text{and} \qquad (\Sigma p)^{\text{CMS}}_{\text{final}} = 0 \tag{2}$$

We can then apply the energy momentum invariant:

$$s = \left[(\Sigma E)^2 - (\Sigma p)^2\right]^{\text{lab}}_{\text{final}} = \left[(\Sigma E)^2 - (\Sigma p)^2\right]^{\text{final}}_{\text{CMS}} \tag{3}$$

Inserting 1, 2 into 3:

$$\left(E_p + m_p c^2\right)^2 - p_p^2 c^2 = \left(2m_p + m_\pi\right)^2 c^4$$
$$\implies E_p^2 + m_p^2 c^4 + m_p c^2 E_p - p_p^2 c^2 = \left(2m_p + m_\pi\right)^2 c^4$$

For a single particle, $E_p^2 - p_p^2 c^2 = m_p^2 c^4$, hence:

$$2m_p^2 c^4 + 2m_p c^2 E_p = \left(2m_p + m_\pi\right)^2 c^4$$

$$E_p = \frac{\left(2m_p + m_\pi\right)^2 c^4 - 2m_p^2 c^4}{2m_p c^2}$$

And we have known masses m_p, m_π hence (noting that the masses are in units energy per speed of light squared, so the cs cancel out):

$$E_p = \frac{(2 \times 938 + 104)^2 - 2 \times 938^2}{2 \times 938} = 1228\text{MeV}$$

2 Example II

In an LHC experiment, a B_0 meson ($b\bar{d}$) with mass 5GeV/c^2 is produced in a lab frame momentum of 50GeV/c^2. The meson has proper decay time 2×10^{-12}s. How far does the experiment observe the meson fly before decay?

3 Relativistic Longitudinal Doppler Effect

We want to build an analogue for the sonic doppler effect for light. This is especially important in astronomy, i.e. tracking the movement of cosmic spectral lines to determine speeds of distant sources.

We will only considerer the longitudinal effect, i.e. the movement of bodies directly towards or away from an observer in their line of sight. We will not (yet) consider transverse effects in this course.

This comes from two components:

- The standard doppler effect (as for sonic waves).
- Relativistic time dilation.

LC Electric Circuits

Fri 13 Feb 2026 11:00

Lecture 3

Fri 20 Feb 2026 11:00

Lecture 5

LC Electromagnetism I

Mon 19 Jan 2026 11:00

Lecture 1 - Electric Fields I: EM1 Introduction

1 Course Intro

Course Materials:

- Background material and derivations etc on PowerPoint.
- Worked examples etc are handwritten on visualiser, these are the bits we really need to know.

Why is EM important?

- Foundations of modern technology and the modern world.
- What gives elements their properties.
- Responsible for life itself.
- Everyday materials are held together by EM forces.
- Optics can only be understood through EM theory.

The course aim is to lay down the foundations, eventually leading us to Maxell's Laws.

1.1 Maxwell's Laws

Maxwell's four equations are:

$$\nabla \cdot \boldsymbol{E} = \frac{\rho}{\epsilon_0}$$
$$\nabla \wedge \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t}$$
$$\nabla \cdot \boldsymbol{B} = 0$$
$$\nabla \wedge \boldsymbol{B} = \mu_0 \left(\boldsymbol{J} + \epsilon_0 \frac{\partial \boldsymbol{E}}{\partial t} \right)$$

Where:

$$\nabla = \boldsymbol{i}\frac{\partial}{\partial x} + \boldsymbol{j}\frac{\partial}{\partial y} + \boldsymbol{k}\frac{\partial}{\partial z}$$

Together, these show that the electric and magnetic fields are related and are two aspects of a single force, the electromagnetic force. We don't have to properly understand them yet, but cannot learn them in EMII unless we fundamentally understand E and B fields from this module.

1.2 Course Structure

Part I: Electric Fields

- Charge and Coulomb's Law.
- The electric field.
- Gauss' Law.
- Capacitors.

Part II: Magnetic Fields

- Magnetic Fields

- Charged Particles in B-Fields
- Electromagnetic Induction.
- Magnetic Dipoles

In this lecture:

- Introduction to EM.
- Electric charge.
- Force between charges.
- The concept of the Electric Field (E-Field).

2 Electric Charge

First attributed to Thales circa. 624 - 546 BC. Experiments by Franklin and Coulomb expanded and showed that there was two types of charge, which they called positive and negative.

The "positive electricity" came from rubbing a glass rod with silk, and the negative from rubbing an ebonite (early plastic) rod with fur. They found that like charges repel and opposite charges attract.

We know that the elementary unit charge is the magnitude of charge of an electron/proton and everything else is a multiple of this [1]:

$$e = 1.6 \times 10^{-19} C$$

This has units of the Coulomb.

2.1 Charge Conservation

Electrons and protons are both stable (protons decay with a life greater than 10^{31} years). This means that the total charge of an isolated system is constant and can be conserved.

They have the same magnitude of charge, exactly:

$$|q_p| = |q_e| = e$$

2.2 Electrostatic Force

Like charges repeal and opposite charges attract, along the line of action given by a line drawn between the two charges. The force is proportional to the product of charges so:

$$F \propto q_1 q_2$$

Here, a negative force means attraction and a positive force means repulsion. Newton called this "force at a distance". Like gravity, two charges will exert a force on each other at a distance without any contact.

There must, therefore, be something between them that mediates this force. Later physics gives this as "virtual particles" which isn't a Y1 topic, so classically we say that this medium is the Electric Field.

2.3 Electric Field

A charge produces a field around it. Another charge also interacts with this field, and this interaction is what causes a force:

$$\underline{F} = \underline{E} q$$

Where F is the force exerted on a test charge of charge q by a charge Q producing a field E. The magnitude of the electric field has units NC^{-1} (force per units charge).

$$|\underline{E}| \propto Q \qquad |\underline{F}| \propto Qq$$

[1]While quarks have fractional charge, we don't get free quarks

Consider a point charge with a spherical electric field spreading out around it. As the distance from the charge increases, the surface area increases as $4\pi r^2$. Therefore the magnitude of the electric field must decrease with $4\pi r^2$

Therefore:

$$|\underline{E}| \propto \frac{Q}{4\pi r^2}$$

We need a (inverse) constant of proportionality. This depends on the medium, but for a vacuum we call it the permittivity of free space ϵ_0:

$$\epsilon_0 = 8.854 \times 10^{-12} C^2 m^{-2} N^{-1}$$

Hence:

$$\boxed{E = |\underline{E}| = \frac{Q}{4\pi\epsilon_0 r^2}}$$

2.4 Direction of the E-Field

Force is a vector, so the E-field must be too. Consider an E-field from a charge Q at distance r. We give E components E_r, E_θ, E_ϕ, where, since it's a sphere ϕ and θ represent the unit vectors in the two possible tangential directions.

If there was a component in E_θ this would be clockwise from one perspective, but anticlockwise from another (walking behind it). This is not possible, as the field must behave in the same manner from all viewpoints. Therefore:

$$E_\theta = E_\phi = 0$$

$$\underline{E} = \frac{Q}{4\pi\epsilon_0 r^2}\hat{t}$$

2.5 Force between two charges

Consider two charges q_1 and q_2. The force on q_2 due to the e-field from q_1 is:

$$\underline{F_1} = \underline{E_1} q_2$$

This is equal to the force on q_1 due to the e-field from q_2, given by:

$$\underline{F_2} = \underline{E_2} q_1$$

So the force between two charges is:

$$\underline{F} = \frac{q_1 q_2}{4\pi\epsilon_0 r^2}\hat{r}_{12}$$

2.6 Force between many charges

If we have more than two positive charges, we use the "principle of superposition". Effectively, you consider each pair of charges at the same time and vector sum of the forces together. I.e. if we have three points and we care about the net force on one, we take the vector sum of the two vectors from that point to the others.

In general:

$$\begin{aligned}\underline{F} &= \underline{F_1} + \underline{F_2} + \underline{F_3} + \cdots \\ &= q\sum_i \frac{q_i}{4\pi\epsilon_0 r^2}\hat{r}_j\end{aligned}$$

Where r_j is the distance between q_i and q, with unit vector $\hat{r}_j$ between them.

Since $\underline{F} = q\underline{E}$, the electric field at a test charge q must be:

$$\underline{E} = \sum_i \frac{q_i}{4\pi\epsilon_0 r_i^2}\hat{r}_j$$

Thu 22 Jan 2026 11:00

Lecture 2 - Electric Fields II: Field Lines and Continuous Charges

1 Field Lines

To visualise a field, we draw field lines. These are drawn so that the tangent of these lines at any point in spaces gives the direction of $\underline{E}$ at that point.

The lines point radially outwards from a positive charge, and radially towards a negative charge. The density of these lines is proportional to the magnitude of $\underline{E}$, but the absolute number of lines doesn't convey much.

Field lines must start at a positive charge and end on a negative charge. They are continuous and can never cross each other, they only start/stop at charges or proceed to infinity.

2 Continuous Charge Distributions

What do we do if we have a charge which isn't a point particle?

- We break the object up into a distribution of infinitesimal point charges.
- We treat each piece as a point charge and add up the separate $\underline{E}$ fields across an integral.
- This gets messy for anything non-trivial, resulting in a complex 3D integral often requiring computational techniques.

Charges can be distributed across an object in a number of ways:

- Charge per unit length across a line, given by λ:

$$q = \int \lambda dl$$

- Charge per unit area across a surface area, given by σ:

$$q = \int \sigma dA$$

- Charge per unit volume, given by ρ:

$$q = \int \rho dV$$

3 Example: E-Field from a Line of Charge

Consider an E-field from a line of length $2L$, measured at a point along the midpoint's perpendicular. This point sits x length units along the perpendicular.

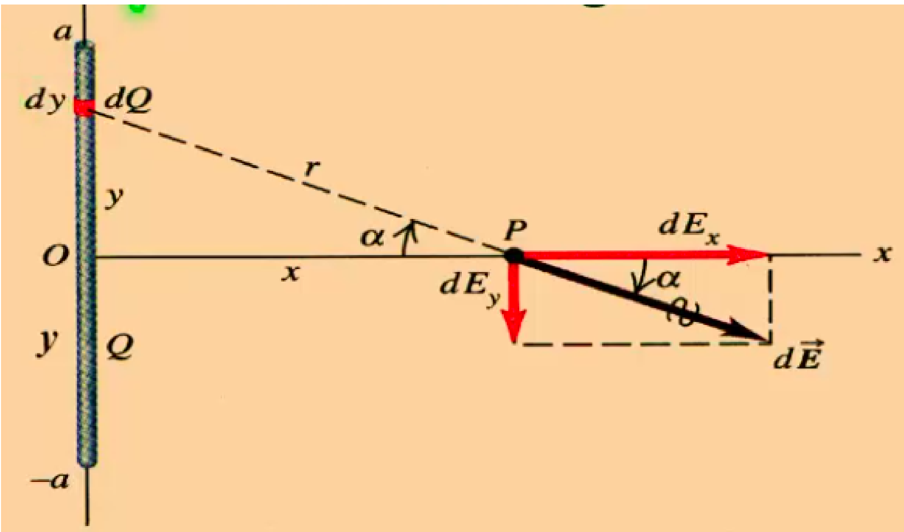


Figure 10.1

We consider some element of charge $\delta Q = \lambda \delta y$ along the line at y. There are two components to the E-field, one pointing in the positive x direction and one in the negative y direction.

When we consider all of the δQs along the length of the charged rod, each point on the positive y-axis along the length will have a corresponding point on the negative y-axis. These two points will have equal and opposite magnitude, so $E_y = 0$ by symmetry. There is therefore no resultant y component and the final field will point along $\hat{x}$.

For some infinitesimal δq generically along the line of charge:

$$\delta E_x = \frac{\delta q}{4\pi\epsilon_0 r^2}\cos\alpha \tag{1}$$

And for each δQ:

$$\delta Q = \lambda \delta y \tag{2}$$

And from trig:

$$r^2 = y^2 + x^2 \tag{3}$$

$$\cos\alpha = \frac{x}{r} \tag{4}$$

Substituting 2, 3, 4 into 1:

$$\delta E_x = \frac{\lambda \delta y}{4\pi\epsilon_0}\frac{x}{[y^2 + x^2]^{3/2}}$$

Hence:

$$E_x = \frac{\lambda x}{4\pi\epsilon 0}\int_{-L}^{L}\frac{1}{[y^2 + x^2]^{3/2}}\,dy$$

To solve this, we would use the standard integral given in the question. Alternatively, if a standard integral isn't given in the question we could express in the integral in terms of α and integrate from the minimum to maximum alpha.

3.1 Example Part B: The Easier Route

As before:

$$\delta E_x = \frac{\lambda \delta y}{4\pi\epsilon_0 r^2}\cos\alpha \tag{1}$$

$$\cos\alpha = \frac{x}{r} \implies r = \frac{x}{\cos\alpha} \tag{2}$$

And:

$$\tan\alpha = \frac{y}{x} \implies y = x\tan\alpha \tag{3}$$

As:

$$\frac{d}{d\alpha}\tan\alpha = \frac{1}{\cos^2\alpha}$$

We have:

$$\delta y = \frac{x\delta\alpha}{\cos^2\alpha} \tag{4}$$

Hence:

$$\begin{aligned}\delta E_x &= \frac{\lambda}{4\pi\epsilon_0}\underbrace{\frac{x\delta x}{\cos^2\alpha}}_{\delta y}\underbrace{\frac{\cos^2\alpha}{x^2}}_{1/r^2}\cos\alpha \\ &= \frac{\lambda}{4\pi\epsilon_0 x}\cos\alpha\end{aligned}$$

Which is much easier to integrate from the minimum to maximum values of α.

$$E_x = \frac{\lambda}{4\pi\epsilon_0 x}\int_{\alpha_{\min}}^{\alpha_{\max}}\cos\alpha\, d\alpha$$

And from the geometry of the problem:

$$\alpha_{\max} = \arctan\left(\frac{L}{x}\right) \qquad \alpha_{\min} = -\arctan\left(\frac{L}{x}\right)$$

Which is easy to integrate.

If the problem was setup such that the rod was infinitely long, then:

$$E_x = \frac{\lambda}{4\pi\epsilon_0 x}\int_{-\pi/2}^{\pi/2}\cos\alpha\, d\alpha$$

$$E_x = \frac{\lambda}{2\pi\epsilon_0 x}$$

This is approximately true for the case where $x \ll L$.

4 Example: Ring of Charge

Consider a ring of charge with radius a. We step out some perpendicular constant distance x from the centre of the ring to point P. What is the resultant field at P?

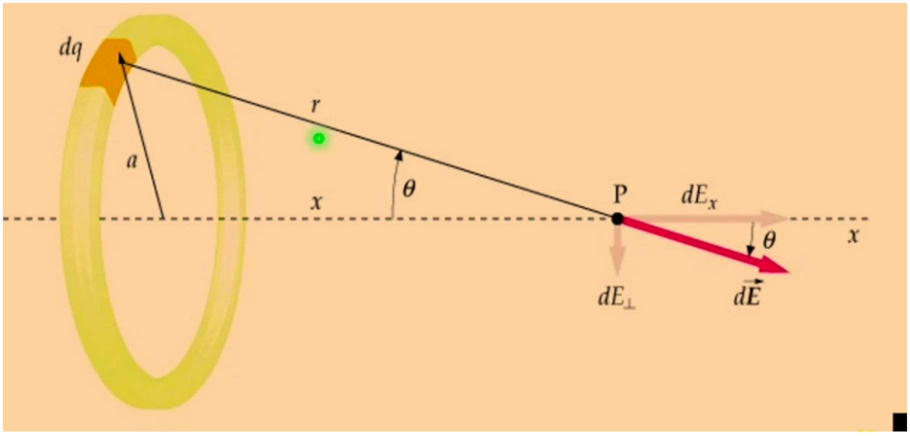


Figure 10.2

The ring has total charge Q, distributed with charge per unit length λ.

Again, we can use symmetry. For every point on the ring, there will be a point one diameter away creating an equal and opposite δE. Hence, the y and z components cancel leaving only E_x, pointing in the $\hat{x}$ direction.

$$E_y = E_z = 0 \text{ by symmetry}$$

We have:

$$\delta E_x = \frac{\delta q}{4\pi\epsilon_0 r^2} \cos\theta$$

$$\cos\theta = \frac{x}{y}$$

$$r = \sqrt{a^2 + x^2}$$

Hence:

$$\delta E_x = \frac{\delta q}{4\pi\epsilon_0} \frac{x}{[a^2 + x^2]^{3/2}}$$

Here, a and x are constants, so the only variable we can integrate across is δq:

$$E_x = \frac{x}{4\pi\epsilon_0} \frac{1}{[a^2 + x^2]^{3/2}} \int dq$$

$$= \frac{xQ}{4\pi\epsilon_0} \frac{1}{[a^2 + x^2]^{3/2}}$$

Thu 22 Jan 2026 11:00

Lecture 3 - Electric Fields III: More Continuous Charges and Gauss' Law

1 Example: Large Circular Plane

Consider a large circular plane with uniform charge density σ. We step out some distance x along the axis perpendicular to the plane, through its centre, to point P.

The plane has radius R and total charge Q. What is the magnitude and direction of the electric field at P?

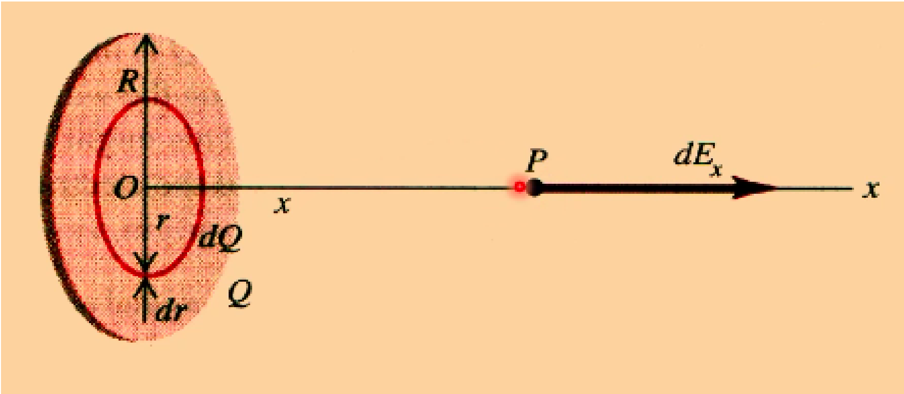


Figure 11.1

Using the same logic as for the ring last lecture, we can see that:

$$E_y = E_z = 0 \text{ by symmetry}$$

So the final E-field lies entirely in the direction of $\hat{x}$.

We want to divide the final E-field into an infinite number of thing rings, each at radius r with infinitesimal thickness δr. We can then integrate across all rings to determine the total E-field.

The distance between P and O is x, the distance between O and the charge element is r, and the distance between P and the charge element is d.

From the previous example (in an exam question we would have to do this too), we have:

$$\delta E_{\text{ring}} = \frac{Q_{\text{ring}}}{4\pi\epsilon_0} \frac{x}{[r^2 + x^2]^{(3/2)}}$$

$$Q_{\text{ring}} = \sigma \times 2\pi r \delta r$$

Hence:

$$\delta E_{\text{ring}} = \frac{\sigma r \delta r}{2\epsilon_0} \frac{x}{(r^2 + x^2)^{(3/2)}}$$

Noting that $r = x \tan\theta$, where θ is the angle between d and x:

$$\delta r = \frac{x}{\cos^2\theta} d\theta$$

$$\implies \cos\theta = \frac{x}{d} = \frac{x}{[r^2+x^2]^(1/2)}$$
$$\frac{1}{[r^2+x^2]^{(3/2)}} = \frac{\cos^3\theta}{x^3}$$

Substituting these in:

$$\delta E_{\text{ring}} = \frac{\sigma}{2\epsilon_0}\underbrace{\left(\frac{x\sin\theta}{\cos\theta}\right)}_{r}\underbrace{\frac{x\delta\theta}{\cos^2\theta}}_{\delta r}\times x\times\frac{\cos^3\theta}{x^3}$$

Which simplifies down to:

$$\frac{\sigma}{2\epsilon_0}\sin\theta\delta\theta$$

Which is again nice and simple to integrate. Integrating across all θ:

$$E_{\text{disc}} = \frac{\sigma}{2\epsilon_0}\int_0^{\theta_{\max}} \sin\theta\, d\theta$$

Where $\theta_{\max} = \arctan(R/x)$. If we wanted a full infinite plane, we could get this as $R \to \infty$:.

$$\theta_{\max} = \pi/2$$
$$E = \frac{\sigma}{2\epsilon_0}\int_0^{\pi/2} \sin\theta d\theta$$
$$\implies E = \frac{\sigma}{2\epsilon_0}$$

Similar to the ring, this is approximately true for a finite plane where $x \ll R$. This is interesting because it implies that (for close distances, where $x \ll R$ still holds), the electric field is constant with respect to distance.

2 E-Field in a Hollow Sphere

Consider a hollow sphere with uniform charged shell density σ. Consider an arbitrary point P inside this spherical shell.

We can use solid angles here, which are effectively three dimensional angles. This is still dimensionless, but is given by a subtended area divided by the radius squared:

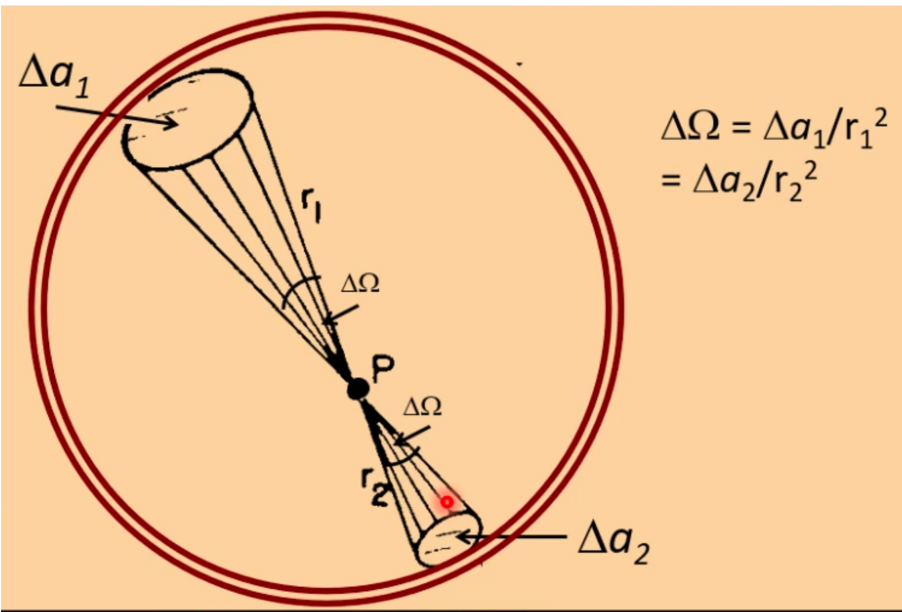


Figure 11.2

While a traditional angle is $\theta =$ arc length / radius in units of radians, the A/r^2 for solid angles has units of Steradians, Sr.

The total surface area of a sphere is $4\pi r^2$, so the total solid angle in all direction is 4π.

As we have defined the two cones to subtend the same solid angle at P, their areas scale as the square of their radii:

$$\frac{\Delta a_2}{\Delta a_1} = \frac{r_2^2}{r_1^2}$$

Hence:

$$\frac{\Delta q_2}{\Delta q_1} = \frac{\sigma \Delta a_2}{\sigma \Delta a_1} = \frac{r_2^2}{r_1^2}$$

The ratio of the two electric fields from each area is given by:

$$\frac{E_2}{E_1} = \frac{\Delta q_2 / r_2^2}{\Delta q_1 / r_1^2} = \frac{\Delta q_2}{\Delta q_1} \frac{r_1^2}{r_2^2} = 1$$

This means that the total E-field at P, and indeed at any point inside the hollow sphere, is zero.

3 Electric Flux

Electric flux is effectively the amount of E-field passing across a surface. It is defined in terms of a normal unit vector perpendicular to the surface, $\hat{n}$.

We get an expression for flux by multiplying the field strength by the area "seen" by the field. This is given by $A\cos\theta$ and for a constant E-field on a flat surface:

$$\text{flux } = \Phi_E = AE \cdot \hat{n} = EA\cos\theta$$

More generally:

$$\Phi_E = \int \underline{E} \cdot \hat{n} dS = \int \underline{E} \cdot d\underline{S}$$

Where dS is an element of surface area.

4 Gauss' Law

Consider an imaginary closed surface, of any shape. Inside this surface contains a charge q.

The E-field at some element of this surface at distance r from the charge is given by:

$$\underline{E} = \frac{q}{4\pi\epsilon_0 r^2}\hat{r}$$

The flux out of this area element is:

$$d\Phi = \underline{E} \cdot d\underline{S} = \underline{E} \cdot \hat{n} dS = E\cos\theta dS$$

Hence:

$$d\Phi = \frac{q}{4\pi\epsilon_0 r^2}\cos\theta dS = \frac{q}{4\pi\epsilon_0}\frac{\cos\theta dS}{r^2} = \frac{q}{4\pi\epsilon_0} d\Omega$$

Where Ω is the solid angle.

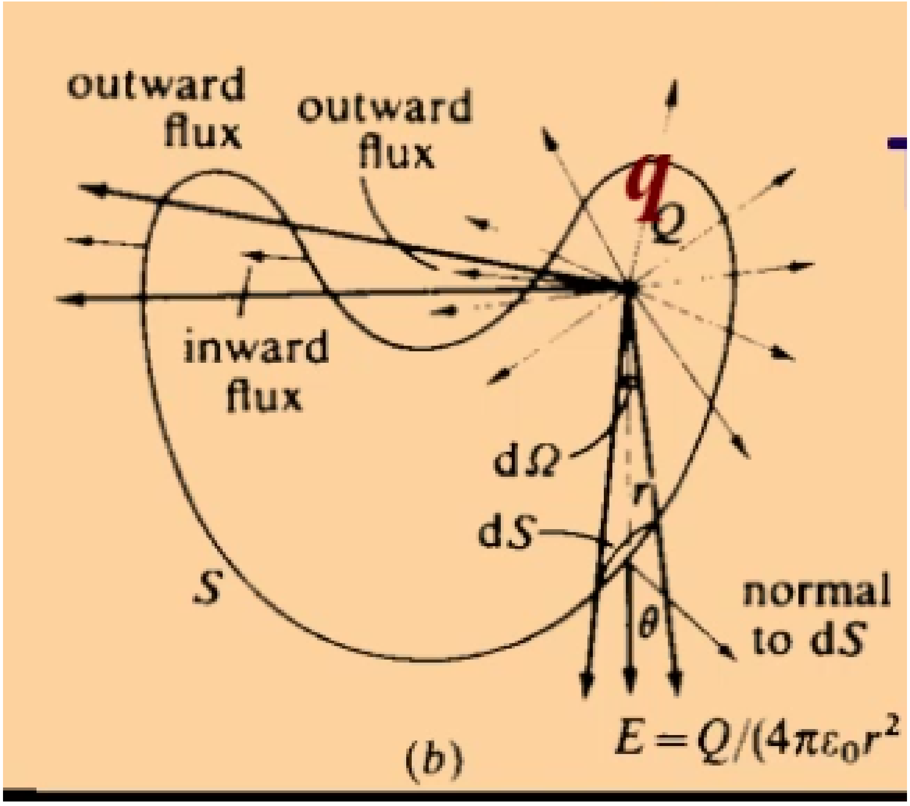


Figure 11.3

The total flux is therefore:

$$\Phi_E = \frac{q}{4\pi\epsilon_0}\int_S d\Omega = \frac{q}{4\pi\epsilon_0}4\pi = \frac{q}{\epsilon_0}$$

Note that the proof/derivation is not needed, as solid angles are off-spec. Only the final result is needed.

This gives us Gauss' Law:

The net electric flux of $\underline{E}$ out of *any* closed surface, enclosing a total charge of Q_{enclosed} is:

$$\Phi_E = \int_S \underline{E}\cdot d\underline{S} = \frac{Q_{\text{enclosed}}}{\epsilon_0}$$

Where $Q_{\text{enclosed}} = \sum_i q_i$ for discrete charges or $Q_{\text{enclosed}} = \int_V \rho dV$ for distributed charge.

This becomes incredibly useful for problems where there is symmetry.

4.1 Example: Solid Sphere of Uniform Charge Density

Consider a sphere of radius R with charge Q distributed uniformly. We want to find the E-field inside the sphere, and the E-field outside the sphere.

We will set up an imaginary spherical Gauss surface of radius r and use symmetry:

$$\int_S \underline{E}\cdot d\underline{s} = \frac{Q_{\text{enclosed}}}{\epsilon_0}$$

Outside the Sphere: $r > R$

The E-field from a sphere and the normal vector of the surface are both radial. Hence, they are parallel. This means that $\theta = 0$ and:

$$\underline{E}\cdot d\underline{S} = EdS$$

Since the sphere is symmetrical, the E-field at a constant r must be constant anywhere around the sphere. Hence, E is constant and can be pulled out of the integral:

$$E\int_S dS = \frac{q_{\text{enclosed}}}{\epsilon_0}$$
$$E\left(4\pi r^2\right) = \frac{q_{\text{enclosed}}}{\epsilon_0}$$
$$E = \frac{Q}{4\pi\epsilon_0 r^2}$$

Which is Coulomb's law! Lets now consider the E-field inside the sphere.

Inside the Sphere: $r < R$

$$\int_S \underline{E}\cdot d\underline{S} = \frac{q_{\text{enclosed}}}{\epsilon_0}$$

As before, the LHS is $E4\pi r^2$. The RHS is now different, as a smaller volume of charge is enclosed by the Gauss surface. This new charge is given by the total charge times the ratio of the Gauss surface volume to the total volume:

$$\frac{q_{\text{enclosed}}}{\epsilon_0} = \frac{1}{\epsilon_0}\left(\frac{Qr^3}{R^3}\right)$$

Hence:

$$E4\pi r^2 = \frac{Qr^3}{\epsilon_0 R^3}$$
$$\implies E = \frac{Qr}{4\pi\epsilon_0 R^3}$$

Mon 29 Jan 2026 11:00

Lecture 4 - Electric Fields IV: Gauss' Law Examples

1 Solid Sphere with Non-Uniform Charge

What if we have a solid sphere with distributed charge where the charge distribution is a function of radius?

$$\rho(r) = \rho_0 r$$

Where r_0 is a constant (note: does not have dimensions of charge density).

We now have:

$$Q_{\text{enclosed}} = \int_0^r \rho(r) dV$$

Hence:

$$\int_S \underline{E} \cdot d\underline{S} = \frac{1}{\epsilon_0} \int_0^r \rho(r) dV$$

We divide the sphere into thin shells of infinitesimal thickness δr. The volume of each shell is:

$$\delta V = 4\pi r^2 \delta r$$

Hence:

$$Q_{\text{enclosed}} = \int_0^r \rho(r)\, dV = \int_0^r \rho_0 r 4\pi r^2\, dr$$
$$= 4\pi\rho_0 \int_0^r r^3\, dr = \pi\rho_0 r^4$$

Using Gauss' law:

$$\int_S \underline{E} \cdot d\underline{S} = \frac{\pi\rho_0 r^4}{\epsilon_0}$$

Considering the LHS and using symmetry:

$$\int_S \underline{E} \cdot d\underline{S} = \int_S E dS = E \int_S dS = E 4\pi r^2$$

Hence:

$$4\pi E r^2 = \frac{\pi\rho_0 r^4}{\epsilon_0}$$
$$\implies E = \frac{\rho_0 r^2}{4\epsilon_0}$$

2 E-Field Inside Uniform Charged Shell

We've already done this with solid angles, but it becomes much easier with Gauss' law. Since the Gauss surface sits inside the shell, none of the charged distributed across the shell is inside the surface. Therefore, $Q_{\text{enclosed}} = 0 \implies E = 0$

3 E-Fields in Conductors

A simple model of a conductor is where each atom has a free valance electron which is free to move throughout the conductor, in vast quantities.

If you place a conductive box inside a conductor, the electrons want to travel against the field. This leaves a positively charged nuclei, which drift in the opposite direction.

This causes the outside of the metal box to pick up a small charge, creating its own electric field which opposes the initial field. This continues until it perfectly cancels out the initial field, leaving no resultant field inside a conductor.

This is called a "Faraday Cage" and can be used to shield the interior of the conducting box from the E-field, i.e. to protect sensitive electronics.

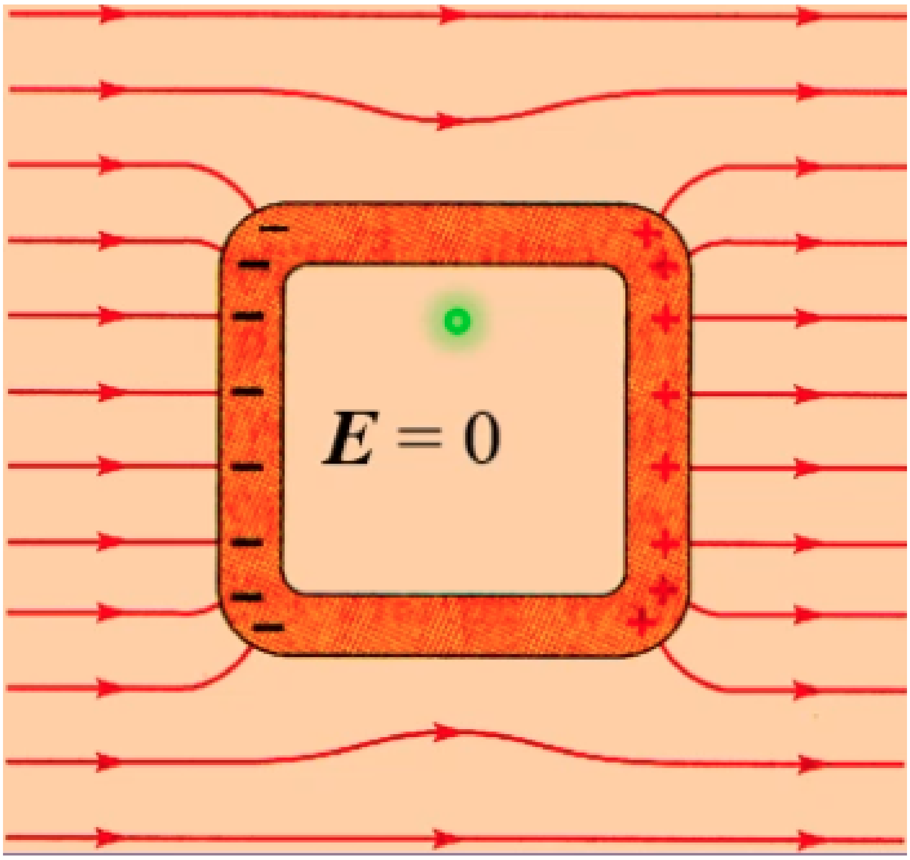


Figure 12.1

4 Infinite Charged Sheet

4.1 Non-Conducting Sheet

Consider an infinitely large plane of surface charge density σ. As the sheet is infinitely large, there's no edge effects and the E-field exists exclusively $\perp$ to the surface of the sheet.

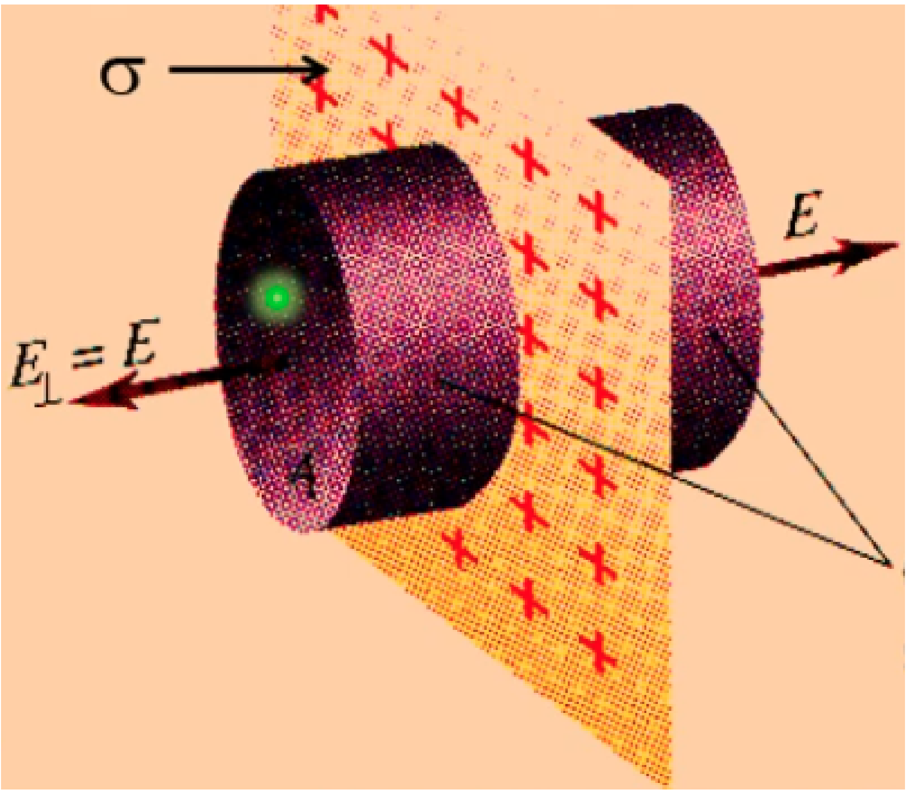


Figure 12.2

We can therefore take a cylindrical Gauss surface, and the only E-field will exist normally to the end circular faces (of cross-sectional area A). Hence, $\underline{E}$ and $d\underline{S}$ are parallel, and:

$$\int_S \underline{E} \cdot d\underline{S} = E \int_S dS$$

So:

$$\begin{aligned} E \int_S dS &= \frac{Q_{\text{enclosed}}}{\epsilon_0} \\ E\,(2A) &= \frac{\sigma A}{\epsilon_0} \\ \implies E &= \frac{\sigma}{2\epsilon_0} \end{aligned}$$

As previously derived! Again, this works for a finite plane where the plane is much larger (relatively) than the distance from the plane, so edge effects cn be disregarded.

4.2 Conducting Sheet

As the sheet is now conducting, there is no E-field inside the conductor, and charge is distributed to build up a charge density (σ) on both surfaces of the sheet.

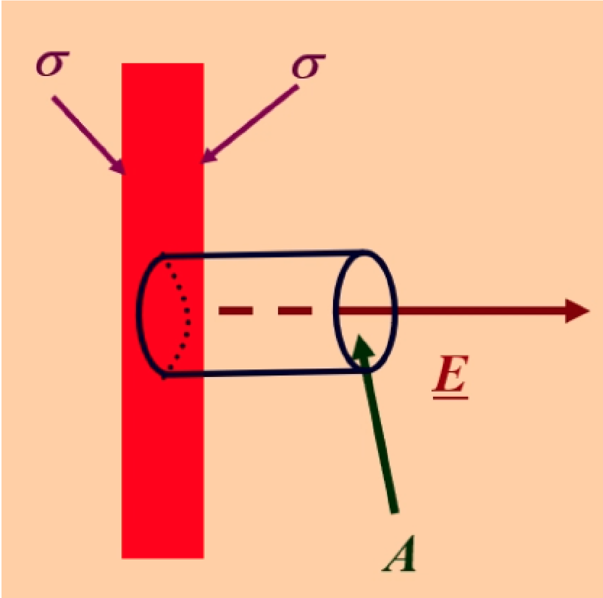


Figure 12.3

Applying Gauss' law gives:

$$EA = \frac{\sigma A}{\epsilon_0}$$
$$\implies E = \frac{\sigma}{\epsilon_0}$$

Which is a factor of two different from the non-conducting case. Note that this doesn't arise because we only took a Gauss surface on one side. If we took a Gauss surface through the sheet (in the same manner as the non-conducting case), we would have two sets of charge distributed, one on each side, both with charge density σ. We would therefore have:

$$E(2A) = \frac{2\sigma A}{\epsilon_0}$$

Which yields the same result.

5 E-Field Between Charged Conducting Plates

Consider a capacitor-like setup with two parallel charged metal plates.

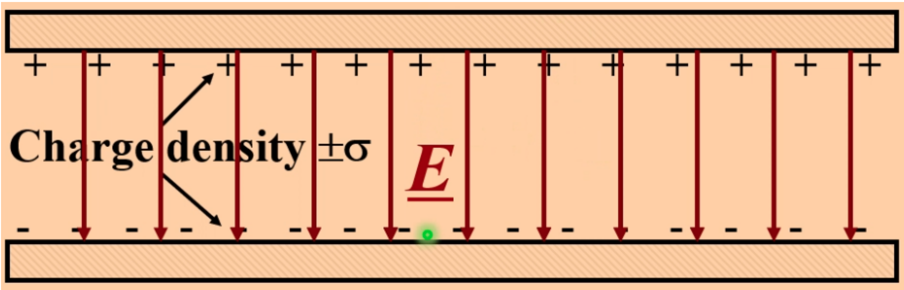


Figure 12.4

While a full derivation will be covered later, the E-field given by this is the same as for an infinite sheet:

$$E = \frac{\sigma}{\epsilon_0}$$

6 Infinite Line of Charge

Consider an infinitely long line charge, of charge density λ. As its infinitely long, there's no edge effects and the electric field must be radial.

We can place a cylindrical Gaussian surface where the axis is parallel to the line, of length L and radius r.

Since the field is radial, it acts in the same direction as the normal vector to the cylinder surface. By symmetry, the E-field is constant for fixed radius, so:

$$\int_S \underline{E} \cdot d\underline{S} = E \int_S dS = E(2\pi r L)$$

$$\frac{Q_{\text{enclosed}}}{\epsilon_0} = \frac{\lambda L}{\epsilon_0}$$

Hence:

$$E(2\pi r L) = \frac{\lambda L}{\epsilon_0}$$

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

7 No Contained Charge

What if the Gaussian surface sits within an E-field, but the charged creating the field is not enclosed in the surface?

This could be thought of as implying by Gauss' law that:

$$\oint \underline{E} \cdot d\underline{S} = O$$

$$\implies E = 0?$$

But this is not the case. Here, the symmetry that we've relied upon isn't here, as $E \nparallel dS$ and is not constant around the surface. Hence, we can't nicely simplify the problem and the first integral becomes unsolvable (by us).

If the charge falls outside of the Gauss surface, the net flux through the surface is zero.

Thu 02 Feb 2026 11:00

Lecture 5 - Electric Fields IV: Gauss Examples and Electric Potential

1 Non-Conducting Infinite Charged Parallel Plates

Consider two non-conducting infinite planes of charge with charge density σ.

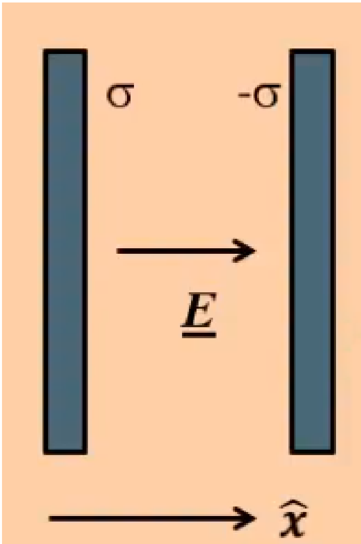


Figure 13.1

We have derived that the surface charge from the LHS plate gives field:

$$\underline{E}_L = \frac{\sigma}{2\epsilon_0}\underline{\hat{x}}$$

The RHS plate gives:

$$\underline{E}_R = \frac{-\sigma}{2\epsilon_0}\underline{\hat{n}} = \frac{\sigma}{2\epsilon_0}\underline{\hat{x}}$$

Where n is the unit vector normal to the RHS surface. By superposition, the total field between the plates is:

$$\underline{E} = \underline{E}_L + \underline{E}_R = \frac{\sigma}{\epsilon_0}\underline{\hat{x}}$$

This is exactly the same as for two conducting plates, per last lecture.

1.1 Back to Conducting Plates

Consider two oppositely charged conducting plates a long way apart. They have charges on each surface.

As the two plates move closer together, their E-fields attract the charges on each other. This causes movement of the charges, until all the charges move towards the front faces. This causes the effective surface charge density to double, hence:

$$E = \frac{\sigma_i}{\epsilon_0} + \frac{\sigma_i}{\epsilon_0} = \frac{2\sigma_i}{\epsilon_0} = \frac{\sigma}{\epsilon_0}$$

Where σ_i is the initial charge densities on the surface of each plate, and σ is the final charge density effective on the inner surfaces. There is therefore no charges (and hence $E = 0$) on the outer surfaces. Note that this is only true when the plates have equal and opposite charge densities.

2 Work Done by an E-Fields

Consider a particle moving in an E-field. Across some infinitesimal distance $\delta \underline{l}$, over which the particle experiences force $\underline{F}$, the infinitesimal element of work done is:

$$\delta W = \underline{F} \cdot \delta \underline{l}$$

As the E-field by definition carries units of force per unit charge, the force is given by:

$$\delta W = q\underline{E} \cdot \delta \underline{L}$$

The work done by an E-field represents a decrease in electric potential energy, $\delta U = -\delta W$:

$$\delta U = -q\underline{E} \cdot d\underline{L}$$

If the charge is positive, δU is negative and energy decreases. If the charge is negative, δU is positive and energy increases.

If the E-field varies wrt space, and the particle moves from point A to point B, we need to integrate to find the total change:

$$U_B - U_A = -q \int_a^b \underline{E} \cdot d\underline{l}$$

We want to be able to define this on a per unit charge basis.

V is the potential energy of the system *per unit charge*.

$$V = \frac{U}{q}$$

$$V_b - V_A = -\int_a^b \underline{E} \cdot d\underline{l}$$

It is a property of a point everywhere in an E-field. **It is a scalar**. The units are Joules per Coulomb, i.e. Volts.

The electric potential at distance r from a point charge q is:

$$\underline{E} = \frac{q}{4\pi\epsilon_0 r^2}\underline{r}$$

$$V(\infty) - V(r) = -\int_\infty^r \underline{E} \cdot d\underline{r} = \frac{q}{4\pi\epsilon_0 r}$$

This is the Coulomb potential, which can be positive or negative (as q can be positive or negative).

As it's potential per unit charge, the U of a charge q_0 at distance r from charge q is:

$$U(r) = V(r)q = \frac{1}{4\pi\epsilon_0}\frac{qq_0}{r}$$

And:

$$F = -\frac{dU(r)}{dr} = \frac{1}{4\pi\epsilon_0}\frac{qq_0}{r^2}$$

3 Multiple Charges

What if we have multiple charges? We simply sum across them:

$$V = \frac{1}{4\pi\epsilon_0}\sum_i \frac{q_i}{r_i}$$

Where r_i is the distance from the ith charge q_i to the point at which we're evaluating the potential.

4 Electric Potential Difference

The electric potential difference (AKA voltage) between two points is:

$$V_B - V_A = \frac{U_B - U_A}{q} = -\int_a^b \underline{E}\cdot d\underline{l}$$

We can use this to get that Newtons per Coulomb and Volts per Metre are dimensionally equivalent units, and we commonly use the latter for describing electric fields.

5 E-Field Lines

The lines of an E-field trace the path that a positive test charge would follow under the action of the electrostatic force. If the positive test charge were released, it would accelerate along the field lines, from high potential to low potential.

We commonly overlay perpendicular (to $\underline{E}$) lines on a field diagram, representing surfaces where V is a constant. These are called equipotentials and are always perpendicular to $\underline{E}$.

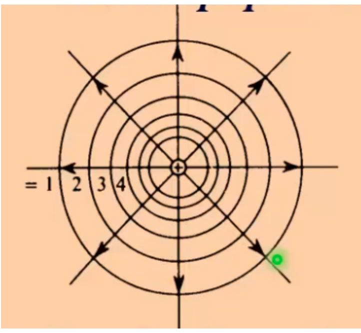


Figure 13.2

6 E from V

Because V is a scalar quantity, its often easier to calculate than E. There are some cases where its therefore easiest to find V and use this to determine E. We start from:

$$V = -\int \underline{E}\cdot d\underline{l}$$

And differentiate both sides:

$$\underline{E} = -\left(\frac{\partial V}{\partial x}\hat{\imath} + \frac{\partial V}{\partial y}\hat{\jmath} + \frac{\partial V}{\partial z}\hat{k}\right) = -\underline{\nabla}V$$

Or in polar coordinates:

$$\underline{E} = -\left(\frac{\partial V}{\partial r}\hat{r} + \frac{1}{r}\frac{\partial V}{\partial \theta}\hat{\theta}\right)$$

Where the $1/r$ term arises to ensure dimensional correctness. Often, we have nice examples where we're only dealing with the radial component, so have:

$$E(r) = -\frac{dV}{dr}\hat{r} = \frac{q}{4\pi\epsilon_0 r^2}\hat{r}$$

7 Conservatism

The $\underline{E}$-field is a conservative field. This means that if the charge returns to its original position, regardless of the route taken, no net work is done:

$$\oint \underline{E} \cdot d\underline{l} = 0$$

This equivalently means that the change in potential between two points is the same whichever path between the two points is taken.

Thu 26 Feb 2026 11:00

Lecture 12 - Currents and Magnetic Force

Today we start Part II, Magnetism:

- Definition of current and current density.
- Magnetic force on a moving charge.
- The Lorentz Force.
- Field lines for a magnetic field.

1 Current

1.1 Definition of Current

Suppose a current carries a current I, this is defined in terms of the rate of flow of charge past a given cross section:

$$I = \frac{dQ}{dt}$$

The SI unit of current is the *ampere*, defined as the flow of one coulomb per second.

1.2 Nature of Current

There's two experimentally observed effects of a current flow:

- Heating
- Creation of magnetic fields.

Consider a current through a conductor. This is caused by electrons moving (with average drift velocity v) as a result of the induced E-field.

In some time, they travel $v\Delta t$ through the cross section, and if this cross section has area A they sweep out an area of $Av\Delta t$.

If a unit volume has n conducting free electrons, the number of charges is given by:

$$N = n(Av\Delta t)$$

And hence the total charge is:

$$\Delta Q = nAv\Delta T(-e)$$

Hence:

$$\hat{\imath} = \frac{\Delta Q}{\Delta t} = -nAe\underline{v}$$

1.3 Current Density

More usefully, we talk about the current per unit cross-section, which is the current density J:

$$\hat{\jmath} = \frac{\hat{\imath}}{\underline{A}} = -ne\underline{v}$$

The negative sign arises as we define current as flowing from positive to negative. Electrons, being negatively charged, actually physically flow from negative to positive. Current flow and actual electron flow are therefore in the opposite direction. This is because current was initially understood and defined before the electron was defined.

Typically, $v < 1\text{mms}^{-1}$ when a current is flowing.

2 Magnetism

2.1 Magnetic Force on a Charge

A magnetic field exerts a force on a moving charge that is present in the magnetic field.

Suppose a particle of charge $+q$ moving with some velocity $\underline{v}$ in a magnetic field $\underline{B}$ experiences some force $\underline{F}_m$.

We cannot derive from first principles, but experimentally have observed:

- $\underline{F}_m \perp \underline{v}, \underline{B}$.
- $\underline{F}_m \propto \underline{v}$.
- $\underline{F}_m \propto \underline{B}$.
- $\underline{F}_m \propto q$

If θ is the angle between the velocity and the B-field direction, we hae:

$$\boxed{\underline{F}_m = Bqv \sin\theta}$$

In vector form:

$$\boxed{\underline{F}_m = q\underline{v} \wedge \underline{B}}$$

Here, $\underline{B}$ has the unit Tesla, T. In base units this is $\text{NC}^{-1}\text{m}^{-1}\text{s}$. A one Tesla magnetic field is really quite powerful, so we also define the Gauss, $1\text{G} = 10^{-4}\text{T}$ to move to a nicer scale.

By definition, if a 1C charge moving at 1m/s perpendicular to a magnetic field experiences a force 1N, the field is 1T.

- Earths magnetic field is ~ 0.5G
- Poles of a large electromagnetic: ~ 2T.
- Surface of a neutron star: $\sim 10^8$T.
- The maximum pulsed magnetic field that can create in a lab: ~ 50T.

The Earth's field is believed to be generated by electron currents in the iron alloys in its core. It's not completely understood yet, but convection currents causing these conductive alloys to flow and move is the working theory. This movement causes the North and South poles to swap places on average every $300,000$ years.

2.2 The Lorentz Force

In a region where we have both an E-field and a B-field, we have the total force as the vector sum of both:

$$\underline{F} = q\left(\underline{E} + \underline{v} \wedge B\right)$$

This is called the Lorentz force.

Suppose Earth's magnetic field is given by:

$$\underline{B} = B\cos 70°\hat{\jmath} - B\sin 70°\hat{k} \qquad \text{where: } B = 5 \times 10^{-5}\text{T}$$

A proton moves in this field with:

$$\underline{v} = 10^7\hat{\jmath}\text{ms}^{-1}$$

$$\underline{F}_m = +e\underline{v} \wedge \underline{B} = 1.6 \times 10^{19} \times 10^7 \times \det \begin{bmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 0 & 1 & 0 \\ 0 & B_y & B_z \end{bmatrix}$$
$$= 1.6 \times 10^{-12} \left[\hat{\imath}\,(B_z - 0) + \hat{\jmath}\,(0-0) + \hat{k}\,(0-0) \right]$$
$$= 1.6 \times 10^{-12} B_Z \hat{\imath}$$
$$= -1.6 \times 10^{-12} \times 5 \times 10^{-5} \sin 70° \hat{\imath}$$
$$-7.5 \times 10^{-17} \hat{\imath} \mathrm{N}$$

2.3 Magnetic Field Lines

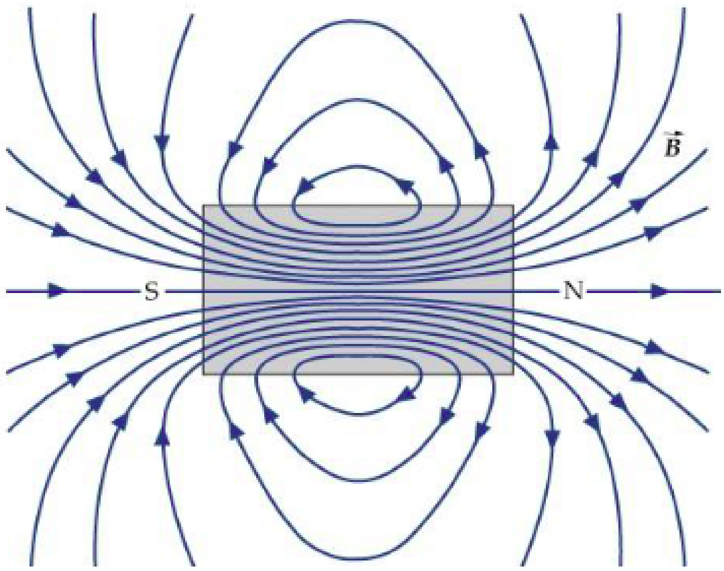


Figure 14.1: Magnetic Field Lines Around a Bar Magnetic

Magnetic field lines are different to electrical field lines. They do not point in the direction of force and travel out of North poles and into South poles. Magnetic field lines are always continuous so there is no magnetic monopoles.

Physicists have searched for a potential magnetic monopoles, and the current upper limit for the number of magnetic monopoles per body is $< 10^{-29}$.

The tangent to a field line at some point gives the direction of B at that point. The number of field lines drawn per unit cross section $\propto$ B.

2.4 Magnetic Flux

The magnetic flux $\Delta\phi_B$ has the same definition as electric flux in an E-field. For a small area ΔA:

$$\Delta\phi_B = B\cos\theta\Delta A$$

Where θ is the angle between the B-field and the surface normal vector.

The total magnetic flux is therefore the integral of this:

$$\phi_B = \int_S \underline{B} \cdot d\underline{S}$$

As there is no monopoles, and exiting flux must also return. Therefore, the net flux over any enclosed surface is zero:

$$\int_S \underline{B} \cdot d\underline{S} = 0$$

While this does form one of Maxwell's equations and is very useful for EMII next year, it's not useful in EMI...

Thu 05 Mar 2026 11:00

Lecture 14 - Magnetic Fields from Currents

Recap of last lecture:

- Special cases of the magnetic force.
- Force on a current carrying conductor:
$$\underline{F} = I\underline{L} \wedge \underline{B}$$
- Current loops and magnetic dipoles:
$$\underline{\mu} = I\underline{A}$$
- Torque on a magnetic dipole in a B-field
$$\underline{\tau} = \underline{\mu} \wedge \underline{B}$$
- Potential energy of a magnetic dipole in a B-field:
$$U = -\underline{\mu} \cdot \underline{B}$$

1 Magnetic Fields from a Moving Charge (Current)

Danish physicist Hans Christian Oersted in 1820 demonstrated that a magnetic field exists around a current-carrying wire.

This was the first connection between electric and magnetic phenomena.

Consider a B-field at point $\underline{r}$ from a charge q, moving with velocity $\underline{v}$. The angle between $\underline{r}$ and $\underline{v}$ is given by ϕ.

The induced B-field is perpendicular to both $\underline{u}$ and $\underline{r}$ and is given by:

$$B \propto \frac{qv \sin\phi}{r^2}$$

In vector form, we have:

$$\underline{B} \propto \frac{q}{r^2}\underline{u} \wedge \hat{r}$$

In SI units, the constant of proportionality (involving "permeability of free space", μ_0) is such:

$$\underline{B} = \frac{\mu_0}{4\pi}\frac{q}{r^2}\underline{u} \wedge \hat{r} \qquad \mu_0 = 4\pi \times 10^{-7}\text{TmA}^{-1}$$

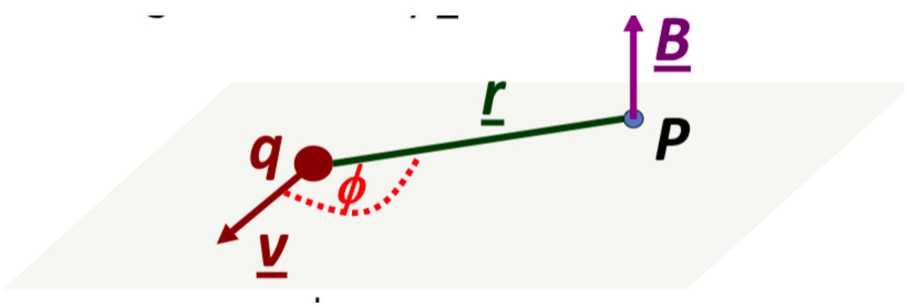


Figure 15.1

In a more complex 3D case, this looks as such:

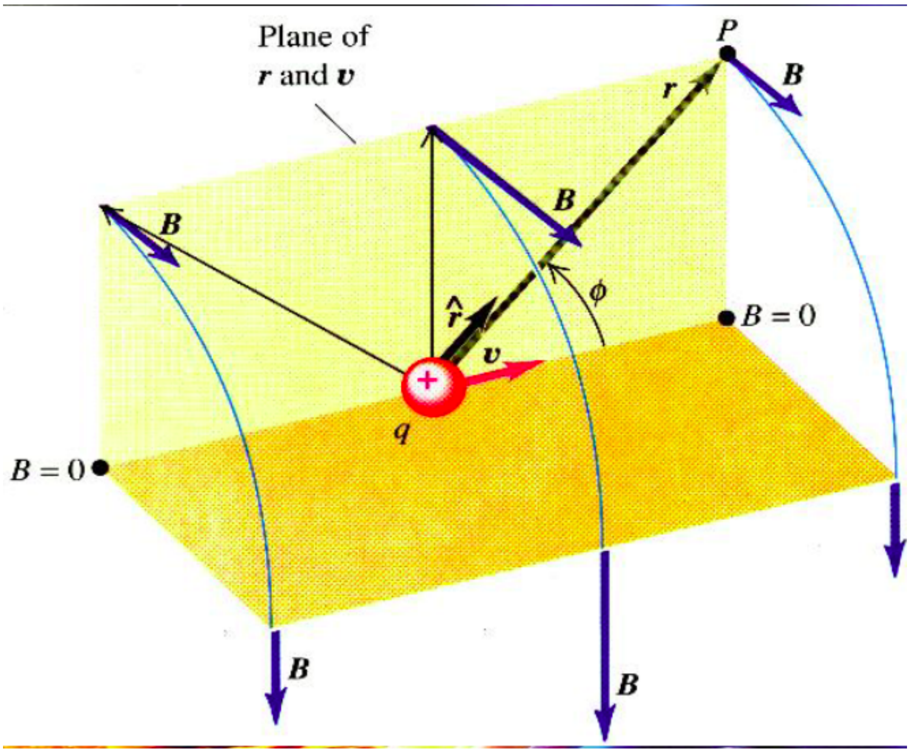


Figure 15.2

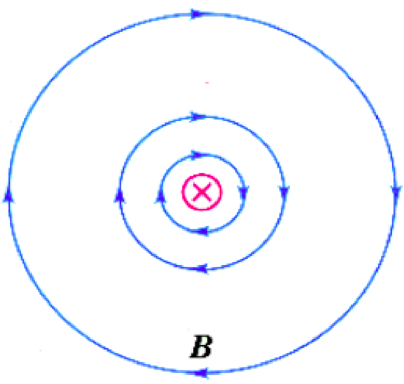


Figure 15.3

The B-field sweeps out a screw shape around the motion of direction. The direction is given by the right hand rule, where a "thumbs up" with the thumb pointing in the direction of the motion of the charge indicates the direction with the curl of the fingers.

1.1 Currents

Consider a wire with a current flow through it. The total charge flowing through some infinitesimal element in a unit time is:

$$I = \frac{qv}{\delta L}$$

Hence:

$$q\underline{v} = \frac{qv}{\delta L}\delta\underline{L} = I\delta\underline{L} \qquad \text{where } \delta L \text{ is in the direction of current.}$$

Plugging this in, we get the Biot-Savat Law:

$$\delta\underline{B} = \frac{\mu_0}{4\pi}\frac{I\delta L \wedge \hat{r}}{r^2}$$

We can split our object into infinitely many infinitesimal elements and integrate across them all to find the total B-field. We want to:

- Define $d\underline{B}$ in terms of a single variable.
- Integrate between the relevant limits.
- Be careful about the directions of the included vector quantities.
- Use symmetry where possible.

2 Current Loops

Consider a current loop with a clockwise current flow. The centre of the loop's circle is P, with a radius from P of R.

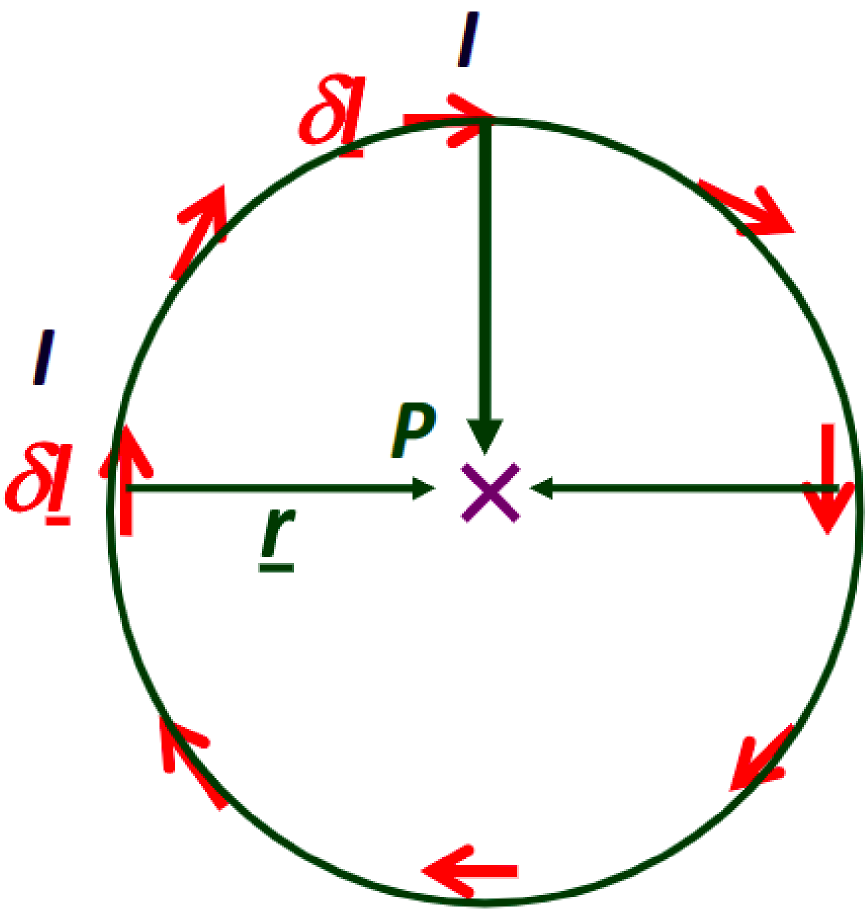


Figure 15.4

The Biot-Savat law gives:

$$\delta \underline{B} = \frac{\mu_0}{4\pi} I \frac{\delta l \wedge \hat{r}}{r^2}$$

$$\delta B = \frac{\mu_0}{4\pi} I \frac{\delta \underline{L}}{r^2}$$

$$B = \frac{\mu_0}{4\pi} \frac{I}{r^2} \oint dL = \frac{\mu_0}{4\pi} \frac{I}{4\pi} \times 2\pi r$$

$$B = \frac{\mu_0 I}{2r}$$

3 Finite Wire

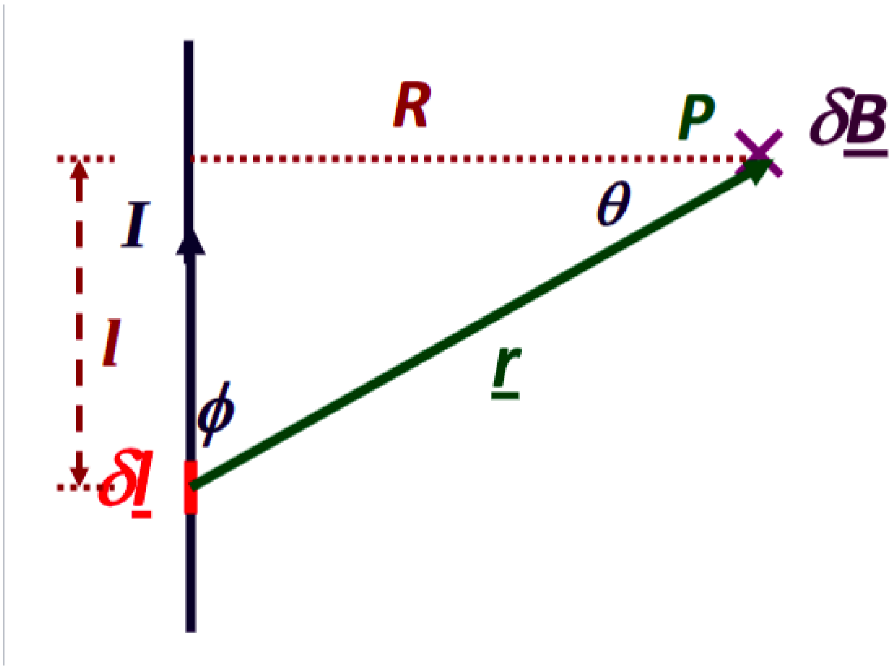


Figure 15.5

We want to find the B-field from the lime of current. The B-field is going into the page at P, given by the cross. The Biot Savat Law again gives:

$$\delta \underline{B} = \frac{\mu_0 I}{4\pi r^2} \delta \underline{L} \wedge \hat{r}$$

As $\hat{r}$ is a unit vector, the magnitude is given with the magnitude of L and the sine of phi:

$$\delta B = \frac{\mu_0 I}{4\pi r^2} \delta L \sin \phi \tag{1}$$

We want this in terms of θ instead, and given these two angles form a right angled triangle we have:

$$\sin \phi = \sin \left(\frac{\pi}{2} - \theta \right) = \cos \theta \tag{2}$$

Hence:

$$\delta B = \frac{\mu_0}{4\pi r^2} \delta L \cos \theta$$

By trig we simply have:

$$r \cos \theta = R \implies r = \frac{R}{\cos \theta} \tag{3}$$

$$\tan \theta = \frac{L}{R} \implies L = R \tan \theta \implies \frac{dL}{d\theta} = \frac{R}{\cos^2 \theta}$$

$$\implies \delta L = \frac{R}{\cos^2 \theta} \delta \theta \tag{4}$$

Inserting (2), (3), (4) back into (1) gives:

$$\delta B = \frac{\mu_0 I}{4\pi} \times \frac{\cos^2 \theta}{R^2} \times \cos \theta \times \frac{R}{\cos^2 \theta} \delta \theta$$

Hence:

$$\delta B = \frac{\mu_0 I}{4\pi} \frac{1}{R} \cos \theta \delta \theta$$

$$B = \frac{\mu_0 I}{4\pi R} \oint_{\theta_1}^{\theta_2} \cos\theta d\theta$$

Where θ_1, θ_2 depend the angles given by the length of the wire.

$$\boxed{B = \frac{\mu_0 I}{4\pi R}(\sin\theta_2 - \sin\theta_1)}$$

If the wire was infinitely long, $\theta_2 = \pi/2$ and $\theta_1 = -\pi/2$ (technically a limit as $L \to \infty$). This gives us:

$$B = \frac{\mu_0 I}{2\pi R}$$

4 More Complex Loops

Consider a current loop, with a set of axes centred at the middle of the loop. The loop sits in the yz plane, and we step out some perpendicular distance x in the x plane.

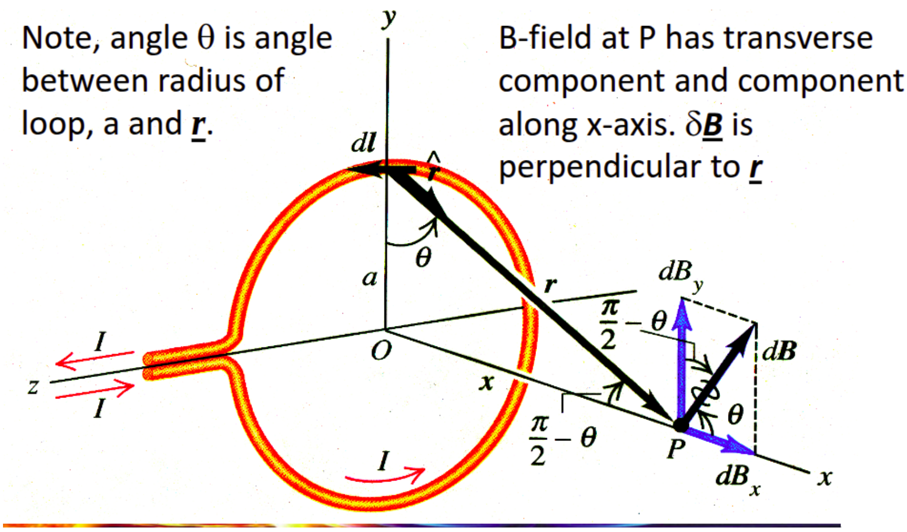


Figure 15.6

We step some distance δL along this loop, and the vector from the point we're investigating to B-field at from this δL is $\underline{r}$, at angle θ.

We can consider symmetry to see that transverse components to the axis will cancel each other out and therefore do not need considering:

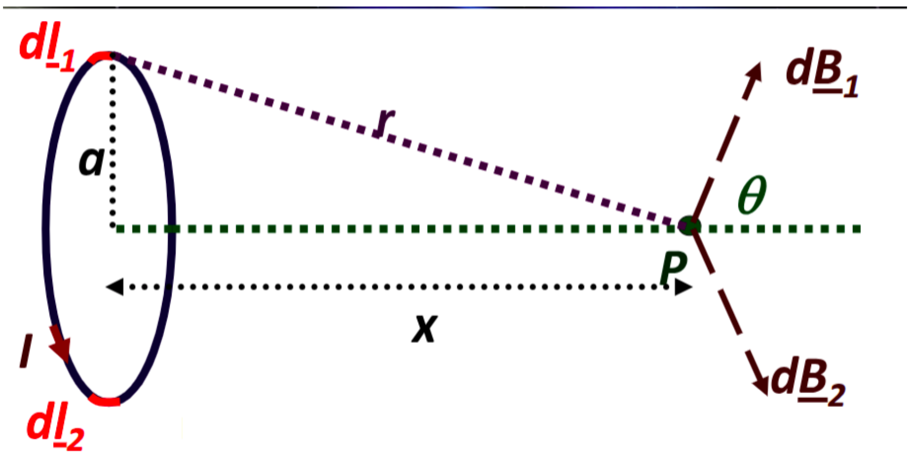


Figure 15.7

Therefore all of the components of the B-field in the *yz* planes will cancel out, and the only resultant B-field is in the x-direction. We will fully evaluate this on Monday.

Mon 09 Mar 2026 11:00

Lecture 15 - Ampere's Law

1 Biot-Savat: Complex Loops Continued

Continuing from the end of last lecture, we found by symmetry that the yz components cancel, leaving only an x component for any point P along the x-axis.

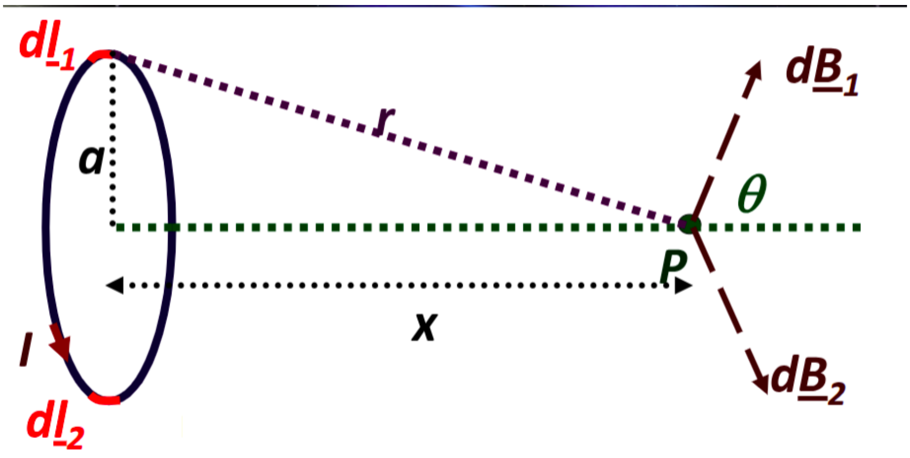


Figure 16.1

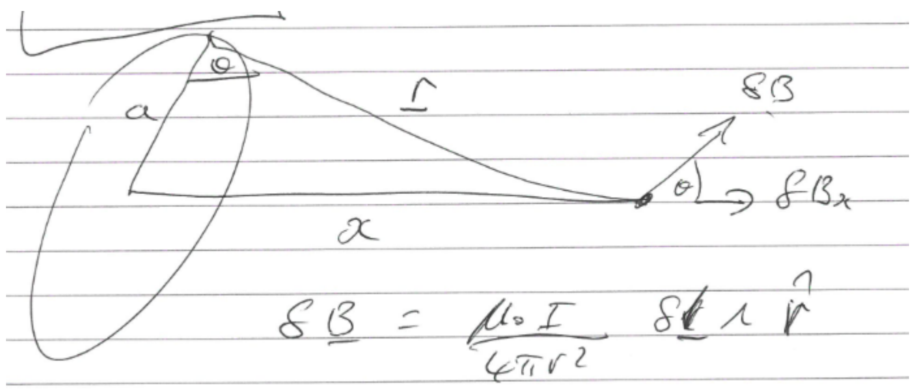


Figure 16.2

Taking magnitudes (and noting that $\hat{r}$ is a unit vector):

$$\delta B = \frac{\mu_0 I}{4\pi r^2}\delta l$$

Resolving into the x-direction, we get that:

$$\delta B_x = \frac{\mu_0 I}{4\pi r^2}\delta l \cos\theta \tag{1}$$

And as we have a right angled triangle:

$$r^2 + a^2 + x^2 \tag{2}$$

$$\cos\theta = \frac{a}{r} \implies \cos\theta = \frac{a}{(a^2+x^2)^{(1/2)}} \tag{3}$$

Inserting 2 and 3 into 1:

$$\delta B_x = \frac{\mu_0 I}{4\pi} \times \frac{a}{(a^2+x^2)^{(3/2)}}\delta l$$

$$B_x = \frac{\mu_0 I}{4\pi} \times \frac{a}{(a^2+x^2)^{(3/2)}}\oint \delta l$$

$$B_x = \frac{\mu_0 I}{4\pi} \times \frac{a}{(a^2+x^2)^{(3/2)}} 2\pi a$$

$$B_x = \frac{\mu_0 I}{2}\frac{a^2}{(a^2+x^2)^{(3/2)}}$$

Taking limits, we can observe that at the centre of the loop (where $x = 0$):

$$B_0 = \frac{\mu_0 I}{2}\frac{a^2}{(a^2)^{(3/2)}} = \frac{\mu_0 I}{2a}$$

And when $x \gg a$, we have:

$$B_x = \frac{\mu_0 I a^2}{2x^3}$$

However, we can simplify using the magnetic dipole moment $\mu_m = I \times \text{area} = I\pi a^2$, hence:

$$B_x = \frac{\mu_0}{4\pi}\frac{2\mu_m}{x^3}$$

This is strikingly similar for the expression for an electric dipole:

$$E_x = \frac{1}{4\pi\epsilon_0}\frac{2p}{x^3}$$

2 E- and B-Fields

We have already determined the following, for E- and B-fields respectively:

$$\int_S \underline{E} \cdot d\underline{S} = \frac{Q}{\epsilon_0} \tag{E}$$

$$\int_S \underline{B} \cdot d\underline{S} = 0 \tag{B}$$

We also know that the E-field is conservative, hence no work is done if a charge returns to its original position:

$$\oint \underline{E} \cdot d\underline{l} = 0$$

Is there an equivalent of this for B-fields?

Consider a circiular path of a B-field created around an infinite line of current at radial distance r. By symmetry, $\underline{B}$ and $d\underline{l}$ are parallel, and are constant for a fixed r. Hence:

$$\oint \underline{B} \cdot d\underline{l} = \oint B dl = B\oint dL = B2\pi r$$

We have also determined that:

$$B = \frac{\mu_0 I}{2\pi r} \implies \oint \underline{B} \cdot d\underline{l} = B2\pi r = \frac{\mu_0 I}{2\pi r} \times 2\pi r = \mu_0 I$$

This gives us Ampere's Law:

$$\boxed{\oint \underline{B} \cdot d\underline{l} = \mu_0 I_{\text{enclosed}}}$$

2.1 Example

Consider a long solid cylinder carrying a uniformly distributed current. The cylinder has radius R and length

Thu 12 Mar 2026 11:00

Lecture 16 - Electromagnetic Induction

1 Ampere's Law Continued

2 Magnetic Inductance

Faraday suggested that, if currents induce magnetic fields, that the opposite might be possible. Could a magnetic field be used to induce a current? He found that whenever field lines move or change, an electric field is induced, which charges in the field respond to.

3 Motion of a Conductor in Magnetic Field

4 Induced E-Fields

5 Faraday's Law

LC Introduction to Particle Physics and Cosmology

Thu 22 Jan 2026 16:00

Lecture 1 - Particle Physics I: The Standard Model

1 Course Introduction

Course Structure

- Particle Physics: 6 lectures in weeks 1 to 6.
  1. Introduction and the standard model.
  2. Experimental measurements.
  3. Interactions with matter
  4. Tracking detectors I
  5. Tracking detectors II
  6. Calorimeters and Particle Identification.
- Cosmology: 4 lectures in weeks 7 to 11.

Course Aims

- Overview of current methods in Particle Physics experiments.
- An emphasis placed on the questions and challenges.

For example, the LHC has already been programmed with experiments all the way up to 2041. Therefore, any detectors we design today only become relevant in over a decade, which makes good detector design decisions incredibly important. This course will equip us to understand what drives those design choices.

The course is assessed by a single one hour long exam, weighted half particle physics and half cosmology.

Recommended Texts

- Detectors for particle radiation (2nd edition), K. Kleinknecht (1998)
- Particle Physics, Martin and Shaw.
- High Energy Physics, D. H. Perkins (2nd through 4th editions)
- Feynman Lectures.
- Modern Particle Physics, M. Thomson.

2 Matter Particles

Fermions all have quantum spin 1/2. Spin is a purely inherent quantum property (like mass or charge) and has no classical representation, but is analogous to angular momentum. They are subject to Fermi-Dirac statistics, which means that no identical fermion in a system of fermions can have the same quantum number as any other. Fermions are divided into two types, quarks and leptons.

2.1 Quarks

There are three generations of quarks:

First Generation

- Up Quark (u), mass of ≈ 0.001GeV
- Down Quark (d), mass of ≈ 0.001GeV

Second Generation

- Charm Quark (c), mass of ≈ 1.3GeV
- Strange Quark (s) mass of ≈ 95MeV

Third Generation

- Top Quark (t), mass of ≈ 175GeV
- Bottom Quark (b), mass of ≈ 4.3GeV

"up-type quarks", u, c, t have electromagnetic charge $+2/3$ and "down-type quarks", d, s, b, have electromagnetic charge $-1/3$ (charges in units of e). Quarks do not ever exist alone in isolation.

2.2 Leptons

There are three generations of leptons, given by the electron e^-, muon, μ^- and tau τ^-. These all have charge of -1 (in units of e). These have masses (in MeV) of approximately $0.5, 105, 1800$.

These also have associated neutrinos, the electron neutrino, the mu neutrino and the tau neutrino, ν_e, ν_μ, ν_τ. These are not massless, but have a tiny mass many orders of magnitude smaller than their corresponding non-neutrino counterparts. These are neutrally charged.

Leptons are not subject to the strong interaction.

2.3 Hadrons

Quarks do not exist in isolation, but form bound states subject to the strong force called hadrons. There are two types of hadrons - baryons and mesons.

Baryons are formed from three quarks, which may be the same or different $q_1 q_2 q_3$.

Mesons are formed from a quark-antiquark pair, $q_1 \bar{q}_2$. Examples of baryons include the proton and the neutron, given by uud and udd.

3 Forces

As far as we know, there are four fundamental forces:

- Gravity
- Electromagnetism
- Strong
- Weak

For considering particle interactions, we disregard gravity as it becomes incredibly weak for small masses. Creating a complete theory that incorporates all four is an open question in physics. It's okay to neglect it, but it is unsatisfying.

We consider these forces as arising by the exchange (between two particles subject to a force between them) of particles called bosons. These have spin-1, so are called "gauge bosons". These are subject to Bose-Einstein statistics, which does not impose the same restriction as Fermi-Dirac for quantum numbers in a system.

For EM: The exchange particle is a photon, γ. This is represented on a Feynman diagram as a wiggly line. They are massless and couple to electric charge.

For Weak: The exchange particle is a $W^\pm$ or Z^0 boson. This is represented by a wiggly line or a dotted straight line. They are not massless, and have masses of approx. 80 and 90GeV respectively.

For Strong: The exchange particle is a gluon g. This is represented by a series of curls on a Feynman diagram. They are massless. They couple to "colour charge" which is just another quantum number analogous to electric charge. Just like electric charge has values $\pm$, colour charge has values we denote r, g, b

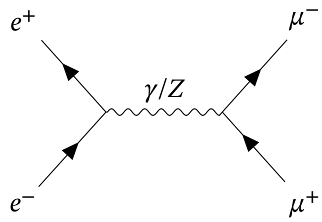
Quarks are subject to the strong, electromagnetic and weak interactions

Leptons are not subject to the strong interaction, but the e, μ, τ are subject to EM (ν is not as it is neutrally charged), and all are subject to the weak interaction. This makes neutrinos very difficult to detect as they are only affected by the weak interaction.

4 Feynman Diagrams

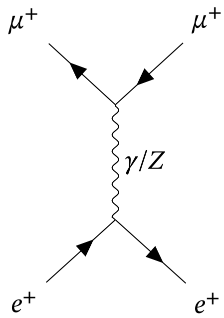
Feynman diagrams are space (y-axis), time (x-axis) diagrams to show allowed interactions between particles.

Consider a simple example of electron-positron annihilation. They travel towards each other, meeting and annihilating into either a photon or a Z boson. This is called a 'time-like exchange'. The boson then decays and we see pair production of two muons (one μ^- muon and one μ^+ antimuon).



Arrows in Feynman diagrams convey "fermion flow". This means that for a matter particle, the arrows aligns to the time axis. For antimatter particles, they antialign. Some conservation laws (i.e. charge) apply at the vertex level, while others only apply across whole processes.

We now consider a space-like exchange where the exchange is aligned with the vertical (space) axis. An antimuon scatters off a position like such:



5 Luminosity

We can determine the rate of interactions with the following:

$$W = \mathcal{L}\sigma$$

Where $W\,(\mathrm{s}^{-1})$ is interaction rate, $\mathcal{L}\,(\mathrm{cm}^{-2}\mathrm{s}^{-1})$ represents the luminosity (an attribute of the accelerator being used), and $\sigma\,(\mathrm{cm}^{-2})$ is the cross section, representing the underlying physics of the interaction.

These are investigated in greater detail in Lecture 03.

Mon 02 Feb 2026 12:00

Lecture 2 - Particle Physics II: Luminosity and Particle Signatures

1 Cross Sections

Consider a proton-proton interaction, producing some unknown particle X:

$$pp \to ppX$$

We have said that the rate of interaction is given by:

$$W = \mathcal{L}\sigma$$

Where $\mathcal{L}$ in $\text{cm}^{-2}\text{s}^{-1}$ is (coarsely) a parameter of the accelerator, describing its ability to produce collisions, and σ in cm^{-2} is a measure of interaction probability. Even though the particles are point-like, we treat them as having an effective area, and the magnitude of that area dictates how likely an interaction is to take place.

In this interaction, we have two protons (modelled as solid balls) passing immediately next to each other (one travelling clockwise and one counter-clockwise) around the accelerator. Assuming they pass immediately next to each other, and we model them as having radius 10^{-15}m, we have a a separation between the centres of each proton as $2 \times 10^{-15}\text{m}$, therefore a cross section of:

$$\pi \left(2 \times 10^{-15}\text{m}\right)^2 \sim 0.125 \times 10^{-28}\text{m}^2$$

To move this to a less annoying length scale, we define a new unit, the barn:

$$1\text{barn} \equiv 10^{-28}\text{m}^2 = 10^{-24}cm^2$$

In reality, this model may approximate a cross section, but it's not accurate. In reality, there's a much wider variation:

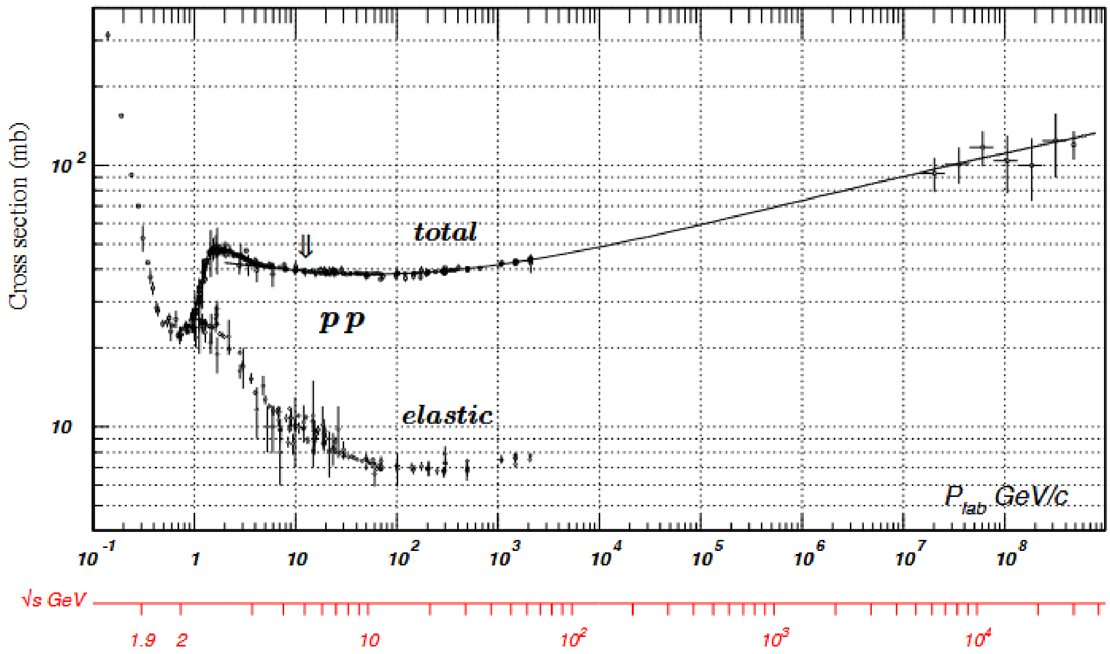


Figure 19.1

Here we have two x-axes running in parallel - the black axis is the momentum in a lab frame (as it hits some fixed target) while the red axis is the corresponding "centre of mass energy". How do we relate these two?

We want to know what the maximum mass of the particle we can generate is. In the lab frame, this requires us to take the momentum of the incoming and generated particle into account. We then have to take the final momentum of the system into account to conserve momentum, as the whole system must continue moving in the direction of motion for conservation.

Translating into the frame of reference given by the system centre of mass gives us a system where the two masses can be thought of as approaching each other with equal and opposite momenta. Since the total momentum is zero, the objects (incoming particle and the target) can theoretically hit each other and come to a complete stop. Because the system does not have to keep moving after the collision, all of the energy in this frame is free to be converted into the mass of a new particle.

While this may not be accurate in practice, it gives us a hard upper maximum for the possible energy available for production.

We can show relativistically that the energy in the centre of mass frame (labelled $\sqrt{s}$, the energy available for particle production) is:

$$E_{\text{com}} = \sqrt{s} \propto \sqrt{p_{\text{lab}}}$$

This can be seen in the final values of the x-axis, which (both starting approx. 1) are 10^8 and 10^4. This tells us that we reach diminishing returns with a fixed target collider - increasing energies by 8 orders of magnitude only increases the energy available for particle production by 4 orders. This is an inherent inefficiency of a fixed target collider.

This is much cheaper as its easier to align, we just fire a beam at a block of (for example) lead. It also makes it quite easy to change the target material. Changing materials in a colliding-beam collider (where two beams are fired in opposite directions, one clockwise and one anticlockwise, and collide with each other), i.e. to fire lead nuclei requires an extensive recalibration process.

In a dual-beam collider, only a very small proportion of the accelerated material from each beam actually interact with each other. In a fixed target collider, the target is much more dense, so we see a higher rate of interactions.

In summary, the advantages of a fixed target collider are:

1. Easier to collide.
2. Easier to change the target.

3. Very high density

2 Luminosity

2.1 Fixed Target Case

Consider a fixed target collider. We want to build an expression for the luminosity of this setup.

We have some incoming flux of particles (per second per unit area), J, incident on the block of material with density ρ, thickness t and mass of one nucleus m_A. The beam, modelled by a cylinder, illuminates some circular portion of the block, with area A.

Consider our interaction rate W. This is given by (where V is the volume of a cylinder from the illuminated beam, of thickness equal to the target):

$$W = \underbrace{JA}_{\substack{\text{incident particles}\\ \text{per unit time}}} \times \underbrace{\frac{\rho}{m_A}V}_{\substack{\text{total number of}\\ \text{target particles}}} \times \underbrace{\frac{\sigma}{A}}_{\substack{\text{probablity of}\\ \text{interaction}}}$$

$$W = JA \times \frac{\rho}{m_A}At \times \frac{\sigma}{A}$$

$$W = J \times \frac{\rho}{m_A}At \times \sigma$$

$$W = \frac{J\rho At}{m_A}\sigma$$

Comparing to $W = \mathcal{L}\sigma$ gives the luminosity as:

$$\mathcal{L} = \frac{J\rho At}{m_A}$$

2.2 Colliding Beam Case

In a colliding beam case, the derivation is more (and too) complex. It is equal to:

$$\mathcal{L} = \frac{f_{\text{rep}} n_b N_1 N_2}{4\pi\sigma_x\sigma_y}$$

Where:

- f_{rep} is the repetition frequency, i.e the rate of the beam passing the collision point.
- n_b is the number of bunches in the beam.
  - A beam can be thought of as as string of pearls, rather than a single discrete constant beam - i.e. clusters of particles “bunches”, followed by empty space between them.
- N_1, N_2 are the number of particles per bunch for each beam
- σ_x, σ_y are the dimensions of the beam in the x- and y-direction, not a cross section as previously.

3 Examples of Detectors

We have some interaction point producing a spray of particles, and surround this with a series of different detector layers. Each produced particle will trigger a different subset of these layers, defining a unique signature we can use to identify produced particles.

Broadly, in some generic detector, we have:

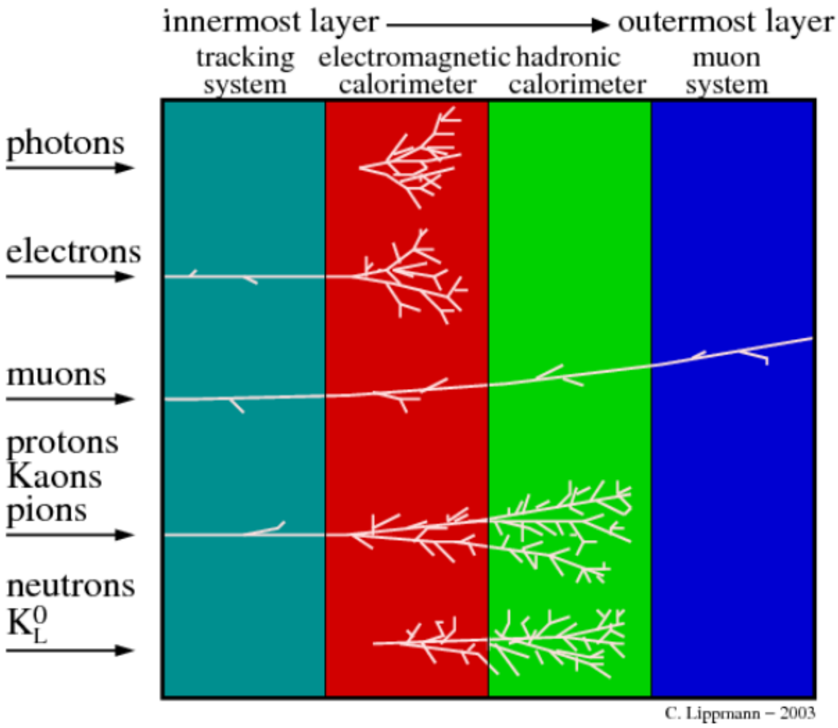


Figure 19.2

The tracking system is non-destructive. Formed of layers of silicon, a charged particle with ionise small portions of each layer. This can be turned into a signal. Neutral hadrons and photons will pass straight through, but charged particles will leave a deposit of charge and pass through unaffected.

We then move to destructive layers. Particles leave tree-like structures as they pass through these layers and create a shower of particles.

3.1 Decays

Most decays take place over a very low time scale, $< 10^{-8}$s and produce a final state made up of some subset of the following: $\gamma, \pi^+, \pi^-, \kappa^+, \kappa^-, p, n, \pi^0, e^+, e^-, \mu^+, \mu^-$.

So, in order to detect some exotic particle, we assume that it will either persist long enough to be detected itself, or decay into some subset of these known particles which will reach out detector.

Consider a parent particle A, for example B^0 ($\bar{b}d$) decaying into two child particles B, C, given by a "J Psi" ($c\bar{c}$), J/ψ and a "K Short", K_s^0 ($d\bar{s}$):

$$\begin{aligned} B &\rightarrow J/\psi \quad K_s^0 \\ \bar{b}d &\rightarrow c\bar{c} \quad d\bar{s} \end{aligned}$$

This decay has a lifetime of 10^{-12}s, via the weak interaction due to the change in quark flavour. The J Psi decays into e^+e^- or $\mu^+\mu^-$ via the EM interaction very rapidly in 10^{-21}s (its lifetime is governed by the strong interaction, which it may also use to decay via, even though we detect it via the EM decay path). The K Short decays into $\pi^+\pi^-$ or $\pi^0\pi^0$ again via the weak interaction with lifetime in 10^{-10}s.

The range of a particle is given by:

$$\text{Range} = \beta\gamma c\tau$$

Where τ is a time scale (lifetime), c is the speed of light, γ is the Lorentz Factor and β scales the range based on the speed actually being travelled. We also have:

$$E = \gamma m$$

$$p = \beta\gamma m$$

And familiarly:

$$E^2 - p^2 = m^2$$

If the B^0 has energy 20GeV and mass 5GeV, we have $\gamma = 4$ and this gives a range of $\approx$ 1mm. This is so small we will never observe it directly. The K_s^0 however has range $\approx$ 30cm, so is detectable.

Crucially:

- SI lifetimes: $\sim 10^{-21} - 10^{-24}$s
- EM lifetimes: $\sim 10^{-16} - 10^{-20}$s
- WI lifetimes: $\sim 10^{-12}$s

Thu 05 Feb 2026 16:00

Lecture 3 - Particle Physics III: Particle/Matter Interactions

In order to look at specific detectors and how they work, we need to consider how particles interact with matter. We'll consider categories of particles and their standard interactions, and use this to build a model for how we can build a detector.

We've looked at this previously:

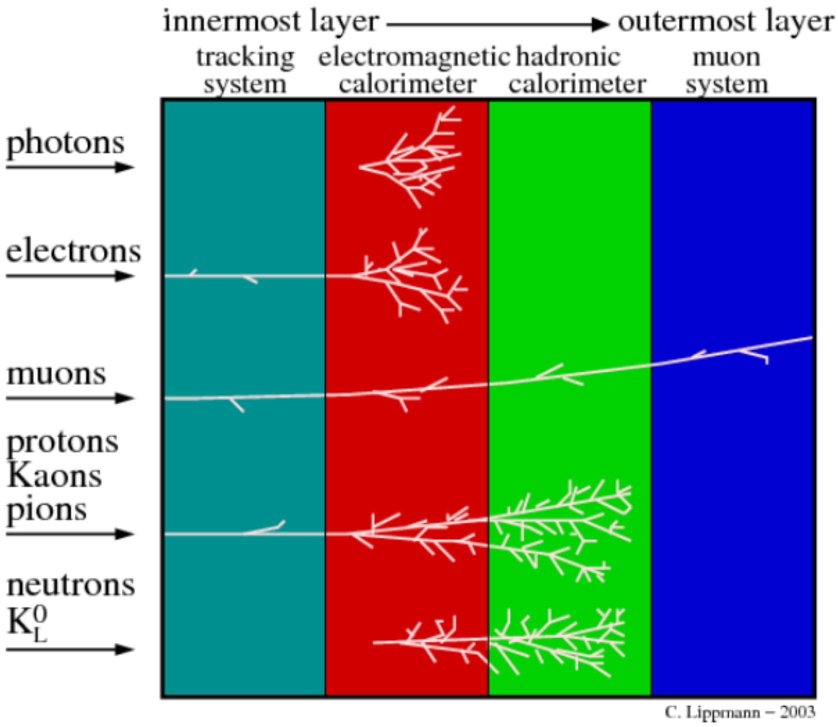


Figure 20.1

For example:

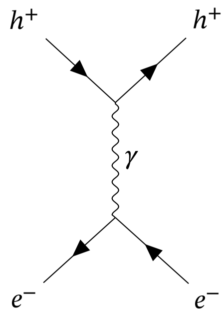
- Charged particles produce ionisation in the non-destructive tracking system, leaving deposits we can detect as signal.
- Electrons and photons leave distinctive signatures (shape of the shower of particles) in the electromagnetic calorimeters.
- Hadrons leave deposits in the electromagnetic calorimeter, but they dump all their energy (and are mostly identifiable by) the hadronic calorimeter.
- Muons make it through all of the previous layers, and are picked up at the very end by the outermost muon system.

1 Charged Particles

1.1 Ionisation

This is the process that powers the tracking system. A charged particle, h comes into the system (it could be a hadron or a lepton). It exchanges a photon with an electron in the system and excites the electron, ejecting it from the atom.

Strictly:



We can to try to characterise the rate of energy loss of the charged particle. The average rate of energy loss per unit distance is:

$$-\left\langle \frac{dE}{dl} \right\rangle \propto \ln E$$

Where E is energy, and the sign is negative as energy is being lost here.

For example, a real detector here might be two charged plates with a high voltage sandwiching a gas mixture. As the gas mixture is ionised, the ionised portion drifts towards one of the electrodes where it induces a detectable current.

1.2 Bremsstrahlung

Translates to "braking radiation". Consider a free electron radiating a photon. This is impossible for a free particle (as if we consider the electron's rest frame, emission would violate conservation of energy). We therefore need a source of external interference, in this case matter.

An electron is accelerated by nuclear change as it passes through material and is scattered. This scattering causes bremsstrahlung photon emission. The average rate of energy loss is given by:

$$-\left\langle \frac{dE}{dx} \right\rangle \propto \frac{E}{m^2} \propto \frac{E}{X_0}$$

An electron will then generate more bremsstrahlung than a muon, due to its much smaller mass. X_0 is called radiation length, and is covered in future lectures.

1.3 Cherenkov Radiation

Consider a charged particle moving through a (non-vacuum) material. It emits photons at some angle θ_c if the particle is moving faster than the speed of light in the medium (note this does not violate relativity, as the speed of light in a medium is less than the speed of light in a vacuum).

These emitted photons cause a coherent wavefront to form around the particle's trajectory, forming a cone around the direction of travel. The angle θ_c is given by:

Geometrically, after time t, the emitted photon has travelled ct/n and the particle vt, hence:

$$\cos\theta_c = \frac{ct/n}{vt} = \frac{c}{nv} = \frac{1}{n\beta}$$

Where n is the refractive index of the material. This is analogous to shock waves forming when an object goes faster than the speed of sound.

A planar detector will take a single cross-section through this cone, detecting rings around the point the charged particle passed through the material. By measuring this ring, we can work out the speed of the particle, and use this along with a measured momentum (in a tracking detector) to work out the mass of the particle.

Again:

$$-\left\langle \frac{dE}{dx} \right\rangle \propto z^2 \sin^2 \theta_c$$

Where z is the particle charge in units of $|e|$. It is important to note that this is a very small energy loss for the particle. It may emit $10^3\gamma$/cm, and only lose a few keV/cm

2 Photons

2.1 Photoelectric Effect

We have a photon strike a atom, transferring energy and forcing an electron to be ejected. The max kinetic energy of the electron is given by the photon energy minus some amount of work to eject it:

$$E_{kmax} = hf - \phi$$

Where ϕ, the work function, is the energy required to liberate one electron from the atom's surface.

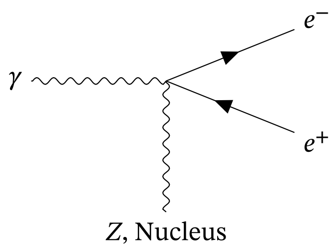
2.2 Compton Scattering

See QM1. A photon scatters off a quasi-free electron in an atom. The photon and electron are scattered with a change in energy:

$$\gamma + e^- \to \gamma' + e^{-\prime}$$

Thomson scattering is a low-energy form of scattering where the energies do not change, and Rayleigh scattering is a very low energy limit, where the interaction takes place between a photon and multiple atomic electrons.

2.3 Pair Production



A photon interacting with a nucleus can (leaving the nucleus unscathed) produce a particle and the corresponding antiparticle, typically a positron and an electron. This has a minimum photon energy of $E_\gamma > 2m_ec^2$[1] to ensure conservation of energy isn't violated and must occur in the presence of matter (for the same reason as Bremsstrahlung). The photon is not present in the final state.

However, say a photon has precisely the energy required to create a pair. This would (as it stands) create a pair of electrons with zero momentum (or very small momentum). However, a photon always has a momentum given by its energy, so momentum is not conserved. If the pair travel to attempt to resolve this, they now have some kinetic energy too, which means the photon must have a higher energy, and hence a higher momentum. This creates mismatch, we cannot conserve both energy and momentum in this situation.

By this interaction taking place in the presence of a nucleus, the nucleus can absorb some recoil (via exchange of a virtual photon) to ensure conservation of energy and momentum are satisfied.

This must take place in a Coulomb field to contribute this photon. The present nucleus has charge $z|e|$.

Lets consider the cross section of this interaction:

$$\sigma_{\text{pprod}} \propto \frac{7}{9} \frac{A}{N_A} \frac{1}{X_0}$$

[1] $m_ec^2 = 0.511\text{MeV}$

X_0 is the radiation length. It is a complex property of the material but is proportional to $1/z^2$.

3 Neutrinos

Neutrinos are really difficult to detect, as they are only effected by the weak interaction and have no ionisation/pair production etc - they are effectively non-interacting. We detect them by detecting the products of a weak force interaction.

For example:

- A muon neutrino can exchange a W boson with a proton, which becomes a neutron. The muon neutrino becomes a muon and the up quark in the proton becomes a down quark in the neutron.
  - We can detect the muon that's produced as the neutrino passes through the detector and interacts.
  - We build a massive multi-tonne detector that puts a large amount of mass in the way of the neutrino, often water.
  - We do this in the hope that it will interact with some of the matter and produce the more-detectable muon.
- The neutrino can scatter off a nucleus.
  - A mall amount of energy is exchanged with the nucleus, depositing a small amount of heat energy in the detector's matter.
  - Nothing changes/decays/is produced etc other than a small amount of heat.
  - We build a very sensitive detector capable of detecting this very small amount of heat.

4 Hadrons

Hadrons are really complex in their interactions. They can have inelastic nuclear interactions, with large energy deposits at a small number of sites (compared to an ECal shower where we'd expect to see many smaller deposits). This arises as a result of the nucleus becoming excited or breaking apart entirely and producing a chain of secondary hadrons/particles and a change in the nucleus.

These secondary particles can go on to interact again. This are complex and messy objects which can fragment off to cause many secondary impacts.

5 Conclusion

In summary:

- The key interactions we care about are ionisation, Bremsstrahlung and Cherenkov radiation for charged particles.
- For photons, we care about pair production.
- For hadrons, hadron showers are messy and complex. We add inelastic nuclear interactions on top of an electromagnetic component from ionisation and everything becomes rather tricky.

Mon 09 Feb 2026 12:00

Lecture 4 - Particle Physics IV: Vertexing and Tracking Systems

In our familiar layered model:

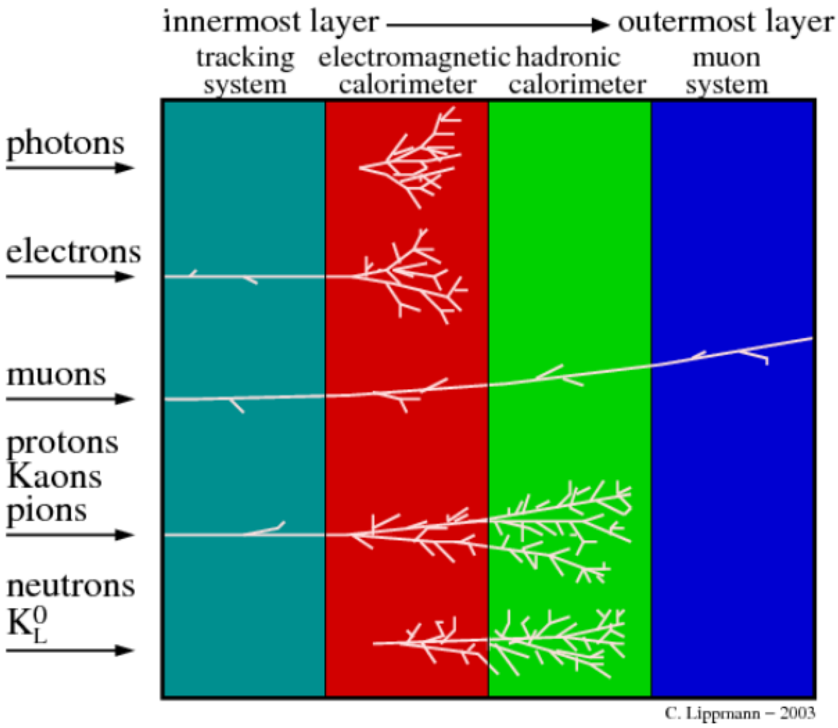


Figure 21.1

The first subdetector type is the initial tracking system which provides a non-destructive estimate of a particle's trajectory.

1 Tracking Systems

Purpose: To determine the trajectory of charged particles (usually in the presence of a magnetic field), in order to infer the momentum of the particle. This happens by ionisation, whereby the particle leaves small deposits of charge in either a gas or layers of a semiconductor sensor.

Position: As the following layers are destructive (i.e. calorimeters will absorb the particle entirely in order to make an energy measurement, while a muon system will block any other particle), we need to place the tracking system first for it to be effective.

We also want it placed near the primary interaction point. Since we want to use position measurements in the tracking system to extrapolate backwards and determine the origin point (where the collision took place), keeping them as close as possible to this origin point reduces the overall uncertainty of the track.

For a long path, a small uncertainty in position points propagates to a much larger uncertainty in track the further away we are from the initial collision point.

1.1 Material Budget

- In an ideal world, the tracking detector would have a perfectly massless and lightweight material, in order to reduce the risk of scattering. However, we need some mass in order to measure ionisation, so it's a constant trade off between the two.
- We want as small a number of radiation lengths X_0s within / upstream of the tracking system as possible.
- Some material is unavoidable, for example in the LHC there needs to be a conductive shield around the beam to prevent the large magnetic fields from inducing currents that would interfere with the sensitive measurement equipment. It also separates the highly pure vacuum of the beam pipe from the slightly less pure vacuum of the outer portion containing the LHCs electronics.

Radiation Length This is an inherent property of each material and is a measure of energy loss. A particle of energy E_0 passes through a distance of one radiation length and loses a factor of energy $1/e$.

$$E(x) = E_o \exp\left(\frac{-x}{X_0}\right)$$

For example:

- For Cu: $X_0 = 15\text{mm}$.
- For Be: $X_0 = 35.2\text{cm}$.

The processes by which energy is lost depends on the particle species in question. For example, an electron loses energy by Bremsstrahlung much more rapidly compared to a muon, due to the differences in mass (average energy loss $\propto 1/m^2$). To define radiation length, we use an electron as a scale.

This lets us talk about "material budget", as adding more material causes a higher uncertainty. For each particle path we care about, we want the fewest radiation lengths possible.

2 Measuring Trajectories

For a particle passing through a tracking system, we reconstruct its trajectory by measuring individual energy deposits (called "hits") caused by ionisation in a large volume (typically $1\text{m} \times 1\text{m} \times 1\text{m}$)

This volume is made up often of many layers, so we can gain an idea of the particles position as it passes through each layer and use this to extrapolate. A tracking system has a resolution of a few $100\mu\text{m}$ and may leave $10 - 100$ hits.

In a "vertex detector", we use a smaller system to reconstruct tracks at/around the interaction point with a higher precision - typically aiming for $\sim 10\mu\text{m}$. It is typically a silicon detector (layers of silicon detector sheets) and generally records fewer hits (< 10).

In a "tracking detector" we use a much larger system commonly further away from the primary interaction point. In a vertex system, there's no magnetic field so while we can use it to extrapolate back to find the PIP, we cannot use it to estimate momentum.

The LHC-b experiment for example has both a vertexer very close to the PIP (to determine the PIP location), and a much more substantial tracking system behind a magnet further away (to determine the deflected track in a magnetic field and hence the momentum.)

3 Measuring Momentum

The motion of a charged particle in a magnetic arises due to the Lorentz force and is proportional to charge:

$$\underline{F} = q(\underline{E} + \underline{v} \times \underline{B})$$

Where $\underline{F}$ is the force, $\underline{E}$ is the electric field, $\underline{v}$ is the particle velocity and $\underline{B}$ is the magnetic field. Assuming we are only dealing with a magnetic field, and taking magnitudes, we have:

$$\boxed{\underbrace{p}_{\text{GeV}} = 0.3 \times \underbrace{B}_{\text{T}} \times \underbrace{q}_{|e|} \times \underbrace{r}_{\text{m}}}$$

Where the magnetic field produces curvature of radius r in the plane perpendicular to the B-field. Motion parallel to $\underline{B}$ is unchanged.

In 3D, and assuming we are contained within the magnetic field, the particle will follow a helix of constant radius of curvature. This assumes that there are no non-conservative forces (i.e. scattering). In the ideal case, there is no work done, meaning in our force:

$$\underline{F} = q\underline{v} \times \underline{B}$$

The force and the velocity are perpendicular, so $\underline{F} \cdot \underline{v} = 0$.

3.1 Quantitatively

Consider a particle path with a constant radius of curvature R. Consider three hits, with vertical separation between the top and the bottom being $L/2$. The distance between the top/bottom hit and the origin is l, and the final distance between this straight line and the deflected path is the "sagitta" s.

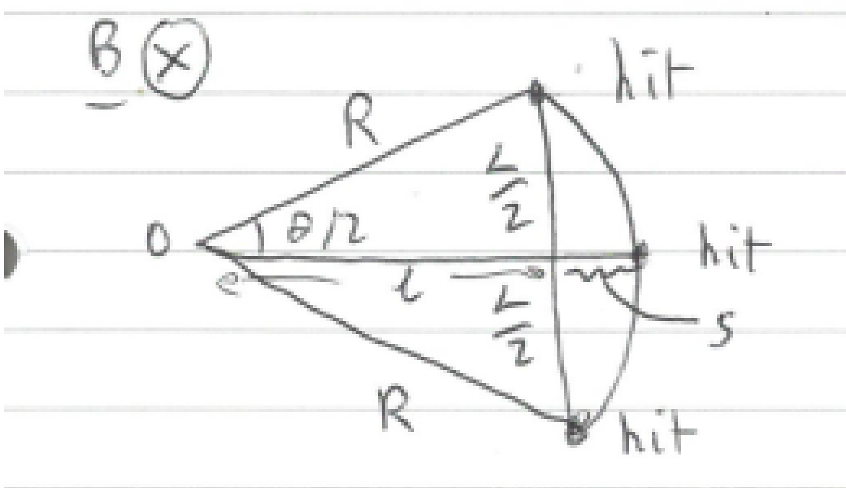


Figure 21.2

We know $L/2$ (the vertical separation between hits) as we've build the detector so know the resolution, and we want to measure the sagitta, the deviation from a straight line.

We know that:

$$p = 0.3BqR$$

And from the diagram:

$$R = l + s$$
$$l = R\cos\left(\frac{\theta}{2}\right)$$

Putting these together:

$$s = R\left(1 - \cos\frac{\theta}{2}\right)$$

Assuming the bending is gentle in a detector, so θ and s are small. Hence we apply a small angle approximation:

$$\cos\frac{\theta}{2} \approxeq 1 - \left(\frac{\theta}{2}\right)^2 \frac{1}{2!} + \left(\frac{\theta}{2}\right)^4 \frac{1}{4!} + \ldots$$

Taking the first order:

$$s \approx \left(\left(\frac{\theta}{2}\right)^2 \frac{1}{2!}\right) = \frac{R\theta^2}{8}$$

We can also use the small angle approximation for sine:

$$\sin\frac{\theta}{2} = \frac{L}{2R}$$

For small theta:

$$\sin\frac{\theta}{2} \approx \frac{\theta}{2}$$

$$\theta \approx \frac{L}{R}$$

So:

$$s = \frac{L^2}{8R}$$

Hence, finally, we have:

$$p = 0.3Bq\frac{L^2}{8s}$$

3.2 Uncertainties

We want to find the uncertainty on momentum, σ_p/p:

$$\frac{\sigma_p}{p} = \frac{\sigma_s}{s} = \frac{8p}{0.3BqL^2}\sigma_s$$

However we want to use our uncertainties on individual hits, x_1, x_2, x_3 etc. We have (and will not derive):

$$\frac{\sigma_p}{p} = \frac{\sigma_{xy}P}{0.3BL^2}\sqrt{\frac{720}{N+4}}$$

Mon 09 Feb 2026 12:00

Lecture 5 - Particle Physics V: Calorimetry

Now that we've finished talking about determining the principal interaction point and the momentum of produced particles in a tracking detector, we move onto calorimeters:

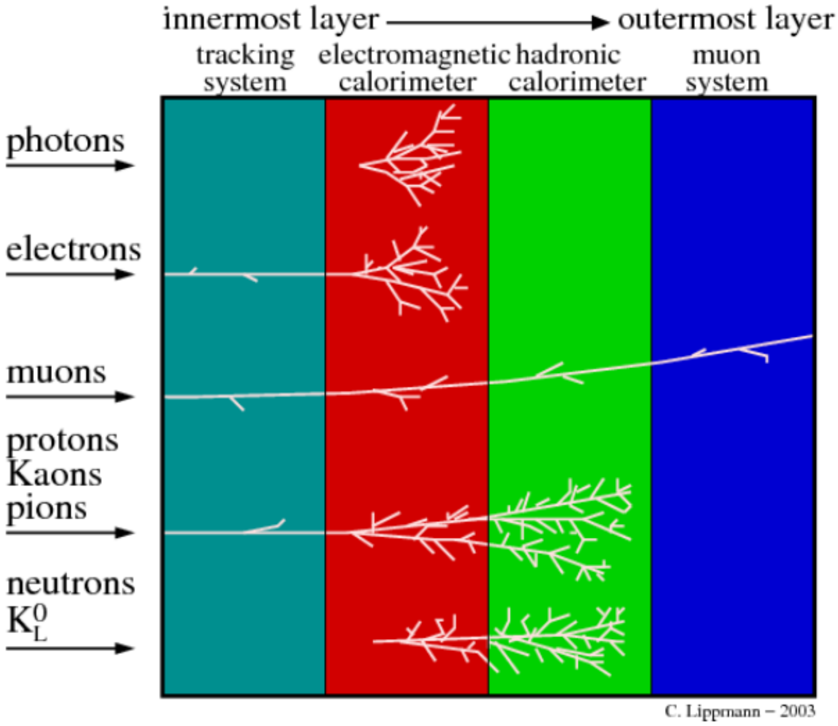


Figure 22.1

1 Materials and Energy Loss

Energy loss in a material is described by the Bethe formula, which gives mean energy loss by ionisation for unit path length:

$$\left\langle -\frac{dE}{dx} \right\rangle = Kz^2 \frac{1}{\beta^2} \left[\frac{1}{2} \ln \frac{2m_e c^2 \beta^2 \gamma^2 W_{\text{max}}}{I^2} - \beta^2 - \frac{\delta(\beta\gamma)}{2} \right]$$

This is:

- Valid for $\beta\gamma < 1000$ within a few percent precision.
- Includes dependence on the medium, I, Z, A etc.

The good news is that we do not need to know this formula. We do however need to know some key features:

- $\frac{dE}{dx} \propto \frac{1}{\beta^2}$ below some minimal value of $\frac{dE}{dx}$
- $\frac{dE}{dx} \propto \ln(\beta^2\gamma^2)$ above this minimal value of $\frac{dE}{dx}$. This is called "relativistic rise".
- This minima happens at approximately $\beta\gamma \sim 3 - 4$.

- At large $\beta\gamma$, polarisation of the medium causes saturation (effectively a plateau).

Broadly, we expect to see a fall up until some minimal value, then a relativistic rise, and then a plateau:

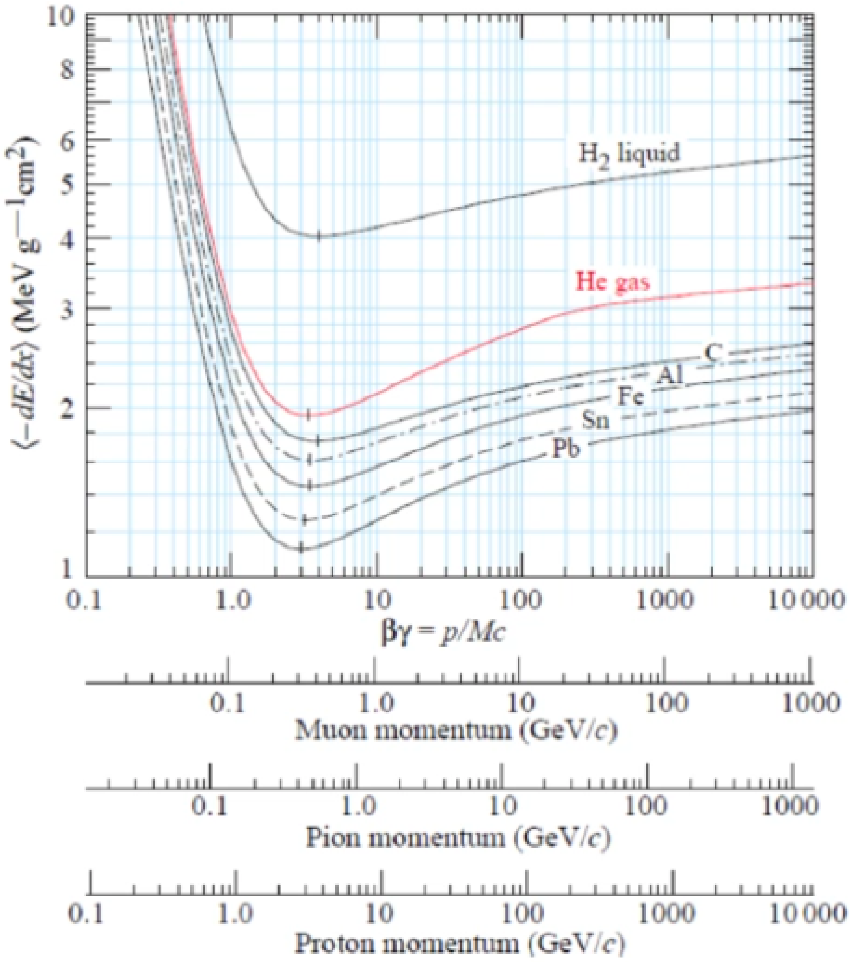


Figure 22.2

2 Calorimeters

The purpose of a calorimeter is to determine the total energy of an incident particle. It does so by absorbing the particle entirely, creating a measurable signal which is proportional to the energy of the incident particle. As it is destructive, it is always located after the non-destructive tracking system. It has no need for the B-field of the tracking detector, and may be called a "calo", "ECAL" or "HCAL".

There are two types of calorimeter:

- Electromagnetic Calorimeter: ECAL.
- Hadronic Calorimeter: HCAL.

We characterise the length of an ECAL in terms of radiation length X_0, typically $15X_0 - 30X_0$. It works by bremsstrahlung, and emitted bremsstrahlung photons causing pair production and more bremsstrahlung emission etc.

A HCAL works on nuclear hadronic interactions, so we instead define an interaction length λ_{int}, typically $5\lambda_{\text{int}} - 8\lambda_{\text{int}}$.

They don't measure the total energy of a particle if particles produced in the process pass straight through, so not all the energy in an ECAL will be captured, as some will pass straight through. Muons, neutrinos and sometimes pions pass straight through and escape, they are not (may not for a pion) be detected by the calorimeters.

An ideal calorimeter has an output signal "response" which is proportional to the energy of the input particle. We would ideally like a directly proportional linear relationship, but this may not always be realistic.

If we already know mass (from Cherenkov rings, to be discussed later in the course) and momentum from the tracking detector, why do we need a calo when we can just use $E^2 - p^2 = m^2$ to determine energy?

- Cherenkov rings and tracking systems only work with a charged particle. The calculation method is not sensitive enough, as the Cherenkov+tracker setup cannot detect γ, ν or neutral hadrons.
- Having a direct energy measurement adds another constraint to our system - the more information we can get the better. A direct energy measurement improves resolution and is more likely to be accurate than determining it from two other uncertain quantities.
- At very high momentum, (with little Coulomb scattering) the relative uncertainty of the transverse momentum of a tracking system is given by:

  $$\frac{\sigma_{p_\perp}}{p_\perp} \propto \frac{p_\perp}{BL^2}$$

  Therefore at a higher momentum, we have a higher uncertainty. At high momenta, there is a smaller deflection, so a high momentum track will just pass rapidly through the field with very little deflection and is therefore much more difficult to measure accurately - the tracking system degrades at higher momenta.

  For a calorimeter, we have nicer behaviour at high momenta:

  $$\frac{\sigma_E}{E} = \frac{a}{\sqrt{E}} \oplus b \oplus \frac{e}{C}$$

  Where:

  - a arises statistical fluctuations inherent to the measurement.
  - b arises from calibration effects (i.e. from non-linearity)
  - c arises from electronic noise.
  - $\oplus$ means to "add in quadrature", $(p \oplus q \oplus r) = \sqrt{p^2 + q^2 + r^2}$
- Trackers are relatively slow as they require ionisation clouds drifting towards a collection point and involve quite a computationally complex problem to recognise the patterns and reconstruct the track. A calorimeter relies on "scintillation" to convert energy to a response, which is much faster.

  This is important for triggering. Every 25ns, we need to decide whether or not to keep an event (as there is nowhere near enough storage/processing power) to perform detailed track reconstruction for every single event. We need a calorimeter to participate in this.

2.1 EM Calorimeters

There are two types of electromagnetic calorimeters, sampling or homogeneous. A sampling calorimeter is layered, with a layer of absorber producing a shower of particles (i.e. lead) in front of a collector layer with a scintillator that generates a signal when particles pass through.

HCALs are always sampling-type calorimeters, as it takes a lot more mass to slow down/break apart a hadron compared to a lighter charged particle.

Lets build a simple model of an electromagnetic shower that might be detected by a calorimeter:

We have a high energy photon (so we can mostly disregard scattering etc) enter the detector. This pair produces (i.e. an electron and a positron) which emits photons via bremsstrahlung emission. These emitted photons can then go on to pair produce themselves, generating (i.e.) an electron and a positron again, which emit more photons via bremsstrahlung, etc.

The maximum energy is deposited when the average particle energy (of particles developing in the shower) is the "critical energy".

Here, a photon has entered from the left, and the first few radiation lengths have left the material intact.

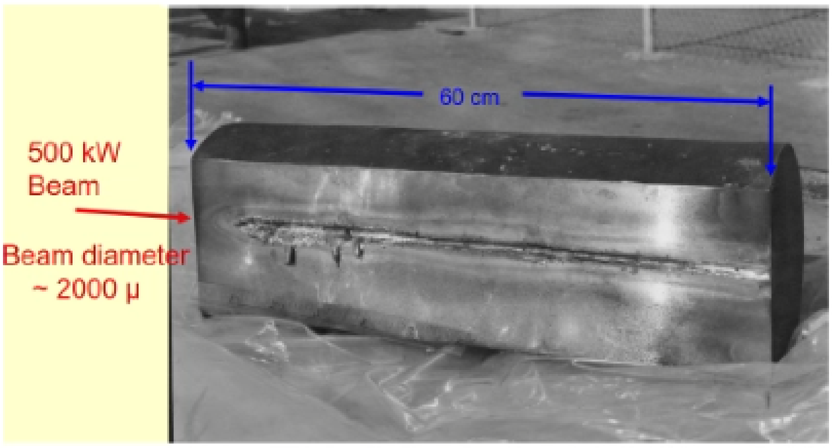


Figure 22.3: Damage to a copper block that has had energy dumped into it via this process.

After the first few radiation lengths, we have the "critical point", where a large amount of energy has been dumped. This point is defined as the point where the probability of bremsstrahlung and ionisation are equal.

The shower deposits a constant amount of energy per X_0 travelled (by definition of radiation length). Each step in path length, in terms of X_0 has a doubling number of particles and a halving individual particle energy. The angle of photon emission is fairly small, so the shower is narrow. This is unlike a hadronic calorimeter that has a wider shower. The width of the shower is given by the multiple scattering of the e^+ and e^-.

When the energy of the particle is less than the critical energy, ionisation dominates the interactions and the shower development stops rapidly. When the particle energy is $> E_c$, pair production and bremsstrahlung dominate.

After depth t (in units of X_0), the number of particles is 2^t, and the average particle energy is $E_0/2t$.

The shower stops developing after $E = \frac{E_0}{2t} < E_c$. This happens when $E = 2^t = E_0/E_c$. Hence:

$$t_{\max} \log 2 = \log\left(\frac{E_0}{E_c}\right)$$

$$t_{\max} = \frac{\log\left(E_0/E_c\right)}{\log 2}$$

This is what defines how deep our calorimeter needs to be in order to collect the energies we are expecting to require.

Mon 16 Feb 2026 16:00

Lecture 6 - Particle Physics VI: Calorimetry II and PID

1 EM Calorimetry: Shower Shape

The energy deposited vs the depth in material can be expressed as:

$$\frac{dE}{dt} = E_0 b \frac{(bt)^{a-1} e^{-bt}}{\Gamma(a)}$$

Where t is expressed as a length in units of radiation length $t = x/X_0$ and $\Gamma(a)$ is the standard gamma function $\Gamma(x) = \int_0^\infty t^{x-1} e^{-x}\, dx$. a, b are both constants.

For small t (early in the detector): t^{a-1} dominates as $e^{-bt} \approx 1$

For large t (later in the detector): e^{-bt} dominates.

Considering a plot of $\frac{dE}{dt}$ against t, we initially have polynomial growth, reaching a peak. We then from that peak decay exponentially with an infinite tail.

The maximum point of the shower is given at $t = t_{\text{max}}$ and is achieved by further differentiating dE/dt and is given by:

$$t_{\text{max}} = \frac{a-1}{b} = \ln\left(\frac{E_0}{E_c}\right) + C_{\gamma \text{ or } e^-}$$

For a photon, $C_\gamma = 0.5$ and for an electron, $C_e = -1$. Hence, the shower maximum is at a smaller value of t for an electron/positron compared to photon.

Crucially, we want to find the total energy deposited by the particle. We get that by integrating dE/dt:

$$\int_0^{\text{calo thickness}} \frac{dE}{dt}\, dt$$

While the tail is an infinitely long decaying exponential, we only care about the portion of the tail which is inside the calorimeter. Any energy that would be deposited past this point is lost:

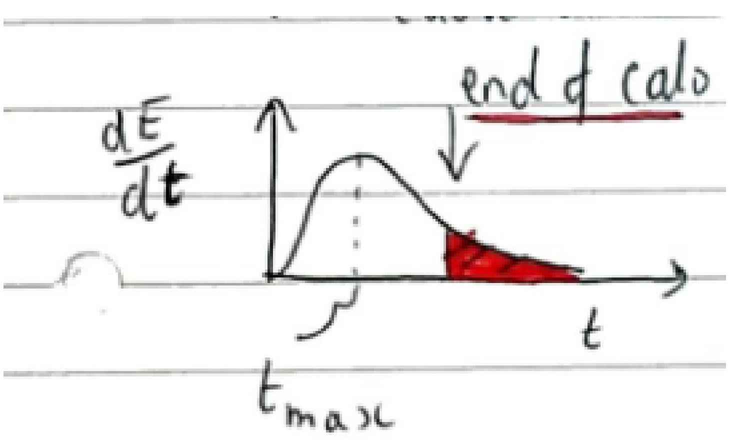


Figure 23.1

If the calorimeter is too thin, we need to make a large correction to adjust for this and we therefore have a bigger uncertainty. We want to ensure that the calorimeter is thick enough to capture the vast majority of deposited energy with minimal loss.

1.1 Resolution

The resolution for the calorimeter varies heavily based on the material used to construct it. Typically, however:

- ECAL resolution: $2 - 10\%$
- HCAL resolution: $50 - 100\%$

Different materials will have different characteristic radiation lengths,

For example:

- W has $X_0 = 0.35$cm, $\lambda_{\text{int}} = 9.9$cm, $R_M = 0.93$cm
- Pb has $X_0 = 0.56$cm, $\lambda_{\text{int}} = 17.6$cm, $R_M = 1.6$cm
- Fe has $X_0 = 1.8$cm, $\lambda_{\text{int}} = 16.7$cm, $R_M = 1.7$cm
- Cu has $X_0 = 1.4$cm, $\lambda_{\text{int}} = 15.3$cm, $R_M = 1.8$cm

R_M is the "Moliere Radius" and is the radial width in which 90% of the shower's energy deposit is confined. Note that we have different design considerations for an HCAL vs an ECAL, for example the size of an HCAL is generally much larger than that of an ECAL.

2 Particle Identification (PID)

We can now measure a particle's momentum and energy, and we now want to determine the mass of the particle. Surely we can do that with:

$$E^2 - p^2 = m^2$$

Unfortunately not...the resolution (especially for energy on a calo) is pretty poor, so we would end up with an unreasonably large uncertainty on our mass, too large to do reasonable particle identification with. Instead, we:

- Use information from the whole ensemble of detectors.
- Use hypothesis tests: proposing a specific particle type and then using all the information we have (i.e. momentum) see if this is a reasonably likely outcome.
- For example, we can use information on whether or not the particle reached the muon chambers to help narrow down what it could be. Alternatively, what is the pattern of energy deposit observed in the ECAL or HCAL?
- We can also use dedicated PID, such as:
  - A time of flight detector, determining the particle velocity (hence mass, with momentum measurements from a tracker).
  - Cherenkov detectors (not useful at high momenta, i.e >100GeV): As a recap, the particle emits a cone of photons at angle θ_c which depends on the refractive index of the material and β. Measuring this cone allows us to determine θ_c hence β hence v.

Here is an example of a detector setup:

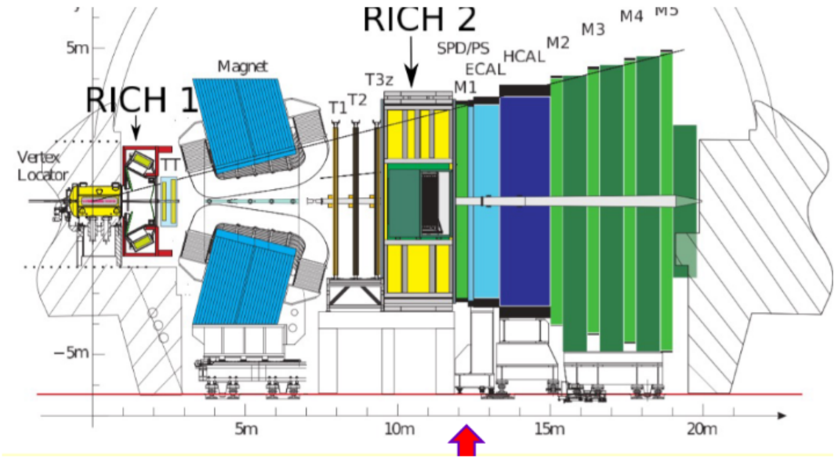


Figure 23.2

We have covered the vertex locator, magnet, tracking stations T1, T2, T3, the HCAL, ECAL and (while we haven't yet touched on them), we have 5 muon systems.

We are yet to discuss RICH1 and RICH2. These are called "Ring Imaging Cherenkov Detectors". These cones of Cherenkov emissions are focussed using a magnet onto a plane, where they form rings. The radii of these rings can be measured and the detector geometry can be used to determine the Cherenkov angle. The two detectors work similarly, but cover different ranges of measurable momenta.

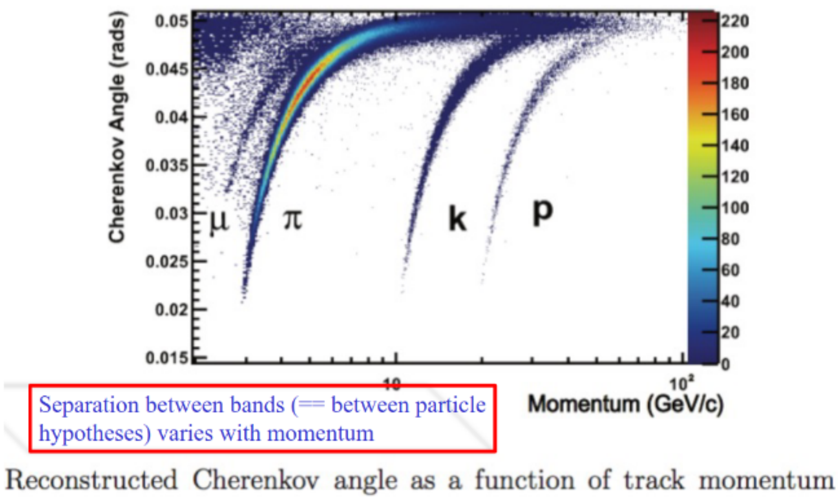


Figure 23.3

This Cherenkov angle can therefore be used to discriminate against particles, up to a limit of momentum, past which they become fairly useless.

NOTE: There was a practice exam question here that would be good to come back to during revision.

End of Particle Physics

Mon 23 Feb 2026 12:00

Lecture 7 - Cosmology I: A Brief Introduction to Cosmology

Primary Textbook: "An Introduction to Modern Cosmology", Andrew Liddle. Each lecture will have a chapter or two as recommended reading from this book and is available as an ePDF from the library.

1 The origin of Cosmology

Comes from the Ancient Greek word $\kappa o \sigma \mu o \sigma$, "kosmos" meaning order. Cosmology is a branch of astronomy concerned with studying the origin and evolution of the universe. We won't really consider anything on a smaller scale than clusters of galaxies.

The earliest form of this is *celestial mechanics* which goes back to Ancient Greek philosophers sch as Aristotle and Ptolemy.

1.1 Geocentric Ptolemaic System

This was the prevailing theory until the 16th century where the earth was at the centre of the universe and other stars/planet etc revolved around the earth in "epicycles". Here, the planets sat on rotating spheres, which orbited the earth.

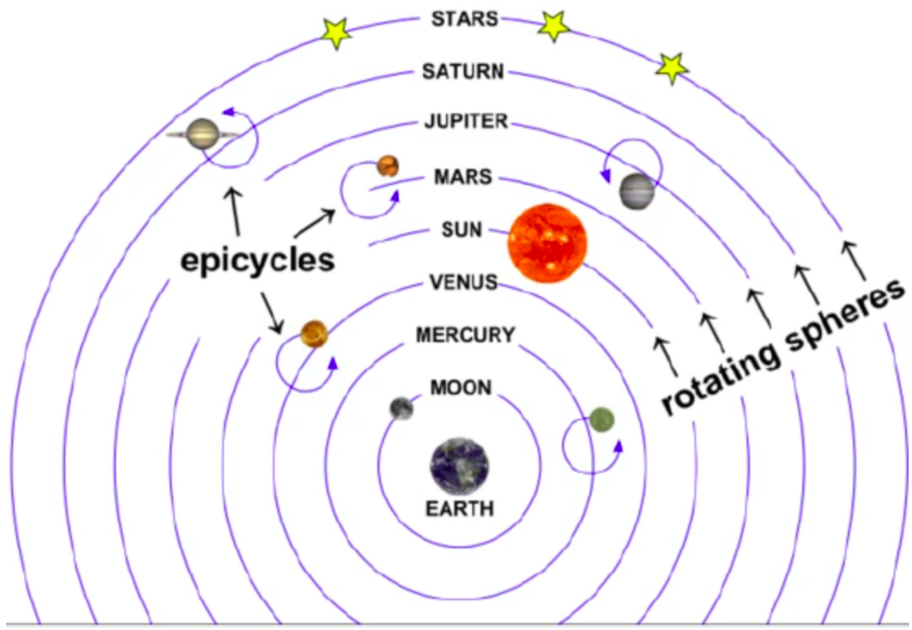


Figure 24.1

The epicycles explain retrograde motion. Observational power was too limited to get more accurate measurements, and they noticed that some objects appeared to go backwards as they were moving forwards overall in the orbit. We can solve this by placing another smaller circular orbit superimposed on the larger orbit.

We now know that this is not accurate, but it explained the motion at the time.

1.2 Heliocentric System

Copernicus in 1543 radically changed the understanding of orbits. He suggested that the sun, not the earth, was near the centre of the universe. The planets, including the earth revolved in circles around the sun, with other stars orbiting much further away.

In order to match the observed motion, we still need epicycles in this initial version of the theory.

This was the foundation of modern astronomy, and lead to:

- Kepler discovering elliptical orbits. This got rid of the need for epicentres and described the motion of planets with the data collected at the time.
- Galilei discovering the moons of Jupiter and phases of Venus. Before Galilei, these telescopes were not sufficiently powerful to observe these phases. The existence of these moons provided additional evidence for the heliocentric system as we now had a celestial body which did not orbit the Earth. This suggested that the Earth wasn't special, which was one of the beliefs that drove the geocentric model.
- Newton's theory of universal gravity in 1678.

This theory, and the implications it had on humanity's role in the universe was deeply unpopular with religious leaders and this caused opposition to its widespread acceptance.

This was a better way to think about the solar system (which was about the limit of what was practically observable). Copernicus therefore needed to provide evidence to justify this. These later from Kepler, Galilei and Newton acted as this evidence.

2 From Herschel to Hubble

In order to go from scales on a solar system level to a cosmological level, we need more powerful measurement equipment to be able to gather as much data as possible on much larger scales. William and Caroline Herschel pioneered this by creating the most powerful telescopes to date in 1877.

They probed deeper and deeper past the solar system, into the "deep sky". They investigated the contents of the Milky Way to build a catalogue of objects and try to find the edge of the galaxy.

This was in an attempt to work out whether or not there was some kind of edge to the universe. They discovered diffuse objects ("nebulae") beyond the milky way, and began to be able to resolve individual stars with improved technology.

This lead to the question of whether these nebulae (named "island galaxies" at the time) belong as components of the milky way or existed in their own right as galaxies.

2.1 Shapley-Curtis Debate

In the 1920s, the "Great Debate of Astronomy" investigated whether these spiral shape nebulae were small object on the outskirts of the Milky Way (advocated for by Harlow Shapley) or were independent small galaxies outside the gravitational attraction of the MIlky Way (advocated for by Heber Curtis).

In 1924, Edwin Hubble established that these bright nebulae were not part of the Milky Way and were instead of extragalactic origin. This told us that the observable universe extended beyond the Milky Way.

He discovered Cepheid Variable Stars, which are stars of periodic varying brightness. The period of pulsation is linked to intrinsic luminosity, and they act as distance indicators based on the perceived brightness and period on Earth.

These observations estimated the distance of these CVSes away from the Earth, and proved that they were too distant to sit within the Milky Way. This profoundly changed our understanding of the scale of brightness.

These CVSes are called "standard candles" which are astronomical objects with known intrinsic luminosity, M. We observe some different apparent brightness m, and can measure the pulsation period to determine the actual intrinsic luminosity.

Using M (determined from period) and m (measured) directly, we can use this formula:

$$5 \log D = m - M - 10$$

$$\implies D = 10^{(m-M-10)/5}$$

This made it apparent that these distances were much larger than our understanding of the size of the Milky Way, by in excess of 2 orders of magnitude.

We can also know the intrinsic luminosity of a Type 1a supernovae, when a star explodes and collapses into a white dwarf.

2.2 Cosmic Ladder

We can form the "Cosmic Distance Ladder". Since traditional distance measurements become infeasible as scales get larger, we need to use different ranges for different distances. For example:

- Cepheid Variable Stars are not that bright, so stop being useful at longer ranges.
- Supernovae are brighter, so stay useful for longer.

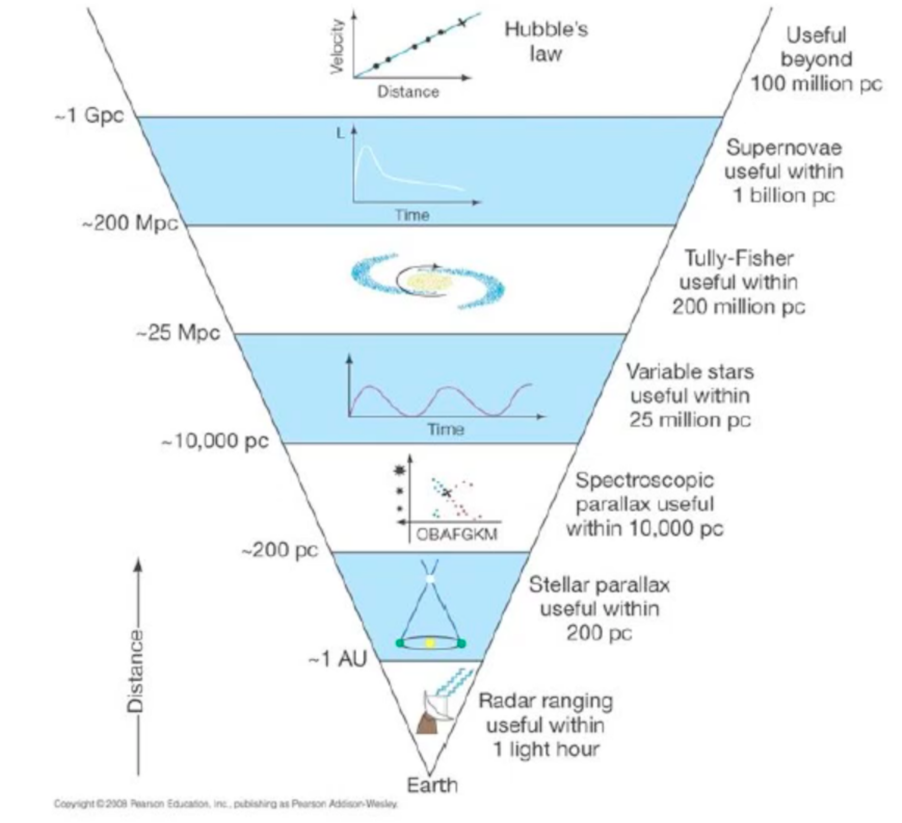


Figure 24.2: The Cosmic Distance Ladder

3 Expansion of the Universe

In 1929, Hubble analysed the emission spectra of galaxies. As a source moves, wavelengths (hence the positions of spectral lines) are impacted by redshift:

$$z = \frac{\lambda_o - \lambda_e}{\lambda_e} \approx \frac{v}{c} \qquad \text{approximation valid for non-relativistic limits.}$$

From this, noting the changes of the spectral lines vs known non-redshifted calibration points on earth allowed for the velocity of each galaxy to be determined.

Plotting v against r yields a straight line, suggesting $v \propto r$. Hence galaxies further away are moving faster. The sign of the redshift (i.e. is it indeed redshift and not blueshift) confirms that they are moving away from us.

This is very strong evidence that the universe is not static and is actually expanding. This gives us a dramatically new understanding of the universe, where it grows in size as time goes on.

4 Key Observations

We have a number of key objectives, and want to build a theory that fits these and makes predictions. Most of these observations are made using electromagnetic radiation, although additional information is provided by particle physics (and more recently gravitational wave astronomy).

Some of these key observations are:

- The universe is expanding (Hubble).
- There is an abundance of light elements (nucleosynthesis).
- Large-scale structure (LSS).
- Cosmic microwave background (CMB).
- Galaxy rotation curves (implying the existence of dark matter).
- Acceleration of the rate of expansion of the universe (dark energy).

4.1 Nucleosynthesis

Stars are very good and producing heavy elements via nuclear fusion. Here, stars start with lighter elements (H, He) and fuse upwards through increasing mass numbers up to Iron. Heavier elements require a stellar explosion to form.

It turns out that young stars have an abundance of light elements. These start from gas clouds that have collapsed gravitationally. In this very early stage, they started out with some contamination of heavier lighter elements (Helium isotopes and other Hydrogen isotopes) and couldn't just start with Hydrogen. They need some of these other elements to kickstart the process.

There has to be some process that took place before stars were formed that provided these elements to kickstart star formation. Stars could not have formed without this.

Theorists (George Gamov 1948) hypothesised reversing the expanding universe, i.e. stepping back in time to a much smaller, denser, hotter early universe. This would have formed a plasma-like state which could have provided these elements.

Nuclear fusion requires a very large energy, i.e. nuclear binding energy for these interactions is $\sim$ 1MeV. This plasma state lasted about 3 minutes, and allows for the formation up to Hydrogen-4. Heavier elements are still formed in stars.

4.2 Large Scale Structure

The fundamental building block of the the universe is galaxies. This cluster themselves together into filaments. While on a local scale, we see very large gaps between these, on a large enough scale (i.e. zoomed out enough) we see that the universe is very smooth on average. We say that, on cosmological scales, the distribution of matter is homogeneous and isotopic (has identical physical properties in all directions with uniform structure).

4.3 Cosmic Microwave Background

There is a nearly isotropic low frequency radiation background that penetrates the entire universe. It has an almost perfect black body spectrum with $T \sim 2.73$K. This is in the microwave region.

4.4 Discovery of Dark Matter

Fritz Zwicky and Vera Rubin discovered strange velocities in galaxy clusters and in the rotational curves of galaxies. This suggests that there is some extra mass that only interacts via gravity.

This is also potentially (dark energy) a driving force for the increasing rate of the expansion of the universe. It implies that we may need an extra particle added to the standard model to explain what actually makes this up.

Thu 26 Feb 2026 14:00

Lecture 8 - Cosmology II: Expansion of the Universe and Key Principles

1 Expansion of the Universe

Data from supernovae found that the universe was not expanding at a constant rate. The rate of expansion is increasing and accelerating.

The best current model we have is the Hot Big Bang Model:

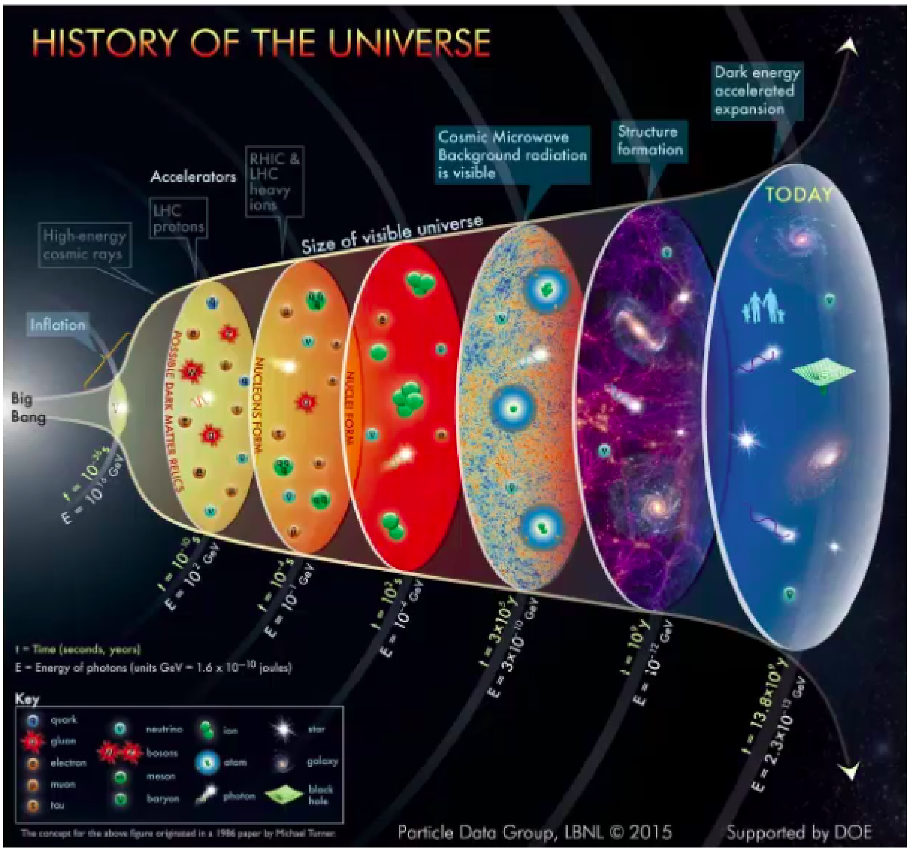


Figure 25.1

This is where the universe was created from an initial singularity, resulting in the big bang - an explosion.

Space itself is expanding and the universe is shaped by gravity as a fundamental force. The fundamental particles were created first in a dense and hot universe. As the universe expanded, it cooled down allowing for the propagation of CMB photons and the creation of meaningful cosmic structure.

1.1 Big Bang Cosmology

For the rest of the course we will:

- Explore big-bang centric cosmology, in the context of Newtonian gravity.

- We'll build and discuss the main equations that model the evolution of the universe. Using these we'll explore:
  - The history of the universe.
  - The composition of the universe.
  - The eventual fate of the universe.
- We'll discuss key observations and any problems where these do not fit the models we're applying them to.

2 Key Principles

2.1 The Cosmological Principle

The distribution of matter is:

- **Homogeneous**, meaning that the universe looks the same at every point. There is no special point in the universe, and the average matter density is the same everywhere.
- **Isotropic**, meaning the universe looks the same in all directions

However this isn't perfectly fulfilled in our universe. It's a valid approximation on sufficiently large scales $> 100\text{Mpc}$, but it isn't perfect.

2.2 Distance Measures - The Parsec

The parsec is a unit of distance useful on cosmological scales:

$$1\text{pc} = 3.086 \times 10^{16}\text{m}$$

Suppose an observer on Earth measures a very distant star. We take two measurements six months apart. As the Earth is moving relative to the star, it appears to us that the star has shifted position (in the Earth's rest frame). The total observed angular shift is $2p$, where p is the parallax angle — the angle at the star in the right triangle formed by the Sun, Earth, and star, with 1 AU as the opposite side.

The parsec is defined as the distance at which $p = 1$ arcsecond (4.848×10^{-6} rad), giving the relationship:

$$d\,[\text{pc}] = \frac{1}{p\,[\text{arcsec}]}$$

2.3 Redshift

This has already been covered in other modules, but to recap the fractional shift is:

$$z = \frac{\lambda_0 - \lambda_e}{\lambda_e} \approx \frac{v}{c}$$

for non-relativistic sources where $v \ll c$.

3 Expansion of the Universe

Hubble found that there was a linear relationship between the recessional velocity of a galaxy and its distance.

$$\boldsymbol{v} = H_0\boldsymbol{r}$$

Where $H \approx 68\text{kms}^{-1}\text{Mpc}^{-1}$. Note that the parameter H varies with time. H_0 represents the value at the current epoch.

This isn't exact, and does not hold for nearby galaxies where other forces dominate over the expansion of the universe. These galaxies have some peculiar velocity which forms the total velocity:

$$v_{\text{total}} = v_{\text{pec}} + v_H$$

For distant galaxies, the recessional velocity dominates. This is called Hubble flow.

3.1 Comoving Coordinates

As the universe expands, the physical (proper) distance between two galaxies grows. This means that, in a coordinate system centred on the earth, the coordinates of other galaxies are increasing.

Instead, we define a new coordinate system where the coordinate system itself expands alongside the universe. This means that galaxies remain at fixed coordinates as the universe expands, and the coordinate grid expands at the same rate as te universe. The distance between galaxies in these coordinates is therefore constant.

This rate of increase is given by the universal expansion rate (a scale factor) $a(t)$. Due to homogeneity, this scale factor is only time-dependant, as the rate of expansion is the same everywhere in the universe.

This means that we can relate proper distance $\boldsymbol{r}(t)$ and the fixed comoving distance x with:

$$\boldsymbol{r}(t) = a(t)\boldsymbol{x}$$

Mon 02 Mar 2026 12:00

Lecture 9 - Cosmology III: Friedmann Equation

1 Equation of Motion for the Universe

Goal: To derive an equation of motion for the dynamics of the Universe.

Assumptions:

- We will assume Newtonian Gravity.
- We will assume The Cosmological Principle holds on the scales we're considering.
- We will assume uniform expansion.
- We will assume no dissipation. This means that total energy for a particle of mass m will be conserved.
- As a result of these assumptions, we wil be using comoving coordinates.

We'll be using a non-vectorised form of the comoving coordinates:

$$\boxed{r(t) = a(t)x}$$

Assume at some time t_1 we have some spherical volume of radius $r(t_1)$, containing mass M. As time progresses and the universe expands, this radius will increase. This is given by some generic $r(t)$ and volume $V(t)$

As a result of our assumptions, we assume that we have uniform mass density given by $\rho(t)$ for some t:

$$\rho(t) = \frac{M}{V(t)}$$

Consider a particle of mass m on the boundary of our initial t_1 universe. This particle will have some gravitational potential energy, and some kinetic energy as it moves on expansion:

$$\text{total energy } = \text{potential energy } + \text{ kinetic energy}$$

$$U = V + T$$

For potential energy:

$$V = \frac{-GMm}{r(t)} = -\frac{4\pi}{3}Gm\rho(t)r^2(t) \qquad \text{assuming a spherical universe.}$$

For kinetic energy:

$$T = \frac{1}{2}mv^2 = \frac{1}{2}m\left(\frac{dr(t)}{dt}\right)^2 = \frac{1}{2}m\dot{r}^2$$

Hence for total energy:

$$U = T + V = \frac{1}{2}m\dot{r}^2 - \frac{4\pi}{3}Gm\rho(t)r^2(t)$$

In doing this, we've had to implicitly assume that we can treat the mass M distributed throughout the sphere as if it were a point mass at the centre of the sphere. This gives us an ordinary differential equation in terms of $r(t)$.

1.1 Comoving Coordinates

We now want to transform this into comoving coordinates.

$$\dot{r(t)} = \frac{d}{dt}(a(t)x) = \dot{a}(t)x$$

$$\implies U = \frac{-4\pi}{3}Gm\rho(t)a^2(t)x^2 + \frac{1}{2}m\dot{a}^2(t)x^2$$

$$\frac{2U}{ma^2(t)x^2} = \left(\frac{\dot{a}(t)}{a(t)}\right)^2 - \frac{8\pi G}{3}\rho(t)$$

Let $\frac{2U}{mx^2} \equiv -kc^2$, hence[1]:

$$\boxed{\left(\frac{\dot{a}(t)}{a(t)}\right)^2 = \frac{8\pi G}{3}\rho(t) - \frac{kc^2}{a(t)^2}}$$

This is called the Friedmann Equation. This is now an ODE in terms of scale factor and not in terms of radius. Before we can solve it however, we need to deal with the $\rho(t)$ term.

Considering the second term, we have:

$$[k] = \frac{1}{L^2}$$

This k has different implications on the geometry of the universe, and we consider 3 cases:

- $k = 0$: A flat universe.
- $k > 0$: A positive-curvature (closed) universe with spherical geometry, analogous to the surface of a sphere.
- $k < 0$: A negative-curvature (open) universe with hyperbolic geometry, analogous to the surface of a saddle.

Current observations suggest that our universe is very close to the $k = 0$ case, so we will disregard the second term in first year cosmology.

1.2 The Fluid Equation

In order to solve the Friedmann Equation, we need to build something to describe $\rho(t)$, called the "fluid equation". We will assume the following:

- We will assume that the universe is a perfect/ideal fluid. This means that it is characterised by a density ρ and an isotropic pressure p.
- We will assume that the 1st Law of Thermodynamics holds:

$$dE = -pdV + TdS$$

- We will assume that the process is adiabatic, $dS = 0$.

Considering a comoving unit volume, where:

$$x = 1 \implies r(t) = a(t)$$

Hence:

$$V(t) = \frac{4\pi}{3}a^3(t)$$

The internal energy of our fluid is given by:

$$E = mc^2 = \frac{4\pi}{3}\rho(t)a^3(t)c^2$$

[1]This seems very random, but all we are doing is setting a constant equal to another constant, so it does work!

$$\begin{aligned}\frac{dE}{dt} &= \frac{4\pi}{3}c^2\frac{d}{dt}\left(\rho(t)a^3(t)\right)\\ &= \frac{4\pi c^2}{3}\left(\dot{\rho}(t)a^3(t) + \rho(t)3a^2(t)\dot{a}(t)\right)\end{aligned}$$

And:

$$V(t) = \frac{4\pi}{3}a^3(t) \implies \frac{dV}{dt} = 4\pi a^2(t)\dot{a}(t)$$

Hence:

$$\begin{aligned}\frac{4\pi c^2}{3}\left(\dot{\rho}(t)a^3(t) + \rho(t)3a^2(t)\dot{a}(t)\right) + p4\pi a^2(t)\dot{a}(t) = 0\\ \dot{\rho}(t) + 3\frac{\dot{a}(t)}{a(t)}\left(\rho(t) + \frac{p}{c^2}\right) = 0\end{aligned}$$

This is now much easier to solve!

Thu 05 Mar 2026 14:00

Lecture 10 - Cosmology IV: Friedmann Equation, Hubble and Redshift Combined

We have the Friedmann Equation from last lecture:

$$\boxed{\left(\frac{\dot{a}(t)}{a(t)}\right)^2 = \frac{8\pi G}{3}\rho(t) - \frac{kc^2}{a(t)^2}}$$

Where $a(t)$ is the scale factor relating the comoving scaled coordinates to the initial coordinate system. As observations suggest our universe has little to no curvature, hence $k \approx 0$, we will generally ignore the second term at this level.

We also have the fluid equation:

$$\boxed{\dot{\rho}(t) + 3\frac{\dot{a}(t)}{a(t)}\left(\rho(t) + \frac{p(t)}{c^2}\right) = 0}$$

1 Acceleration Equation

We want some kind of equation to describe the acceleration of the scale factor, i.e $\ddot{a}(t)$.

We can get this through a combination of:

1. The Friedmann Equation → differentiate it with respect to time.
2. Simplify with the fluid equation.
3. Use the Friedmann equation again to get the acceleration equation in the form:

$$\frac{\ddot{a}(t)}{a(t)} = \frac{-4\pi G}{3}\left(\rho(t) + \frac{3p}{c^2}\right)$$

Note the sign here, if the contents of the brackets are positive, then there's an overall global negative sign, which suggests that the whole expression is negative. This suggests that the rate of universal expansion is slowing... this contradicts observations.

For now, we accept this, but we'll make a modification to the Friedmann equation to correct for this next week.

1.1 Hubble and Friedmann

How is Hubble's observation and Friedmann's equation connected?

We have the Friedmann eqn:

$$\left(\frac{\dot{a}(t)}{a(t)}\right)^2 = \frac{8\pi G}{3}\rho(t) - \frac{kc^2}{a(t)^2}$$

And Hubble's observation:

$$\boldsymbol{v}_h = H \cdot \boldsymbol{r}$$

Where $\boldsymbol{v}_h$ represents the radial recessional velocity only, excluding peculiar motion. But we want to find a form of the Hubble observation which includes a term in a. We consider the recessional velocity, starting from our definition of comoving coordinates:

$$\boldsymbol{r}(t) = a(t)\boldsymbol{x}$$

$$\boldsymbol{v}_H = \frac{d\boldsymbol{r}}{dt} = \dot{a}(t)\boldsymbol{x} = \dot{a}(t)\frac{\boldsymbol{r}(t)}{a(t)} = \frac{\dot{a}(t)}{a(t)}\boldsymbol{r}(t)$$

And comparing with Hubble's law gives:

$$H(t)\cdot\boldsymbol{r} = \frac{\dot{a}(t)}{a(t)}\boldsymbol{r}$$

$$H(t) = \frac{\dot{a}(t)}{a(t)}$$

While commonly called the Hubble Constant, because it's constant wrt space, it's a function of time which means it's not a constant and is properly referred to as the Hubble Parameter. H_0 is $H(t = t_0)$ where t_0 denotes right now.

2 Expansion and Redshift

Consider two galaxies separated by a (cosmologically) small distance dr. A photon is emitted at one galaxy with λ_{em} and travels towards an observer on the second galaxy.

According to Hubble's law, there will be a small recessional velocity between the two (disregarding peculiar velocity):

$$dv = H\,dr = \frac{\dot{a}(t)}{a(t)}\,dr$$

The very small change in wavelength will be:

$$d\lambda = \lambda_{\text{obs}} - \lambda_{\text{em}}$$

And in the non-relativistic limit:

$$\frac{d\lambda}{\lambda_{\text{em}}} \approx \frac{dv}{c}$$

The time dt for the photon to travel distance dr is:

$$dt = \frac{dr}{c}$$

Hence, substituting $dr = c\,dt$:

$$\frac{d\lambda}{\lambda_{\text{em}}} = \frac{dv}{c} = \frac{\dot{a}(t)}{a(t)}\frac{dr}{c} = \frac{\dot{a}(t)}{a(t)}\,dt = \frac{1}{a}\frac{da}{dt}\,dt$$

$$\frac{d\lambda}{\lambda_{\text{em}}} = \frac{da}{a}$$

This is a separable differential equation. Integrating both sides from emission to observation:

$$\int_{\lambda_{\text{em}}}^{\lambda_{\text{obs}}} \frac{d\lambda}{\lambda} = \int_{a_{\text{em}}}^{a_{\text{obs}}} \frac{da}{a}$$

$$\ln\left(\frac{\lambda_{\text{obs}}}{\lambda_{\text{em}}}\right) = \ln\left(\frac{a_{\text{obs}}}{a_{\text{em}}}\right)$$

$$\implies \boxed{\lambda \propto a}$$

This means that, as the universe expands, the wavelength of these photons changes as well.

Going back to the definition of z:

$$1 + z = \frac{\lambda_{\text{obs}}}{\lambda_{\text{em}}} = \frac{a(t_{\text{obs}})}{a(t_{\text{em}})}$$

Effectively, photons have their wavelength stretched proportional to the scale factor of the universes growth between emission and observation. As the universe grows, wavelengths become stretched and energy decreases.

3 Simple Cosmological Models

In order to solve the Friedmann equation, we make assumptions about the contents and geometry of the universe, i.e. $\rho(t)$ and k.

For our universe, where $k \approx 0$, we assume $k = 0$ at a first year level as this is what is currently favoured by observations. We assume a flat universe.

For relativistic particles, we have (and will investigate in more depth in future lectures):

$$p = \frac{\rho c^2}{3}$$

If we assume non-relativistic particles, we have:

$$p = 0$$

In practice, we'd use a model that combines these, however in order to get something we can easily solve analytically at our level, we assume that one is true. We start with a flat, matter-dominated universe. This means that the ρ we're considering is associated with non-relativistic particles.

If we go back to the fluid equation:

$$\dot{\rho}(t) + 3\frac{\dot{a}(t)}{a(t)}\left(\rho(t) + \frac{p(t)}{c^2}\right) = 0$$

It now becomes:

$$\dot{\rho}(t) + 3\frac{\dot{a}(t)}{a(t)}\rho(t) = 0$$

This can be manipulated to get to:

$$\frac{1}{a^3(t)}\frac{d}{dt}\left(\rho(t)a^3(t)\right) = 0$$

And hence:

$$\rho(t)a^3(t) = \text{const.}$$

$$\rho(t) = \frac{\text{const.}}{a^3(t)}$$

To find this constant, we need to use boundary conditions. We say that the scale factor at the current epoch is equal to one, to normalise. Hence: $a_0 \equiv a(t_0) = 1$.

We therefore call the density at the current epoch ρ_0, and this gives the final form:

$$\rho(t) = \frac{\rho_0}{a^3(t)}$$

Fri 01 Mar 2026 14:00

Lecture 11 - Cosmology V:

LC Mathematics for Physicists 1B

Wed 21 Jan 2026 11:00

Lecture 1 - Partial Differentiation I: Intro and Welcome

1 Course Welcome

1.1 Recommended books:

- Mathematical Techniques 4e, Jordan & Smith
- Engineering Mathematics 8e, Stroud
- Calculus (Schaum), 6e, Ayres & Mendelson
- Advanced Calculus (Schaum), 6e, Ayres & Mendelson

1.2 Assessment details:

- Maths 1A/1B form a single 20 credit module.
- 80% assessed by a 3 hour exam - Section 1 is 36% with 6 short questions and Section 2 is 64% with 4 long questions.
- 20% assessed by problem sheets.

1.3 Course structure:

1. Partial Differentiation
    - Definition, total differential, chain rule, gradient.
    - Taylor series, stationary points, Lagrange multipliers.
2. Differential Equations
    - Definition, 1st order separable, exact and homogenous.
    - Linear equations: general solution, 1st order and constant coefficients.
3. Integration
    - Definition as area under the curve, fundamental theorem of calculus.
    - Integration by: substitution, parts, partial fractions and tricks.
4. Multiple Integrals
    - Multiple and repeated integrals. Change of order of integration.
    - Change of variables and the Jacobian. Arc length. Solids of revolution.

2 Multivariate Functions

Lots of physics involves functions of more than one variable. A physical quantity defined at every point in space is called a field. We can have both scalar fields and vector fields.

For example, some scalar fields are:

- $V(x, y, z)$: Electrostatic potential. This is often easier to work with compared to the full electric (vector) field.

- $T(x, y, z)$: Temperature.
- $p(x, y, z)$: Pressure.

While some vector fields are:

- $\underline{E}(x, y, z)$: Electric Field.
- $\underline{B}(x, y, z)$: Magnetic Field.
- $\underline{v}(x, y, z)$: Velocity Field (i.e in fluid mechanics).

2.1 Partial Derivatives

Consider a function of two variables. The partial derivative of a function with respect to one variable is the rate of change of a function wrt that variable, while keeping other variables constant. Effectively, we carry out a derivative while treating the other variables as if they were constants.

Suppose we have a function $f(x, y)$. The definition of a partial derivative is:

$$\frac{\partial f}{\partial x}(x_0, y_0) = \lim_{h\to 0} \frac{f((x_0 + h), y_0) - f(x_0, y_0)}{h}$$
$$\frac{\partial f}{\partial y}(x_0, y_0) = \lim_{k\to 0} \frac{f(x_0, (y_0 + k)) - f(x_0, y_0)}{k}$$

Just like we denote $\frac{df}{dx}$ as the derivative of a function of a single variable, we denote $\frac{\partial f}{\partial x}$ as the partial derivative of a function of several variables.

Note that this is not delta f and delta y, i.e. not $\frac{\delta f}{\delta x}$

In theory, we'd explicitly notate:

$$\left(\frac{\partial f}{\partial x}\right)_y$$

With the subscript y explicitly stating that y is being kept constant. This is rarely, but sometimes, needed.

Consider $f(x, y, z) = x^2 \sin yz$. We have:

$$\frac{\partial f}{\partial x} = 2x \sin yz$$
$$\frac{\partial f}{\partial y} = x^2 z \cos yz$$
$$\frac{\partial f}{\partial z} = x^2 y \cos yz$$

2.2 Higher Orders

Higher derivatives are defined as they were previously, but they can now be mixed. For example, with $f(x, y) = x^2 \sin y$, we can write:

$$\frac{\partial f}{\partial x} = 2x \sin y \qquad \frac{\partial f}{\partial y} = x^2 \cos y$$
$$\frac{\partial^2 f}{\partial x^2} = 2 \sin y$$

We can also have:

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = 2x \cos y$$

Shorthand notation exists, i.e. $f_{xx} = \frac{\partial^2 f}{\partial x^2}$ or $f_{yx} = \frac{\partial^2 f}{\partial y \partial x}$

For most cases, but not all, mixed derivatives are often independent of the order of partial derivatives, so: $f_{xy} = f_{yx}$

Thu 22 Jan 2026 09:00

Lecture 2 - Partial Differentiation II: The Total Differential

1 The Total Differential

In order to generalise the chain rule, we need to define the total differential. Consider the change in a function of two variables, $f(x, y)$ when we move from some point (x, y) to some point $(x + dx, y + dy)$.

The partial derivative only tells us what happens when we change one variable, but we're changing two here. The total differential sums these two in order to get the full total change.

$$df = f(x + dx, y + dy) - f(x, y)$$

We have to look at the change in a single variable at a time, so we split it into two pieces where only x changes in the first, and only y changes in the second.

$$df = \underbrace{[f(x + dx, y + dy) - f(x, y + dy)]}_{\text{isolates change in } x} + \underbrace{[f(x, y + dy) - f(x, y)]}_{\text{isolates change in } y}$$

$$df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy$$

More generally for a function of $f(x_1, x_1, \dots, x_n)$:

$$df = \frac{\partial f}{x_1}dx_1 + \frac{\partial f}{x_2}dx_2 + \cdots + \frac{\partial f}{x_n}dx_n = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}dx_i$$

The total change in the function $f(x_1, x_1, \dots, x_n)$ is the sum of partial changes due to changing a single variable.

2 The Chain Rule

Recall that if $y = y(x)$, $x = x(t)$, then:

$$dy = \frac{dy}{dx}dx = \frac{dy}{dx}\frac{dx}{dt}dt \implies \frac{dy}{dt} = \frac{dy}{dx}\frac{dx}{dt}$$

Now, if $f = f(x, y)$ and $x = x(t)$, $y = y(t)$, we can adjust the chain rule to say:

$$df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy$$

$$= \frac{\partial f}{\partial x}\frac{dx}{dt}dt + \frac{\partial f}{\partial y}\frac{dy}{dt}dt$$

$$\implies \frac{df}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}$$

Example

$$f(x, y) = x^2 + y^2, \qquad x(t) = t^2, \qquad y(t) = t^3$$

Hence:

$$f(t) = t^4 + t^5 \implies \frac{df}{dt} = 4t^3 + 6t^5$$

By rewriting in terms of one variable:

$$\frac{\partial f}{\partial x} = 2x = 2t^2 \qquad \frac{\partial f}{\partial y} = 2y = 2t^3$$

$$\frac{dx}{dt} = 2t \qquad \frac{dy}{dt} = 3t^2$$

And instead using the new chain rule:

$$\begin{aligned} &\frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt} \\ &= (2t^2 + 2t) + (2t^3)(3t^2) \\ &= 4t^3 + 6t^5 \end{aligned}$$

Hence the new chain rule works!

2.1 Polar Coordinates

Suppose our x and y are now functions of two different variables themselves, so:

$$f = f(x, y) \qquad x = x(r, \theta) \qquad y = y(r, \theta)$$

From r, θ we want to calculate x, y and then from x, y we want to calculate f.

$$\begin{aligned} df &= \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy \\ dx &= \frac{\partial x}{\partial r}dr + \frac{\partial x}{\partial \theta}d\theta \\ dy &= \frac{\partial y}{\partial r}dr + \frac{\partial y}{\partial \theta}d\theta \end{aligned}$$

Hence:

$$df = \frac{\partial f}{\partial x}\left(\frac{\partial x}{\partial r}\,dr + \frac{\partial x}{\partial \theta}\,d\theta\right) + \frac{\partial f}{\partial y}\left(\frac{\partial y}{\partial r}\,dr + \frac{\partial y}{\partial \theta}\,d\theta\right)$$

$$df = \left(\frac{\partial f}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial r}\right)dr + \left(\frac{\partial f}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial \theta}\right)d\theta$$

We also know (if we substitute x, y into the original function to get a function of r, θ):

$$df = \frac{\partial f}{\partial r}dr + \frac{\partial f}{\partial \theta}d\theta$$

We can read off the final partial derivatives:

$$\frac{\partial f}{\partial r} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial x}{\partial r}, \qquad \frac{\partial f}{\partial \theta} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial \theta}$$

As expected!

2.2 Generalising

Suppose we have two functions which map $\mathbb{R}^m \to \mathbb{R}^p$, and $\mathbb{R}^p \to \mathbb{R}^n$, respectively:

$$(x_1, x_2, \dots, x_m) \to (y_1, y_2, \dots, y_p) \to (z_1, z_2, \dots, z_n)$$

Then we have:

$$dz_i = \sum_{k=1}^{p} \frac{\partial z_i}{\partial y_k} dy_k$$
$$dy_k = \sum_{l=1}^{m} \frac{\partial y_k}{\partial x_l} dx_l$$

And substituting:

$$dz_i = \sum_{k=1}^{p} \sum_{l=1}^{m} \frac{\partial z_i}{\partial y_k} \frac{\partial y_k}{\partial x_l} dx_l$$

And (as the two sums are independent), we can pull out the inner sum:

$$= \sum_{l=1}^{m} \left[\sum_{k=1}^{p} \frac{\partial z_i}{\partial y_k} \frac{\partial y_k}{\partial x_l} \right] dx_l$$
$$= \sum_{l=1}^{m} \frac{\partial z_i}{\partial x_l} dx_l$$

The partial derivatives $\partial z_i / \partial x_j$ are therefore given by:

$$\frac{\partial z_i}{\partial x_j} = \sum_{k=1}^{p} \frac{\partial z_i}{\partial y_k} \frac{\partial y_k}{\partial x_j}$$

Fri 23 Jan 2026 12:00

Lecture 3 - Partial Differentiation III: Directional Derivatives and Gradients

1 Directional Derivatives and Gradients

Consider a scalar field $f(\underline{r}) = f(x, y, z)$. We can define derivatives along the x, y, z planes respectively:

$$\frac{\partial f}{\partial x} \qquad \frac{\partial f}{\partial y} \qquad \frac{\partial f}{\partial z}$$

Here, x, y, z aren't inherently special, they just form the standard basis we generally take. What if we want to compute the derivative at a point $\underline{r}$ with respect to a generic unit vector, $\hat{n}$, i.e. $\frac{\partial f}{\partial \hat{n}}$?

At our point $\underline{r}$, we are distance ϵ away from the direction formed by the $\hat{n}$ vector.

$$\frac{\partial f}{\partial \hat{n}} = \lim_{\epsilon \to 0} \frac{f(\underline{r} + \epsilon \hat{n}) - f(\underline{r})}{\epsilon}$$

This is a general form, called the directional derivative, giving the rate of change of the function in the direction $\hat{n}$. If we take the unit vector $\hat{n}$ to be $\hat{x}, \hat{y}, \hat{z}$ etc, this gives the standard derivatives in these directions.

If we know the derivatives in the x, y, z directions (or in any three directions which form an orthonormal basis), we can calculate the gradient of the function in any arbitrary direction.

$$\begin{aligned}\frac{\partial f}{\partial n} &= \lim_{\epsilon \to 0} \frac{f(x + \epsilon n_x, y + \epsilon n_y, z + \epsilon n_z) - f(x, y, z)}{\epsilon} \\ &= \frac{df}{d\epsilon}(x + \epsilon n_x, y + \epsilon n_y, z + \epsilon n_z)\Big|_{\epsilon=0}\end{aligned}$$

This is just the chain rule:

$$= \frac{\partial f}{\partial x}\frac{d}{d\epsilon}(x + \epsilon n_x) + \frac{\partial f}{\partial y}\frac{d}{d\epsilon}(y + \epsilon n_y) + \frac{\partial f}{\partial z}\frac{d}{d\epsilon}(z + \epsilon n_z)$$

$$= n_x \frac{\partial f}{\partial x} + n_y \frac{\partial f}{\partial y} + n_Z \frac{\partial f}{\partial z}$$

Suppose we set $\hat{n}$ along x:

$$\hat{n} = \hat{\imath} \implies n_x = 1, n_y = 0, n_z = 0$$

$$\frac{\partial f}{\partial n} = \frac{\partial f}{\partial x}$$

As expected!

It might be noticed that the expression is just a scalar product:

$$n_x \frac{\partial f}{\partial x} + n_y \frac{\partial f}{\partial y} + n_Z \frac{\partial f}{\partial z} = \hat{n} \cdot \underline{\nabla} f$$

1.1 The Gradient

This ∇ "nabla" is called the gradient operator, it is the vector whose components are the partial derivatives in each direction:

$$\underline{\nabla} f \equiv \frac{\partial f}{\partial x}\hat{x} + \frac{\partial f}{\partial y}\hat{y} + \frac{\partial f}{\partial z} \equiv \underline{\text{grad}}(f)$$

We can also think of it as a general operator:

$$\underline{\nabla} \equiv \frac{\partial}{\partial x}\hat{x} + \frac{\partial}{\partial y}\hat{y} + \frac{\partial}{\partial z}\hat{z}$$

The directional derivative $\partial f/\partial n$ has the maximum magnitude when $\hat{n} \parallel \underline{\nabla} f$. The direction of $\underline{\nabla} f$ is the direction in which f increases most rapidly, and the magnitude is this maximum rate of increase.

1.2 Example

Suppose we have a particle moving in one dimension. Newton's second law says:

$$m\frac{d^2x}{dt^2} = F$$

Assuming that the motion arises from a conservative force, the force is defined as:

$$F = -\frac{dV}{dx}$$

Substituting this into Newton's law, we obtain the equation of motion:

$$m\frac{d^2x}{dt^2} + \frac{dV}{dx} = 0$$

We want to show that energy is conserved here. We know that total energy is:

$$E = \frac{1}{2}mv^2 + V(x(t)) = \frac{1}{2}m\left(\frac{dx}{dt}\right)^2 + V(x(t))$$

To show that total energy is constant, differentiate with respect to t:

$$\frac{dE}{dt} = \frac{d}{dt}\left[\frac{1}{2}m\left(\frac{dx}{dt}\right)^2 + V(x(t))\right]$$

Using the chain rule:

$$\frac{dE}{dt} = m\frac{dx}{dt}\frac{d^2x}{dt^2} + \frac{dV}{dx}\frac{dx}{dt}$$

Factoring out the velocity:

$$\frac{dE}{dt} = \frac{dx}{dt}\left[m\frac{d^2x}{dt^2} + \frac{dV}{dx}\right]$$

Using the equation of motion, the bracketed term is zero:

$$\frac{dE}{dt} = \frac{dx}{dt}[0] = 0$$

Therefore, E is does not change wrt time so is conserved.

1.3 Example in 3D

In 3D, we consider the components of force in each direction, again a conservative force (i.e. the electric field):

$$m\frac{d^2x}{dt^2} = F_x = -\frac{\partial V}{\partial X} \implies F_x + V_x = 0$$
$$m\frac{d^2y}{dt^2} = F_y = -\frac{\partial V}{\partial y} \implies F_y + V_y = 0$$
$$m\frac{d^2z}{dt^2} = F_z = -\frac{\partial V}{\partial z} \implies F_z + V_z = 0$$

Here, total energy is given by:

$$E(t) = \frac{1}{2}m\left[\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2\right] + V(x(t), y(t), z(t))$$

$$\frac{dE}{dt} = m\frac{dx}{dt}\frac{d^2x}{dt^2} + m\frac{dy}{dt}\frac{d^2y}{dt^2} + m\frac{dz}{dt}\frac{d^2z}{dt^2} + \frac{\partial V}{\partial x}\frac{dx}{dt} + \frac{\partial V}{\partial y}\frac{dy}{dt} + \frac{\partial V}{\partial z}\frac{dz}{dt}$$

This gives:

$$\frac{dx}{dt}\left[m\frac{d^2x}{dt^2} + \frac{\partial V}{\partial x}\right] + \frac{dy}{dt}\left[m\frac{d^2y}{dt^2} + \frac{\partial V}{\partial y}\right] + \frac{dz}{dt}\left[m\frac{d^2z}{dt^2} + \frac{\partial V}{\partial z}\right] = 0$$
$$\frac{\partial x}{\partial t}\left[F_x + V_x\right] + \frac{\partial y}{\partial t}\left[F_y + V_y\right] + \frac{\partial z}{\partial t}\left[F_z + V_z\right] = 0$$

This is zero as each of the bracketed terms is zero by N2L. We've done this laboriously component wise, but we can use the gradient to do it all at once:

$$m\frac{d\underline{r}}{dt} \cdot \frac{d^2\underline{r}}{dt^2} + \underline{\nabla}V \cdot \frac{d\underline{r}}{dt}$$

Etc.

1.4 Numerical Example

Given $f(x, y, z) = x^3y^2z$. Find the rate of increase at $(1, 1, 1)$ in the direction given by $\hat{\imath} + 2\hat{\jmath} + 2\hat{k}$.

$$\underline{\nabla}f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right) = (3x^2y^2z, 2x^3yz, x^3y^2) = (3, 2, 1) \text{ at } \underline{r} = (1, 1, 1)$$

And the direction is:

$$n = (1, 2, 2)$$

However this does not have unit length, hence:

$$\hat{n} = \frac{1}{\sqrt{1^2 + 2^2 + 2^2}}(1, 2, 2) = \frac{1}{3}(1, 2, 2)$$

So:

$$\frac{\partial f}{\partial n} = \underline{\nabla}f \cdot \hat{n} = \frac{1}{3}(3, 2, 1) \cdot (1, 2, 2) = \frac{1}{3}(3 + 4 + 2) = 3$$

2 Gradient and Vector Fields

We can also use Nabla on a vector field $\underline{E} = E_x\hat{\imath} + E_y\hat{\jmath} + E_z\hat{k}$.

We have two possible ways of doing this, using the scalar product or the vector product. $\underline{\nabla} \cdot \underline{E}$ is called the divergence and $\underline{\nabla} \times \underline{E}$ is called the curl.

2.1 Divergence

$$\begin{aligned}\underline{\nabla} \cdot \underline{E} &= \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right) \cdot \left(E_x \hat{\imath} + E_y \hat{\jmath} + E_z \hat{k}\right) \\ &= \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z}\end{aligned}$$

This does have a geometric interpretation. Consider a small volume and calculate the flux of the field through that volume. The divergence gives the flux per unit volume.

2.2 Curl

$$\begin{aligned}\underline{\nabla} \times \underline{E} &= \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right) \times \left(E_x \hat{\imath} + E_y \hat{\jmath} + E_z \hat{k}\right) = \det \begin{bmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & E_z \end{bmatrix} \\ &= \left(\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z}\right)\hat{\imath} + \left(\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x}\right)\hat{\jmath} + \left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y}\right)\hat{k}\end{aligned}$$

In a static steady state, the curl of the electric field is zero.

The proper study of gradient, divergence and curl is a second year topic (vector analysis). It's needed in Maxwell's equations for electromagnetism.

Wed 28 Jan 2026 11:00

Lecture 4 - Partial Differentiation IV: Total Time Derivative and Multivariate Taylor Series

1 Total Time Derivative

Consider a fluid with a velocity field:

$$\underline{v}(x, y, z, t) = v_x(x, y, z, t)\hat{\imath} + v_y(x, y, z, t)\hat{\jmath} + v_z(x, y, z, t)\hat{k}$$

We could follow a small piece of this fluid, and use Newton's Second Law, per unit volume. This gives mass per unit volume (i.e. density) times acceleration. Assuming a non-compressible fluid with constant density:

$$\underline{f} = \rho \frac{d\underline{v}}{dt}$$

Where $\underline{f}$ is force per unit volume. Beginning by considering acceleration in only the x direction, there is two sources of time dependence. One from the fluid element itself moving, and one from the velocity of the piece changing with time.

This gives the total acceleration as:

$$\begin{aligned}\frac{dv_x}{dt} &= \frac{\partial v_x}{\partial x}\frac{dx}{dt} + \frac{\partial v_x}{\partial y}\frac{dy}{dt} + \frac{\partial v_x}{\partial z}\frac{dz}{dt} + \frac{\partial v_x}{\partial t} \\ &= v_x\frac{\partial v_x}{\partial x} + v_y\frac{\partial v_x}{\partial y} + v_z\frac{\partial v_x}{\partial z} + \frac{\partial v_x}{\partial t}\end{aligned}$$

We can simplify this with nabla:

$$\frac{dv_x}{dt} = \left(v_x\frac{\partial}{\partial x} + v_y\frac{\partial}{\partial y} + v_z\frac{\partial}{\partial z}\right)v_x + \frac{\partial v_x}{\partial t} = \left(\underline{v}\cdot\underline{\nabla}\right)v_x + \frac{\partial v_x}{\partial t}$$

The total time derivative or advective derivative is given by:

$$\frac{d\underline{u}}{dt} = \left(\underline{v}\cdot\nabla\right)\underline{u} + \frac{\partial \underline{u}}{\partial t}$$

Where the first term gives the contribution from the element of fluid moving from the fluid flow pattern and the second term gives the contribution from the pattern itself changing.

The first term is called the advective term. A fluid element will accelerate even in a constant velocity field, because the velocity changes at it moves through the field.

The advective term is non-linear in the velocity field, which makes fluid mechanics difficult to solve.

2 Multivariate Taylor Series

For a single variable, we want to expand a function $f(x)$ around a point $x = a$ in a series of powers of h, where $h = x - a$:

$$f(a+h) = c_0 + c_1 h + c_2 h^2 + \cdots = \sum_{m=0}^{\infty} c_m h^m$$

We find c_m by differentiating m times wrt h and setting $h = 0$:

$$f^m(a) = m! c_m \implies c_m = \frac{f^m(a)}{m!}$$

Hence:

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!} f''(a) + \cdots = \sum_{m=0}^{\infty} \frac{h^m}{m!} f^m(a)$$

Or in operator form:

$$f(a+h) = \exp\left[h \frac{\partial}{\partial x}\right] f(a)$$

Noting that the exponential arises due to the Taylor series for the exponential function:

$$e^x = 1 + u + \frac{u^2}{2!} + \cdots = \sum_{m=0}^{\infty} \frac{u_m}{m!}$$

$$\begin{aligned} f(a+h) = &\ f(x)\big|_{x=a} \\ &+ h\frac{d}{dx} f(x)\Big|_{x=a} \\ &+ \frac{h^2}{2!} \frac{d^2}{dx^2} f(x)\Big|_{x=a} \\ &\vdots \\ &= \left[1 + h\frac{d}{dx} + \frac{h^2}{2!}\frac{d^2}{dx^2} + \frac{h^3}{3!}\frac{d^3}{dx^3} \cdots\right] f(x)\Big|_{x=a} \\ &= \exp\left[h\frac{d}{dx}\right] f(x)\Big|_{x=a} \end{aligned}$$

2.1 Multiple Variables

We want to expand a function $f(x, y)$ around the point $(x, y) = (a, b)$ as a power series.

This will take the form:

$$f(a+h, b+k) = c_{00} + c_{10}h + c_{01}k + c_{20}h^2 + c_{11}hk + c_{02}k^2 + \cdots \sum_{m=0}^{\infty}\sum_{n=0}^{\infty} c_{mn} h^m k^n$$

We complete the same process to find coefficients, to find c_{mn} we differentiate m times wrt k, and n times wrt h and set $h = k = 0$ to get:

$$\frac{\partial^{m+n}}{\partial x^m \partial y^n}(a, b) = m!n!c_{mn}$$

Hence:

$$f(a+h, b+k) = \sum_{m=0}^{\infty}\sum_{n=0}^{\infty} \frac{h^m k^n}{m!n!} \frac{\partial^{m+n} f}{\partial x^m \partial y^n}(a, b)$$

Again in operator form this gives:

$$\exp\left[h\frac{\partial}{\partial x}\right] \exp\left[k\frac{\partial}{\partial y}\right] f(x, y)\Big|_{x=a, y=b}$$

2.2 Generalising to Many Variables

Consider a function of p variables $(x_1, x_2, \dots, x_p)$:

$$f(a_1 + h_1, \dots, a_p + h_p) = \sum_{n_1=0}^{\infty} \sum_{n_2=0}^{\infty} \cdots \sum_{n_p=0}^{\infty} c_{n_1 n_2 \dots n_p} h_1^{n_1} h_2^{n_2} \cdots h_p^{n_p}$$

Again to find the coefficients we differentiate n_1 times wrt h_1, n_2 times wrt h_2 etc. We set $h_1 = h_2 = \cdots = h_p = 0$ to get:

$$n_1! n_2! \cdots n_p! c_{n_1 n_2 \dots n_p} = \frac{\partial^{n_1+n_2+\cdots+n_p} f}{\partial x_1^{n_1} \partial x_2^{n_2} \cdots \partial x_p^{n_p}}(a_1, \dots, a_p)$$

Hence:

$$f(a_1 + h_1, \dots, a_p + h_p) = \sum_{n_1=0}^{\infty} \cdots \sum_{n_p=0}^{\infty} \frac{h_1^{n_1} \cdots h_p^{n_p}}{n_1! \cdots n_p!} \frac{\partial^{n_1+\cdots+n_p} f}{\partial x_1^{n_1} \cdots \partial x_p^{n_p}}(a_1, \dots, a_p)$$

In compact vector notation, where $\underline{h} = (h_1, \dots, h_p)$, this is:

$$f(\underline{a} + \underline{h}) = \exp\left[\sum_{i=1}^{p} h_i \frac{\partial}{\partial x_i}\right] f(\underline{a})$$

$$f(\underline{a} + \underline{h}) = \exp\left[\underline{h} \cdot \underline{\nabla}\right] f(\underline{a})$$

Thu 29 Jan 2026 09:00

Lecture 5 - Partial Differentiation V: 3D Stationary Points and Lagrange Multipliers

1 2D Recap

In 2D, a stationary point of a function $f(x)$ occurs at $x = a$ where $f(x)$ is neither increasing or decreasing such that $f'(a) = 0$.

Consider the Taylor series about $x = a$, where $f'(a) = 0$:

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!}f''(a) + \cdots = f(a) + \frac{h^2}{2!}f''(a) + \cdots$$

If $f''(a) > 0$, the function looks like $+h^2$ near the stationary point. This gives us a minima. If the second derivative is negative the function looks like $-h^2$, giving a maximum point.

If the second derivative is zero, we need to find the first non-zero derivative $f^n(a) \neq 0$ where $n > 2$.

- If $n = 2k+1$ (odd), the function looks like $\pm h^{2k+1}$ near the stationary point. This results in a point of inflexion.
- If $n = 2k$ (even) and $f^{2k}(a) > 0$, the function looks like $+h^{2k}$ near the stationary point so that we have a minimum.
- If $n = 2k$ (even) and $f^{2k}(a) < 0$, the function looks like $-h^{2k}$ near the stationary point so that we have a maximum.

2 Stationary Points of Several Variables

2.1 Two Variables

Consider a function $f(x, y)$ at a point (a, b). For a stationary point to occur here, the function must be flat in all directions at that point, so all the partial derivatives must be zero:

$$f_x(a, b) = f_y(a, b) = 0$$

To classify it, we again consider the Taylor series:

$$f(a+h, b+k) \approx f(a, b) + \frac{1}{2}\left(Ah^2 + 2Bhk + Ck^2\right)$$

Where $A = f_{xx}(a, b)$, $B = f_{xy}(a, b)$ and $C = f_{yy}(a, b)$. We can then complete the square:

$$Ah^2 + Bhk + Ck^2 = A\left(h + \frac{B}{A}k\right)^2 + \left(C - \frac{B^2}{A}\right)k^2$$

- For the stationary point to be a minimum, we need $A > 0$ and $C - B^2/A = AC - B^2 > 0$.
- For a maximum, we need $A < 0$ and $C - B^2/A = AC - B^2 < 0$.
- For a saddle point, with a minimum in one direction and a maximum in the other, we need A and $C - B^2/A$ to be oppositely signed. This corresponds to the case where $AC - B^2 < 0$.

Considering the second derivatives directly:

- For a minimum we need $f_{xx} > 0$ and the Hessian determinant must be positive:

$$\Delta = \det H = \det \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} > 0$$

- For a maximum, we need $f_{xx} < 0$ and the Hessian determinant must be positive:

$$\Delta = \det H = \det \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} > 0$$

- For a saddle point, we need the Hessian determinant to be negative:

$$\Delta = \det H = \det \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} < 0$$

2.2 Generalising To 3D

For a function $f(x, y, z)$ to have a stationary point at (a, b, c), all first derivatives must be zero. The Taylor expansion then has no linear term, and performing the Taylor Expansion gives:

$$f(a+h, b+k, c+l) - f(a, b, c) = \frac{1}{2}\left(Ah^2 + 2Bhk + 2Chl + Dk^2 + 2Ekl + Fl^2\right)$$

where $A = f_{xx}$, $B = f_{xy}$, $C = f_{xz}$, $D = f_{yy}$, $E = f_{yz}$, $F = f_{zz}$, all evaluated at (a, b, c). We define the quadratic:

$$Q = Ah^2 + 2Bhk + 2Chl + Dk^2 + 2Ekl + Fl^2$$

as the extra factor of 1/2 does not change the classification of the point. The stationary point is a minimum $Q > 0$ for all $(h, k, l) \neq (0, 0, 0)$, because a positive Q means that the function increases in all directions away from the point.

Completing the square in h (treating k and l as fixed):

$$Ah^2 + 2Bhk + 2Chl = A\left(h + \frac{B}{A}k + \frac{C}{A}l\right)^2 - A\left(\frac{B}{A}k + \frac{C}{A}l\right)^2$$

Expanding the subtracted term:

$$-A\left(\frac{B}{A}k + \frac{C}{A}l\right)^2 = -A\left(\frac{B^2}{A^2}k^2 + \frac{2BC}{A^2}kl + \frac{C^2}{A^2}l^2\right) = -\frac{B^2}{A}k^2 - \frac{2BC}{A}kl - \frac{C^2}{A}l^2$$

Substituting back and collecting the k- and l-terms:

$$Q = A\left(h + \frac{B}{A}k + \frac{C}{A}l\right)^2 + \underbrace{\left(D - \frac{B^2}{A}\right)}_{A'}k^2 + 2\underbrace{\left(E - \frac{BC}{A}\right)}_{B'}kl + \underbrace{\left(F - \frac{C^2}{A}\right)}_{C'}l^2$$

The second quadratic in $A', 2B', C'$ is just our previous 2D example, so that condition holds. For the quadratic, we require:

$$A > 0$$
$$D - \frac{B^2}{A} > 0$$
$$\left(D - \frac{B^2}{A}\right)\left(F - \frac{C^2}{A}\right) - \left(E - \frac{BC}{A}\right)^2 > 0$$

Expanding the last condition we get:

$$A^2DF - ADC^2 - AB^2F + B^2C^2 - E^2A^2 - B^2C^2 + 2EABC > 0$$

$$A\left(ADF - DC^2 - B^2F - E^2A + 2EBC\right) > 0$$
$$\left(ADF - DC^2 - B^2F - E^2A + 2EBC\right) > 0 \qquad \text{as } A > 0$$
$$A(DF - E^2) - B(BF - EC) + C(BE - CD) > 0$$
$$\implies \det\begin{bmatrix} A & B & C \\ B & D & E \\ C & E & F \end{bmatrix} > 0$$

Finally, our conditions for a minimum are:

- $$f_{xx} > 0$$
  .
- $$\det\begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} > 0$$
- $$\det\begin{bmatrix} f_{xx} & f_{xy} & f_{xz} \\ f_{yx} & f_{yy} & f_{yz} \\ f_{zx} & f_{zy} & f_{zz} \end{bmatrix} > 0$$

And for a maximum:

- $$f_{xx} < 0$$
  .
- $$\det\begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} > 0$$
- $$\det\begin{bmatrix} f_{xx} & f_{xy} & f_{xz} \\ f_{yx} & f_{yy} & f_{yz} \\ f_{zx} & f_{zy} & f_{zz} \end{bmatrix} < 0$$

More generally, for a minimum we need these matrices, called the leading minors of the Hessian, to be positive and for a maximum we need them to alternate in sign.

3 Lagrange Multipliers

Suppose we wanted to minimise a function $f(x, y) = x^2 + y^2$ subject to a constraint, i.e. saying $x + y = 1$ (and therefore excluding the $(0, 0)$ trivial case).

While we could do this by writing $y = 1 - x$, substituting and minimising solely in terms of x, there is a more sophisticated method that becomes useful in more complex cases where the constraint is not invertable.

We consider the function:

$$F(x, y, \lambda) = x^2 + y^2 + \lambda(x + y - 1)$$

And minimise with respect to all three variables. When this happens, $F(x, y, \lambda) = f(x, y)$ at the minimised point.

Generally, to find the extrema of $f(x_1, x_2, \dots, x_n)$ subject to p constraints $g_i(x_1, x_2, \dots, x_n) = 0$, where $p < n$, we introduce a new function of $n + p$ variables:

$$F(x_1, x_2, \dots, x_n, \lambda_1, \lambda_2, \dots, \lambda_p) = f(x_1, x_2, \dots, x_n) + \sum_{i=1}^{p} \lambda_i g_i(x_1, x_2, \dots, x_p)$$

We then extremise F wrt all $n + p$ variables. Extremising with respect to λ_i enforces the constraint:

$$F_{\lambda_i} = g_i(x_1, x_2, \dots, x_n) = 0$$

So the the lambda term ends up dropping out and F and f have the same numerical value at extrema.

Fri 30 Jan 2026 12:00

Lecture 6 - Partial Differentiation VI: 3D Stationary Points

1 2D Recap

In 2D, a stationary point of a function $f(x)$ occurs at $x = a$ where $f(x)$ is neither increasing or decreasing such that $f'(a) = 0$.

Consider the Taylor series about $x = a$, where $f'(a) = 0$:

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!}f''(a) + \cdots = f(a) + \frac{h^2}{2!}f''(a) + \cdots$$

If $f''(a) > 0$, the function looks like $+h^2$ near the stationary point. This gives us a minima. If the second derivative is negative the function looks like $-h^2$, giving a maximum point.

If the second derivative is zero, we need to find the first non-zero derivative $f^n(a) \neq 0$ where $n > 2$.

- If $n = 2k + 1$ (odd), the function looks like $\pm h^{2k+1}$ near the stationary point. This results in a point of inflexion.
- If $n = 2k$ (even) and $f^{2k}(a) > 0$, the function looks like $+h^{2k}$ near the stationary point so that we have a minimum.
- If $n = 2k$ (even) and $f^{2k}(a) < 0$, the function looks like $-h^{2k}$ near the stationary point so that we have a maximum.

2 Stationary Points of Several Variables

2.1 Two Variables

Consider a function $f(x, y)$ at a point (a, b). For a stationary point to occur here, the function must be flat in all directions at that point, so all the partial derivatives must be zero:

$$f_x(a, b) = f_y(a, b) = 0$$

To classify it, we again consider the Taylor series:

$$f(a+h, b+k) \approx f(a, b) + \frac{1}{2}\left(Ah^2 + 2Bhk + Ck^2\right)$$

Where $A = f_{xx}(a, b)$, $B = f_{xy}(a, b)$ and $C = f_{yy}(a, b)$. We can then complete the square:

$$Ah^2 + Bhk + Ck^2 = A\left(h + \frac{B}{A}k\right)^2 + \left(C - \frac{B^2}{A}\right)k^2$$

- For the stationary point to be a minimum, we need $A > 0$ and $C - B^2/A = AC - B^2 > 0$.
- For a maximum, we need $A < 0$ and $C - B^2/A = AC - B^2 < 0$.
- For a saddle point, with a minimum in one direction and a maximum in the other, we need A and $C - B^2/A$ to be oppositely signed. This corresponds to the case where $AC - B^2 > 0$.

Considering the second derivatives directly:

- For a minimum we need $f_{xx} > 0$ and the Hessian determinant must be positive:

$$\Delta = \det H = \det \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} > 0$$

- For a maximum, we need $f_{xx} < 0$ and the Hessian determinant must be positive:

$$\Delta = \det H = \det \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} > 0$$

- For a saddle point, we need the Hessian determinant to be negative:

$$\Delta = \det H = \det \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} < 0$$

2.2 Generalising To 3D

For a function $f(x, y, z)$ to have a stationary point at point (a, b, c), it must be flat in all directions around that point. Therefore, all the first derivatives must be zero:

$$f_x(a, b, c) = f_y(a, b, c) = f_z(a, b, c) = 0$$

The Taylor Series about the point (a, b, c) is:

$$f(a+h, b+k, c+l) \approx f(a, b, c) + \frac{1}{2}\left(Ah^2 + 2Bhk + 2Chl + Dk^2 + 2Ekl + Fl^2\right)$$

Where $A = f_{xx}, B = f_{xy}, C = f_{xz}, D = f_{yy}, E = f_{yz}, F = f_{zz}$. Again we complete the square:

$$A\left(h + \frac{B}{A}k + \frac{C}{A}l\right)^2 + \left(D - \frac{B^2}{A}\right)k^2 + 2\left(E - \frac{BC}{A}\right)kl + \left(F - \frac{C^2}{A}\right)l^2$$

Fri 13 Feb 2026 12:00

Lecture 12 - End of Partial Differentiation & Start of ODEs

Recap of lecture 11:

- The tangent plane to a surface $f(x, y, z) = 0$ at (x_0, y_0, z_0) is given by:

$$\left(\frac{\partial f}{\partial x}\right)_0 (x - x_0) + \left(\frac{\partial f}{\partial y}\right)_0 (y - y_0) + \left(\frac{\partial f}{\partial z}\right)_0 (z - z_0) = 0$$

  Such that $\underline{\nabla} f(x_0, y_0, z_0)$ is the normal vector to the plane
- The parametric representation of a curve $\underline{r}(t)$ has:
  - Unit tangent: $\underline{\hat{T}} = \frac{dr}{dt} / \left|\frac{dr}{dt}\right|$
  - Arc length $s(t)$: $\frac{ds}{dt} = \left|\frac{dr}{dt}\right| \to \underline{\hat{T}} = \frac{dr}{ds}$.
  - Unit normal and curvature:

1 Orthonormal Triads

We can create an *orthonormal triad* by introducing a new normal vector called the unit binormal, $\underline{\hat{B}} = \underline{\hat{T}} \times \underline{\hat{N}}$.
Since $\underline{\hat{N}} \times \underline{\hat{N}}$, differentiating wrt s gives:

$$\underline{\hat{N}} \cdot \frac{d\underline{\hat{N}}}{ds} = 0$$

TODO

We have:

$$\frac{d\underline{\hat{T}}}{ds} = \kappa \underline{\hat{N}}$$

$$\frac{d\underline{\hat{N}}}{ds} = -\kappa \underline{\hat{T}} + \tau \underline{\hat{B}}$$

Hence:

$$\begin{aligned}
\frac{d\underline{\hat{B}}}{ds} &= \frac{d}{ds}\left(\underline{T} \hat{\times} \underline{\hat{N}}\right) \\
&= \frac{d\underline{\hat{T}}}{ds} \times \underline{\hat{N}} + \underline{\hat{T}} \times \frac{d\underline{\hat{N}}}{ds} \\
&= \kappa \underline{\hat{N}} \times \underline{\hat{N}} + \underline{\hat{T}} \times \left(-\kappa \underline{\hat{T}} + \tau \underline{\hat{B}}\right) \\
&= \tau \underline{\hat{T}} \times \underline{\hat{B}} = \tau \underline{\hat{T}} \times \left(\underline{\hat{T}} \times \underline{\hat{N}}\right) \\
&= \tau \left[\left(\underline{\hat{T}} \cdot \underline{\hat{N}}\right) \underline{\hat{T}} - \left(\underline{\hat{T}} \cdot \underline{\hat{T}}\right) \underline{\hat{N}}\right]
\end{aligned}$$

$$= \tau \underline{\hat{N}}$$

This gives the Frenet-Serret Formulae:

This concludes partial differentiation! :D

2 Ordinary Differential Equations

A differential equation is any equation that involves derivatives. We care, because most laws of physics manifest themselves in the form of differential equations. For example:

$$\text{Newton's Second Law:} \qquad \underline{F} = m\frac{d^2\underline{r}}{dt^2}$$

$$\text{3D Time-Independent Schrödinger Eqn:} \qquad -\frac{\hbar}{2m}\left(\frac{\partial^2\psi}{dx^2}+\frac{\partial^2\psi}{dy^2}+\frac{\partial^2\psi}{dz^2}\right)+V(x,y,z)\psi = E\psi$$

$$\text{3D Wave Eqn:} \qquad \frac{\partial^2\phi}{\partial x^2}+\frac{\partial^2\phi}{\partial y^2}+\frac{\partial^2\phi}{\partial z^2} = \frac{1}{c^2}\frac{\partial^2\phi}{\partial t^2}$$
$$= \nabla^2\phi = \frac{1}{c^2}\frac{\partial^2\phi}{\partial t^2}$$

$$\text{Gauss' Law:} \qquad \frac{\partial E_x}{\partial x}+\frac{\partial E_y}{\partial y}+\frac{\partial E_z}{\partial z} = \frac{\rho}{\epsilon_0}$$

$$\text{Navier-Stokes Eqn:} \qquad \rho\left(\frac{\partial\underline{v}}{\partial t}+\left(\underline{v}\cdot\underline{\nabla}\right)\underline{v}\right) = -\underline{\nabla}p+\rho\underline{g}+\mu\nabla^2\underline{v}$$

In this course, we will only solve DEs of a single variable, i.e. Ordinary Differential Equations (ODEs). We don't look at Partial DEs of multiple variables yet.

In order to think about solving these, we need to classify them. Most DEs aren't soluble in closed form with elementary functions and need to be solved numerically. Here, we only consider nice soluble functions, but this is a vast minority in reality. We want to identify classes of DEs we can reasonable solve with a method for each.

We can generally solve linear equations by breaking them into small chunks and solving them individually, for example.

3 Types of DEs

3.1 Partial vs. Ordinary

In the examples above, only the first was an ODE, and the rest PDEs. Ordinary Differential Equations (ODEs) involve only a single variable.

Consider a vector $\underline{r}(t) = (x(t), y(t), z(t))$. t is called the independent variable, with x, y, z being dependant variables. While we have 3 dependant variables, we only have one independent variable (so only one thing to differentiate wrt), so this would end up being ordinary.

PDEs involve equations of two or more variables and hence involve partial derivatives.

3.2 Order

The order of a DE is given by the order of the highest derivative involved, so Newton's 2nd Law is a second order DE, as the highest order derivative is a second derivative.

3.3 Degree

The degree of a DE is a less important measure than the others. It is given by the highest power of the highest order derivative. For example, Newton's 2nd is a first degree, while an equation containing a^3 would be third degree (and second order, as a is a second derivative).

Ideally, we want this to be 1 for ease of solving, and higher degrees are rare but they do exist. For example, from Lagrangian Mechanics we have:

$$\frac{1}{2m}\left[\left(\frac{\partial s}{\partial x}\right)^2+\left(\frac{\partial s}{\partial y}\right)^2+\left(\frac{\partial s}{\partial z}\right)^2\right]+V(x,y,z) = \frac{ds}{dt}$$

3.4 Homogenous and Inhomogeneous

A homogenous DE is a DE that does not have any terms of only the independent variable(s), while an inhomogeneous DE does.

For example, Newton's 2nd is homogenous as there is no term that involves t alone. This would be inhomogeneous:

$$\frac{\partial^2 x}{\partial t^2} = t + x$$

While this would be homogenous:

$$\frac{\partial^2 x}{\partial t^2} = tx$$

As t is a coefficient and not a pure term in its own right.

3.5 Linear and Non-Linear

A DE is linear if the dependant variable(s) and all of its/their derivatives occur purely as linear functions. For example:

$$\left(1 - x^2\right) \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + n(n+1)y = 0$$

This is linear, as the dependant variable y never has a power greater than 1.

$$\frac{dy}{dx} + xy = 0$$

Is also linear, while this is not:

$$\frac{dy}{dx} + xy^2 = 0$$

This is also non-linear (as shown by the Taylor Expansion of sine):

$$\frac{d^2 \theta}{dt^2} = -\frac{g}{l} \sin\theta$$

3.6 Examples

$$(1) \quad \left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 = u^2$$

Homogenous first-order second-degree non-linear PDE.

$$(2) \quad \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = x^2 + y^2 + z^2$$

Inhomogeneous second-order first-degree linear PDE.

$$(3) \quad \frac{\partial y}{\partial x} + y^2 = x$$

Inhomogeneous first-order second-degree non-linear ODE.

Wed 18 Feb 2026 11:00

Lecture 13 - ODEs II

Summary of Lecture 12:

- Frenet-Serret Equations for a curve $\underline{r}(s)$, where $\hat{T} = d\underline{r}/ds$.
- Classification of differential equations:
  - Partial vs Ordinary
  - Order
  - Degree
  - Homogeneous vs Inhomogeneous
  - Linear vs Non-Linear

1 Equations Soluble By Direct Integration

Given some:

$$\frac{dy}{dx} = f(x) \implies y(x) = \int^x f(x')dx' + c$$

We can generalise this to some repeated derivative:

$$\frac{d^n y}{dx^n} = f(x)$$

Instead of having one undetermined constant here, we now have n. As each round of integration picks up a factor of the integration subject, these constants will form an nth degree polynomial.

1.1 Example: Particle Falling

$$\frac{d^2 z}{dt^2} = -g$$
$$\implies \frac{dz}{dt} = -gt + v_0$$
$$z = -\frac{1}{2}gt^2 + v_0 t + z_0$$

Here we consider our unknown constants as boundary conditions, i.e. the initial velocity and initial height.

2 Separable Equations

These are equations in the form:

$$\frac{dy}{dx} = \frac{f(x)}{g(x)}$$

$$g(y)dy = f(x)dx \implies \int g(y)dy = \int f(x)dx + c$$

2.1 Example I: Falling Particle with Air Resistance

$$\frac{dv}{dt} = -g - kv$$
$$\int \frac{dv}{g + kv} = -\int dt$$
$$\implies \frac{1}{k}\ln(g + kv) = -t + c$$
$$\implies k + gv = Ae^{-kt}$$
$$\implies v(t) = -\frac{g}{k} + \left(v_0 + \frac{g}{k}\right)e^{-kt}$$

2.2 Example II

$$\frac{dy}{dx} - x^2y^2 = x^2$$
$$\frac{dy}{dx} = x^2 + x^2y^2 = x^2(y^2 + 1)$$
$$\implies \frac{1}{y^2 + 1}dy = x^2dx$$
$$\implies \arctan y = \frac{1}{3}x^3 + c$$
$$\implies y = \tan\left(\frac{1}{3}x^3 + c\right)$$

2.3 Example III

$$\frac{dy}{dx} = -\frac{x}{y}$$
$$ydy = -xdx$$
$$\frac{1}{2}y^2 = -\frac{1}{2}x^2 + c$$
$$y^2 + x^2 = 2c$$

This is the equation for a circle.

3 Exact Equations

Suppose a function $y(x)$ is implicitly defined such that $f(x, y) = c$. It follows that the total differential:

$$df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy = 0$$

As $f(x, y)$ is a constant for all values of x, y, the constant differentiates to zero.

We can set:

$$M(x, y) = \frac{\partial f}{\partial x} \qquad N(x, y) = \frac{\partial f}{\partial y}$$

To try to solve this equation:

$$M(x, y)dx + N(x, y)dy = 0$$

If we can do this (i.e. if the function can be decomposed into gradient components M and N), the equation is called "exact". For this to be true, we need:

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = \frac{\partial^2 f}{\partial x \partial y}$$

If this relationship holds, we have an exact equation and can integrate M and N to get $f(x, y)$ to solve the equation in the form $f(x, y) = c$

3.1 Example

$$\frac{dy}{dx} = -\frac{2x+y}{x+2y}$$
$$(2x+y)dx + (x+2y)dy = 0$$

So:

$$M(x,y) = 2x+y \qquad N(x,y) = x+2y$$

And:

$$\frac{\partial M}{\partial y} = 1 \qquad \frac{\partial N}{\partial x} = 1$$

So yes, it is an exact function. It follows that:

$$\frac{\partial f}{\partial x} = 2x+y \implies f(x,y) = x^2 + xy + c(y)$$
$$\frac{\partial f}{\partial y} = x+2y \implies f(x,y) = xy + y^2 + d(x)$$
$$\implies f(x,y) = x^2 + xy + y^2 = c$$

Thu 19 Feb 2026 09:00

Lecture 14

Recap of last lecture:

- Soluble by direct integration:

$$\frac{dy}{dx} = f(x) \implies y(x) = \int f(x)dx + c$$

- Separable equations:

$$\frac{dy}{dx} = \frac{f(x)}{g(y)} \implies \int f(x)dx = \int g(y)dy$$

- Exact equations:

  Given in the form (or re-arrangeable to the form):

$$M(x, y)dx + N(x, y)dy = 0$$

  These are soluble if:

$$M(x, y) = \frac{df}{dx} \qquad N(x, y) = \frac{df}{dy}$$

  With $f(x, y) = c$.

  The condition for an equation being exact is:

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

1 "Homogeneous" Equations

Here, the word homogenous has a different meaning to classification of differentiating equations. If we have some function that satisfied this:

$$g(\lambda x, \lambda y) = \lambda^p g(x, y)$$

We call it a homogeneous function of order p.

"Homogeneous" differential equations in this case are functions of the form:

$$\frac{dy}{dx} = f\left(\frac{y}{x}\right)$$

Here, we make the substitution $v(x) = y(x)/x$. We can write:

$$y(x) = v(x)x$$

And substitute back in. We use the product rule where:

$$\frac{dy}{dx} = x\frac{dv}{dx} + v = f(v)$$

$$x\frac{dv}{dx} = f(v) - v$$

This is now separable, so:

$$\int \frac{dv}{f(v) - v} = \int \frac{dx}{x}$$

1.1 Example

$$2xydy - (x^2 + y^2)dx = 0$$

This is not exact, so we put it into a standard form and substitute:

$$2xydy = (x^2 + y^2)dx$$

$$\frac{dy}{dx} = \frac{x^2 + y^2}{2xy} = \frac{(y/x)^2 + 1}{2(y/x)}$$

Let $u(x) = y/x$:

Fri 20 Feb 2026 11:56

Lecture 15 - Differential Equations III

Summary of last lecture:

- "Homogeneous" Equations
$$\frac{dy}{dx} = f\left(\frac{x}{y}\right) \to x\frac{dv}{dx} = f(v) - v \text{ where } y(x) = xv(x)$$
- Linear Equations
$$\sum_{k=0}^{n} a_k(x)\frac{d^k y}{dx^k} = f(x) \to y(x) = \sum_{k=1}^{n} \alpha_k y_k(x) + y_{PI}(x)$$
The solution is a sum of the complementary function (the general solution of the homogenous equation) and the particular integral (one solution of the inhomogeneous equation).
- First Order Linear Equations
$$\frac{dy}{dx} = P(x)y = Q(x) \to \frac{d}{dx}\left[y(x)e^{\int P(x)dx}\right] = Q(x)e^{\int P(x)dx}$$

1 Examples

1.1 Example I

$$(1-x^2)\frac{dy}{dx} - xy = 1$$

Rewriting in the standard form:
$$\frac{dy}{dx} - \frac{x}{1-x^2}y = \frac{1}{1-x^2}$$

Hence:
$$P(x) = \frac{-x}{1-x^2}$$
$$I(x) = \exp\left(-\int \frac{xdx}{1-x^2}\right) = \exp\left(\frac{1}{2}\ln(1-x^2)\right) = \sqrt{1-x^2}$$

Multiplying through by the integrating factor:
$$\sqrt{1-x^2}\frac{dy}{dx} - \frac{x}{\sqrt{1-x^2}}y = \frac{1}{\sqrt{1-x^2}}$$

The L.H.S is now the derivative of a product:
$$\frac{d}{dx}\left(y\sqrt{1-x^2}\right) = \frac{1}{\sqrt{1-x^2}}$$

And integrating both sides:
$$y\sqrt{1-x^2} = \arcsin x + c$$
$$\implies \boxed{y = \frac{c}{\sqrt{1-x^2}} + \frac{\arcsin x}{\sqrt{1-x^2}}}$$

1.2 Example II

2 Linear ODEs with Constant Coefficients

In the most general form, we have:

$$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1 \frac{dy}{dx} + a_0 y = f(x)$$

Firstly, we solve the homogenous equation:

$$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots a_1 \frac{dy}{dx} + a_0 y = 0$$

Let $y(x) = e^{\lambda x}$:

$$e^{\lambda x} \left(a_n \lambda^n + a_{n-1} \lambda^{n-1} + \cdots + a_1 \lambda + a_0 \right) = 0$$

This reduces to finding the zeroes of the corresponding nth order polynomial. If this polynomial has n distinct zeroes, then the complimentary function is given by:

$$y_{CF}(x) = \alpha_1 e^{\lambda_1 x} + \alpha_2 e^{\lambda_2 x} + \cdots + \alpha_n e^{\lambda_n x}$$

If these roots contain a repeated root, this will reduce the number of unique solutions by one. If we have any complex solutions, they wil come in complex conjugate pairs:

$$a^{(\pm ib)x} = e^{ax} (\cos bx \pm i \sin bx)$$

This therefore has independent real solutions:

$$e^{ax} \cos bx$$

$$e^{ax} \sin bx$$

3 Equidimensional Equations

These have coefficients which do depend on x, but where the coefficients are functions of x such that the nth derivative has a coefficient of $a_n x^n$

The general form of a homogeneous equidimensional equation is:

$$a_n x^n \frac{d^n y}{dx^n} + a_{n-1} x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1 x \frac{dy}{dx} + a_0 y = f(x)$$

This has a solution in the general form $y(x) = x^\lambda$. When we differentiate k times with respect to x we lose k powers of x, as each differentiation decreases the power. We need to restore this therefore with a x^k prefactor.

4 Mass on a Spring (SHM)

A mass, m on a spring is displaced from its equilibrium position by some distance x. There is a restoring force given by $F = -kx$.

$$F = ma \implies F = m \frac{d^2}{dt}$$

$$m \frac{d^2 x}{dt^2} + kx = 0$$

$$\frac{d^2 x}{dt^2} + \omega^2 X = 0 \qquad \text{where } \omega^2 = \frac{k}{m}$$

We have:

$$x(t) = e^{\lambda t}$$

$$(\lambda^2 + \omega^2)e^{\lambda t} = 0$$
$$\lambda^2 + \omega^2 = 0$$

Hence:

$$\lambda = e^{\pm i\omega t}$$

Thu 05 Mar 2026 09:00

Lecture 20 - Methods of Integration

The derivative of an elementary function is always an elementary function. The integral of an elementary function is not always elementary however.
As a result, differentiation is always purely algorithmic churn, integration is often tricky.

Methods of integration are often based on methods of differnetiation, for example:

- Integration by inspection: just spotting the answer.
- Integration by parts: Inverse of the product rule of differentiation.
- Integration by substitution: Inverse of the chain rule of differentiation.

1 Integration By Inspection

This is the most difficult to codify and can be easy for some people and very easy for others.

This might involve:

- Trigonometric Identities, for example replacing an expression whose integral is unknown with an expression whose integral is, using a trig sub.

LC Temperature and Matter

Tue 20 Jan 2026 12:00

Lecture 1 - Matter I: Welcome and Intro

1 Course Welcome

- Study guide and full revision notes (from Prof. Mike Gunn) on Canvas.
  - The course has changed a bit recently, so the full revision notes have additional theory/content which isn't examinable.
- Series of non-assessed problems also on Canvas.

1.1 Course Aims

We want to be able to explain the properties of matter (in the macroscopic world) of various kinds, based on their microscopic behaviour. We also want to be able to understand more about the microscopic behaviour of matter from macroscopic observables.

1.2 Recommended Reading

- Young and Freedman, Ch14 and Ch17-19.
- Properties of Matter, Flowers and Mendoza.
- Introduction to Thermal Physics, Adkins.

2 Intro to Matter

2.1 States of Matter

We want to move away from a simple "cold stuff is a solid, warmer stuff is a liquid and hotter stuff is a gas" by also considering how different materials change state as a function of both pressure and temperature:

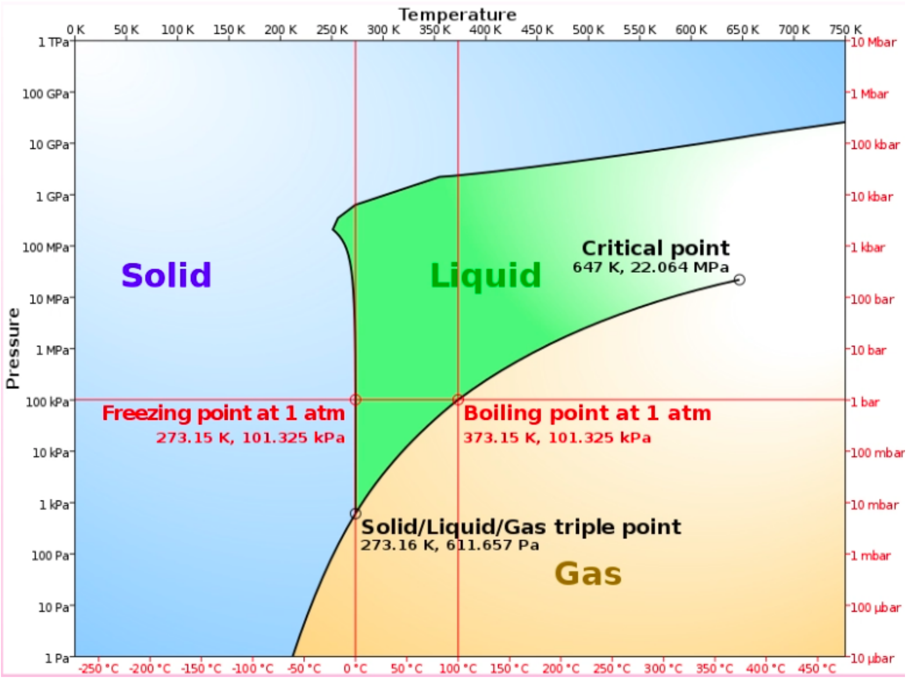


Figure 40.1: Phase Diagram for Water

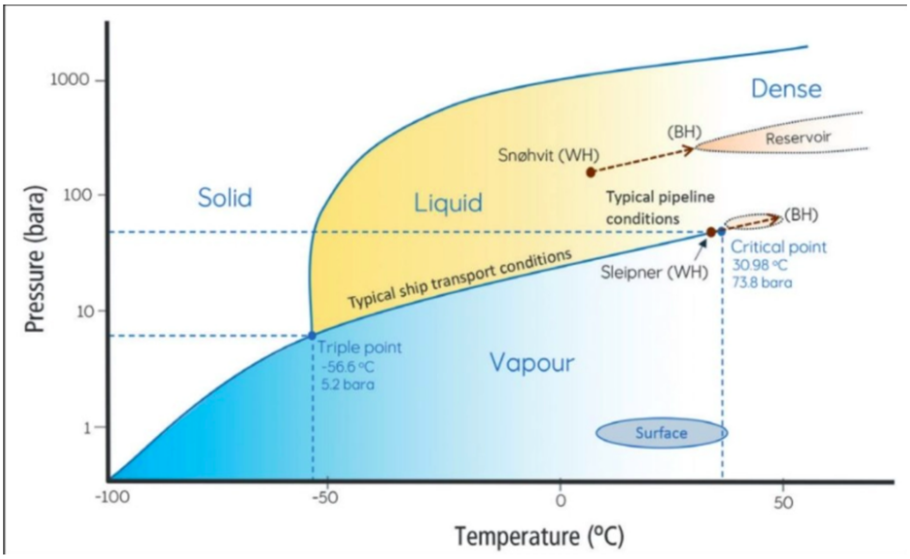


Figure 40.2: Phase Diagram for CO2

At "Standard Pressure and Temperature" (at 20°C and 1ATM), we see that increasing pressure of C02 takes it to a liquid state. However, at lower temperatures, decreasing the pressure can cause it to sublimate straight from a solid to a gas, without a liquid intermediary.

Taking water as an example, pushing the pressure/temperature causes another high-energy state altogether called a "supercritical fluid". The gradient for the initial vertical portion of the solid/liquid boundary around the freezing point is slightly negative, so applying pressure to ice on this boundary causes it to become liquid. This is uncommon for many materials.

This is because water is weird, where the solid form is less dense than the liquid form. This is how ice skating works, where applying pressure (from a skate) causes a thin upper layer of ice to melt. This thin water layer is what the ice skate can travel on.

Every material also has a "triple point" where the material can act simultaneously in all three standard states of matter.

Thu 22 Jan 2026 13:00

Lecture 2 - Matter II: Atomic Physics and LJP

1 Atomic Sizes

As we increase proton number, we might expect that the atomic radius increases (as there's physically more stuff). We might expect that adding more electrons would expect a corresponding increase in the atomic radius to fit these extra electrons.

This doesn't actually prove to be true, and atomic size and properties are generally determined by the outermost electrons, and atomic radius stays mostly constant.

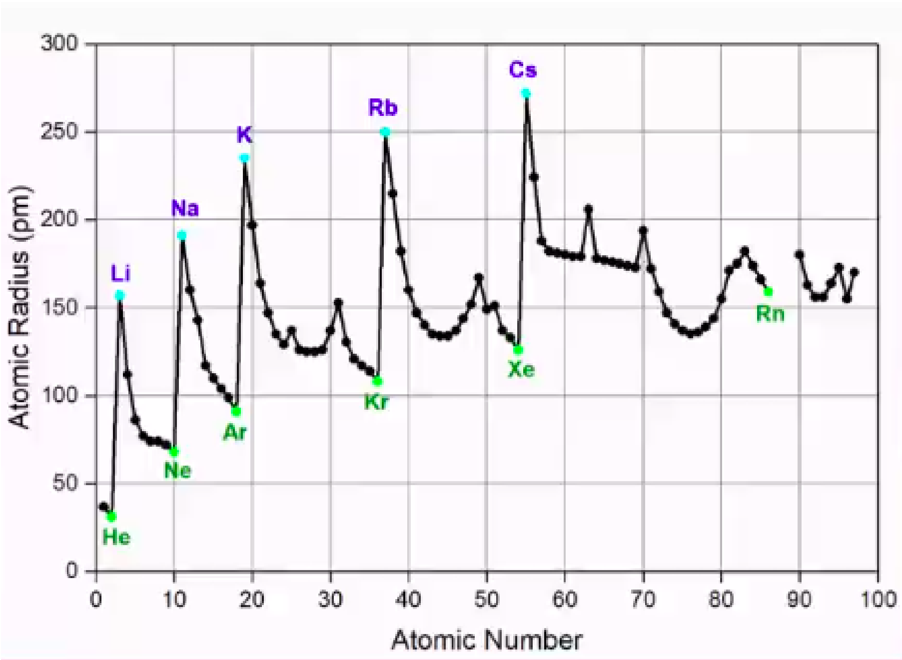


Figure 41.1: Variation of Atomic Radius with Atomic Number

We do see a large increase in atomic radius for the group one metals, where the outermost electron is very weakly attracted to the nucleus (is easily ionised). For an exam, we can consider atomic radius to be approximately 10^{-10}.

We see a similar trend when considering ionisation energy (the energy required to remove the outermost electron):

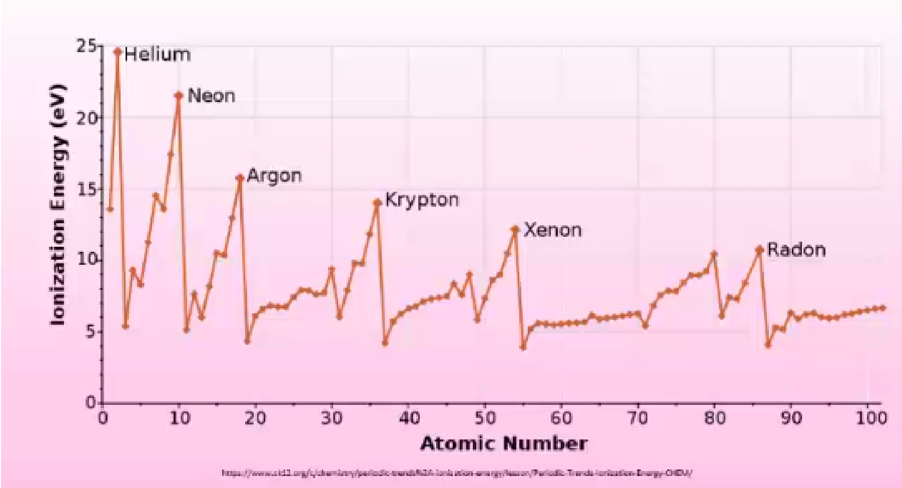


Figure 41.2

This is approximately the same for the all atomic numbers, with the exception of noble gases (which are extremely difficult to ionise as a result of having full capacity outer shells).

This doesn't apply however to the nuclear radius, R, as the nucleus is compressed to a much higher degree, so there's not space to fit additional nucleons without increasing radius. For atomic radius A, and modelling the nucleus as a sphere, we get:

$$R = r_0 A^{\frac{1}{3}}$$

Where $r_0 = 1.2 \times 10^{-15} m = 1.2$fm is a constant.

2 Atomic Masses

Given the extremely small mass scales, we generally give masses of atoms or molecules in terms of atomic mass units, amu, where:

$$1\text{amu} = 1.66 \times 10^{-27}\text{kg}$$

This is defined as 1/12th the mass of a Carbon-12, ^{12}C atom. The atomic mass of Carbon-12 is thus 12amu, however a proton and neutron each have mass 1.007amu and 1.009amu respectively. Why is there a discrepancy if we add the masses of the individual nucleons?

This is because, when combining these nucleons into a nucleus, some of this mass is converted into energy. This is called "binding energy" and is used to hold the nucleus together.

2.1 Moles

A mole (mol) is an SI base unit used to described the amount of a substance. 1 mole of a substance is defined by Avagadro's number, $N_A = 6.022 \times 10^{23}$, where one mole has N_A atoms/molecules. We again use Carbon 12, as 12 grams of Carbon-12 has N_A atoms.

We further have:

$$\text{amount, mol} = \frac{N}{N_A}$$

$$\text{amount, mol} = \frac{\text{Mass}}{\text{Atomic Mass}}$$

For gasses, one mole occupies exactly 22.4L or 22.4dm^3 at standard temperature and pressure (0°C, 10^5Pa = 1atm), regardless of the gas in consideration.

3 Rayleigh's Oil Drop Measurement

Rayleigh conducted an experiment where he took a large volume of water, and placed a drop of olive oil (of known volume V) on top. He postulated that, due to the nature of olive oil being long-chained hydrocarbons, the oil would spread out perfectly thinly, and produce a layer exactly one atom tall.

He said that this would form a circular layer of radius R and height d. If we take the film to be a cylinder (and d is too small to measure directly), we can measure R and use the known volume to determine d:

$$V = \pi R^2 d \implies d = \frac{V}{\pi R^2}$$

When he carried this out, he determined the atomic radius [1], as $\sim$ 1nm, i.e. being one order of magnitude out. This wasn't bad for how rudimentary the experiment was!

4 Latent Heat

We know that there has to be some kind of force between two atoms. If we consider a diatomic molecule, separating these atoms from each other to infinity requires some amount of energy ϵ. Because there are two atoms, the work done per atom is $\epsilon/2$.

Lets consider a more complex example with many more atoms (i.e. converting a solid to liquid):

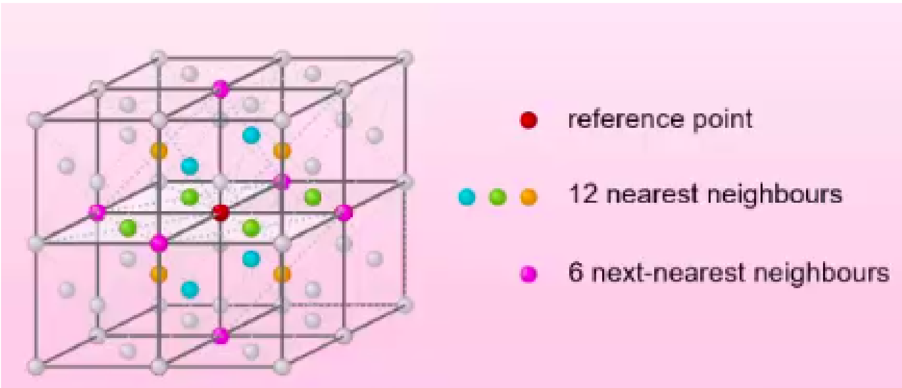


Figure 41.3

In this case, extracting the red atom from all its 12 nearest neighbours will take $12\epsilon/2$. We should additionally consider the 6 next nearest neighbours and so and so on. Each increase in number of nth next nearest neighbours has a decreasing impact in the force required, so leads to diminishing returns. In this course, we often ignore he next nearest neighbours and further. If asked to estimate, we use $6-10$ for a liquid, and $8-12$ for a solid.

While we're considering this in a solid, lets use it as an approximation for a liquid. The latent heat of vaporisation L(Jkg) is the amount of heat required to completely vaporise a liquid into a gas.

$$L = 12\epsilon/2 \times \frac{\text{number of atoms}}{\text{mass of material, kg}}$$

And the number/mass is given by:

$$\frac{\text{kg}}{\text{atoms}} = \rho \times d^3 \implies \frac{\text{atoms}}{\text{kg}} = \frac{1}{\rho d^3}$$

Hence:

$$d = \sqrt[3]{\frac{1}{\text{atoms per kg} \times \rho}}$$

[1]Strictly, an upper limit for the radius

5 Forces Between Atoms

What causes the forces between atoms? It can't be the electrostatic (coulomb) force, as atoms are electrically neutral.

It can't be the nuclear force, as the range is too small. Likewise, it can't be the weak force as this mediates the wrong particles and has an even smaller range.

Finally, it can't be gravitational as atoms have far too low a mass for the gravitational force to have any real impact.

Even though the electrostatic component has no impact, the electromagnetic force as a whole can.

5.1 Water Example

Oxygen is slightly electronegative, while hydrogen is not. This means that if we have a H_2O water molecule, the electrons will be pulled ever so slightly closer to the oxygen and away from the hydrogen. Across the whole atom, this causes a slight charge on each side of the atom:

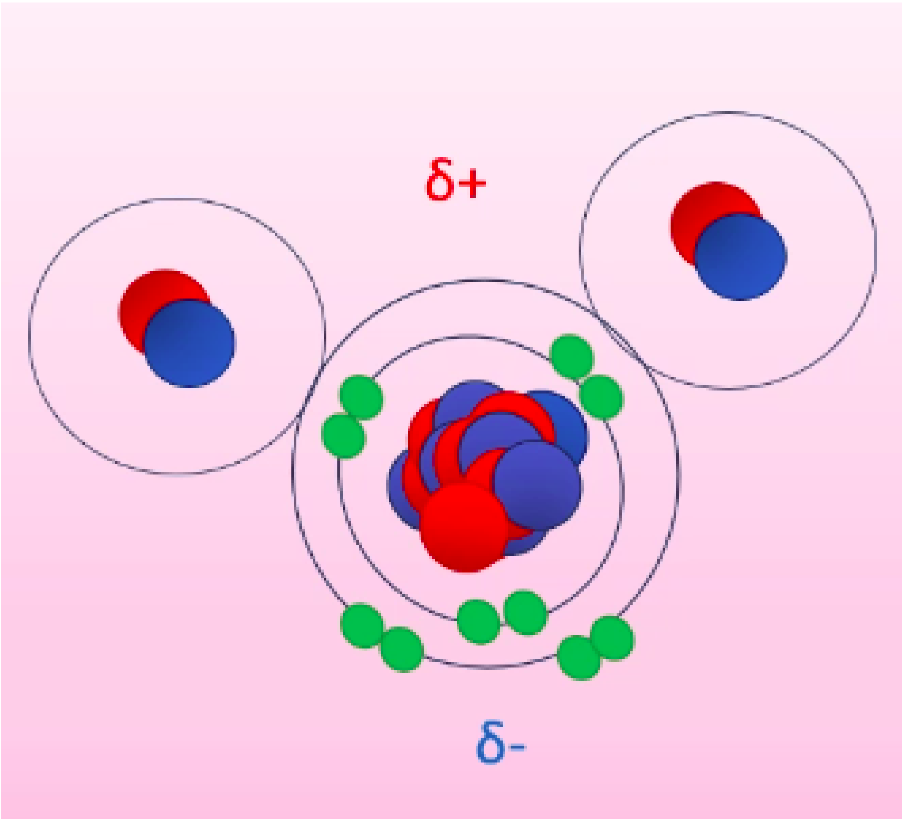


Figure 41.4

Water is "polar molecule" and forms a permanent dipole, where the positive and negative poles of molecules attract.

6 "Normal" Atoms

This only works for some atoms. A block of iron for example cannot be explained by this, as all the atoms are the same, so have equivalent electronegativity.

Here, fluctuations in the electron cloud cause an instantaneous dipole to form. This may be very brief, but lasts for sufficiently long to impact another atom. This atom has electron distribution impacted by this force, causing it to be become a dipole. This creates a field which impacts the original atom etc. This causes the atoms to be stuck as instantaneous dipoles, resulting in a net attractive force called the Van der Waal's force.

As the electric field due to a dipole goes as $|E| \propto 1/r^3$, the induced dipole in the firs atom depends on the strength of its dipole field and the size of the induced dipole in the second atom:

$$|E| = \frac{1}{r^3}\frac{1}{r^3} = \frac{1}{r^6}$$

The force induced by these dipoles explains attraction for non-polar molecules. Force between two atoms is considered positive (by convention) if they repel and negative if they attract.

However, the fact that atoms have a consistent interatomic spacing and do not collapse into black holes suggests that there is some kind of repulsive force that keeps them at fixed distances.

We can infer that this force must be very strong at small distances, but much weaker at larger distances.

7 LJ Potential

We can divide particles into fermions (half integer spin) and bosons (integer spin). The wave fuction of a boson is symmetric about y, so overlapping two bosons causes superposition. Fermions are antisymmetric (i.e. a sine wave). This means that two fermions in the same position causes destructive interference, with a net wave function of 0.

This means that two fermions in the same quantum state (same wave function) cannot share a position, and have a force repelling each other should they try. We approximate this as a potential of the form:

$$V \propto \frac{1}{r^{12}}$$

Lennard-Jones combined these, describing the net potential of the form:

$$V_{LJ} = \frac{A}{r^{12}} - \frac{B}{r^6}$$

Tue 27 Jan 2026 12:00

Lecture 3 - Matter III: Lennard-Jones Potential

1 Potential Distance Plots

If the values of the constants A, B are known, the general form of a Lennard-Jones potential plot is:

$$V_{\text{LJ}} = \underbrace{\frac{A}{r^{12}}}_{\text{+ so repulsive}} - \underbrace{\frac{B}{r^{6}}}_{\text{- so attractive}}$$

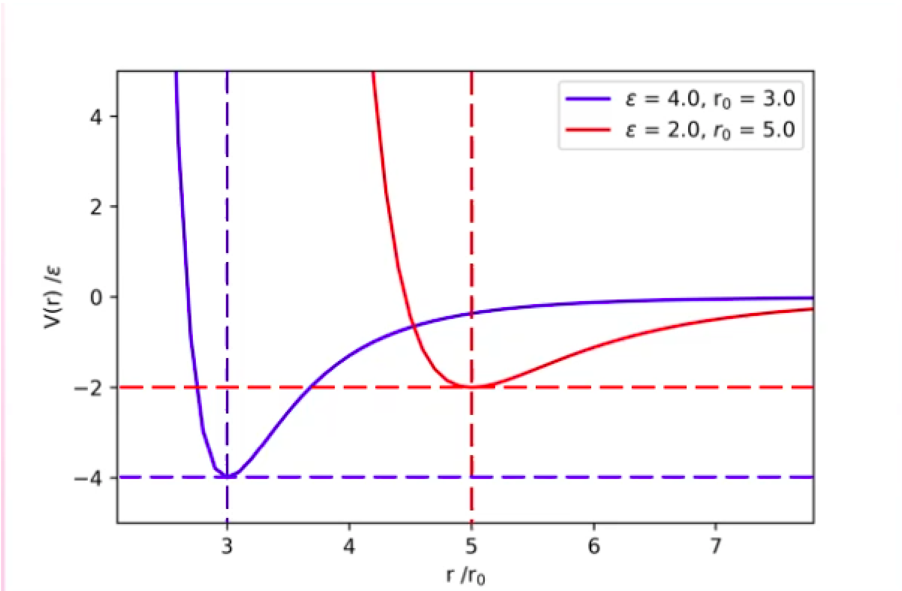


Figure 42.1

The potential is at some baseline minimum value $-\epsilon$ at distance r_0.

For our LJP force, we require:

- Repulsive at very short distances (smaller than interatomic spacing).
- Attractive force for intermediate distances (larger than interatomic spacing).
- No force as $r \to \infty$.
- The atoms equilibrium point is roughly the size of the interatomic spacing in liquids and solids.

In order to describe this more effectively, we need to find expressions for A, B. We do this based on the earlier initial conditions of $V_{LJ} = -\epsilon$ at $r = r_0$

2 Deriving A and B

At the equilibrium point $r = r_0$, we have:

$$V_{LJ} = -\epsilon$$

$$F = 0$$

We can get an expression for force by $F = -dV/dr$:

$$F = -\left(-\frac{12A}{r^{13}} + \frac{60}{r^7}\right) = \frac{12A}{r^13} - \frac{6B}{r^7}$$

At equilibrium:

$$F = 0 = \frac{12A}{r_0^{13}} - \frac{6B}{r_0^7}$$

Disregarding trivial case where $r_0 = 0$, as there cannot be any atoms in this case. Therefore, $r_0 \neq 0$:

$$12A = 6Br_0^6$$

$$\frac{2A}{B} = r_0^6 \implies r_0 = \sqrt[6]{\frac{2A}{B}}$$

And considering potential at equilibrium:

$$-\epsilon = \frac{A}{r_0^{12}} - \frac{B}{r_0^6}$$

$$\implies -\epsilon = \frac{A}{\left(\frac{2A}{B}\right)^2} - \frac{B}{\left(\frac{2A}{B}\right)}$$

$$\implies -\epsilon = \frac{B^2}{4A} - \frac{B^2}{2A}$$

$$\epsilon = \frac{B^2}{2A} - \frac{B^2}{4A} = \frac{B^2}{4A}$$

Putting it all together:

$$\epsilon = \frac{B^2}{4A} \tag{1}$$

$$r_0^6 = \frac{2A}{B} \tag{2}$$

Hence:

$$\frac{2A}{r_0^6} = B \implies \epsilon = \frac{(2A)^2/(r_0^6)^2}{4A}$$

$$A = \epsilon r_0^{12}$$

$$B = 2\epsilon r_0^6$$

So the final functional form of the LJP is:

$$V_{LJ} = \epsilon\left[\left(\frac{r_0}{r}\right)^{12} - 2\left(\frac{r_0}{r}\right)^6\right]$$

Sometimes in textbooks we see:

$$V_{LJ} = 4\epsilon\left[\left(\frac{\sigma}{r}\right)^{12} - 2\left(\frac{\sigma}{r}\right)^6\right]$$

Here, σ is the distance where potential is zero, rather than r_0 being the distance where force is zero (potential is $-\epsilon$).

3 Full LJP

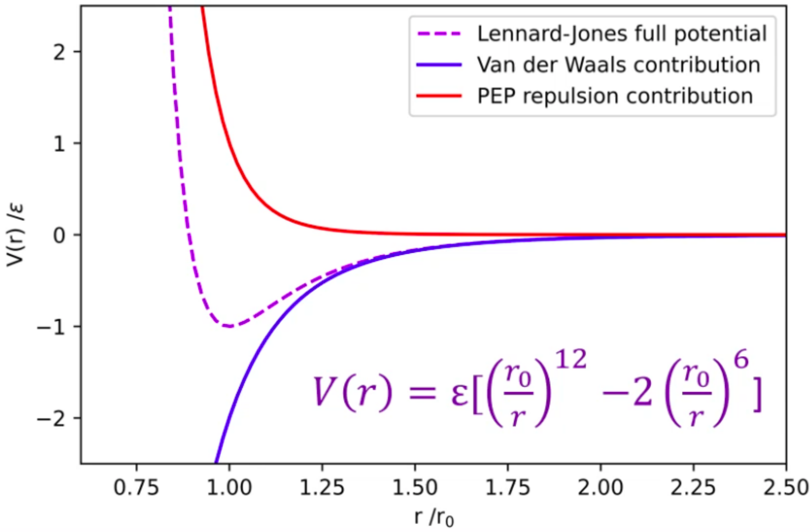


Figure 42.2: The full LJP and the two contributing forces

This gives us the overall LJP that satisfies our three requirements:

- At distances below r_0, the Pauli Exclusion Principle force dominates, causing net repulsion.
- As $r \to \infty$, both components (and the overall potential) tend to zero.
- At distances above, but close to r_0, the Van der Waal's force dominates, causing net attraction.

Where $r/r_0 = 1$, we have a stable equilibrium position, needing $-V(r_0) = -\epsilon = U_0$ to break these apart.

3.1 Implications of LJP

Consider a two dimensional atomic lattice:

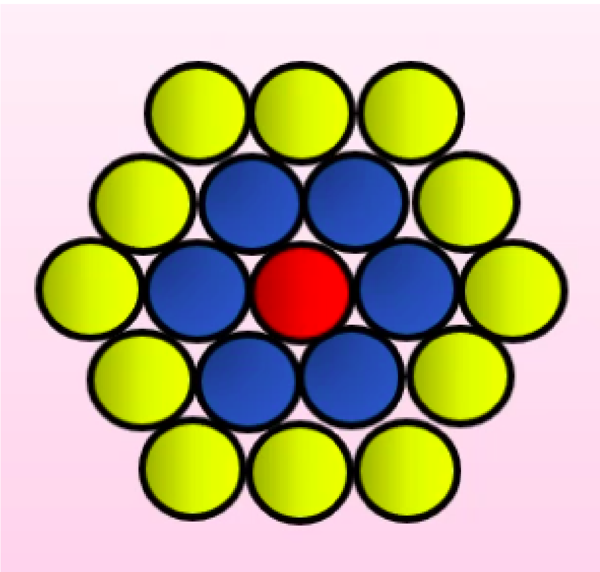


Figure 42.3

The central atom sits in a potential well of approximate depth $N\epsilon$, where N is the number of nearest neighbours. In the equilibrium position, all of these atoms are separated by a distance $r = r_0$.

The next nearest neighbours and beyond do contribute, but to a much smaller degree. It's not zero, but the next nearest neighbours have approximate potential $\epsilon/30$.

We have:

$$V_{LJ} = \epsilon\left[\left(\frac{r_0}{r}\right)^{12} - 2\left(\frac{r_0}{r}\right)^{6}\right]$$

Hence, using $F(r) = -\frac{dV(r)}{dr}$:

$$F(r) = 12\epsilon\left(\frac{r_0^{12}}{r^{13}} - \frac{r_0^6}{r^7}\right)$$

This has a very similar form to the potential:

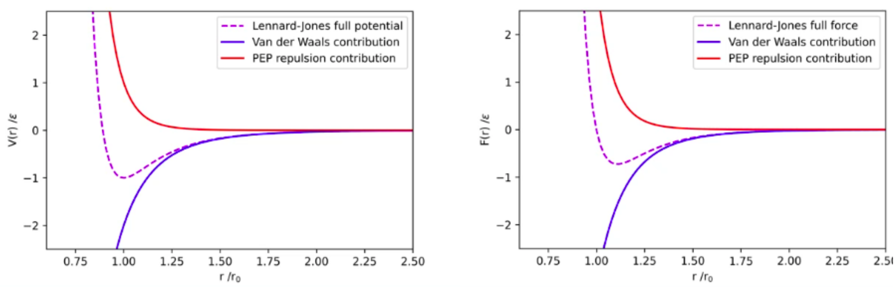


Figure 42.4

Just noting that while potential is $-\epsilon$ at $r_0 = r$, the force is zero.

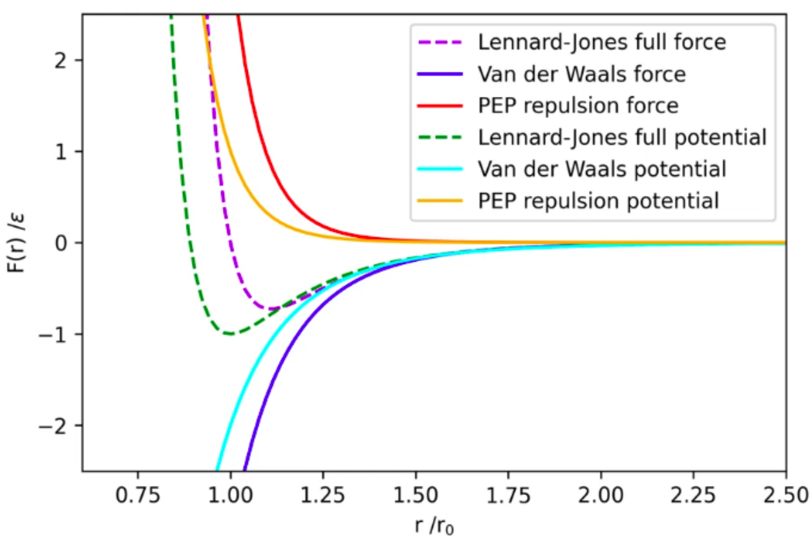


Figure 42.5

3.2 Empirical Data

The LJP provides a surprisingly accurate model for the genuine empirically measured data:

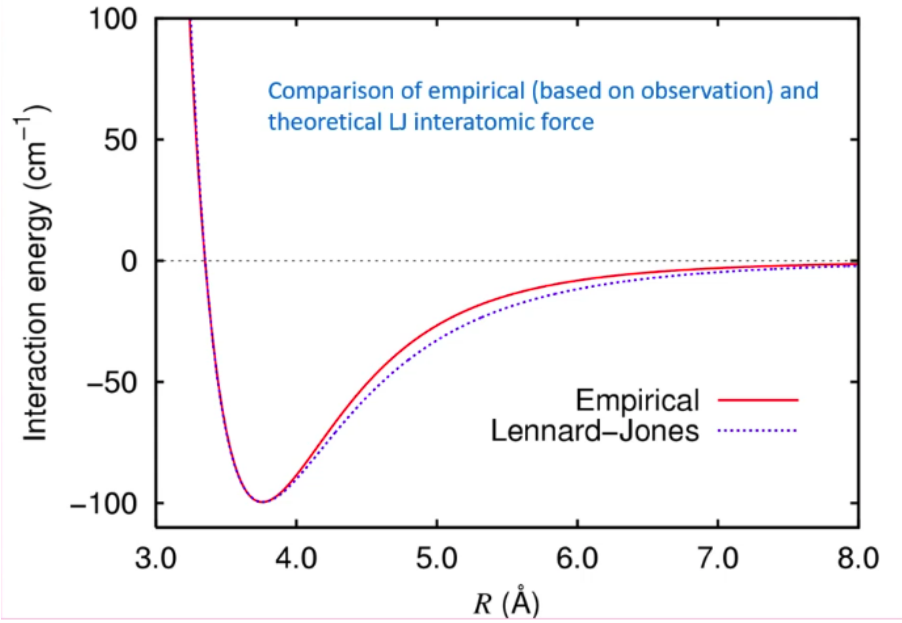


Figure 42.6

It does have limitations though:

- The fact its an infinite range potential means it's very computationally expensive to calculate. We need to truncate the potential once it shrinks below a value below which its negligible.
- For large molecules (such as an amino acid), they are often too large. The distance between the centres of adjacent molecules is too large for the LJP as LJP assumes that they are effectively point like. Amino acids can have forces all the way along their chains and this is not well modelled by the LJP.

Thu 29 Jan 2026 13:00

Lecture 4 - Matter IV: Surface Tension and Capiliary Action

Recap from last time:

- Force and potential are related, with $F = \frac{-dV}{dx}$, with x relating to position.
- At equilibrium, neighbouring atoms sit at distance r_0 (average bond length) from each other. They sit in a potential well of depth $N\epsilon$ where N is the number of nearest neighbours.
- At $x < r_0$, the force is +'ve (repulsive), at $x > r_0$ the force is -'ve (attractive).

1 Coordination Numbers

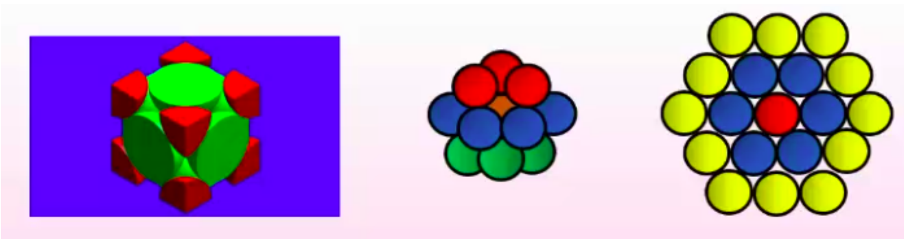


Figure 43.1

In a densely-packed Face-Centred Cubic (left) or Hexagonally Close-Packed Solid (3D, middle - 2D, right) configuration, the maximum coordination number (number of nearest neighbours) is 12. The outermost.

For solids, we expect an interatomic spacing of $\sim 2 \times 10^{-12}$m, for liquids we expect 3×10^{-10}m and for gases at STP we expect about 40×10^{-10}m.

For liquids, we expect a coordination number $n < 10$.

2 Molecular/Structural Binding Energies

To first order (nearest neighbours only), how much energy do we need to remove the innermost atom (orange in the middle figure) from the centre of the structure.

We might expect 12ϵ, the total binding energies of all of the nearest neighbours. Instead however, we are leaving the 12 nearest neighbours in place and only taking one atom to infinity. Since moving both atoms to infinity is given by ϵ, we have:

$$E = \frac{n\epsilon}{2}$$

Where n is the number of nearest neighbours.

To remove one mole of connected atoms, we have:

$$E = \frac{nN_A\epsilon}{2}$$

Again, the factor of two arises to avoid double-counting bonds, i.e. a mole consists of $N_A/2$ pairs of atoms.

3 Latent Heat

Moving macroscopically, we have seen:

$$L = 10\epsilon/2 \times \frac{\text{number of atoms}}{\text{mass of material, kg}}$$

For a liquid of coordination number 10. This is very similar to the just seen:

$$E = \frac{nN_A\epsilon}{2}$$

As N_A gives us atoms per mol, this is effectively identical - one just considers moles and one considers mass in kg.

The molar latent heat equation assume however that the molecular energy comes exclusively from potential energy. This is an approximation that works when kinetic energy is much less than potential energy, i.e. at low temperatures. We can empirically see that molar latent heat varies with temperature:

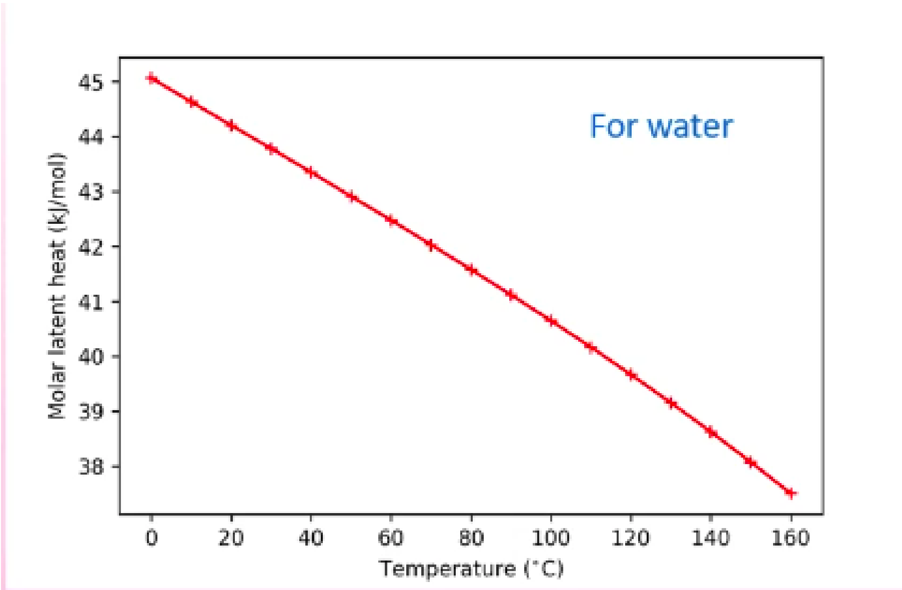


Figure 43.2

This is because, at higher temperatures, the average bond length increases (as the particles have a greater kinetic energy, so vibrate with a larger magnitude):

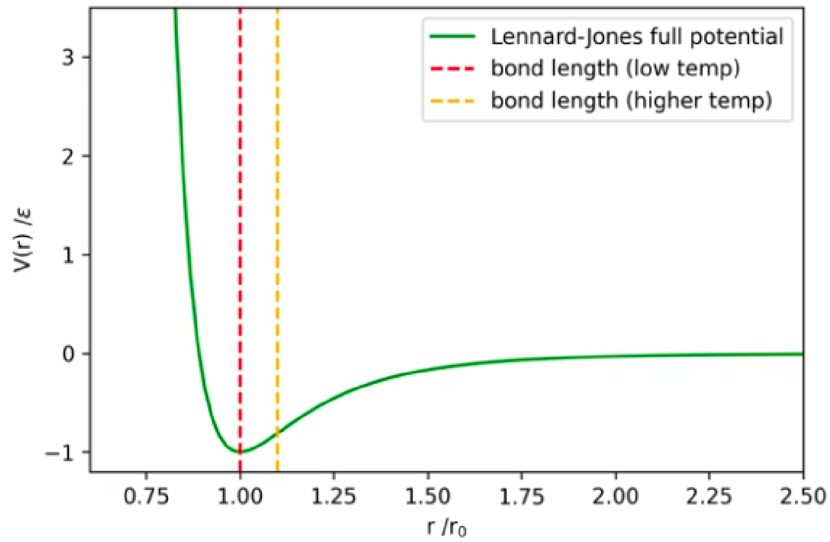


Figure 43.3

Because the LJP is antisymmetric, at higher temperatures the molecules spend less vibrational time in the contracted position and more time in the expanded position, resulting in a larger average bond length. Effectively, they experience more repulsion when they are together than they do attraction when apart, hence a larger bond length on average. This also explains thermal contraction.

We can see from the potential graph that this results in a lower ϵ, which is why we see a change in latent heat at different temperatures.

4 Surface Tension

Consider a (literal) sea of particles. Does a particle surrounded on all sides deep in the liquid experience a lesser or greater potential than a particle on the surface?

The particle deep in the liquid has twice as many nearest neighbours (being surrounded on all sides) so has a greater potential. The particle on the surface will have:

$$V = \frac{n\epsilon}{4}$$

So requires $n\epsilon/4$ amount of work to take a deeper particle to the surface.

We define a quantity called surface tension γ, the work done to move 1m^2 of particles, N to the surface from deeper inside the liquid:

$$\gamma = \left(\frac{n\epsilon}{2} - \frac{n\epsilon}{4}\right) \times N$$

Where N is the number of particles per m^2. It has units of energy over area (force over length).

This allows animals such as small insects who can sit on the surface of the liquid to do so, because allowing the insect to submerge would require an increase in the water's surface area. The work required to generate this new surface area is not achievable by the insects gravitational potential energy.

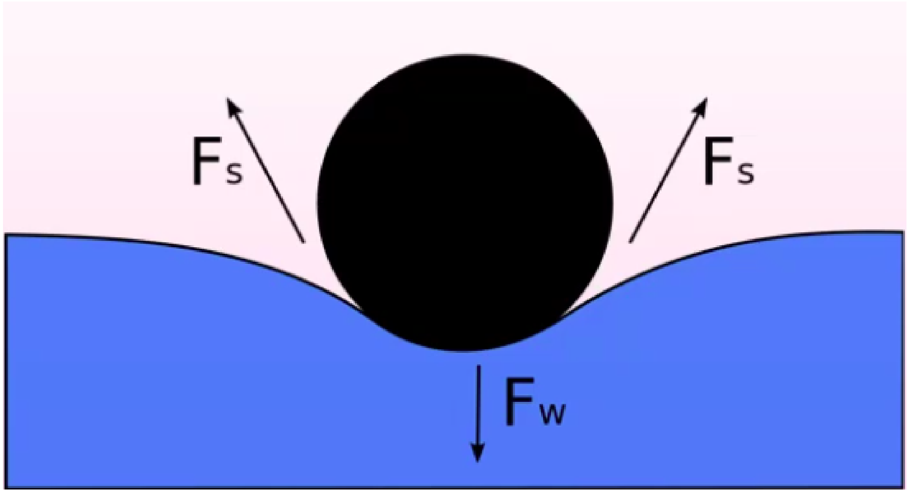


Figure 43.4: Object sinks if $\Sigma F_s < F_w$

What other consequences does this have?

- Given a body of water wants to minimise its surface area (for a constant volume), drops of water will form spheres.

5 Surfaces Revisited

Consider a liquid contained in some beaker. Consider a particle on the surface of the *beaker* (purple):

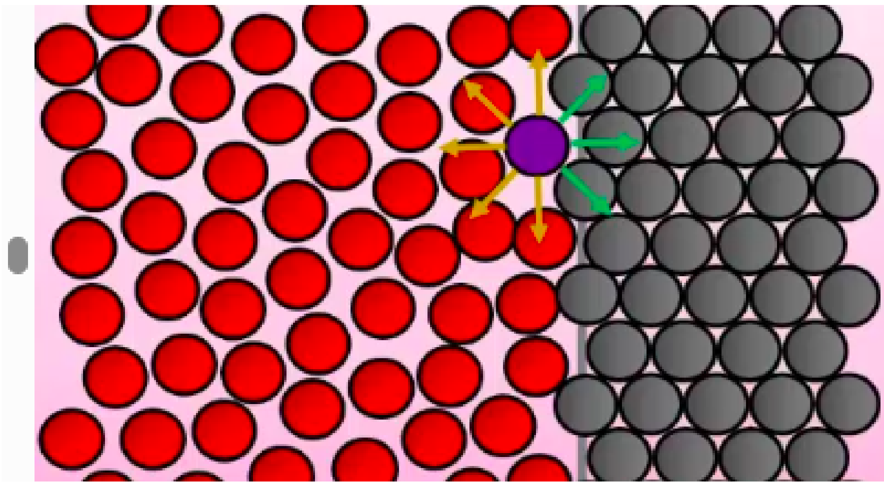


Figure 43.5

There are two forces in effect here, the interatomic forces of the liquid and of the beaker. We call the forces between the liquid itself cohesion, and between the liquid and a foreign body adhesion.

5.1 Wetting

"Wetting" is the ability of a liquid to maintain contact with a solid surface and is inversely proportional to surface tension. Most liquids are "wetter" than water, due to water's high surface tension.

If there are strong adhesive forces, the liquid preferentially sticks to the side of the container, causing the formation of a concave meniscus.

If there are strong cohesive forces, minimisation of surface area causes the formation of a convex meniscus.

6 Capillary Action

If an open tube is placed into contact with the surface of a liquid with strong adhesion, the liquid will rise up the sides of the tube:

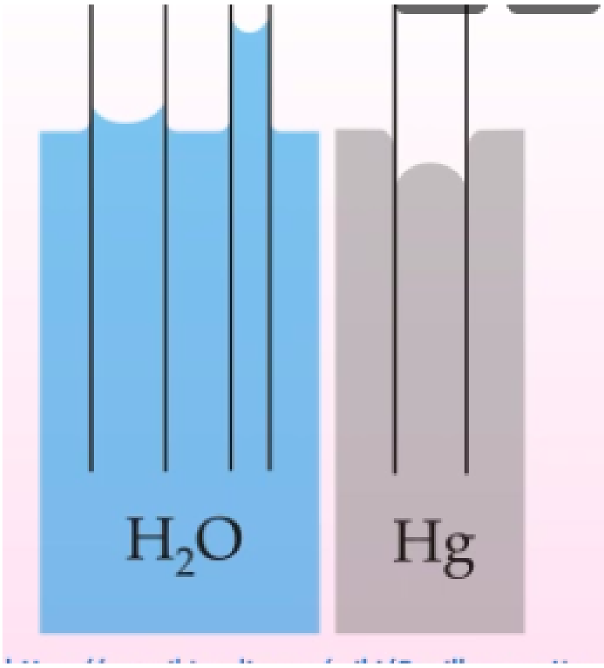


Figure 43.6: Capillary action for water and mercury. Note that mercury, being strongly cohesive, minimises surface area so does not rise up.

We can describe this mathematically in terms of the angle formed by the meniscus, the width of the meniscus and the height reached by the liquid:

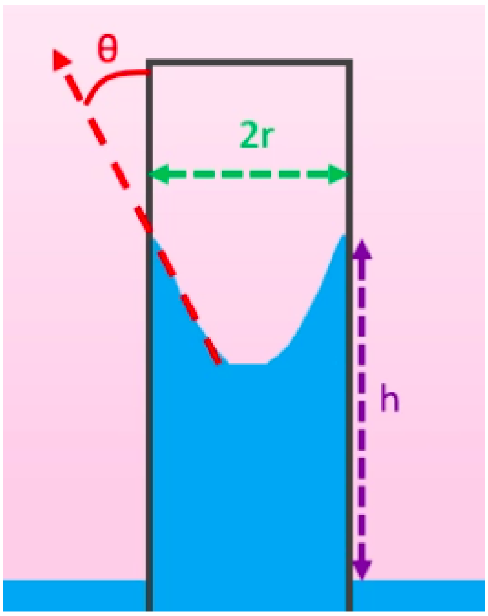


Figure 43.7

There are two competing forces here:

- Surface tension, pulling the liquid up.
- The weight of the liquid, pulling it down.

The weight is given by the volume (if we approximate the meniscus as being small compared to the total volume, hence just a cylinder) and density:

$$F_W = mg = \rho V g = \rho \pi r^2 h g$$

Surface tension has units of force divided by length. We can therefore get the force by the multiplying by the active contact length, which is the circumference of the cylinder. Finally, we only care about the vertical component, so take $\cos\theta$:

$$F_\gamma = \gamma 2\pi r \cos\theta$$

Hence, as equilibrium:

$$F_\gamma = F_W$$

$$2\gamma\pi r \cos\theta = \rho \pi r^2 h g$$
$$2\gamma \cos\theta = \rho g r h$$

This gives:

$$\boxed{h = \frac{2\gamma\cos\theta}{\rho g r}}$$

Tue 03 Feb 2026 12:00

Lecture 5 - Matter V: Stress, Strain and Young's Modulus

1 Definitions

Consider a wire of length L and cross-sectional area A. We apply some force F to the material, causing it to stretch from L to $L + \Delta L$.

Stress is the force applied per unit area on a material, $\text{kgm}^{-1}\text{s}^{-2}$ and strain is the fractional change in length: $\Delta L/L$.

The Young Modulus is the ratio of the two:

$$Y = \frac{\text{stress}}{\text{strain}} = \frac{F/A}{\Delta L/L} = \frac{FL}{\Delta LA}$$

and has units of pressure (Pascals). Different materials have different Young Moduli as inherent material properties.

2 Stress/Strain Graphs

We can consider a plot of stress against strain:

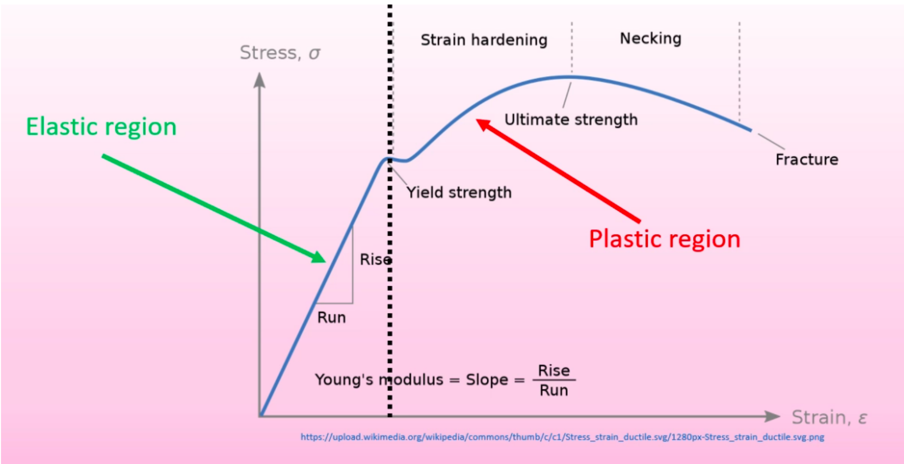


Figure 44.1

The leftmost region is where stress and strain are proportional, i.e:

$$\frac{F}{A} \propto \frac{\Delta L}{L}$$

$$F \propto \Delta L$$

As A and L are constants. Any changes here are reversible, i.e. a rubber band.

$$F = k\Delta L = kx$$

Past a certain extension, we encounter plastic (non-reversible) extension. This is where we've passed the limit of proportionality. There are a few important points here:

- Strain hardening. We apply a force to the object to extend it to its ultimate extension. This means that we can include it in i.e. machinery without worrying about smaller forces regularly encountered causing extensions.
- Necking, where an increase in length causes a decrease in cross-sectional area (as the total volume is held constant)
- Once we get past the strain hardening point, tiny defects cause non-linear necking. A weaker area decreases the cross-sectional area more significantly. This lower A causes the effect to be more significant, and leads to the area becoming thinner and thinner until it fractures.

3 Other Elastic Moduli

3.1 Shear Modulus

For Young's Modulus, we're explicitly considering that the force acts in the direction of the extension. We can have other elastic moduli (still defined as stress over strain) depending on the configuration of the force.

For example, applying a tangential force gives the shear modulus:

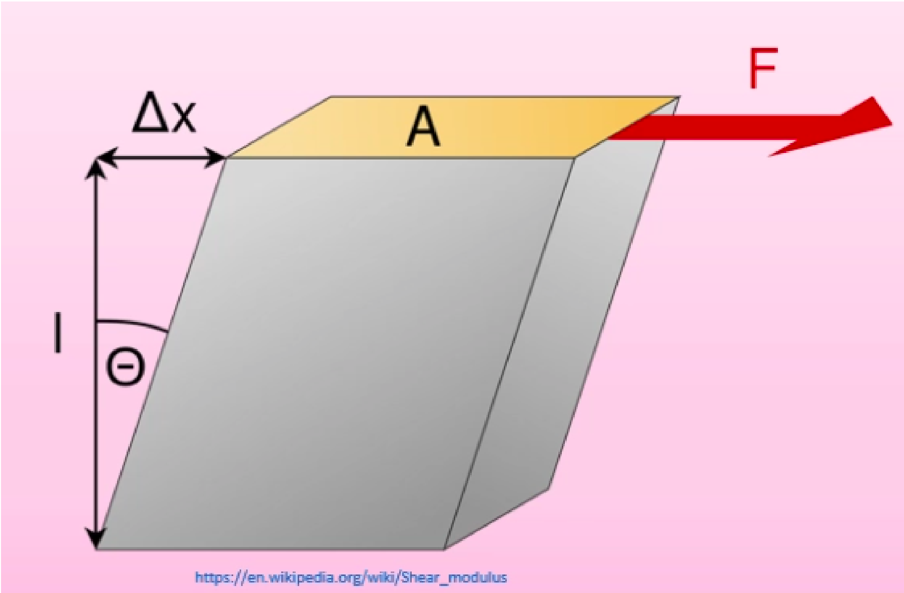


Figure 44.2

The shear stress remains as F/A and the sheer strain is $\Delta x/l$. The sheer modulus is shear stress/shear strain.

3.2 Bulk Modulus

The bulk modulus is for objects deformed under a uniform pressure, for example compressing a liquid from some volume V_0 down to a compressed volume $V_0 - \Delta V$.

The bulk stress is the compressing pressure, F/A and the bulk strain is $\Delta V/V_0$.

So the Bulk Modulus is:

$$\frac{P}{(DeltaV/V_0)} = \frac{PV_0}{\Delta V}$$

Or (using the ideal gas equation we'll encounter later):

$$B = -V_0 \frac{dp}{dV}$$

This is comparable for solids and liquids. We also define the compressibility as the inverse of the bulk modulus.

3.3 Poisson's Ratio

Applying a stress to a force in one axis causes compression in this direction, but expansion in the other:

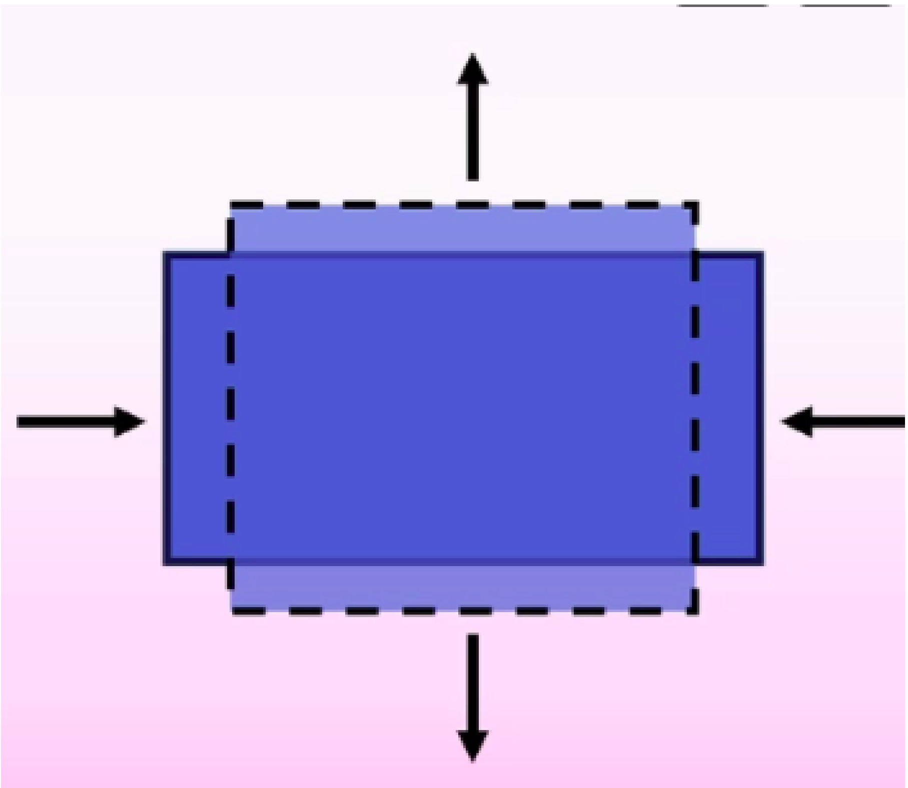


Figure 44.3

This ratio is given by Poisson's Ratio (unitless):

$$\nu = \frac{\text{transverse strain}}{\text{longitudinal strain}}$$

4 Material Values

All of these moduli have varying values as inherent metal properties:

Material	Young's modulus (GPa)	Shear modulus (GPa)	Bulk modulus (GPa)	Poisson's ratio (unitless)
Aluminium	70	30	70	0.33
Brass	91	36	61	0.36
Copper	110	42	140	0.36
Glass	55	23	37	0.22
Iron	190	70	100	0.26
Lead	16	5.6	7.7	0.44
Nickel	210	77	260	0.31
Steel	200	84	160	0.28
Tungsten	360	150	200	0.28

Figure 44.4: Elastic Moduli Values

Note that while the Young and Bulk Modulus values are generally similar, the sheer modulus is often 2-3x smaller.

Thu 05 Feb 2026 13:00

Lecture 6 - Matter VI: LJF Meets Young Modulus and Breaking Materials

1 Young's Modulus Meets Lennard-Jones

On a microscopic scale, applying a linear force extends the distance between neighbouring atoms:

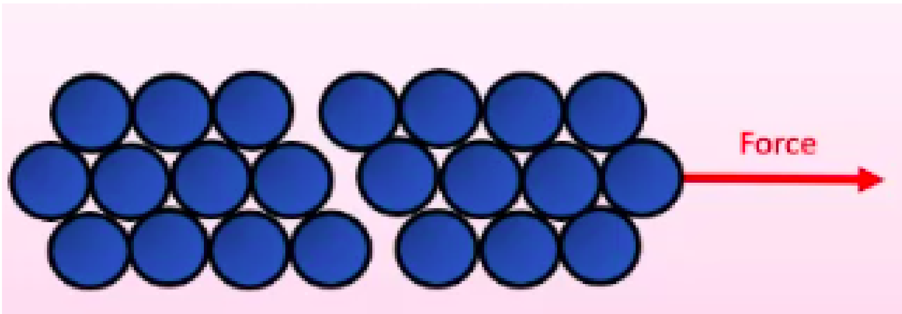


Figure 45.1

The mean interatomic distance changes from $r_0 \to r_0 + dr$ for an infinitesimal force dF. Near the equilibrium point, where the material is in the elastic region, we expect the extension $dr \propto dF$, as in this region $\Delta L \propto F$.

Going back to the Lennard-Jones Force, we have a small region around the equilibrium point:

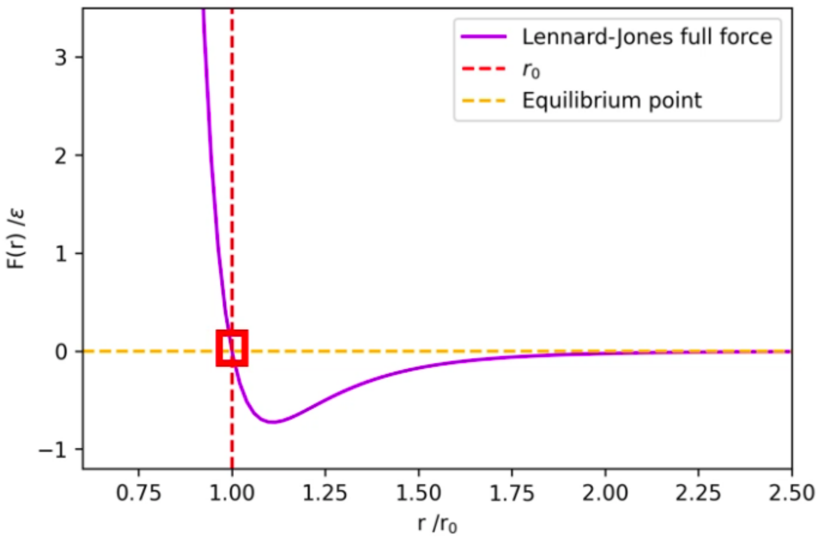


Figure 45.2

If we zoomed into this region, we would see it is approximately linear:

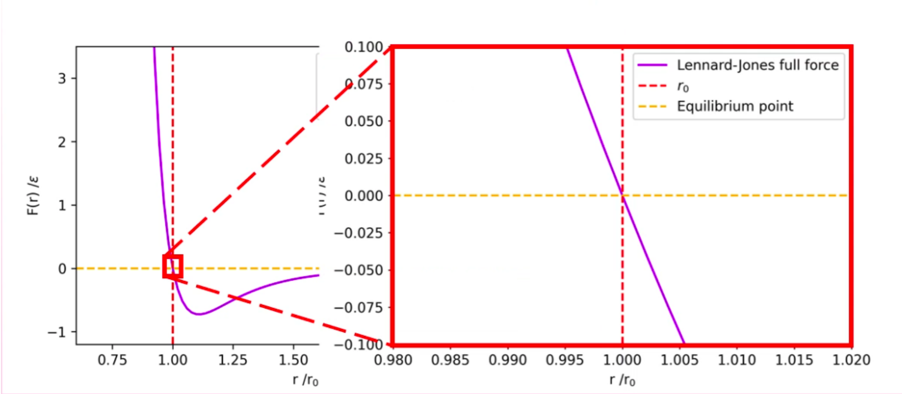


Figure 45.3

Hence, around r_0, dF/dr is constant. Going back to Young's Modulus, we have:

$$Y = \frac{\text{stress}}{\text{strain}} = \frac{F/A}{\delta Y/Y}$$

Here, the force is just dF. The area this force is acting over (microscopically) is actually a square, as the material is a cubic lattice, $A = (r_0 + dr)^2$. Hence stress is:

$$\text{stress} = -\frac{dF}{(r_0 + dr)^2}$$

The negative sign compensates for the fact that the LJF opposes the deformation (is negative under tension), ensuring the stress and strain have the same sign and their ratio is positive.

$$\text{strain} = \frac{dr}{r_0}$$

$$\implies Y = -\frac{\left(\frac{dF}{(r_0+dr)^2}\right)}{\left(\frac{dr}{r_0}\right)}$$

$$= -\frac{dF r_0}{dr(r_0 + dr)^2}$$

We take the quadratic to first order, assuming $dr \ll r_0$:

$$= -\frac{1}{r_0}\frac{dF}{dr}\bigg|_{r=r_0}$$

As force is $-dV/dr$ we can show that:

$$Y = \frac{1}{r_0}\frac{d^2V}{dr^2}\bigg|_{r=r_0}$$

And using the definition:

$$V = \epsilon\left(\frac{r_0^{12}}{r^{12}} - 2\frac{r_0^6}{r^6}\right)$$

Hence:

$$F = -\frac{dV}{dr} = -\epsilon\left(\frac{12r_0^{12}}{r^{13}} - 12\frac{r_0^6}{r^7}\right) = \epsilon\left(12\frac{r_0^6}{r^7} - 12\frac{\epsilon_0^{12}}{r^{13}}\right)$$

$$Y = -\frac{dF}{dr} = \frac{d^2V}{dr^2} = \epsilon\left(\frac{156r_0^{12}}{r^{14}} - \frac{84r_0^6}{r^8}\right)_{r=r_0}$$

$$Y = \frac{156\epsilon}{r_0^2} - \frac{84\epsilon}{r_0^2} = \boxed{\frac{72\epsilon}{r_0^2}}$$

This provides a microscopic definition of the Young's Modulus to complement our previous macroscopic definition.

2 Breaking Materials

Materials break when the attractive force between atoms is less than the force applied to the material. This occurs when:

$$\frac{dF}{dr} = 0 \implies F = F_{\text{max}}$$

If we pull continuously with a force greater than F_{max} we will pull atoms apart and break the material. The maximum definable stress is based on the maximum attractive LJF over the area of a single molecule:

$$\text{Breaking Stress} = \frac{F_{\text{max}}}{r_0^2}$$

2.1 Deriving r at F_{max}

F_{max} occurs where $\frac{dF}{dr} = 0$. Using the earlier result:

$$F(r) = 12\epsilon\left(\frac{r_0^{12}}{r^{13}} - \frac{r_0^6}{r^7}\right)$$

$$\frac{dF}{dr} = 12\epsilon\left(-\frac{13r_0^{12}}{r^{14}} + \frac{7r_0^6}{r^8}\right) = 0$$

$$\frac{7r_0^6}{r^8} = \frac{13r_0^{12}}{r^{14}} \implies 7r^6 = 13r_0^6$$

$$\implies \boxed{r = \sqrt[6]{\frac{13}{7}}\, r_0}$$

2.2 Finding F_{max}

Substituting $r^6 = \frac{13}{7}r_0^6$ into $F(r)$. Simplifying each term:

$$\frac{r_0^{12}}{r^{13}} = \frac{1}{r}\cdot\left(\frac{r_0^6}{r^6}\right)^2 = \frac{1}{r}\cdot\frac{49}{169}, \qquad \frac{r_0^6}{r^7} = \frac{1}{r}\cdot\frac{r_0^6}{r^6} = \frac{1}{r}\cdot\frac{7}{13}$$

$$F_{\text{max}} = \frac{12\epsilon}{r}\left(\frac{49}{169} - \frac{91}{169}\right) = -\frac{504\epsilon}{169\,r}$$

Substituting $r = \left(\frac{13}{7}\right)^{1/6} r_0$:

$$\implies \boxed{|F_{\text{max}}| = \frac{504\epsilon}{169\,r_0}\left(\frac{7}{13}\right)^{1/6} \approx \frac{2.69\,\epsilon}{r_0}}$$

2.3 Breaking Stress

Hence the breaking stress is:

$$BS = \frac{F_{\text{break}}}{A} = \frac{F_{\text{max}}}{r_0^2}$$

Macroscopically and microscopically.

3 Fractures

Say your material has some tiny imperfection of depth d and width $2r$.

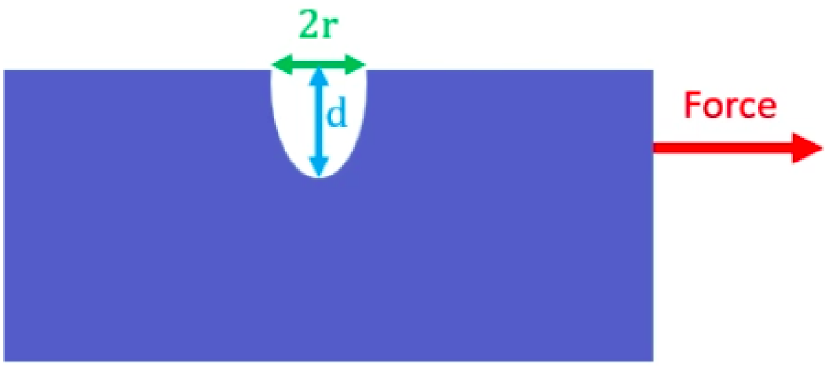


Figure 45.4

This causes stress to increase by a factor of:

$$\left(1 + 2\sqrt{\frac{d}{r}}\right)$$

The breaking force decreases with increasing d, which makes sense as a deeper groove will cause a much smaller cross sectional area and a larger stress.

The breaking force increases with a decreasing r, where grooves on atomic length scales wide causes a huge increase in the stress.

Mon 09 Feb 2026 11:00

Lecture 7 - Thermal Physics I: Introduction to Temperature

We're starting the portion of the module on thermodynamics, looking at temperature, pressure etc and transitions involving these.

1 What is Temperature?

Particle for a single particle isn't well defined for the definition of temperature we want to use. There is two definitions:

- One is statistical, looking at average kinetic energy of particles.
- One is based on entropy and isn't looked at until second year.

We define temperature as *a statistical collection of energies for particles that make up the system for which we are measuring the temperature.* A material will have a range of kinetic energies, some very large, some very small, so we care about a statistical average. Thermalisation happens as a result of these high kinetic energy particles striking lower kinetic energy particles.

> **Temperature:** A statistical collection of energies of the particles that comprise the system in question.

We can express temperature as using the Maxwell-Boltzmann distribution:

$$P(v) = 4\pi \left(\frac{M}{2\pi RT}\right)^{\frac{3}{2}} v^2 \exp\left(\frac{-Mv^2}{2RT}\right)$$

We look at this in more depth when we cover statistical physics later, but for now we care about the key features. If we plot $P(v)$ (the probability of finding a particle at certain velocity) against these velocities for a range of temperatures:

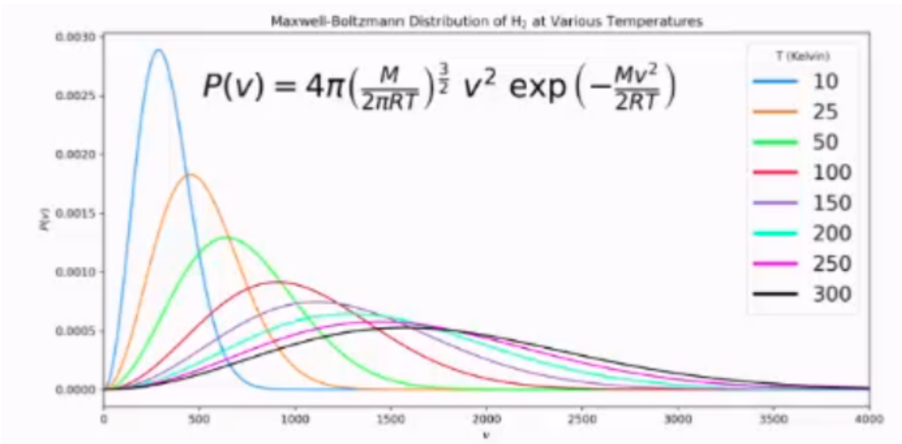


Figure 46.1

This has exponential decay, so theoretically we can find a particle at any velocity within a gas of any temperature. Particles with higher temperatures have a higher peak, so a higher average velocity.

2 Temperature Scales

We can use a number of different temperature scales:

- The Fahrenheit scale is set such that the freezing point of water is 98.6°F, and the boiling point of water at 212°F. It's designed to be based around the range of temperatures that a person could realistically could experience in the core range from 0-200. It's not a useful unit and won't be used in exams.
- Celsius is the standard unit.
- Kelvin is the common unit of thermodynamics. It has the same graduations as degrees celsius but is shifted such that absolute zero is at 0K. Note that Kelvin has no degrees symbol. °C → K can be converted between using:

$$T(\mathrm{K}) = T(°\mathrm{C}) + 273.15$$

3 Thermal Expansion

As a solid is heated, it expands in all directions. Consider a cuboid of height, width, depth W, L, D, we have new dimensions of $D + \Delta W$, $L + \Delta L$, $D + \Delta D$ after a small change in temperature. We can define the linear expansion coefficient (a material property) α_L based on these dimensions and the change in temperature:

$$\alpha_L \Delta T = \frac{\Delta L}{L} = \frac{\Delta W}{W} = \frac{\Delta D}{D}$$

If we consider the area instead, we have the area expansion coefficient:

$$\frac{\Delta A}{A} = \alpha_A \Delta T \approx 2\alpha_L \Delta T$$

The latter approximation can be derived as follows. Consider one dimension, i.e. L:

$$\frac{\Delta}{L} = \alpha_L \Delta T$$
$$\implies \Delta L = L\alpha_L \Delta T$$

So:

$$L_{\mathrm{new}} = L + \Delta L = L\left(1 + \alpha_L \Delta T\right)$$

Pairing this with width to find an area (of a face):

$$W_{\mathrm{new}} = W + \Delta W = W(1 + \alpha_L \Delta T)$$
$$A_{\mathrm{old}} = L \times W$$
$$A_{\mathrm{new}} = L_n \times W_n$$
$$A_n = LW(1 + \alpha_L \Delta T)^2$$
$$= LW\left(1 + 2\alpha_L \Delta T + \alpha_L^2 \Delta T^2\right)$$

For a small change in temperature, the second order term is very small, hence:

$$\frac{\Delta(LW)}{LW} \approx 2\alpha_L \Delta T$$

The linear expansion coefficient has units of K^{-1}.

3.1 Thermal Stress

Consider a rod fixed between two immovable walls. The rod attempts to expand, but is fixed. This provides a thermal stress on the wall. If the rod has cross-sectional area A and Young Modulus Y, the thermal stress is given by:

$$\frac{F}{A} = Y\alpha_L \Delta T$$

This is why bridges etc need to have thermal expansion joints, otherwise they would buckle/break under this thermal stress.

3.2 Lennard Jones and Thermal Expansion

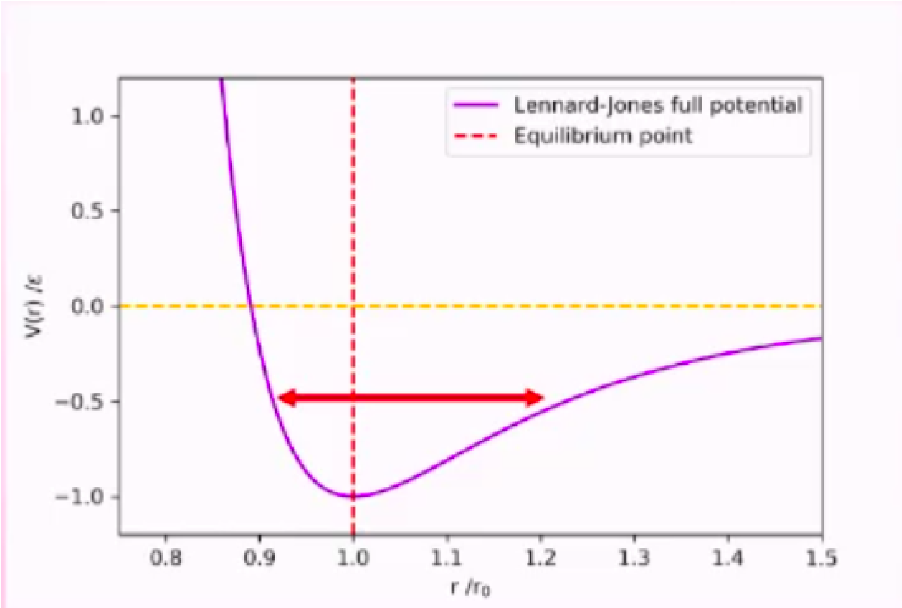


Figure 46.2: LJ Potential

As we increase temperature, the potential decreases and the depth of the potential well decreases. As the LJP is less and less symmetric for increasing r/r_0, this leads to an asymmetric potential where the bonds spend more time on average vibrating in an expanded state than in an equilibrium or contracted state. As energy increases, so does average bond length.

We only see small increases in length as it's an average, some bonds will still be in the equilibrium state or even below it, statistically.

Thu 12 Feb 2026 13:00

Lecture 8 - Thermal Physics II: Thermodynamics Intro

1 Temperature Scales II

In order to measure temperature, we need to define a quantity which directly depends on temperature - ideally linearly.

We can't measure temperature directly, we need this extra intermediary quantity. For example, thermometers use volume and measure the volume of a known quantity of a substance which behaves nicely at known temperatures.

This means that:

$$T + T_0 + k\frac{x - x_0}{x_0}$$

Where T_0 is a calibration point, at which point our quantity has value x_0. For the celsius scale, we use $T_0 = 0°\text{C}$ and we choose a value for the constant k such that when water boils, $T = 100°\text{C}$.

However, $PV = nRT$ says that this is dependant on pressure too which isn't ideal, as for different pressures the relationship between volume and temperature differ and our scale is only consistent for a single pressure....

Instead of using the boiling/freezing point of water at 1ATM has our calibration point, we can use the "triple point" of water. This point only happens at a single pressure/volume, so is more consistent:

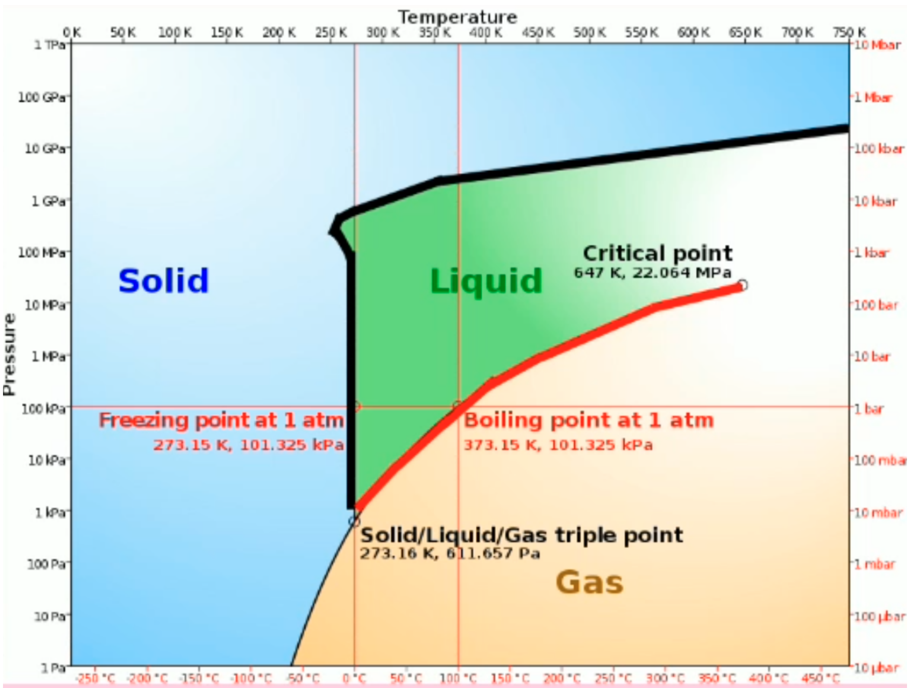


Figure 47.1

Alternatively, we could use the technique employed by the Kelvin scale and use absolute zero as our calibration point. Regardless of pressure/volume/amount of substance (provided its kept constant), they will all reach absolute zero at exactly the same theoretical temperature. We can therefore calibrate our scale off of this point.

2 Thermodynamics

Thermodynamics: The science of the relationship between heat, work, temperature and energy. Thermodynamics broadly deals with the transfer of energy from one place/form to another. Heat is a form of energy corresponding to a definite amount of mechanical work.

2.1 What is heat?

Heat is measure of thermal energy transfer between two systems. This shouldn't be confused with the colloquial understanding of heat i.e. being hot or cold, which is relative.

2.2 Systems

A system is a part of the universe we are investigating or discussing eg a room, bottle, galaxy etc. We often consider the ideal physics case of a closed system where the system doesn't exchange any heat with outside the system however this is not realistic.

We often discuss systems comprised of a container of some kind of fluid which either closed (with a lid) or open (no lid).

We can consider two types of walls:

- Adiabatic walls: No heat transfer possible (does not exist in practice).
- Diathermal walls: Heat transfer is possible through the wall.

2.3 Thermal Equilibrium

A system is in thermal equilibrium if it has a constant energy density throughout, so the system is in internal thermal equilibrium.

Two systems are in thermal equilibrium if there is no net transfer of heat energy between them. There will be some amount of heat transfer between both systems, and thermal equilibrium does not mean that there is no exchange of heat energy between the two. Some of the particles in one part of the system may statistically have higher energy and exchange energy, but there is no *net* transfer.

Consider some gas in a piston with volume, pressure and temperature V_I, P_I, T_I. As we pull out the piston, the volume of the gas changes and we have a new final temperature volume (and as pressure/volume are related to pressure), pressure and temperature V_F, P_F, T_F.

This does not happen instantly and there will be some time required to reach thermal equilibrium (a few seconds) after we have finished pulling out the system.

We can bring a system in contact with another system, i.e. placing a rubber duck (system B) in a bath (system A) or the ocean (system C). Which of these two pairs (A and B vs C and B) will thermalise first? While the duck will reach the temperature of the water faster when in the ocean than the bath, thermal equilibrium requires the entire system to thermalise. Therefore, as the ocean has much higher volume it takes longer for the ocean to entirely thermalise to the newer duck-adjusted temperature.

3 Definitions

- **Isothermal:** A change to a system which takes place at a **constant temperature**.
- **Isobaric:** A change to a system which takes place at a **constant pressure**.
- **Isochoric:** A change to a system which takes place at a **constant volume**.
- **Adiabatic:** A change to a system which takes place **without transfer of heat**. Temperature may change, but heat cannot enter or leave the system.

4 Heat Capacity

What is the relationship between heat energy ΔQ transferred to a system and the corresponding increase in system temperature ΔT? This is given by:

$$\Delta Q = C\Delta T$$

Where C is the heat capacity in J/K, hence:

$$C = \frac{dQ}{dT}$$

C varies with the amount of material, so we additionally define specific and molar heat capacity:

- Specific Heat Capacity J/kgK: $C = mc$
- Molar Heat Capacity $J/molK$: $C = nc$

A higher heat capacity means a smaller increase in temperature for a given heat (i.e. more energy required to raise the temperature of the system by some amount).

To make things more difficult, heat capacity also varies with the system temperature. We also need to specify whether we are measuring isobaric or isochoric, as these will give different values:

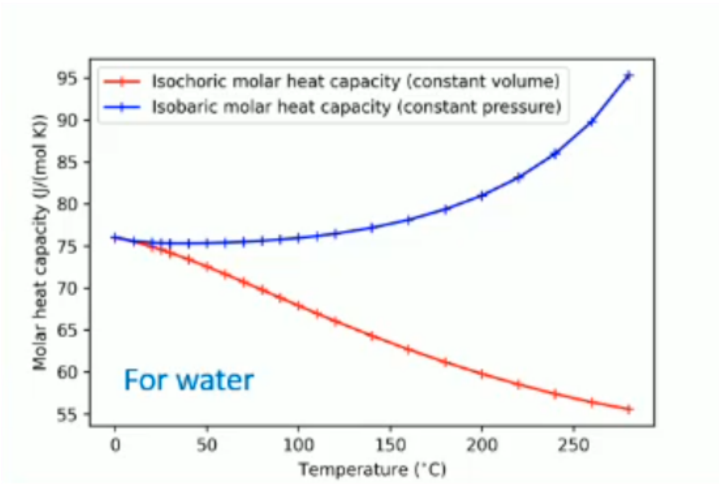


Figure 47.2

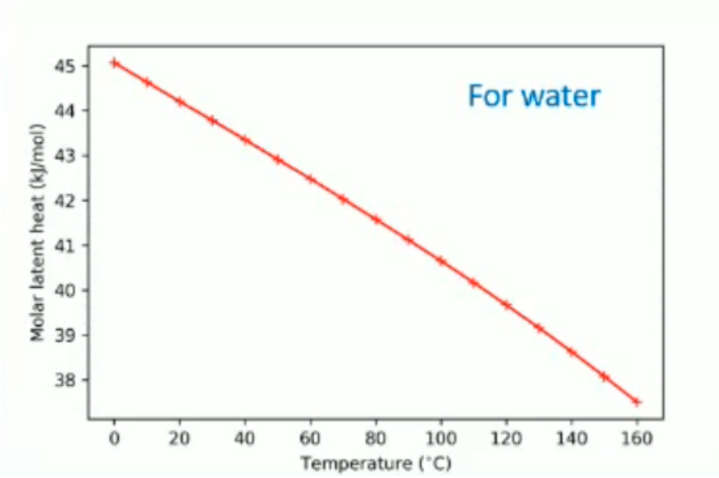


Figure 47.3

Notably, isobaric heat capacity is mostly constant around the standard values of liquid water at standard temperature:

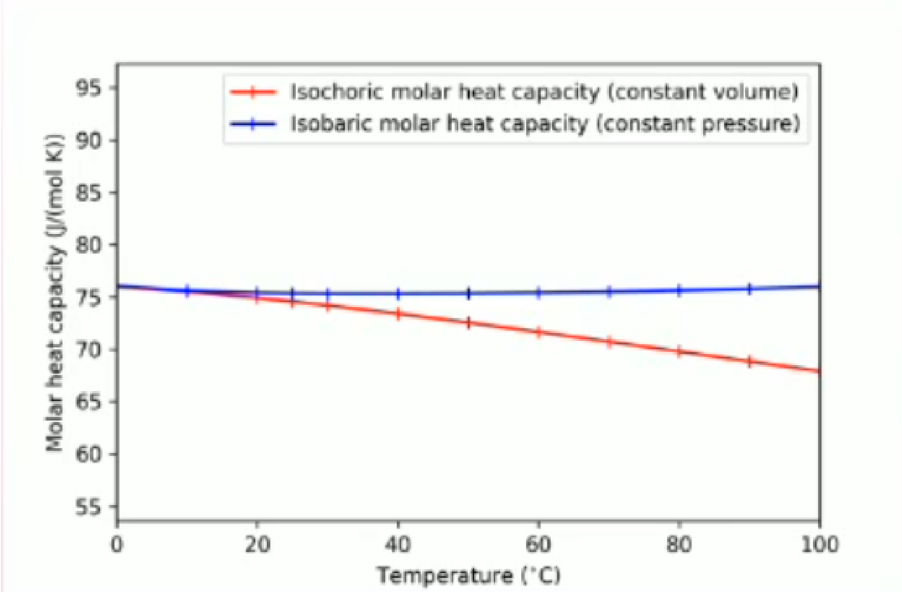


Figure 47.4

It is therefore not unreasonable to treat heat capacity as a constant in some circumstances, provided we state this as an assumption and justify it.

Tue 17 Feb 2026 12:00

Lecture 9 - Thermal Physics III: Laws of Thermodynamics

1 0th Law of Thermodynamics

Consider two blocks next to each other, one with T_h and one with T_c. The blocks with thermalise to their mean $(T_h + T_c)/2$ in the trivial case where the block are identical.

Net heat will flow between thw two systems until they have the same energy density and hence temperature.

If we have three systems: A, B, C. If we know that A and B are in thermal equilibrium, and A and C are in thermal equilibrium, then B and C are also in thermal equilibrium.

It's a fairly trivial axiom, so we denote it the zeroth law. Effectively, all objects in a system in equilibrium share the same temperature.

2 Ideal Gases

An ideal gas is defined as a collection of molecules (or atoms, if monatomic) that are non-interacting with each other (no interatomic forces) and collide elastically with each other.
The internal energy of the gas is dependant on the velocities of the molecules, and hence on the temperature, and not on pressure or volume.

Boyle's Law: "The absolute pressure exerted by a given mass of an ideal gas is inversely proportional to the volume is occupies, if the temperature and amount of gas remain unchanged within a closed system."

Effectively:

$$P \propto \frac{1}{V}$$

Charles' Law: "When the pressure of a sample of an ideal gas is held constant, the Kelvin temperature and volume will be in direct proportion."

Effectively:

$$T \propto V$$

Ideal Gas Law: Since $P \propto \frac{1}{V}$ and $T \propto V$, we have: $PV = kT$, where k varies with context, for example when considering moles:

$$\boxed{PV = nRT}$$

Where n is the number of moles and $R = 8.314\text{Jmol}^{-1}\text{K}^{-1}$ is the gas constant.
Or for molecules:

$$\boxed{PV = Nk_bT}$$

Where $N = N_a n$ is the number of molecules, and $k_b = R/N_a$ is the Boltzmann Constant.

This is called an equation of state and allows us to describe the gas' state macroscopically. We generally express this on a P/V diagram, where each point on the plot represents a specific gas state. If we keep the amount of gas present constant, we can use any two of the variables to determine the third.

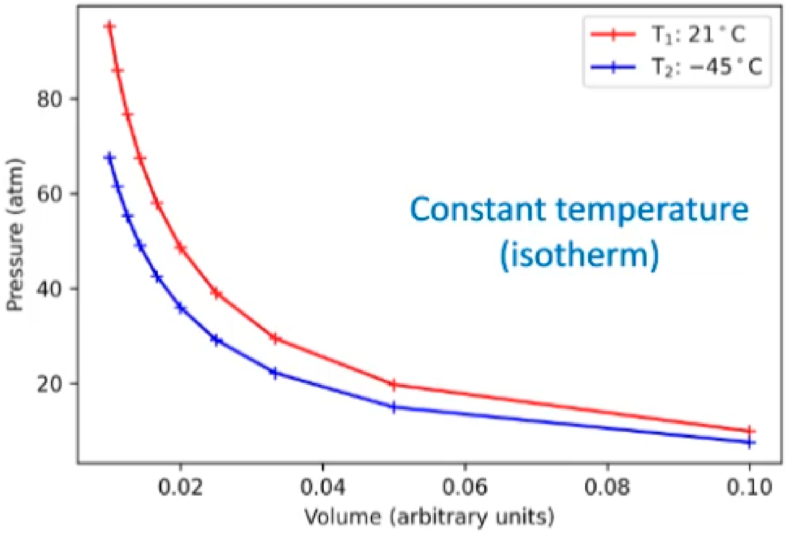


Figure 48.1

2.1 Joule's Second Law

Consider a system with a box divided in two. An ideal gas is contained in the leftmost half, and the rightmost half is a vacuum. We separate the two with a divider constituting an impermeable membrane.

We remove the membrane laterally, doing no work on the gas. Joule observed that the gas stayed at the same temperature as it diffused into the vacuum.

Consider the internal energy of the gas, denoted U, assuming that U is a function of two of the state variables. We know the temperature did not change, and the volume did, so $U(T, V)$.

We would expect:

$$dU = \frac{\partial U}{\partial T}dT + \frac{\partial U}{\partial V}dV$$

Temperature was observed experimentally to not change, hence $dT = 0$, so:

$$dU = 0 + \frac{\partial U}{\partial V}dV$$

Volume did change, so $dV \neq 0$. We did no work on the gas, as the divider was removed laterally, hence there was no change in internal energy, so:

$$dU = \frac{dU}{dV}dV = 0$$

And since $dU = 0$, we must have:

$$\frac{dU}{dV} \neq 0$$

This means that there is no dependence on volume for internal energy, hence U is dependent only on T, as found by Joule. This means that the gas Joule chose was well approximated by an ideal gas.

This is easier at higher temperatures, as a high temperature leads to high kinetic energies, so the interatomic forces becomes less significant and easier to disregard in reality.

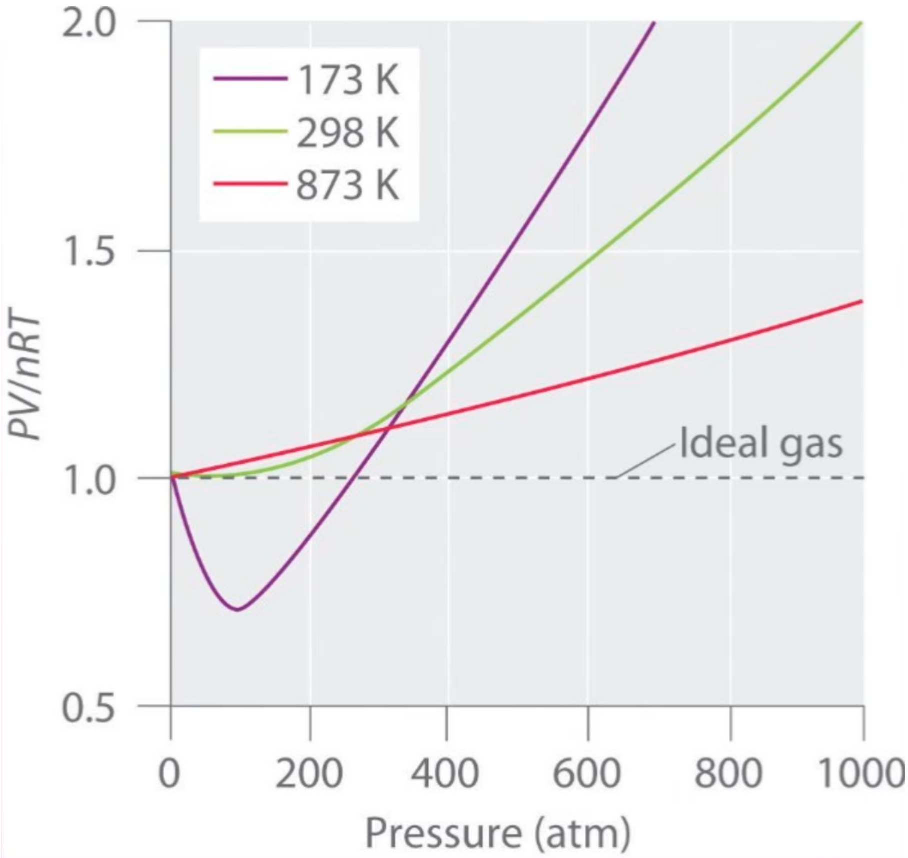


Figure 48.2

We see here that it the ideal gas law becomes a better approximation has we increase temperature, and a worse and worse approximation as pressure increases.

From a LJP perspective, increasing the pressure decreases the average distance between gas molecules. This means that the potential between gas molecules is no longer negligible, and the assumption of zero potential no longer holds.

If we increase pressure even further, the force becomes repulsive and PV/nRT returns to being positive. At higher temperatures, this little dip isn't observed as a higher kinetic energy makes any interatomic forces relatively more negligible.

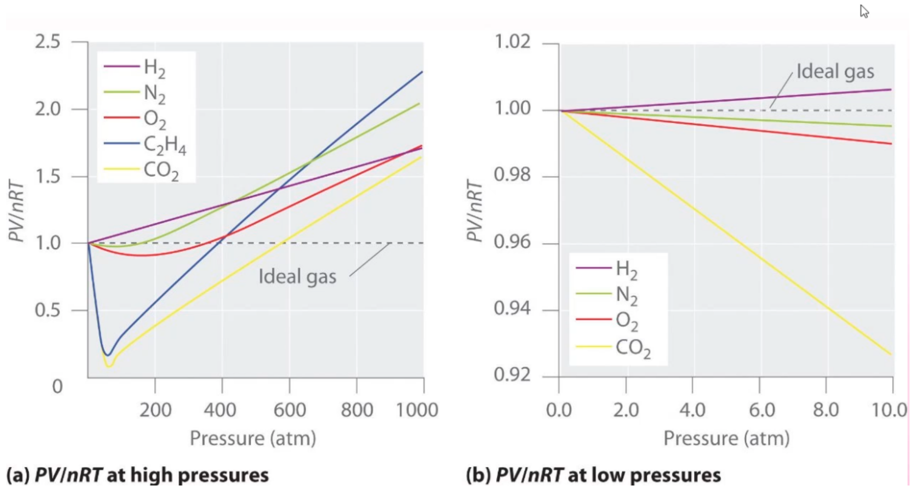


Figure 48.3

At low pressure or high temperature, the ideal gas law is a good assumption as:

- At low pressure, interatomic spacing is large enough to disregard interatomic forces.

- At high temperature, kinetic energy is large enough to comparatively disregard interatomic forces.

We deal solely with Maxwell-Boltzmann gases in this source that follow classical laws. We also have Fermi and Bose gases (made entirely of fermions and bosons respectively), but they're quantum mechanical, exotic and outside of this course.

2.2 Changing Energies

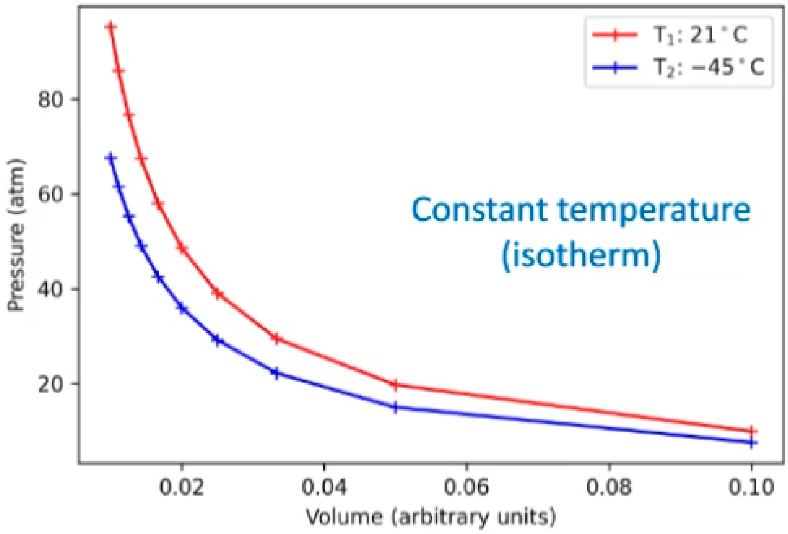


Figure 48.4

Say we want to go from T_1 to T_2. We need to change the internal energy of the gas somehow. We have two general ways to change this:

- Add heat to the system by transferring heat to the gas at a constant volume.
- Do some work on the gas, i.e. by crushing it and reducing volume.

Consider a piston of area A, applying constant force F with extension ΔL. The pressure on the gas is $P = F/A$. The work done on the gas is:

$$\Delta W = F\Delta L$$
$$\implies \Delta W = P(A\Delta L)$$
$$\implies \Delta W = P\Delta V$$

Moving from infinitesimal Delta values:

$$W_{\text{on gas}} = \int dW = \int P\,dV$$

In the opposite case considering work done by the gas, conservation of energy says it must be oppositely signed:

$$W_{\text{by gas}} = -\int P\,dV$$

2.3 Mass Drop Experiment

The falling mass spins the paddles in the water, the work as potential energy is converted to kinetic energy raising the temperature of the water.

This tells us that mechanical work done and heat energy must be related in some manner.

3 1st Law of Thermodynamics

The change in internal energy of a system, ΔU is increased with increasing heat transfer into the system, Q_{in} and work done on the system, W_{on}:

$$\Delta U = Q_{\text{in}} + W_{\text{on}}$$

This is just conservation of energy.

Thu 19 Feb 2026 13:00

Lecture 10 - Thermal Physics IV: Thermodynamic Transitions

1 Thermodynamic Transitions

1.1 Isothermal (Fixed Temperature)

This may look like:

- A curve on a PV plot. As $PV = nRT$, a PV plot shows a $1/x$ relationship for constant T.
- A vertical straight line on a PT plot.
- A vertical straight line on a VT plot.

If the temperature change is 0, ΔU (which we derived last time only depends on temperature) follows:

$$\Delta T = \Delta U = 0$$

Hence:

$$0 = Q_{\text{in}} + W_{\text{on}}$$

1.2 Isochoric (Fixed Volume)

This may look like:

- A vertical straight line on a PV plot.
- A proportional relationship on a PT plot.
- A horizontal straight line on a VT plot.

For an isochoric transition, $\Delta U = Q_i n$, as:

$$W_{\text{on}} = (-) \int P dV$$

And as $dV = 0$, $W_{\text{on}} = 0$

1.3 Isobaric (Fixed Pressure)

This may look like:

- A horizontal straight line on a PV plot.
- A horizontal straight line on a PT plot.
- A proportional relationship on a VT plot.

For an isobaric transition, P is just a constant, so the integral:

$$W_{\text{on}} = \int_{V_1}^{V_2} P\,dV = [PV]_{v_1}^{v_2}$$

Where P is constant, which is nice and easy. Generally:

$$W_{\text{on}} = -\int PdV = (-)P\Delta V$$

$$\Delta U = \Delta Q - P\Delta V$$

2 Heat Capacities of Ideal Gases

Recall that:

$$\Delta Q = C\Delta T \implies C = \frac{dQ}{dT}$$

At a fixed volume:

$$C_V = \left(\frac{dQ_{\text{in}}}{dT}\right)_T$$

And at a fixed pressure:

$$C_P = \left(\frac{dQ_{\text{in}}}{dT}\right)_P$$

Mayer's Relation says that:

$$C_P = C_V + nR$$

This is an examinable derivation.

2.1 Deriving Mayer's Relation

Proof. Consider a gas piston. We have state variables V, P, T, and N is constant as the piston is sealed. The first law of thermodynamics says:

$$\Delta U = Q_{\text{in}} + W_{\text{on}}$$

We can consider this for both a constant volume or a constant pressure. For a constant volume:

$$\Delta U = \Delta Q_1$$

And for a constant pressure:

$$\Delta U = \Delta Q_2 - P\Delta V$$

Noting that the two Qs are different as we have two different transitions. We can create a slightly different expression for constant volume by multiplying by $\Delta T/\Delta T$:

$$\Delta U = \frac{\Delta Q_1}{\Delta T} \times \Delta_T = C_V \Delta T$$

And for constant pressure:

$$\Delta Q_2 = \Delta U + P\Delta V = \frac{\Delta Q_2}{\Delta T}\Delta T = C_P \Delta T$$

Hence:

$$C_P \Delta T = \Delta U + P\Delta V$$
$$C_V \Delta T = \Delta U$$

Setting equal to each other based on ΔU:

$$\Delta U = \boxed{C_V \Delta T = C_P \Delta T - P\Delta V}$$

$$C_P \Delta T - C_V \Delta T = P\Delta V$$

$$\Delta T(C_P - C_V) = P\Delta V$$

$$\implies C_P - C_v = \frac{\Delta V}{\Delta T} \times P \qquad \text{NB: } P \text{ is still constant.}$$

Using $PV = nRT$, for constant pressure we have $\frac{P\Delta V}{\Delta T} = nR$. Hence:

$$C_P - C_V = nR$$

$$C_P = C_V + nR$$

□

It is clear that $C_P > C_V$.

Note: We tend to use heat capacity at a constant volume unless specified.

3 Back to Thermodynamic Transitions

3.1 Adiabatic Transitions (No Heat Transfer)

These seem quite a bit like isothermal transitions, but look different on a PV plot:

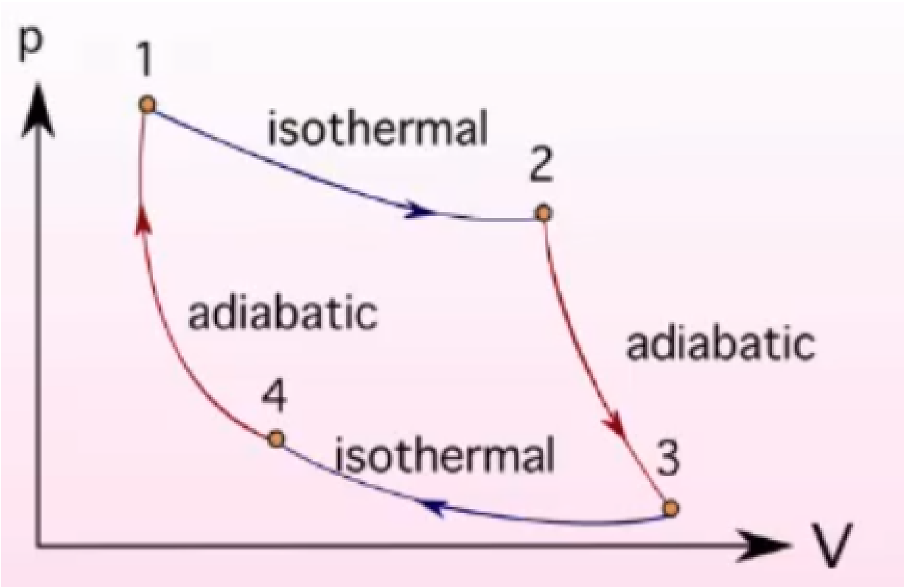


Figure 49.1: Differences in transitions on a PV Plot

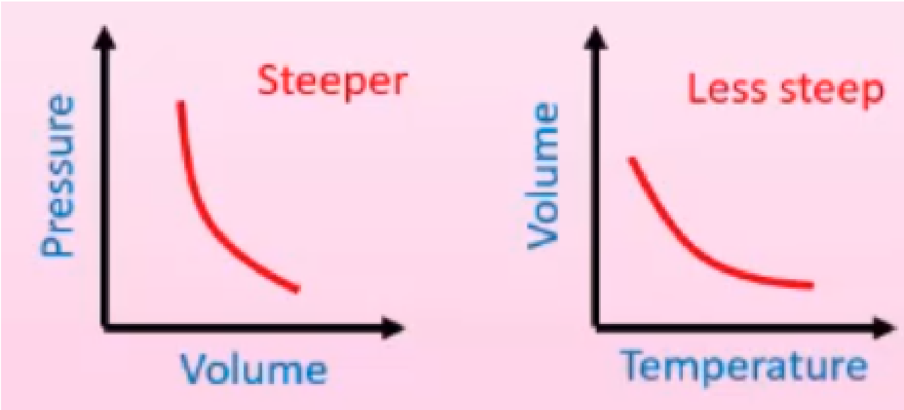


Figure 49.2: Adiabatic Transitions on a PV and VT Plot

For isothermal transitions, $PV = \text{const}$.

For adiabatic transitions, $PV^\gamma = \text{const}$ where $\gamma = \frac{C_P}{C_V}$

In a PV plane, adiabatic processes appear steeper than isothermal processes and adiabatic processing are steeper in the PV plane than the VT plane.

For an adiabatic transition, we can reduce the first law of thermodynamics like so:

$$\Delta U = Q_{\text{in}} + W_{\text{on}}$$

$Q_{\text{in}} = 0$ as there is no heat transfer, hence:

$$\Delta U = W_{\text{on}}$$

3.2 Derivations

Proof. For adiabatic transitions, where $Q = 0$.

The First Law becomes $\Delta U = Q_{\text{in}} + W_{\text{on}}$. Since $Q_{\text{in}} = 0$:

$$dU = 0 + dW_{\text{on}}$$

And assuming work is being done *on* the gas, the sign becomes negative:

$$dU = -PdV$$
$$\frac{dUdT}{dT} = -PdV$$

We slightly abuse the definition of heat capacity here, and this will be elaborated on in a later lecture:

$$C_V dT = -PdV$$

Using $PV = nRT$, we have: $d(PV) = nRdT$:

$$nRdT = PdV + VdP$$
$$dT = \frac{PdV + VdP}{nR}$$
$$C_V dT = -PdV$$
$$0 = C_V \frac{PdV + VdP}{nR} + PdV$$
$$0 = C_V (PdV + VdP) + nRpdV$$
$$0 = PdV(C_v + nR) + C_V VdP$$
$$0 = PdV(C_P) + VdP(C_V)$$
$$0 = \frac{C_P}{C_V} PdV + VdP$$

Let $\gamma = \frac{C_P}{C_V}$, and dividing through by pressure and volume:

$$\gamma \frac{dV}{V} + \frac{dP}{Ps}$$
$$\int \gamma \frac{dV}{V} = -\int \frac{dP}{P}$$
$$\gamma \ln V + c_1 = -\ln(P) + c_2$$

Finally:

$$\gamma \ln(v) + \ln(p) = \text{const}$$
$$\ln(PV^\gamma) = \text{const}$$
$$PV^\gamma = e^{\text{const}} = \text{another constant}$$

Hence:

$$PV^\gamma = \text{const}$$

As required! ☐

Note that in a TV plane, we instead have:

$$TV^{\gamma-1} = \text{const}$$

Tue 24 Feb 2026 12:00

Lecture 11 - Thermal Physics V: PV Planes and Transitions

Note RE: Last Lecture
To avoid confusion, the Physics convention is:

$$W_{\text{by}} = \int PdV$$

$$W_{\text{on}} = -\int PdV$$

1 PV Diagrams

We can move between points on a PV plane through processes which usually involve some kind of heat transfer and work. For example:

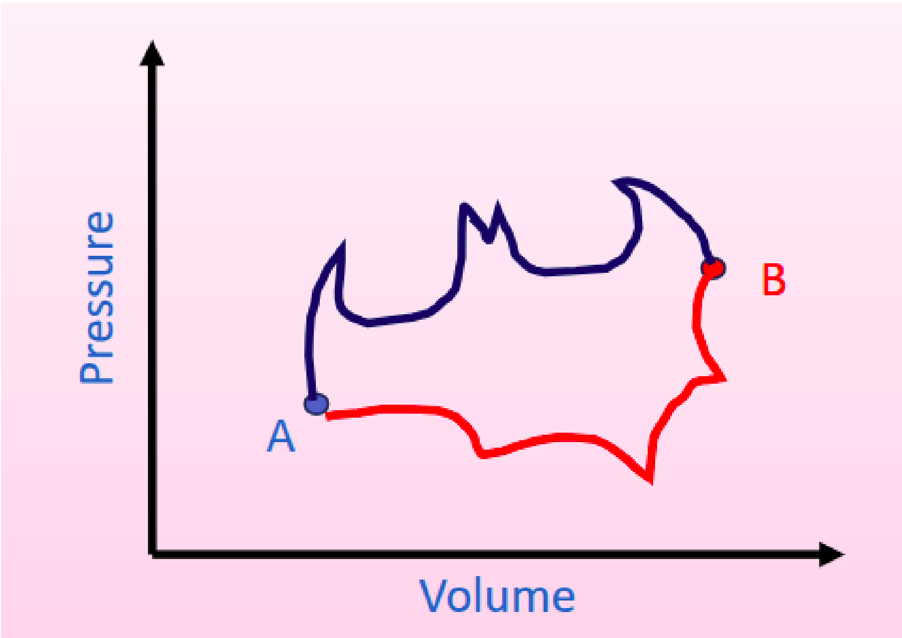


Figure 50.1

The two transitions from $A \to B$ are processes, and the nature of the process depends on the shape of the transition on the PV plot.

- Since the two possible transitions between A and B share the same start and end point, the change in internal energy U_{AB} is the same for both. Since we go from T_A to T_B in both cases (as U depends on T, which in turn depends on P, V so for the same P, V we have the same T), both paths have the same start/end internal energy, so the same change between them.
- The work done by the gas in the two paths is not the same, however. W_{by} is larger for the blue path as W is an integral of $\int PdV$, which represents the area under the curve. The blue line is further up the pressure axis, so the integral is larger and more work is done.

- What about heat input to the gas, Q_{in}? Since the First Law says $\Delta U = Q_{\text{in}} + W_{\text{on}} \implies \Delta U = Q_{\text{in}} - W_{\text{by}}$. ΔU is constant, and $-W_{\text{by}}$ has a larger magnitude for blue, so must Q_{in} to keep ΔU constant.

These transitions are called a *PV Cycle* as we can start at A, go to B and end back up at A.

Consider a simple example with straight lines:

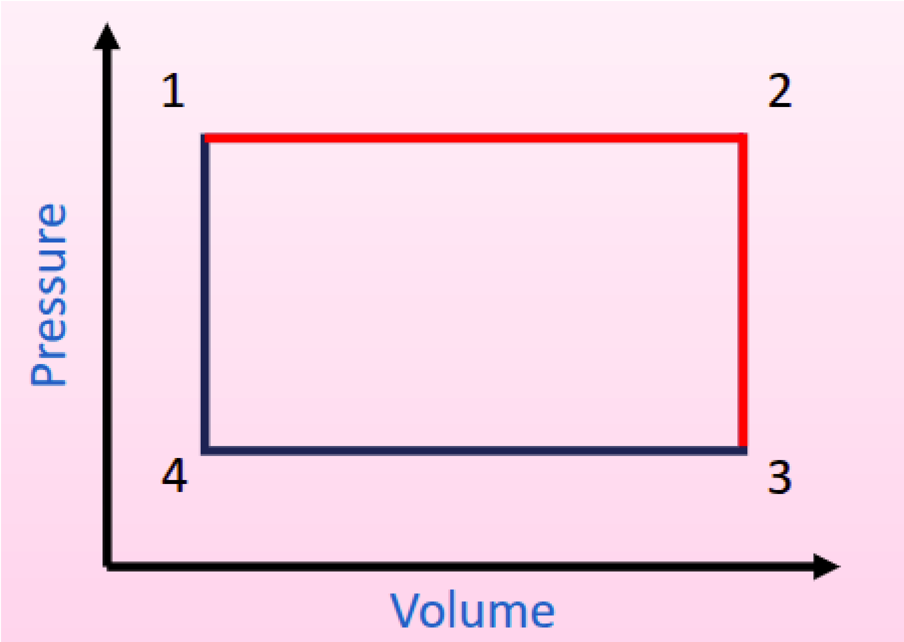


Figure 50.2

Where $1 \to 2$ and $3 \to 4$ are isochoric processes and $2 \to 3$ and $4 \to 1$ are isobaric. We can also look at isothermal and adiabatic processes:

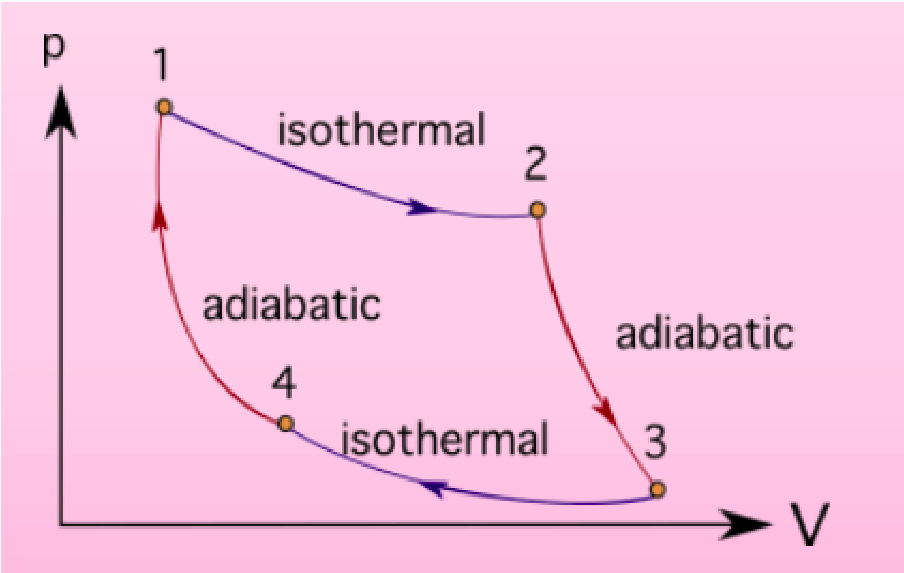


Figure 50.3

A cycle must follow these rules:

1. $\Delta U = 0$, for a full cycle.
2. Total work done is given by the area contained by the cycle.
3. Clockwise cycle for positive work, anticlockwise for negative work.

4. $Q_{\text{in}} + W_{\text{on}} = 0$ as $\Delta U = 0$.

2 Otto Cycle

This describes a petrol internal combustion engine:

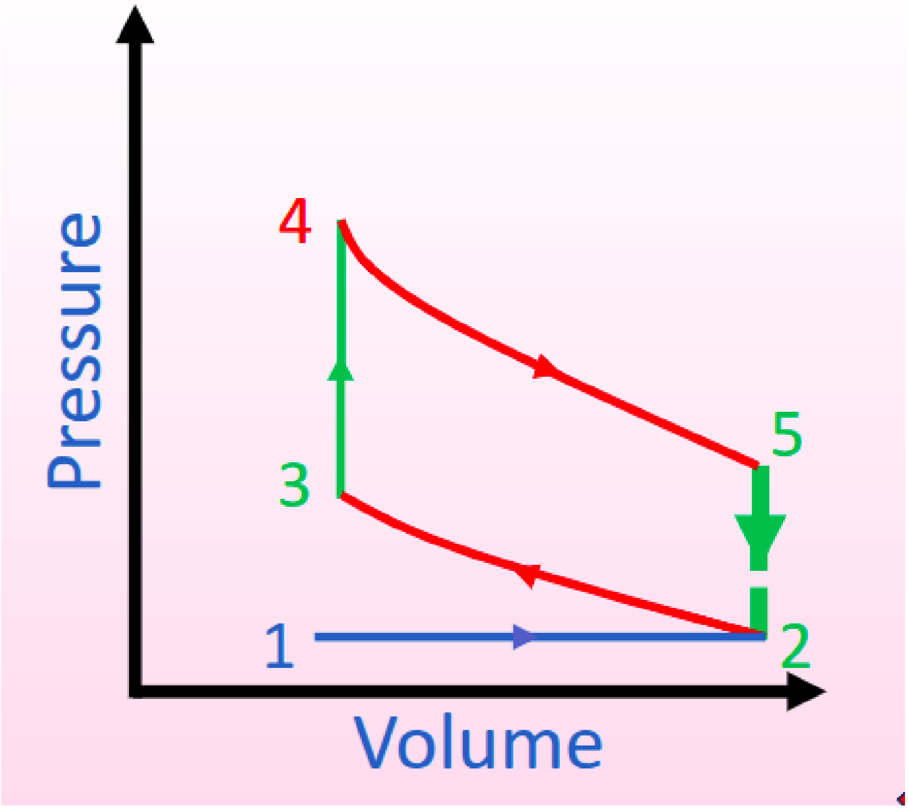


Figure 50.4: Blue: Isobaric, Green: Isochoric, Red: Adiabatic

This has the following components:

1. 1 → 2: Intake stroke: Isobaric introduction of air and fuel to the system.
2. 2 → 3: Adiabatic compression of the fluid as the piston compresses it.
3. 3 → 4: Isochoric introduction of heat energy to the system (ignition of the fluid mixture).
4. 4 → 5: Adiabatic expansion of the fluid mixture as the gas does work on the piston (pushing it down).
5. 5 → 2: Isochoric expulsion of heat from the system.
6. 2 → 1: Mass of air and exhaust gases released at a constant pressure.

3 Diesel Cycle

Here, the ignition and combustion process is performed at a constant pressure where work is done by the gas:

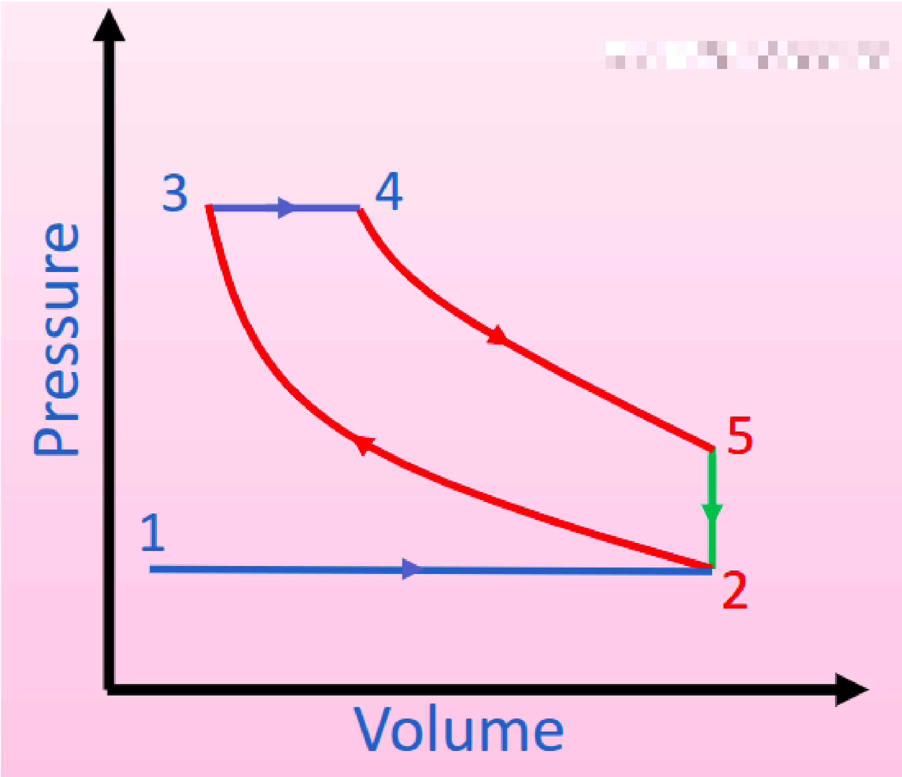


Figure 50.5: Blue: Isobaric, Green: Isochoric, Red: Adiabatic

In both cases, heat is introduced (by combustion) from $3 \to 4$, denoted Q_1 and heat is lost from $5 \to 2$ when heat is expelled from the system. The useful work done is $Q_1 - Q_2$ and is given by the area enclosed by the loop.

3.1 Example

Consider the adiabatic gas compression stroke in a diesel engine. (This is the adiabatic stroke where pressure is increasing/volume is decreasing, i.e. from $2 \to 3$.)

10^{-3}m^3 of nitrogen gas is initially at atmospheric pressure and 298K and is compressed to 1/15th of its original volume.

You may assume that nitrogen is an ideal diatomic gas with $\gamma = 1.4$ and $C_V = 20.85 J/Kmol$.

Calculate:

1. The final pressure.
2. The final temperature.
3. The number of moles in the gas.
4. The change in internal energy.
5. The work done by the gas.

Part I
Atmospheric pressure is $101325 \approx 10^5$Pa. As this is an adiabatic transition, $PV^\gamma = \text{const}$. Hence:

$$P_1 V_1^\gamma = P_2 V_2^\gamma$$

$$\implies P_2 = P_1 \left(\frac{V_1}{V_2}\right)^\gamma$$

$$\implies P_2 = P_1 \left(\frac{V_1}{(V_1/15)}\right)^\gamma$$

$$\implies P_2 = P_1 (15)^\gamma = 4.5\text{MPa}$$

Part II
We can use either $PV = nRT$ twice, using the initial conditions to determine n or we can use:

$$T_1 V_1^{(\gamma-1)} = T_2 V_2^{(\gamma-1)}$$

Using the same process gives:

$$T_2 = T_1 (15)^{(\gamma-1)} = 880\text{K}$$

Part III
Since we know P_1, V_1 or P_2, V_2 we can use $PV = nRT$, as mentioned above. It would be a good sanity check to do both and compare the values.

Part IV
It's an adiabatic transition, we can related $C_V = \frac{\partial U}{\partial T}$.

Integrating both sides gives:

$$\int dT = \int \frac{\partial U}{\partial T}$$

Thu 26 Feb 2026 13:00

Lecture 12 - Thermal Physics VI: Heat Transfer

1 Thermal Transport

Newton's Law of Cooling: "The rate of heat loss of a body is directly proportional to the difference in the temperatures between the body and the environment"

There are three means of thermal transport:

- **Conduction:** Heat is transmitted directly from one material to an adjoining material (or across parts of the same material) if there is a temperature difference between the two. There is no movement of the material.
- **Convection:** Movement of particles through a substances (often particles that comprise a fluid) that transport their heat energy along with them.
  - The Archimedes principle describes buoyancy based on the material's density ρ, g and the volume of the the fluid displaced by the object, V:
  $$F_b = \rho g V$$
  - In convection, heating a particle causes an increase in bond size (by the LJP), hence the density decreases. This causes it to displace gas of a larger density, hence it rises.
- **Radiation:** Emission of electromagnetic radiation by all bodies which have heat (i.e. all bodies full stop), depending on their temperature.

1.1 Heat Baths

A heat bath is an object with a sufficiently large heat capacity, C such that a large change in its heat energy will result in a negligible change in temperature.

2 Conduction

Suppose we have a small uniform rod of length L and cross-sectional area A linking two heat baths. The rod is conducting and the heat baths have temperature $T_1 < T_2$.

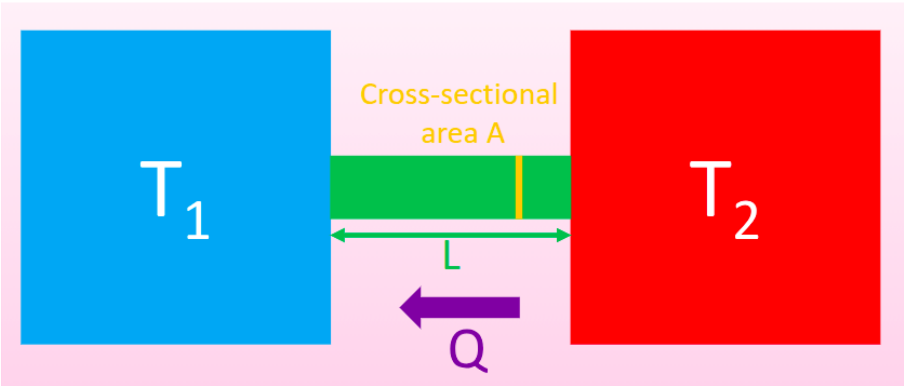


Figure 51.1

Eventually, we reach a "steady state" where all points on the road have a $\dot{Q}$ which does not change with time.

$$\frac{dQ}{dt} = \dot{Q} = \text{constant}, \qquad \forall \text{x along rod (in steady state)}$$

What does this value of $\dot{Q}$ depend on? For a small section dx of the rod:

- Temperature change of the small section, dT.
- Cross-sectional area of the rod A.
- The "thermal conductivity" of the rod, κ[1]. It has units $\mathrm{Wm^{-1}K^{-1}}$.

Putting these together gives us Fourier's Law:

$$\dot{Q} = -\kappa A \frac{dT}{dX}$$

As T_2 is hotter, the flow of heat is from right to left. Therefore until thermalisation occurs, temperature increases from left to right across the rod ($\frac{dT}{dx} > 0$). However, the heat flow is in the opposite direction from right to left, hence the negative sign in the formula for $\dot{Q}$.

We want to show that $\frac{dT}{dx} = \frac{T_2 - T_1}{L}$:

$$\begin{aligned}
\dot{Q} &= -\kappa A \frac{dT}{dX} \\
\frac{dT}{dx} &= -\frac{\dot{Q}}{\kappa A} \\
-\int \frac{\dot{Q}}{\kappa A} dx &= \int dT \\
-\int_0^L \frac{\dot{Q}}{\kappa A} &= \int_{T_1}^{T_2} dT \\
-\frac{\dot{Q}}{\kappa A} [x]_0^L &= T_2 - T_1 \\
\frac{dT}{dx} &= \frac{T_2 - T_1}{L}
\end{aligned}$$

2.1 Thermal Conductivity

A higher value of κ means a material is more thermally conductive, for example:

- Diamond (natural): $\kappa = 2200\mathrm{Wm^{-1}K^{-1}}$.
- Copper: $\kappa = 400\mathrm{Wm^{-1}K^{-1}}$.
- Air: $\kappa = 0.02\mathrm{Wm^{-1}K^{-1}}$.
- Aerogel: $\kappa = 0.003\mathrm{Wm^{-1}K^{-1}}$.

Diamond is highly thermally conductive due to high frequency vibrations ("phonons") being able to propagate through the material as a result of its rigidity.

There are two sources of heat conductivity, one being as a result of electrons and one as a result of phonons.

- Inside the material, we have two bands - valence and conduction. These bands are a thick collection of many energy levels very close together. As the Pauli exclusion principle says that no two electrons can share a quantum number, each level must shift infinitesimally to not overlap. The levels within each band are so close that we treat electron as being able to move freely between them.
- The valence band is where electrons are still bound to an atom, the conduction band is where they are bound to the material as a whole and are free to move through the material.
- For an insulator, the difference between the two bands is $\sim$ 10eV. This represents a very small proportion of particles with sufficient energy and results in very little thermal excitation (hence little thermal conduction). As an electron is excited, it leaves a hole behind (that we treat as a particle h^+ for ease). We treat this hole as being able to move (as electrons themselves move).

[1] We also have another very similar quantity labelled K which will be covered later and may be a source of confusion

- For an electron to fall back to the valence band, it must fall into a hole. In an insulator the conduction band is effectively empty. For a semiconductor, this gap is more like $\sim$ 1eV, so a much higher number of electrons can jump, but this is still temperature dependant.
- In metals, the conduction and valence bands have little to no gap. Therefore, all electrons are free conduction electrons.

Diamonds sit on the boundary between being an insulator and a semiconductor. They work by phonons. As we heat a material, bonds in the material expand and cause a vibration. As diamond is very rigid, vibrating one bond causes a ripple effect - effectively a wave of excitation (where one of these waves is called a phonon). This phonon propagation causes the transfer of vibrations hence the transfer of heat energy.

2.2 Thermal Properties

We can also define the:

- Thermal Resistivity: $\rho = 1/\kappa$.
- Thermal Conductance $K = \kappa A/L$. [2]
- Thermal Resistance: $R = L/(\kappa A)$.

This are the heat analogues of the equivalent terms for electrical conductance in a circuit.

Annoyingly, thermal conductivity also varies with temperature…It also does so in a really weird way depending on the material:

[2] This is the aforementioned annoying symbol overlap between κ and K

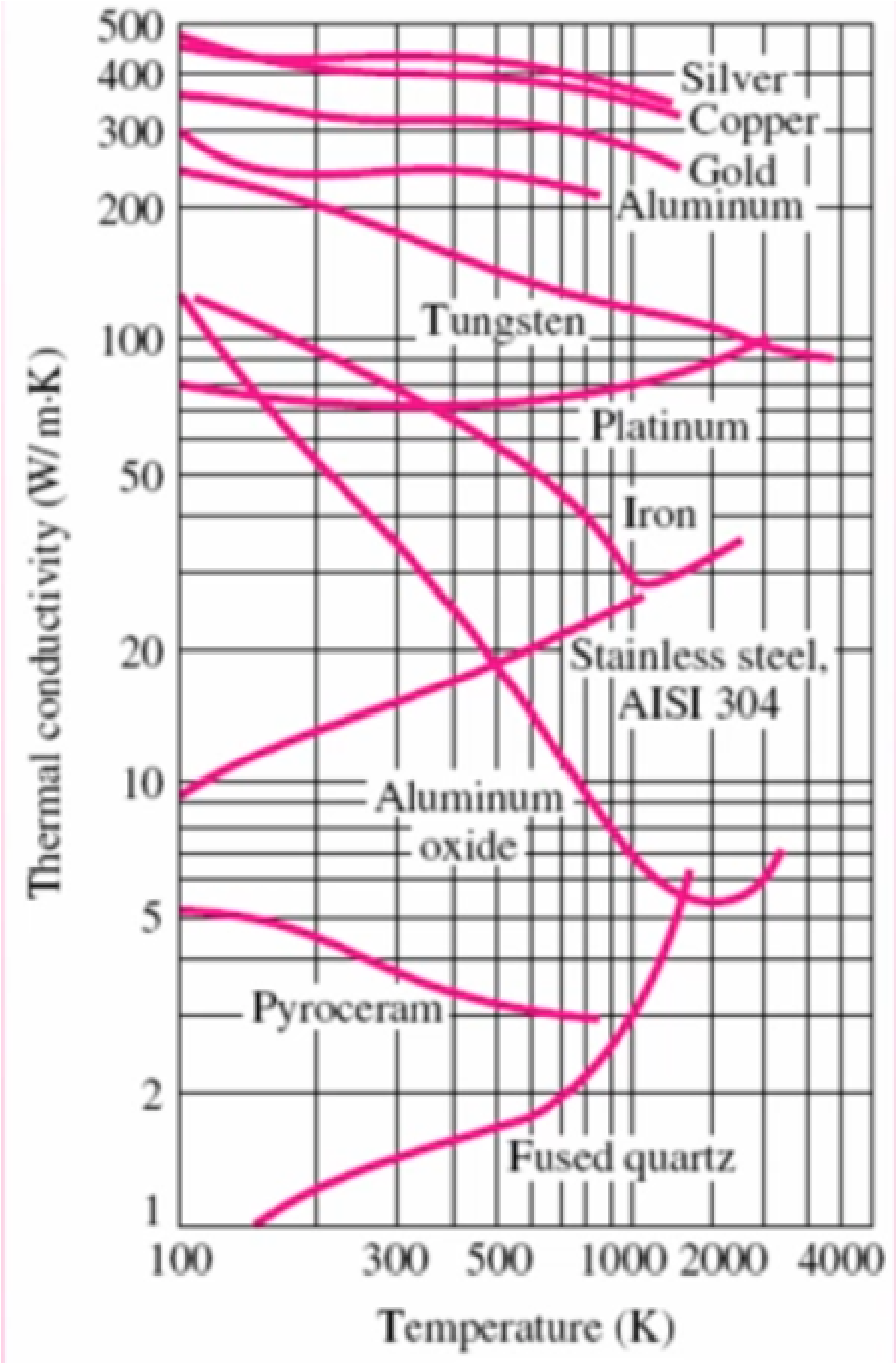


Figure 51.2: κ for various materials, as a function of temperature.

This comes from phonon excitation and electron excitation being two different competing process that both have different degrees of contribution at different temperatures.

This means that Newton's Law of Cooling no longer holds for $\kappa = f(T)$, as the proportionality no longer holds...

Tue 03 Mar 2026 12:00

Lecture 13 - Thermal Physics VII: Convection and Black Bodies

Recap from last lecture

- Three methods of heat transport: conduction, convection and radiation.
- Fourier's Law:
$$\dot{Q} = -\kappa A \frac{dT}{dx}$$
- We can show that:
$$\frac{dT}{dx} = \frac{T_2 - T_1}{L}$$
- Conduction determined by two distinct physical processes that transfer heat:
  - Movement of electrons.
  - Photon transfer.
- Thermal resistivity, thermal conductance and thermal resistance, acting as analogues of their electrical equivalents.

1 Conduction II

1.1 Combinations of Thermal Resistances

A derivation of why isn't examinable (but is shown in the non-assessed problems), but we have the same rules for Thermal Resistances as we do in electrical resistances:

- Thermal resistances add in series, $R_T = R_1 + R_2 + \cdots R_n$.
- Thermal resistances add in inverse when in parallel $1/R_T = 1/R_1 + 1/R_2 + \cdots + 1/R_n$.

1.2 Example: Pipe Lagging

Consider a solid copper pipe of radius r_0, initial temperature T_0 and length L. We surround this pipe with insulation of initial temperature T_1 and resulting in a new radius r_1. The insulation has $\kappa = 0.05\mathrm{Wm}^{-1}\mathrm{K}^{-1}$ and the system is in a steady state.

We start from Fourier's law:
$$\dot{Q} = -\kappa A \frac{dT}{dr}$$

As the pipe is a cylinder:
$$\dot{Q} = -\kappa(2\pi r l) \frac{dT}{dr}$$
$$-\int_{r_0}^{r_1} \frac{\dot{Q}}{2\kappa\pi r l} = -\int_{T_0}^{T_1} dT$$
$$\int_{r_1}^{r_0} \frac{\dot{Q}}{2\kappa\pi l} \frac{dr}{r} = \int_{T_1}^{T_0} dT$$

$$\frac{\dot{Q}}{2\kappa\pi l}[\ln r]_{r_0}^{r_1} = [T]_{T_1}^{T_0}$$
$$\frac{\dot{Q}}{2\kappa\pi l}\ln\left(\frac{r_0}{r_1}\right) = T_0 - T_1$$
$$\dot{Q} = (T_0 - T_1)\frac{2\kappa\pi l}{\ln(r_1/r_0)}$$

2 Convection

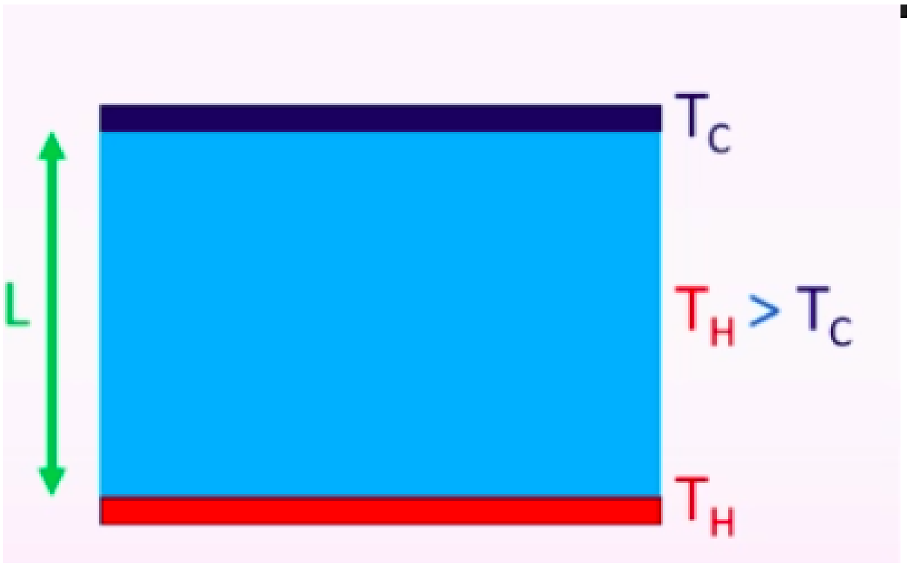


Figure 52.1

Consider a fluid suspended between two plates, distance L apart. The upper plate T_C is colder than the lower plate, T_H.

What governs whether or not convection can occur (and to what extent)?

- Buoyancy of the fluid $\beta = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_P$
- The thermal conductivity, κ.
- The viscosity of the fluid, η.

To decide whether or not convection can occur, we use the Rayleigh number:

$$R_A = \frac{\rho L^3(\beta\{T_H - T_C\})g}{\eta\alpha}$$

Where:

$$\alpha = \frac{\kappa}{\rho C_P}$$

If $R_a > 1,000$ then convection occurs laminarly, and if $R_a > 100,000$ then convection occurs turbulently convection does not occur $R_A < 1,000$. Laminar flow is simple, predictable and consistent flow (i.e. straight out of a well made tap). Turbulent flow is much more complex and is pseudorandom. We can predict statistical effects, but turbulent flow is too complex to predict accurately to a very low level.

The details of convection (except for the basic concept) are not examinable.

3 Thermal Radiation

All bodies emit EM radiation characteristic of the object temperature (and not the temperature of the surroundings). The frequency is temperature dependant. We often use infrared (i.e. thermal imaging) as the frequencies are emitted by an object around the temperature of a human body are IR.

A black body perfectly emits and absorbs radiation of all wavelengths, without any reflection. A while body, on the other hand, perfectly reflects all incident light. A black body in thermal equilibrium with surroundings emits exactly the same power spectrum as it absorbs.

Classical mechanics predicts the Rayleigh-Jeans law for emission:

$$u(\lambda, T) = \frac{8\pi k_B T}{\lambda^4}$$

Where u is power emitted per unit area, T is temperature and λ is the wavelength of emitted light. As this is inversely proportional to lambda, this causes the UV catastrophe, where $u \to \infty$ as $\lambda \to 0$. Max Plank developed an improved version[1].

$$u(\lambda, T) = \frac{8\pi hc}{\lambda^5} \frac{1}{\exp\left(\frac{hc}{\lambda k_B T}\right) - 1}$$

We can characterise non-black bodies by the emissivity, ϵ, the fraction of the black body spectrum which it absorbs. This gives $\epsilon = 1$ for a true black body and $\epsilon = 0$ for a true white body.

4 Wein's Approximation

We want to determine the maximum of the spectrum, to find the wavelength corresponding to the maximal power output.

$$u(\lambda, T) = \frac{8\pi hc}{\lambda^5} \frac{1}{\exp\left(\frac{hc}{\lambda k_B T}\right) - 1}$$

Letting $a = \frac{hc}{\lambda k_B T}$:

$$\begin{aligned} \frac{1}{\lambda} &= \frac{k_B T}{hc} a \\ \implies \frac{1}{\lambda^5} &= \left(\frac{k_B T}{hc}\right)^5 a^5 \\ \implies \frac{8\pi hc}{\lambda^5} &= 8\pi hc \left(\frac{k_B T}{hc}\right)^5 a^5 \end{aligned}$$

And:

$$\frac{1}{\exp\left(\frac{hc}{\lambda k_B T}\right) - 1} = \frac{1}{\exp(a) - 1}$$

$$\frac{\partial a}{\partial \lambda} = \frac{-hc}{\lambda^2 k_B T} = -\frac{1}{\lambda} \frac{hc}{\lambda k_B T} = -\frac{a}{\lambda}$$

Hence:

$$u(\lambda, T) = \underbrace{8\pi hc \left(\frac{k_B T}{hc}\right)^5}_{\text{constant } = C} a^5 \times \frac{1}{\exp(a) - 1}$$

$$u(\lambda, T) = \frac{Ca^5}{e^a - 1}$$

$$\begin{aligned} \frac{\partial u}{\partial \lambda} &= \frac{\partial a}{\partial \lambda} \frac{d}{da} \left[\frac{Ca^5}{e^a - 1}\right] \\ &= \frac{-a}{\lambda} \frac{d}{da} \left[\frac{Ca^5}{e^a - 1}\right] \end{aligned}$$

[1] See QM1

Using the quotient rule:

$$\frac{d}{da}\left[\frac{Ca^5}{e^a-1}\right]=\frac{5Ca^4(e^a-1)-Ca^5e^a}{(e^a-1)^2}$$

Therefore:

$$\frac{\partial u}{\partial \lambda}=\frac{-a}{\lambda}\left[\frac{5Ca^4(e^a-1)-Ca^5e^a}{(e^a-1)^2}\right]$$

Solving for the maxima:

$$\frac{\partial u}{\partial \lambda}=0$$

$$-a\left(5Ca^4(e^a-1)-Ca^5e^a\right)=0$$

As C is a non-zero constant:

$$-a\left(5a^4(e^a-1)-a^5e^a\right)=0$$

As $a\neq 0$:

$$5a^4(e^a-1)-a^5e^a=0$$

Dividing by a^4:

$$5(e^a-1)-ae^a=0$$

$$5e^a-5-ae^a=0$$
$$(5-a)e^a=5$$

Solving numerically:

$$a\approx 4.965$$
$$a=\frac{hc}{\lambda k_BT}$$
$$\lambda_{\max}T=\frac{hc}{k_Ba}$$

Finally:

$$\lambda_{\max}T\approx 2.898\times 10^{-3}\,\mathrm{m\,K}$$

Thu 05 Mar 2026 13:00

Lecture 14 - Statistical Physics I: Introduction

1 Stefan-Boltzmann Law

We can integrate across the whole spectrum to work out total energy emitted per unit time:

$$\dot{Q} = \epsilon \sigma A T^4$$

Where $\sigma = 5.67 \times 10^{-8} \text{Wm}^{-2}\text{K}^{-1}$ is the Stefan-Boltzmann constant, ϵ is the emissivity, A is the surface area of the radiating body and T is the temperature.

2 Kinetic Theory of Gases

We've now finished the module on temperature and move into statistical physics. We start wth a kinetic theory of the motion of gas molecules. Gas molecules move "randomly" (effectively, it's not computationally viable to determine what's actually going on), with vastly different velocities and positions.

We would love to describe a gas using the individual molecules that make it up, but that's really difficult to do for $\sim 10^{23}$ molecules so it's not viable and we have to use statistical physics to describe it.

We've previously touched upon the Maxwell-Boltzmann distribution to describe the velocities of particles as a statistical distribution wrt temperature:

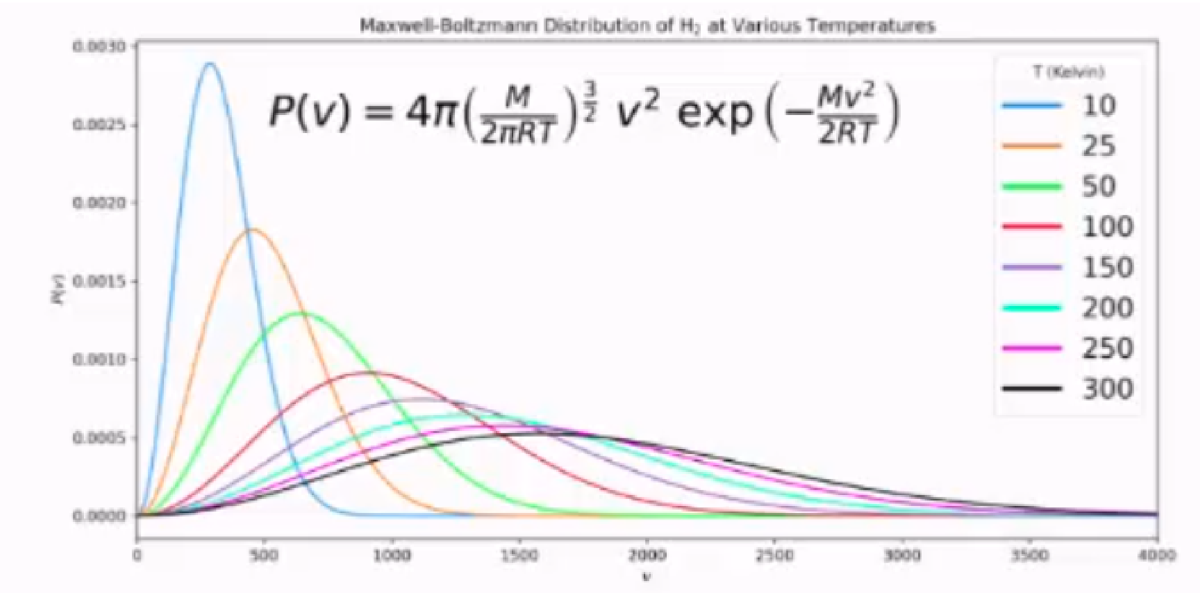


Figure 53.1

These statistical approximations improve as we increase the number of particles, and for very large numbers they become good approximations. We can in statistical physics have a probability > 1. This seems mathematically strange, but we interpret it to mean that (if the probability represents the probability of an event occurring in a measurement), the event will take place multiple times. We still cannot have any probabilities < 0.

The rest of the lecture then turned into a fun tangent on statistics which isn't examinable and needs no notes.

Tue 10 Mar 2026 12:00

Lecture 15 - Statistical Physics II: The Atmosphere

1 The Atmosphere

The very outer part of the atmosphere, from 90km to 350km is the ionosphere. This acts as a mirror which reflects EM radiation

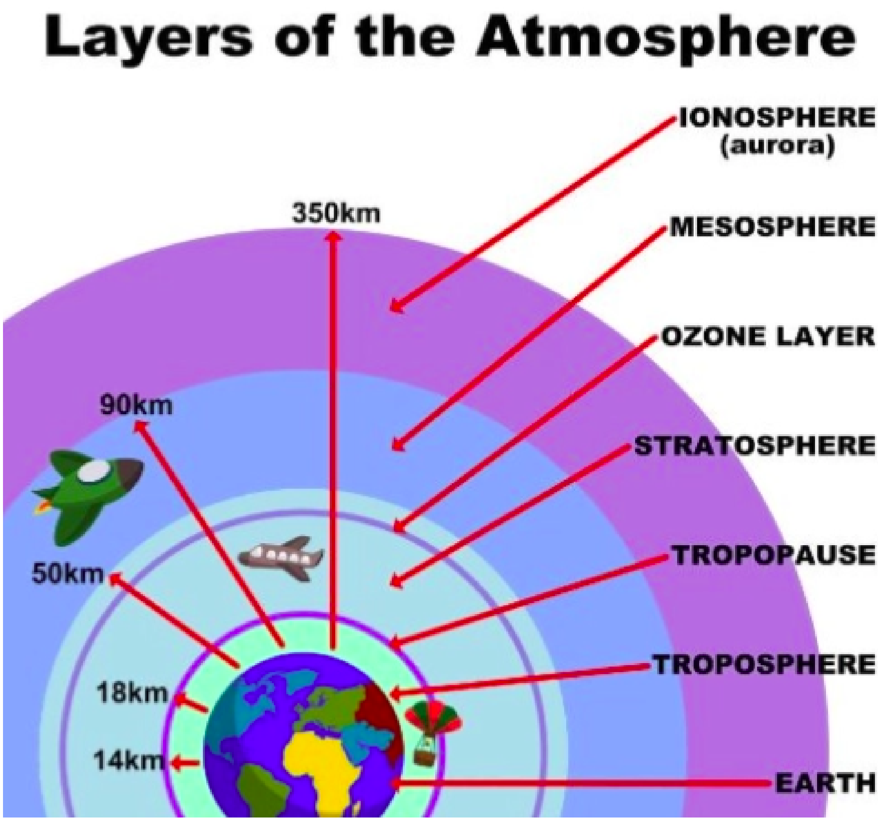


Figure 54.1

This is how we're able to do international radio communications, by bouncing radio waves off of this layer in order to reach continents where there is no direct line of sight.

1.1 Isothermal Model of the Atmosphere

Consider an observer sitting in the troposphere above the earth, looking at a small cuboidal slice of the troposphere.

We will assume that:

- The atmosphere is isothermal. In reality, increasing height reduces temperature by approx. 10K per km.
- The atmosphere is solely comprised of an ideal gas. At the low pressure scales relevant for the atmosphere, this is a reasonable assumption.

- The earth is flat[1]. Technically, assume that we we consider small enough heights and cross-sectional areas that we can disregard the curvature of the earth.
- The atmosphere is stationary and is therefore in mechanical equilibrium. This can be a good approximation depending on weather conditions.

The slab of atmosphere we're considering has area A, height dh and sits at height h above the surface of the earth. From the ideal gas state equation:

$$PV = \underbrace{nRT}_{\text{moles}} = \underbrace{Nk_BT}_{\text{particles}}$$

$$\implies P = \frac{N}{V}k_BT$$

$$\implies P = \rho_N k_B T$$

Where $\rho_N = N/V$ is number density, and is defined as the number of molecules per unit volume.

The slab of atmosphere has two forces acting upon it, a force due to pressure and a force due to gravity. As we assume that slice is in mechanical equilibrium, we can equate these forces.

The pressure is going to vary across the slab. Force due to pressure increases the further down into the atmosphere you go. This means the bottom face of the slab experiences a higher pressure than the top face. Since the slab is in mechanical equilibrium, the force due to gravity must balance the net upwards pressure force.

This gives us:

$$\text{force due to gravity} = \text{force due to pressure}$$

$$mg\rho_N V = AP(h) - AP(h + dh)$$

$$mg\rho_N V = -dP(h)A$$

$$mg\rho_N \times dh \times A = -d\rho(h)A$$

$$mg\rho_N \times dh = -d\rho(h)$$

$$\frac{d\rho(h)}{dh} - mg\rho(N)$$

$$\int d\rho(h) = -\int mg\rho_N dh$$

And considering number density in the density integral:

$$\int_{\rho(0)}^{\rho(h)} \frac{1}{\rho_N} d\rho(h) = -\int_0^h mgdh$$

And substituting $dp = d\rho_N k_B T$ to give a LHS integral entirely in number density:

$$\int_{\rho_N(0)}^{\rho_N(h)} \frac{k_BT}{\rho_N} d\rho_N = -mgh$$

$$\int_{\rho_N(0)}^{\rho_N(h)} \frac{k_BT}{\rho_N} d\rho_N = -\frac{mgh}{k_BT}$$

$$[\ln(\rho_N)]_{\rho_N(0)}^{\rho_N(h)} = -\frac{mgh}{k_BT}$$

$$\frac{\rho_N(h)}{\rho_N(0)} = \exp\left(-\frac{mgh}{k_BT}\right) \implies \rho_N(h) = \rho_N(0)\exp\left(-\frac{mgh}{k_BT}\right)$$

The contents of the exponential have to be unitless, and they are because they're both units of energy. The upper gives gravitational energy, the lower gives thermal energy and the contents of the exponential depends on the ratio between the two of them.

[1] Oh dear...

This gives the final result:

$$\boxed{\rho_N(h) = \rho_N(0)\exp\left(-\frac{mgh}{k_BT}\right)}$$

And as $P = \rho_n k_B T$:

$$P(h)k_BT = P(0)k_BT\exp\left(-\frac{mgh}{k_BT}\right)$$

$$\implies \boxed{P(h) = P(0)\exp\left(-\frac{mgh}{k_BT}\right)}$$

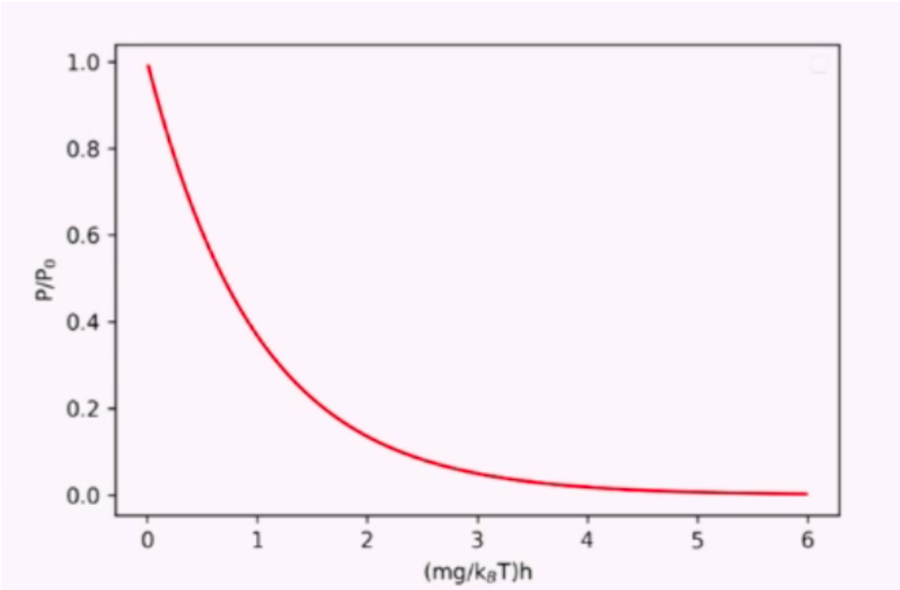


Figure 54.2

1.2 Scale Height

As mgh/k_BT must be dimensionless, mg/k_BT must have units of h. We call this the scale height, h_0. This is the height at which $P(h_0) = P(0)e^{-1}$, i.e. the point at which pressure drops off by a factor of $1/e$.

Particle	Mass	Scale height
O_2 gas	32 amu = 5.3137e-26 kg	~10 km
N_2 gas	28 amu = 4.6495e-26 kg	~10 km
Covid virus	~10^{-18} kg	~0.5 mm
Smallest known virus	~10^{-21} kg	~50 cm
Bacteria	~10^{-15} kg	~5 µm
Dr Stuart Pirrie	~10^{2} kg	~50 ym (10^{-24} m is 1 ym)

Figure 54.3: Example Scale Heights

This helps answer the question "if someone coughs on a desk, are other people in the room safe from Covid?". This implies yes, as the scale heigh is much larger than anyone is likely to be above a desk. This isn't true for all viruses however, i.e. the 50cm scale height for the smallest known virus.

Thu 12 Mar 2026 13:00

Lecture 16 - Statistical Physics III: Atmosphere and Boltzmann Factors

Recap from last lecture:

- We derived:
$$\rho_N(h) = \rho_N(0)\exp\left(-\frac{mgh}{k_BT}\right)$$
$$P(h) = P(0)\exp\left(-\frac{mgh}{k_BT}\right)$$
To give the number density and pressure as an exponential function of height in the atmosphere.
- Since mgh/k_BT is being passed into an exponential, it must be dimensionless. Hence k_BT/mg must have dimensions of h.
- This leads to the scale height h_0 where $P(h_0) = P(0)/e$

1 Atmosphere Continued

We have established that the number density (number of fluid molecules per unit volume), at height 0, $\rho_N(0)$ can be related to atmospheric pressure ($P(0)$) by:

$$\rho_N(0) = \frac{P(0)}{k_BT}$$

To determine the total number of gas molecules in the slab of atmosphere with cross-sectional area A, we can integrate number density across all h:

$$\begin{aligned}
N &= A\int_0^\infty \rho_N(h)\,dh \\
&= A\int_0^\infty \rho_N(0)\exp\left(-\frac{mgh}{k_BT}\right)dh \\
&= A\rho_N(0)\int_0^\infty \exp\left(-\frac{mgh}{k_BT}\right)dh \\
&= A\rho_N(0)\left[-\frac{k_BT}{mg}\exp\left(-\frac{mhg}{k_BT}\right)\right]_0^\infty \\
&= -A\rho_N(0)\frac{k_BT}{mg}\left[0-1\right] \\
N &= A\rho_N(0)\frac{k_BT}{mg} \\
\implies \rho_N(0) &= \frac{Nmg}{Ak_BT}
\end{aligned}$$

And using:

$$h_0 = \frac{k_BT}{mg}$$
$$\rho_N(0) = \frac{N}{Ah_0}$$

This has a physical interpretation. If you took all N molecules and distributed them in a volume Ah_0, i.e. our slab of cross-sectional area A with $h = h_0$, the average density of this would be equal to $\rho_N(0)$. This defines h_0 additionally as the average height of an equivalent uniform atmosphere with the same number of molecules.

We further define a new quantity:

$$\Pr(h) = \frac{A\rho_N(h)}{N}$$

Which gives the probability of (for a given specific particle) finding it at height h

1.1 Probability and Normalisation

We start from:

$$\Pr(h) = \frac{A\rho_N(h)}{N}$$
$$h_0 = \frac{k_BT}{mg}$$
$$\rho_N(h) = \rho_N(0)\exp(-mgh/k_BT)$$

And substitute our previous results for $\rho_N(0)$ and $\rho_N(h)$ in:

$$\Pr(h) = \frac{A}{N} \times \rho_N(0)\exp(-mgh/k_BT)$$
$$\Pr(h) = \frac{A}{N} \times \frac{N}{Ah_0}\exp(-mhg/k_BT)$$
$$\Pr(h) = \frac{mg}{k_BT}\exp(-mhg/k_BT)$$

As this is a probability, we need to check that it is normalised:

$$\int_0^\infty \Pr(h)\,dh$$
$$= \int_0^\infty \frac{A\rho_N(h)}{N}\,dh$$
$$\frac{A}{N}\int_0^\infty \rho_N(h)\,dh$$
$$\int_0^\infty \Pr(h)\,dh = \frac{1}{N}\left(A\int_0^\infty \rho_N(h)\,dh\right)$$

And as previously derived:

$$A\int_0^\infty \rho_N(h)\,dh = N$$

So:

$$\int_0^\infty \Pr(h)\,dh = \frac{1}{N}\left(A\int_0^\infty \rho_N(h)\,dh\right) = \frac{N}{N} = 1$$

So $\Pr(h)$ is normalised, as required!

2 Boltzmann Factors

This is on the exam. We have the quantity mgh/k_BT, which is dimensionless. What does this quantity actually physically mean? mgh is a term for the potential energy (gravitational here), and k_BT is a term for thermal energy.

Boltzmann factors describe the probabilities of measuring a given energy state in a system. In general form:

$$\Pr(E_i) \propto \exp\left(-\frac{E_i}{k_BT}\right)$$

This is true for all energy states, i.e. a mass on a spring oscillating in SHM has:

$$\Pr(\text{state where extension is } x) \propto \exp\left(-\frac{0.5kx^2}{k_BT}\right)$$

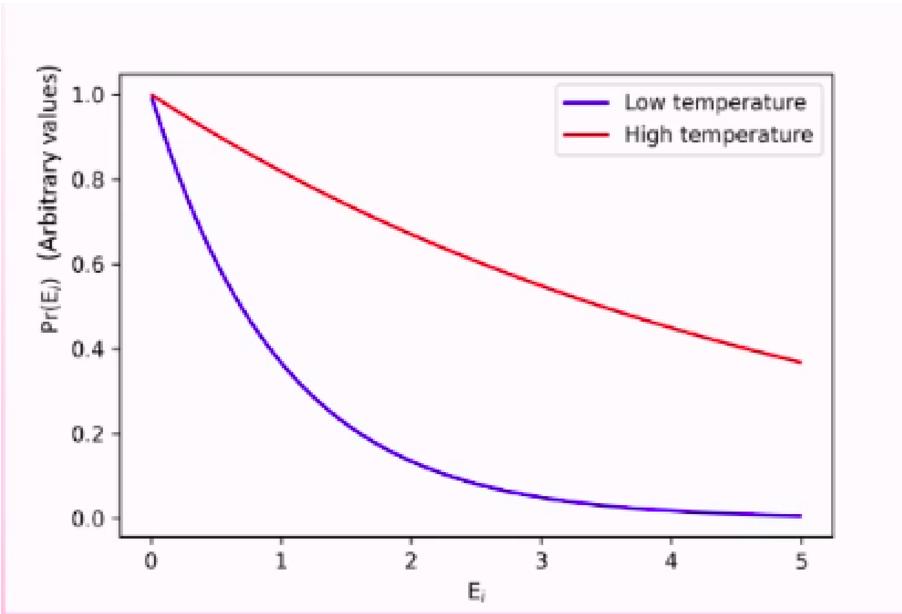


Figure 55.1

From this figure, we can see that higher temperatures have a higher probability of observing higher energy states, which makes sense. We can use this to describe continuous energy distributions, or discrete ones (i.e. energy levels in atoms).

The integral across all energy states must be equal to one.

3 Example: Simple Quantum System

Consider a simple atom with two energy levels E_0 and E_1.

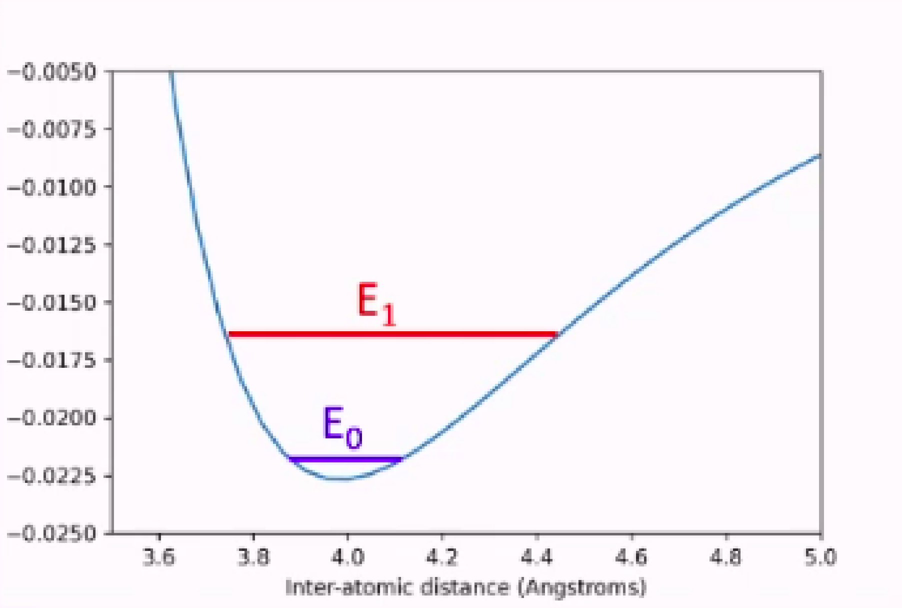


Figure 55.2

What is the probability of finding the atom at energy state E_0?

$$\Pr(E_0) \propto \exp\left(-\frac{E_0}{k_B T}\right)$$

Or adding a constant of proportionality:

$$\Pr(E_0) = C \exp\left(-\frac{E_0}{k_B T}\right)$$

And if both energy states have the same constant of proportionality, which is true here:

$$\Pr(E_1) = C \exp\left(-\frac{E_1}{k_B T}\right)$$

Since the probability must be normalised across all states:

$$\Pr(E_0) + \Pr(E_1) = 1$$

$$C \exp\left(-\frac{E_0}{k_B T}\right) + C \exp\left(-\frac{E_1}{k_B T}\right) = 1$$

$$1 = C\left(\exp\left(-\frac{E_0}{k_B T}\right) + \exp\left(-\frac{E_1}{k_B T}\right)\right)$$

$$C = \frac{1}{\exp(-E_0/k_B T) + \exp(-E_1/k_B T)}$$

Is the constant of proportionality here. Hence:

$$\Pr(E_1) = \frac{\exp(-E_1/k_B T)}{\exp(-E_0/k_B T) + \exp(-E_1/k_B T)}$$

$$\Pr(E_0) = \frac{\exp(-E_0/k_B T)}{\exp(-E_0/k_B T) + \exp(-E_1/k_B T)}$$

We can now do a bit of a maths trick:

$$\Pr(E_1) = \frac{\exp(-E_1/k_B T)}{\exp(-E_0/k_B T) + \exp(-E_1/k_B T)} \times \frac{e^{E_1/k_B T}}{e^{E_1/k_B T}}$$

$$\frac{1}{\exp\left(\frac{E_1 - E_0}{k_B T}\right) + 1}$$

Which lets us start considering limits. If we consider the limit as $T \to 0$:

$$\Pr(E_1) \to 0 \text{ and } \Pr(E_0) \to 1$$

The maths is unpleasant, but we can take it as given! This implies that in the low temperature limit, the atom always sits in the non-excited state.

What about the high temperature limit? We begin with a Taylor expansion:

$$\Pr(E_1) \approx \frac{1}{1 + (E_1 - E_0)/k_B T + 1}$$

Now as $T \to \infty$:

$$\Pr(E_1) \to \frac{1}{1 + 0 + 1} \implies \Pr(E_1) \to \frac{1}{2} \text{ and } \Pr(E_0) \to \frac{1}{2}$$

So in the high temperature limit, we end up with an even distribution between the two states.

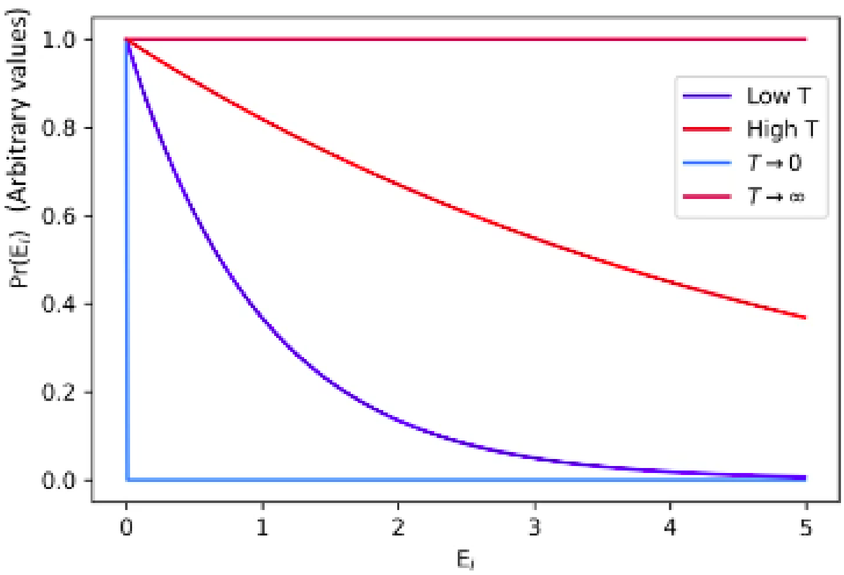


Figure 55.3

4 Negative Temperatures

Tue 17 Mar 2026 12:00

Lecture 17 - Statistical Physics IV: Boltzmann Factors and Degeneracies

Recap from last lecture:

- We can predict the microscopic behaviour of a system using a probability density function defined using macroscopic terms.
- We can predict the likelihood of a given energy state begin populated for a system (discrete or continuous) with the Boltzmann factor.
- For a two level system with $E_1 < E_0$ we have:

$$\Pr\left(E_i\right) = \frac{\exp\left(\frac{-E_i}{k_B T}\right)}{\exp\left(\frac{-E_0}{k_B T}\right) + \exp\left(\frac{-E_1}{k_B T}\right)}$$

- While not explicitly covered last lecture, this trivially generalises to a system with N levels:

$$\Pr\left(E_i\right) = \frac{\exp\left(\frac{-E_i}{k_B T}\right)}{\sum_{j=0}^{N-1} \exp\left(\frac{-E_j}{k_B T}\right)}$$

1 Degeneracies

In an atom, electrons can have the same energy in multiple different combinations (multiple ways of having the same result). For example, considering only one energy level, electrons can be spin up or spin down. This gives two possible options, hence a degeneracy of two.

With electrons in shells, the degeneracy gives the number of electrons which can fit in a shell. The Pauli exclusion principle says that no two atoms can occupy a shell if they share a quantum number, so the number of possible combinations of distinct states is equal to the number of possible electrons in the shell.

Given the degeneracy $g(E)$, we can modify the Boltzmann Factor accordingly:

$$\Pr(E_i) \propto g(E) \exp\left(\frac{-E_i}{k_B T}\right)$$

For $n = 1$, $g(E) = 2$, for $n = 2$, $g(E) = 4$, and so on.

2 Boltzmann Factors and Ideal Gases

For an ideal gas, we have some assumptions (in terms of energy) that have been discussed previously. The most important of these in terms of energy is:

- Internal energy all comes from the average kinetic energy of the molecules (temperature) and not from potential energy.
- The Boltzmann Factor for an ideal gas hence takes the form:

$$\Pr(E_i) \propto \exp\left(\frac{-E_K}{k_B T}\right)$$

Our other assumptions are:

1. The gas is in thermal equilibrium, hence $\langle E_K \rangle$ is constant.
2. No gravity (as molecules are extremely light).
3. Assume that the average separation of each molecule is much larger than the molecule size, hence collisions are infrequent.
4. There is a very large number of molecules $\sim 10^{23}$ to allow us to treat the gas statistically without high statistical uncertainties.
5. All molecules have the same mass and size. This is not a good assumption for gases made up of multiple component gases mixed.
6. We assume that the direction of molecular motion is effectively random, i.e. is unbiased to any particular direction.
7. Molecules can collide, but do so elastically (and as previously mentioned, infrequently).
8. Molecules are not relativistic, so $E_K = 0.5mv^2$ holds.

3 Model of Ideal Gas

We can split the velocity of a gas into three cartesian coordinates by adding in quadrature:

$$v = \sqrt{v_x^2 + v_y^2 + v_z^2}$$

By assumption 6, we assume that all of these components are effectively equal, so we can consider the x-direction alone initially (and extrapolate out).

We want to consider the probability of having a velocity somewhere between v_x and $v_x + dx$.

$$\Pr(v_x \to v_x + dx) = \int_{v_x}^{v_x+dx} \Pr(v_x)\, dv_x$$

$$= \int_{v_x}^{v_x+dx} C \exp\left(-\frac{0.5mv_x^2}{k_B T}\right) dv_x$$

And using the standard integral: $\int_{-\infty}^{\infty} e^{-ax^2}\, dx = \sqrt{\pi/a}$, we have:

$$\int_{-\infty}^{\infty} C \exp\left(-\frac{mv_x^2}{2k_B T}\right) dv_x = 1$$

$$\implies C\sqrt{\frac{\pi}{a}} = 1$$

And as $a = m/2k_B T$:

$$C\sqrt{\frac{2\pi k_B T}{m}} = 1$$

$$\implies C = \sqrt{\frac{m}{2\pi k_B T}}$$

And as the Boltzmann factor for this x-directional motion is $\exp\left(mv_x^2/2k_B T\right)$:

$$\Pr(v_x) = C \exp\left(\frac{mv_x^2}{2k_B T}\right)$$

We finally have:

$$\Pr(v_x) = \sqrt{\frac{m}{2\pi k_B T}} \exp\left(-\frac{0.5mv_x^2}{k_B T}\right)$$

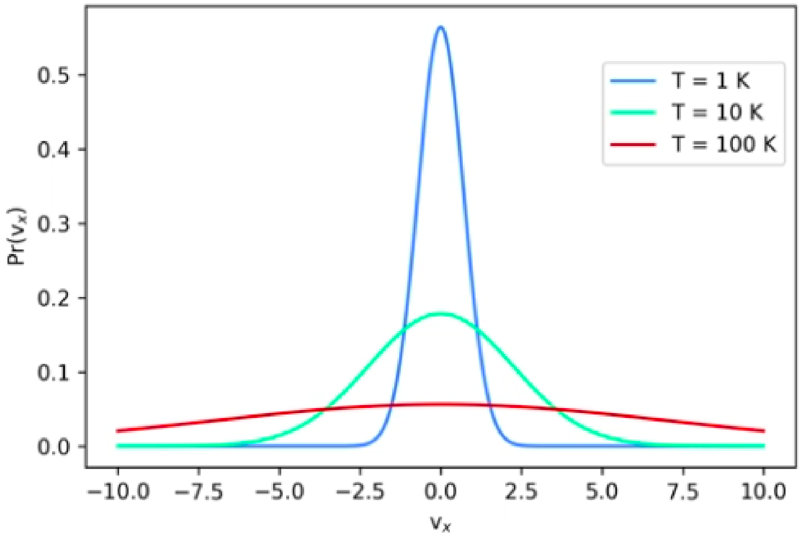


Figure 56.1

As we're experiencing random gas motion, the distribution is centred on 0, as velocity can be positive or negative.

As we increase the temperature, we increase the chance of occupying higher energy states (i.e. higher v_x).

This distribution has no degeneracies, as there's only one way for a particle to have a specific v_x, which is by travelling at that value. It either has a specified velocity, or it simply doesn't.

3.1 3D Model

Now we have a single component:

$$\Pr(v_x) = \sqrt{\frac{m}{2\pi k_B T}} \exp\left(-\frac{0.5 m v_x^2}{k_B T}\right)$$

We can extend this to 3D:

$$\Pr(v) = \Pr(v_x)\Pr(v_y)\Pr(v_z)$$

$$\Pr(v) = \left(\frac{m}{2\pi k_B T}\right)^{(3/2)} \exp\left(-\frac{0.5 m(v_x^2 + v_y^2 + v_z^2)}{k_B T}\right)$$

3.2 Density of States

While we had no degeneracies in each component direction, we do have degeneracies for v itself. For example, $v^2 = 100$ could mean that x, y are zero and $z = 10$, or any other number of combinations.

We therefore want to define the degeneracies as a function of v. As a model, consider a grid of infinitesimally small dots in 2D (but for the actual application we would want a 3D model).

Our final velocity in 2D is $v^2 = v_x^2 + v_y^2$, so if each axis represents possible values for v_x and v_y, v is given as the circumference of a circle with radius v. The density of states $g(v)dv$ is given as the ring formed between v and $v + dv$.

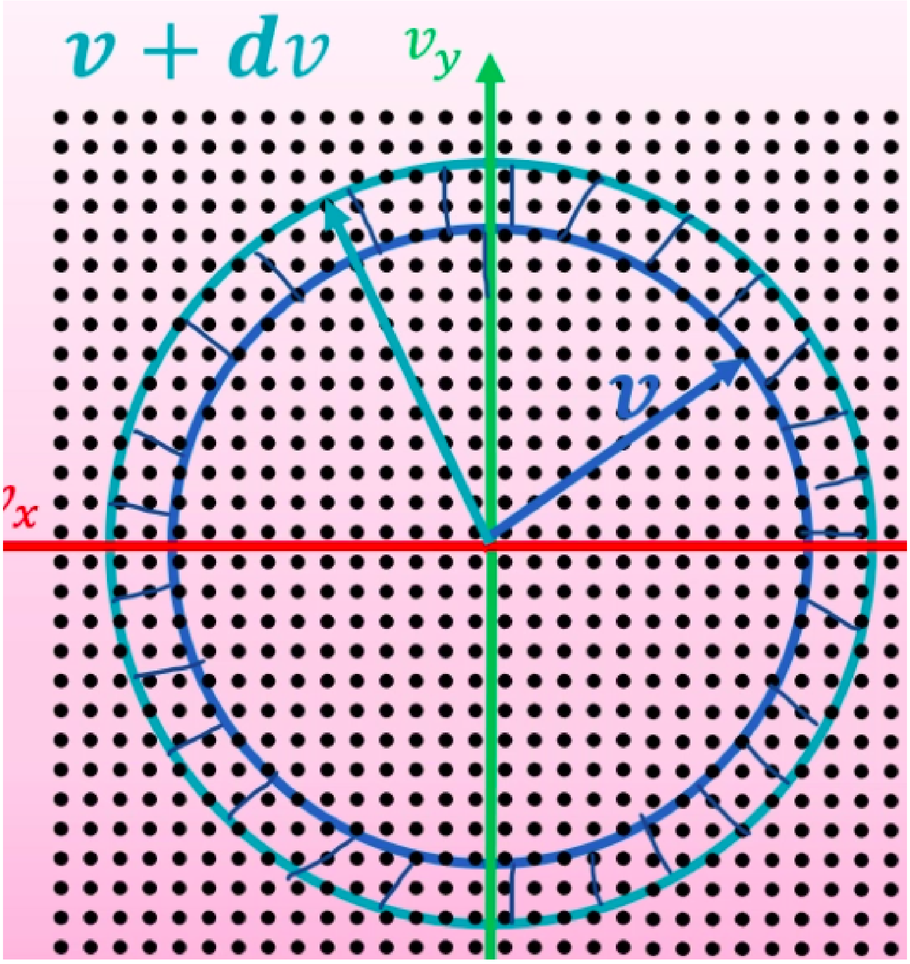


Figure 56.2

In 2D, the degeneracy is therefore given by:

$$g(v)dv = 2\pi v dv$$

In 3D, rather than a ring this would be a spherical shell surface:

$$g(v)dv = 4\pi v^2 dv$$

Applying this to the Maxwell-Boltzmann distribution, we obtain:

$$\Pr(v) = 4\pi \left(\frac{m}{2\pi k_B T}\right)^{3/2} v^2 \exp\left(-\frac{m(v_x^2 + v_y^2 + v_z^2)}{2k_B T}\right) \qquad \text{where: } v \neq 0$$

And as $v^2 = v_x^2 + v_y^2 + v_z^2$:

$$\Pr(v) = 4\pi \left(\frac{m}{2\pi k_B T}\right)^{3/2} v^2 \exp\left(-\frac{mv^2}{2k_B T}\right) \qquad \text{where: } v \neq 0$$

This finally gives us the Maxwell-Boltzmann distribution we encountered before (although we have derived it for molecules vs moles):

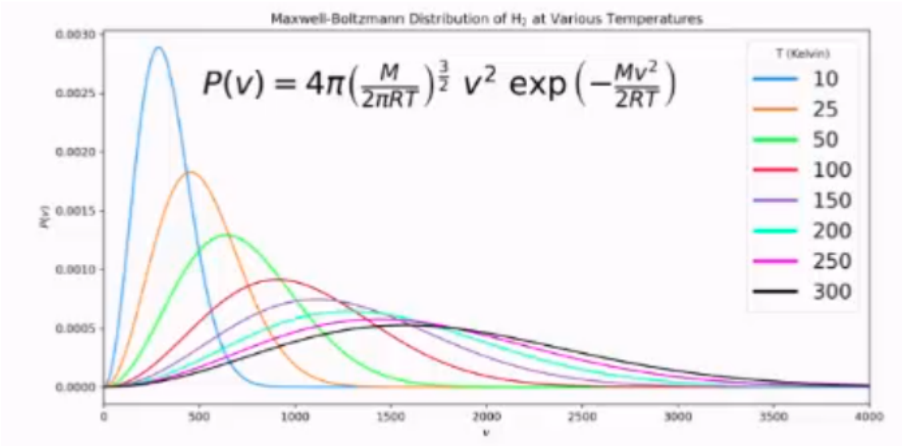

Maxwell-Boltzmann Distribution of H_2 at Various Temperatures
$P(v) = 4\pi\left(\frac{M}{2\pi RT}\right)^{\frac{3}{2}} v^2 \exp\left(-\frac{Mv^2}{2RT}\right)$
T (Kelvin)
10
25
50
100
150
200
250
300
P(v)
v
0.0030
0.0025
0.0020
0.0015
0.0010
0.0005
0.0000
0
500
1000
1500
2000
2500
3000
3500
4000

Figure 56.3

Thu 19 Mar 2026 13:00

Lecture 18 - Statistical Physics V

Recap from last lecture:

Tue 24 Mar 2026 12:00

Lecture 19 - Statistical Physics VI: Something or Other

Thu 26 Mar 2026 13:00

Lecture 20 - Fluid Dynamics

1 Fluid Dynamics

Say we have no applied force, and the pressure at the liquid surface (for this weirdly shaped container) is P_0. What is the pressure P at some depth h?

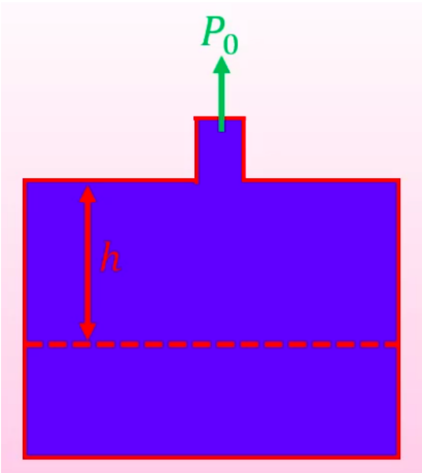


Figure 59.1

$$P = P_0 + \rho g h$$

If, and only if, the fluid has constant density ρ. If ρ varied as a function of height, we'd need to start integrating.

What if we applied a force to the liquid at the top of the spout?

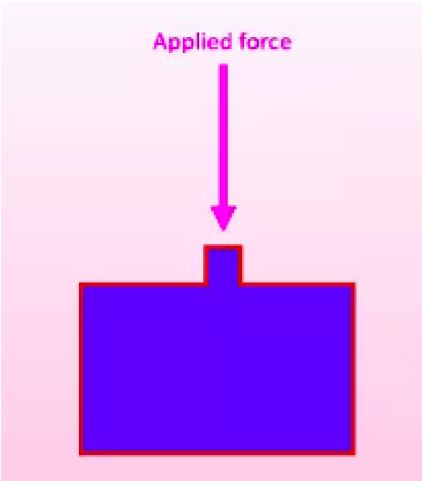


Figure 59.2

Pressure in the liquid will increase, so the applied force to the walls of the container will increase. Pascal's law says that this increase in pressure is applied to every part of the fluid (and the walls of the enclosure) equally and undiminished.

1.1 Example: Hydraulic Presses

If we apply a force F_1 to a liquid in some very small cross-sectional area a_1, then the pressure this results in must be applied to all points in the liquid:

$$P = \frac{F_1}{a_1}$$

Therefore, on the "output" of the press:

$$P = \frac{F_2}{a_2}$$

If a_2 is much larger than a_1, the output force F_2 must be proportionally larger than the input force F_1 such that:

$$F_2 = \frac{F_1 a_2}{a_1}$$

This effectively lets us magnify forces.

1.2 Fluid in a Pipe

In a steady state (a state with does not change with respect to time) the amount of water that passes a given point must also be constant with respect to time.

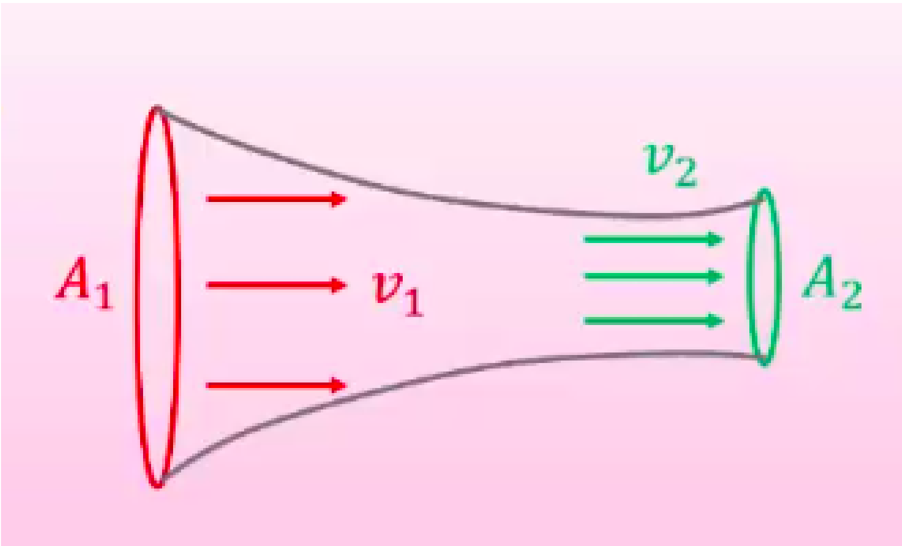


Figure 59.3

For this to be a steady state, the amount of liquid passing both points must be the same (otherwise liquid would get stuck and build up). Therefore, we must see:

$$A_1 v_1 = A_2 v_2$$

For $v_1 \neq v_2$, a force must be applied to the liquid to change velocity. This force comes from the neighbouring fluid molecules.

Previously, we saw:

$$P = P_0 + \rho g h$$

To account for the liquid now being non-stationary and having a velocity, we need to add a term accounting for pressure from kinetic energy:

$$P + \rho g h + \frac{1}{2}\rho v^2$$

Here, P is called static pressure, ρgh is called hydrostatic pressure and $\frac{1}{2}\rho v^2$ is called dynamic pressure. This full equation becomes useful in a context like this:

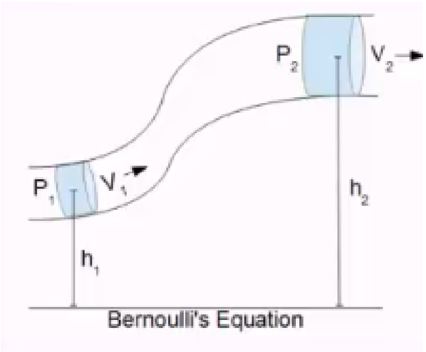


Figure 59.4

End of Module